



Global Market Outlook

For Solar Power 2024-2028

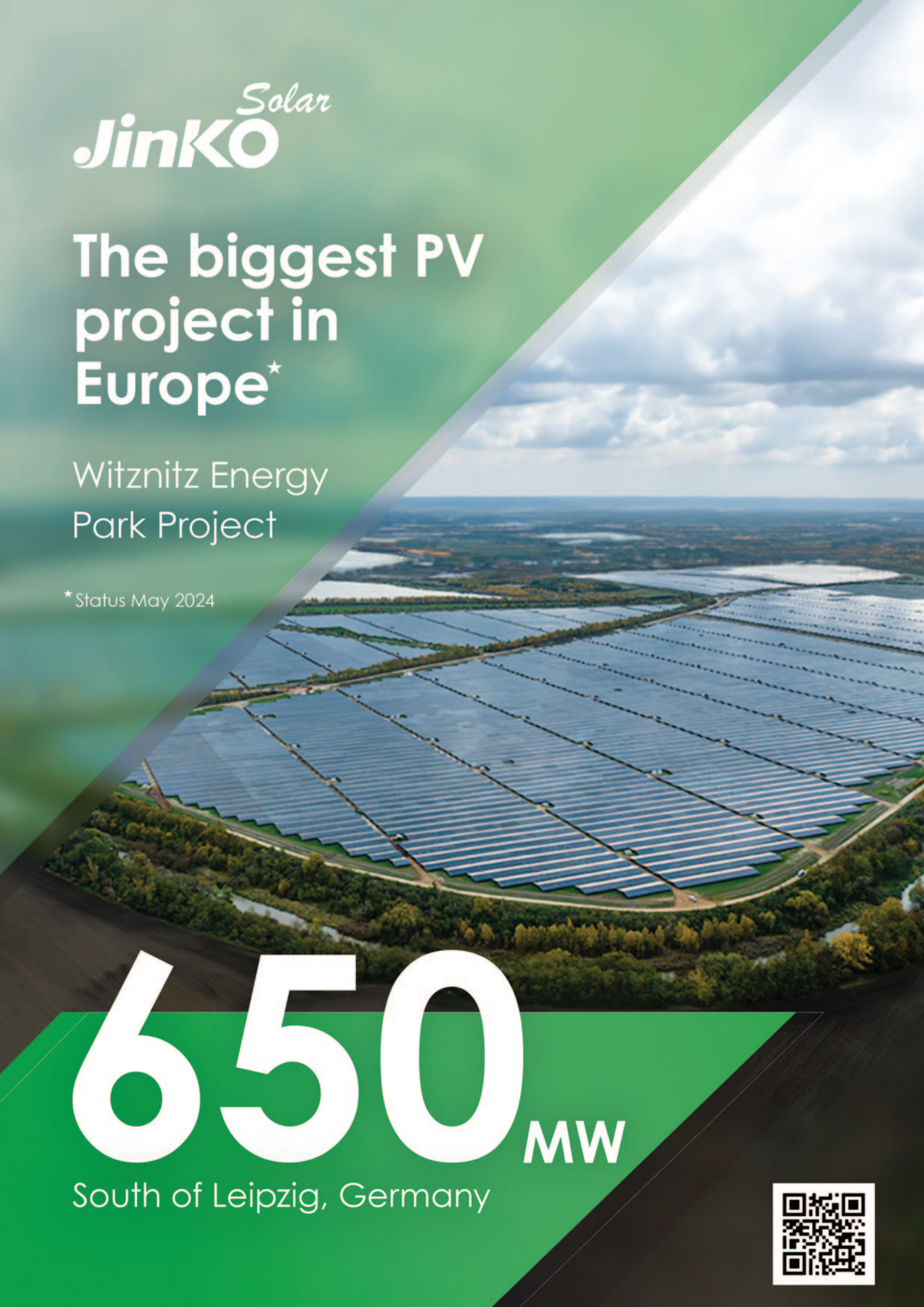
FOCUS ON CHINA



The biggest PV project in Europe[★]

Witznitz Energy Park Project

[★] Status May 2024

An aerial photograph of the Witznitz Energy Park, showing a vast field of solar panels arranged in neat rows, surrounded by greenery and a body of water in the background.

650 MW

South of Leipzig, Germany



Foreword

Welcome to the Global Market Outlook for Solar Power 2024–2028.

For an established sector like solar, approaching double growth in one year was simply not part of any analyst's script. But it happened in 2023. The world deployed 447 GW of new solar PV capacity last year; an incredible 87% more than 2022 and achieving a growth rate we haven't seen since 2010, when the global solar market was only 4% of today's size.

In last year's Global Market Outlook, we actually did anticipate growth for several reasons, but not as high as what was achieved. At that time, a very strong growth in global manufacturing capacities was already visible, making products available again after the pressures of the pandemic and energy crisis. This, of course, also brought overcapacity – effectively doubling supply against demand. Resulting price drops triggered by hefty competition from all these new capacities was expected as well (though not at the severity the industry eventually experienced). In fact, our growth forecast for all global markets combined, other than China, was nearly spot on – we deviated only by 3% absolute.

The accuracy and value of the annual Global Market Outlook for Solar Power would not be possible without the industry stakeholders and all national and regional solar industry associations that provide their input and expertise each year. We would like to extend our gratitude to those experts and their organisations in supporting our mapping and modelling work, and particularly those from the 31 annual GW-scale solar markets that wrote in-depth features of their home markets for this year's edition (see p. 81).

However, the market that is continually difficult to anticipate remains to be China, the world's largest solar market by far, which boomed by 167% in a single year. This country alone was responsible for 57% of global installed capacity in 2023, adding a quarter of a TW, basically the same amount that the entire world had deployed in the previous year. The massive price collapses of around 50% for modules in 2023 alone triggered unforeseen growth, driven by price elasticity on the demand side. To better understand solar developments in China, this year's report covers the country in a dedicated chapter, provided by the Global Solar Council and its partner the Chinese Renewable Energy Industries Association (CREIA) (see p. 61).

So, what's in store for solar this year and beyond? For the supply side, the PV manufacturing industry will continue to suffer from record-low prices and overcapacities – how long will depend on the consolidation speed in the sector. The demand side will further benefit from low product prices that will stay a main driver in a difficult macro-economic environment. We will see the climate emergency continuing to capture governments' attention and energy security remaining the new compelling argument to invest in solar power in a quickly fragmenting world order. Further driven by cost improvements and the numerous benefits the technology provides, we have increased our most likely outlook to 544 GW in 2024, based on a 22% YoY growth, and low two-digit improvements to an annual market of 876 GW in 2028.

We have been hearing doubts from many analysts if solar can grow much beyond today's annual installation levels. While obviously there will be an eventual limit, we don't foresee this any time soon. Who would have thought that only a year ago that China would install over 250 GW last year? Nobody! We should also remember, solar's share in global electricity generation was only at 5.5% in 2023, China's installed watt per capita was 460 W/c, and MEA on average only 27 W/c, compared to 1,359 W/c in Australia – there's clearly quite some room to grow. For the first time in our five-year forecasting period, our High Scenario is touching an annual TW deployment rate in 2028.

While the PV industry is ready for more, it now hinges on policymakers creating the right legal frameworks to enable solar in unfolding its full potential. That's why, for the first time, this edition contains policy recommendations from solar's global industry organisation, the Global Solar Council (see p. 57). This includes asks for setting ambitious national targets, and immediate efforts to unleash the flexibility revolution by scaling up investments in battery storage, grids, and electrification. We also urgently need diversification – on both the supply and demand side, and plenty of financing.

To the sector's gratification, the recent IEA World Energy Investment 2024 report revealed that investment in solar surpassed all other generation technologies combined. However, there remains severe disparity in global renewable investment flows. Two thirds of the world's population, living in emerging and developing economies outside China, only received 15% of global investment. Last year's COP saw world leaders committing to triple world renewables by 2030 to achieve 11 TW, and this year's COP29 must accordingly set the finance plan for the transition for the countries most in need. The climate crisis, and breaching 1.5°C warming, is barreling toward us like a freight train. We hope this year's Global Market Outlook supports the solar sector in building a new track toward more sustainable and secure prosperity for all.

Enjoy reading our Global Market Outlook.



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Methodology: SolarPower Europe's five-year forecast consists of Low, Medium and High Scenarios. The Medium scenario anticipates the most likely development given the current state of play of the market. The Low Scenario forecast is based on the assumption that policymakers halt solar support and other issues arise, including interest rate hikes and severe financial crisis situations. Conversely, the High Scenario forecasts the best optimal case in which policy support, financial conditions and other factors are enhanced.

Segmentation is based on the following system size: Residential (<10 kW); Commercial (<250 kW); Industrial (<1000 kW); Utility-scale (>1000 kW, ground-mounted). SolarPower Europe's methodology includes only grid-connected systems. Installed capacity is always expressed in DC, unless otherwise stated.

All figures are based on SolarPower Europe's best knowledge at the time of publication.

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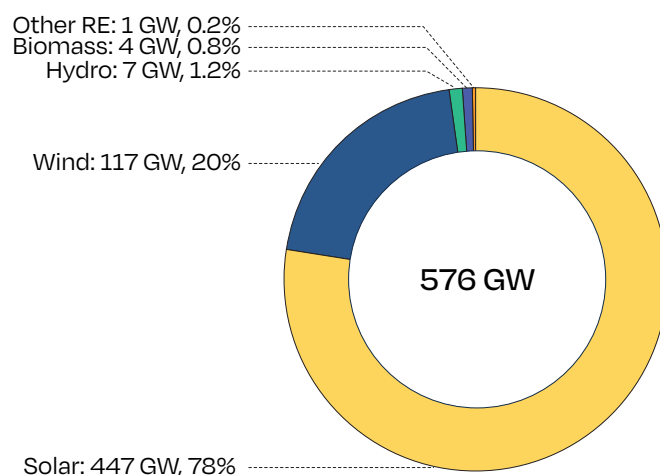
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In 2023, the world grid-connected 447 GW of new solar capacity, which, once again, dominated the newly added global power generation capacity. Solar PV accounted for 78% out of 576 GW of new renewable capacity added last year. Solar's share of new renewable capacity increased considerably, up 12

percentage points from the 66% contributed in 2022 and 22 percentage points more than the 56% in 2021. This new record confirms solar's role as an established and still rising leader of the global energy transition, installing over three times more capacity than all other renewable technologies combined.

FIGURE 1 NET RENEWABLE POWER GENERATING CAPACITY INSTALLED IN 2023



SOURCES: GWEC (2024), IRENA (2024), SolarPower Europe.

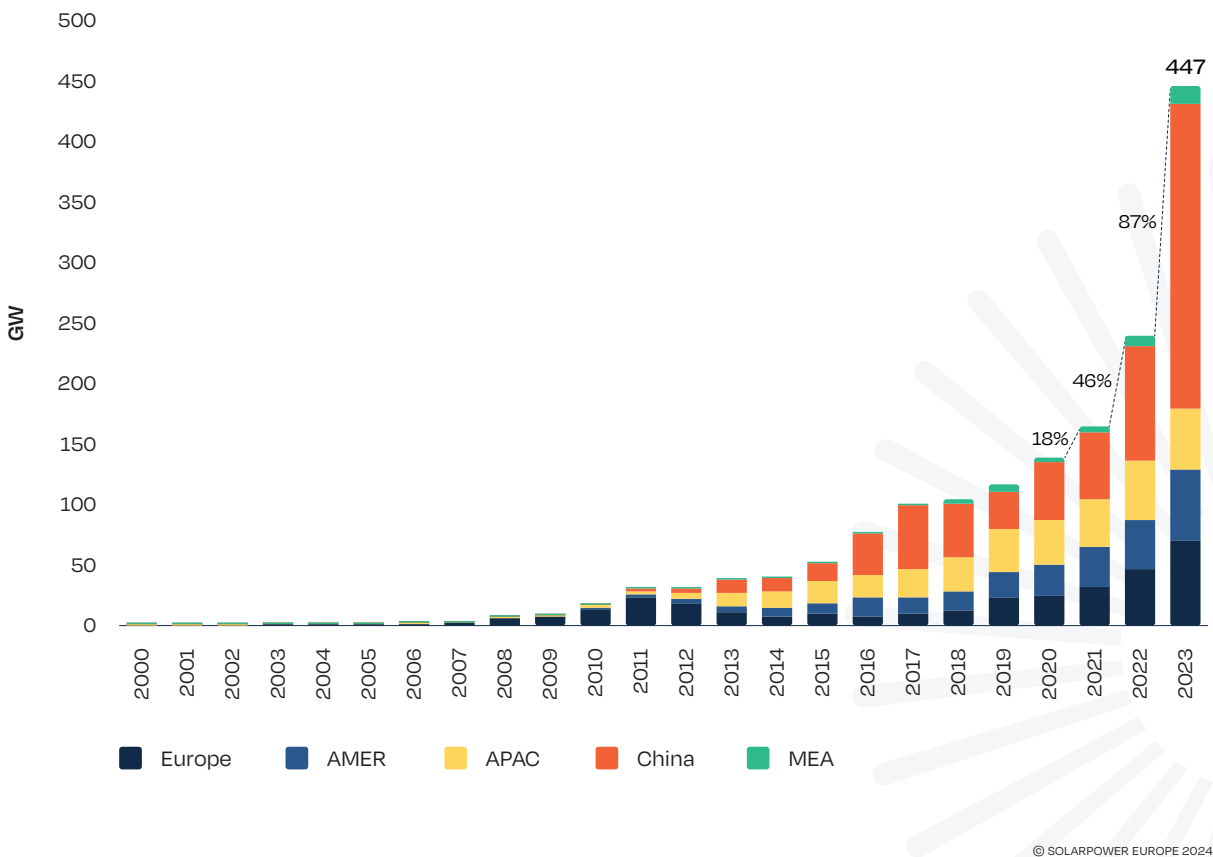
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The 447 GW of new solar capacity broke any previous record and far exceeded any solar analyst's expectations, marking an extraordinary 87% growth rate (see Fig. 2). This compares to 2022's addition of 239 GW and 46% year-on-year growth.

The surge in installed solar capacity can be attributed to several key factors. Firstly, a significant increase in global PV manufacturing capacities greatly improved the availability of solar modules following the supply chain issues experienced during the pandemic. This capacity expansion resulted in overcapacities and severe price drops for system components over the course of 2023, with modules' values falling by around 50%.

In addition, the effects of the 2022 global energy crisis continued to stimulate demand across various regions. Governments, individuals, and businesses turned to solar power as a reliable and cost-effective solution to high energy prices, while accelerated electrification of the heat and transport sectors also gained much more attention. Many orders from 2021 and 2022 were finally installed in 2023, thanks to the increased availability of products and installers at more affordable prices.

FIGURE 2 ANNUAL SOLAR PV INSTALLED CAPACITY 2000-2023

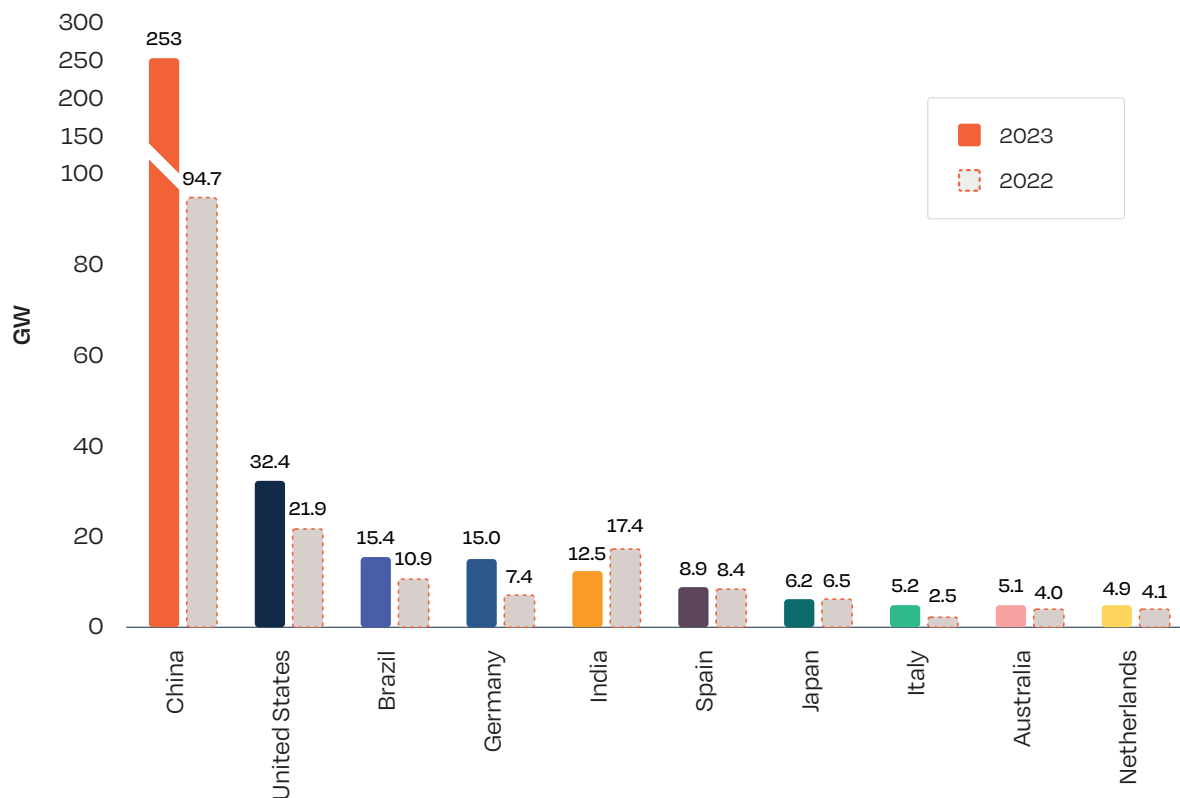


Executive summary / continued

While the scale of 2023's growth was unprecedented, it needs to be highlighted that most of this global market expansion was driven by and in China. The world's largest solar product supplier and market for years, China broke any previous record by adding an incredible 253 GW of new solar PV capacity in 2023, marking a 167% year-on-year growth rate. Meanwhile, the rest of the world installed 'only' 194 GW of new solar PV, reflecting a 35% increase from the 144 GW deployed in 2022 (Fig. 3). In other words, without China's strong solar investments in deployment, the solar sector's growth would have been much more modest.

Eight of the top 10 largest solar markets experienced growth in 2023, with only India and Japan experiencing a downturn in their market. The United States got back on track after a turbulent solar business year in 2022 and recorded a 48% growth rate in 2023, with 32.4 GW of solar PV grid-connected. Germany installed 15 GW in 2023, marking a new record for any European country. The previous record was held by Italy with 9.3 GW installed long back in 2011, during the first boom phase of the European solar story. Comprised together, the top 10 markets represented 80% of the global solar PV market in 2023.

FIGURE 3 TOP 10 SOLAR PV MARKETS, 2022-2023



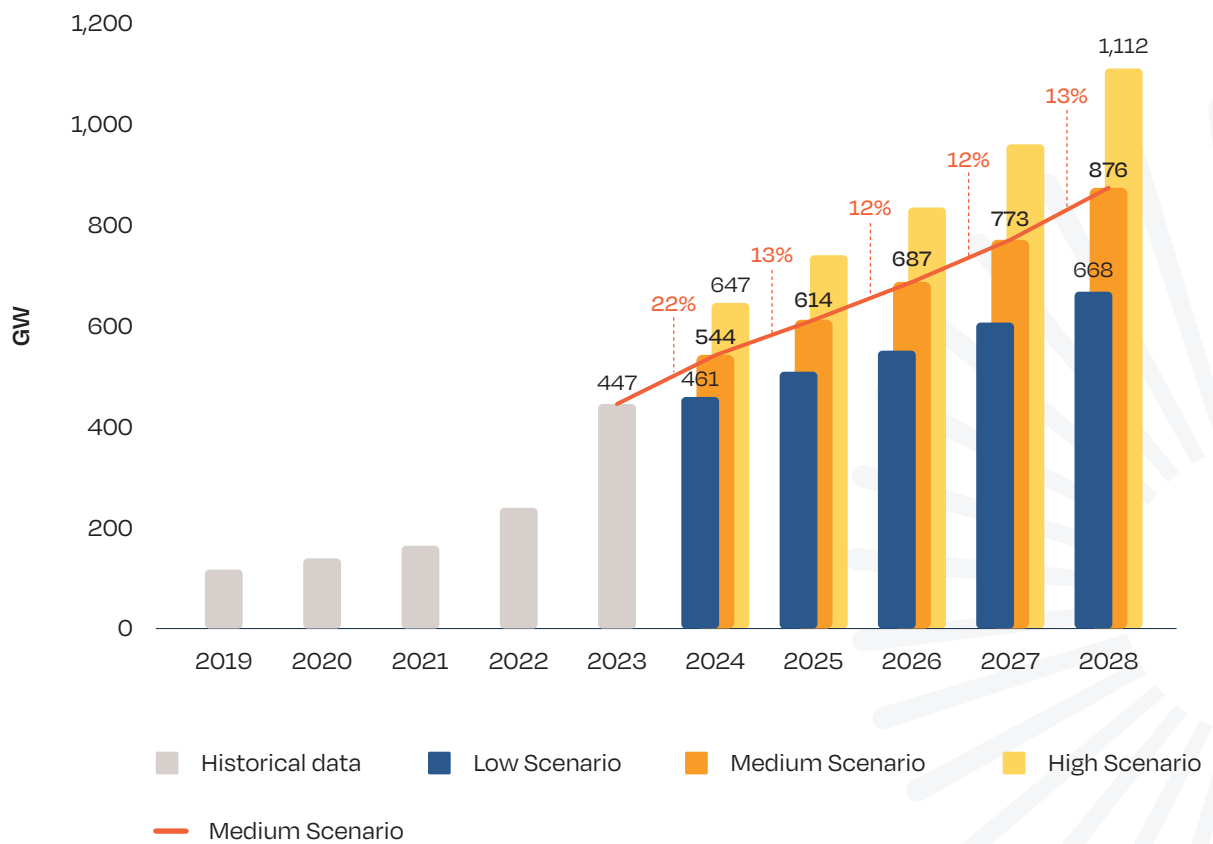
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We anticipate substantial demand growth for solar PV power in the coming years, driven by further cost improvements, product availability, and the numerous benefits the technology provides. Climate emergency will continue to capture governments' attention, while energy security will remain a compelling argument to invest in solar power in a quickly fragmenting world order.

For 2024, our Medium Scenario forecasts a global growth rate of 22% to a market size of 544 GW, approximately 100 GW more than in 2023 (see Fig. 4). As we've seen in recent years, current global market forecasting will depend largely on getting China's solar deployment right due to its outstanding

position in the global PV landscape. With analysts severely under-estimating China's market expansion in the past, and several reasons speaking for further growth, we estimate deployments of 299 GW, a year-on-year uptick of 18%. Modelling variations depending on policy and economic developments, our more optimistic High Scenario projects 45% annual growth to 647 GW, while our Low Scenario anticipates only 3% growth to 461 GW in 2024. In the mid-run, annual global market installations reach 614 GW in 2025 under the Medium Scenario, a 13% increase from 544 GW in 2024, followed by a 12% increase to 687 GW in 2026, 12% to 773 GW in 2027, and 13% to 876 GW in 2028.

FIGURE 4 WORLD ANNUAL SOLAR PV MARKET SCENARIOS 2024-2028



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Executive summary

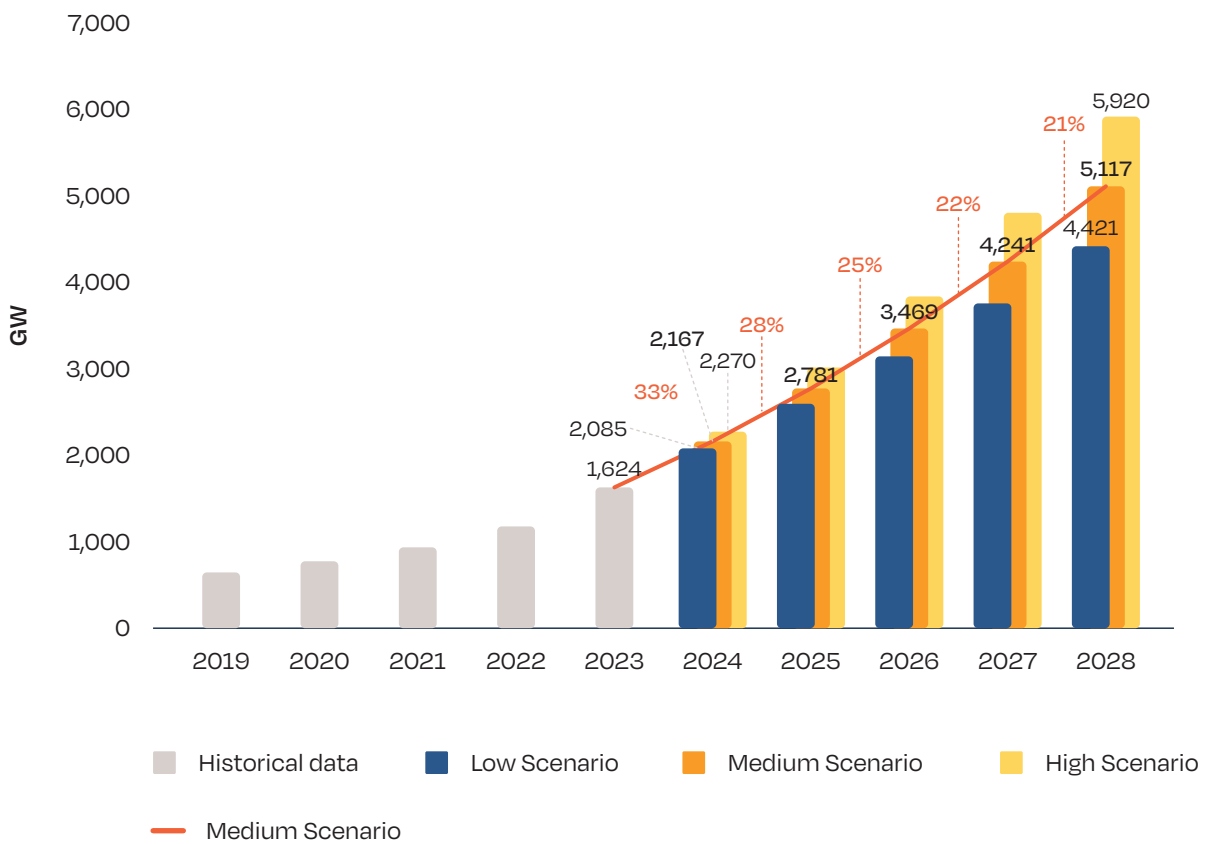
After surpassing the 1.6 TW level in 2023, the global solar power fleet is on track to exceed 2 TW by 2024 (see Fig. 5). Our Medium Scenario estimates 2.2 TW in 2024, 2.8 TW in 2025, 3.5 TW in 2026, 4.2 TW in 2027, and 5.1 TW in 2028 – a forecast that is significantly higher than last year’s Medium Scenario.

Looking beyond our traditional 5-year forecast horizon – to 2030 as a significant milestone year on the path to global net zero – a simple extrapolation of our Medium Scenario 2024–2028 projects 7.4 TW of solar operating capacity worldwide, which is above any of the most recent 2030 estimates of leading solar analysts.

The strong PV expansion in 2023 brought a new record number of sizeable solar markets. The number of GW-scale solar markets – countries installing at least 1 GW – increased from 28 in 2022 to 31 in 2023, 14 of which are located in the European Union. We estimate this group to further augment to 37 GW-scale markets in 2024 and 50 in 2025.

With China’s dominant solar role further increasing to unknown heights, we have picked the country as this year’s focus theme. The chapter was contributed by the Global Solar Council (GSC), which also provided another new addition – Policy Recommendations addressing the key challenges for the global solar sector.

FIGURE 5 WORLD CUMULATIVE SOLAR PV MARKET SCENARIOS 2024-2028



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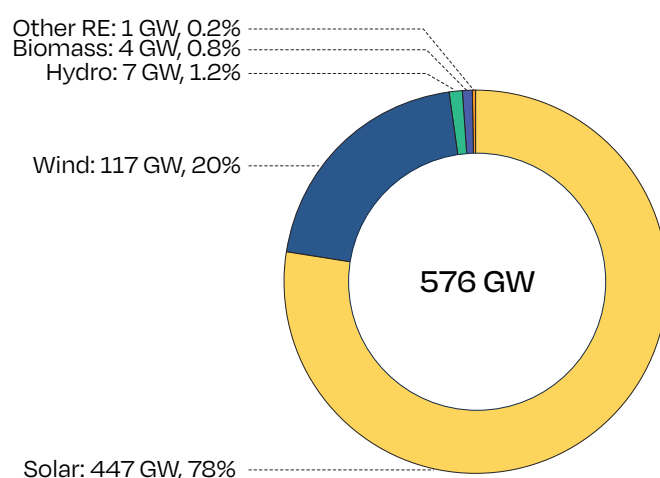


Solar power dominated once again the newly added global power generation capacity in 2023, cementing its status as the fastest growing renewable electricity source for the 19th year in a row. Out of 576 GW of new renewable (RES) capacity added last year, solar PV accounted for 78%, connecting 447 GW to the grid (see Fig. 6). Solar's share of new renewable capacity increased considerably, up 12 percentage points from the 66% contributed in 2022 and 22 percentage points more than the 56% in 2021. This new record confirms solar's role as an established and still rising

leader of the global energy transition, installing over three times more capacity than all other renewable technologies combined.

While the overall trend for renewable capacity additions has been consistently upward in recent years, pushed by the global energy crisis, policy support, and technology and cost advancements, 2023 brought record-breaking growth primarily through developments in the solar and wind segments. Total renewable capacity additions are 214

FIGURE 6 NET RENEWABLE POWER GENERATING CAPACITY INSTALLED IN 2023



SOURCES: GWEC (2024), IRENA (2024), SolarPower Europe.

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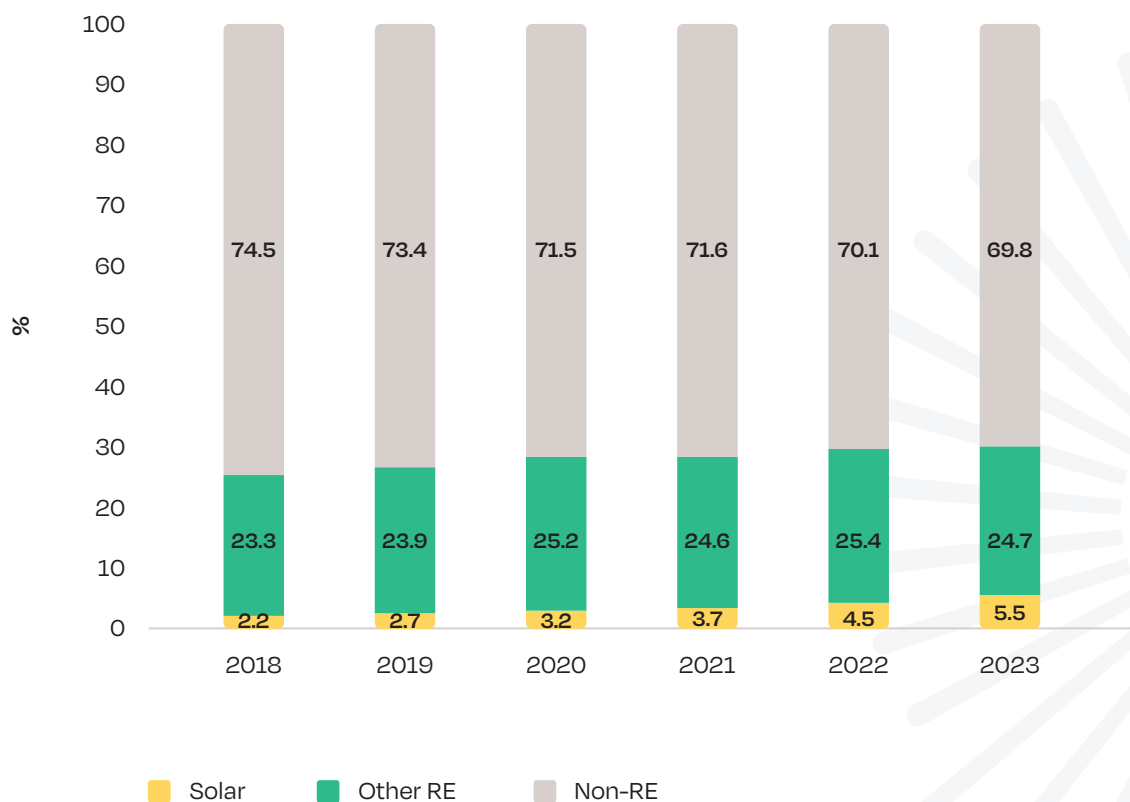
GW higher than last year, a strong 59% increase from 362 GW in 2022. Next to solar PV's extraordinary surge in 2023, the renewables drive was supported by a renewed growth phase in the wind sector, jumping by 50% to a record 117 GW in 2023, up from 78 GW in 2022. The exceptional renewables growth is reflected by reaching a notable milestone: 2023 marks the year renewables climbed to 30%, or nearly one third, of global electricity generation.

These encouraging developments must be kept in perspective. Solar power still contributes a minor share of total electricity demand, accounting for only 5.5% of global power production in 2023, but increasing from 4.5% in the previous year (see Fig. 7). Though still relatively small, we observe a clear growth acceleration – solar's share increased by 1 percentage

point in 2023, compared to 0.8 percentage points in 2022 and 0.5 between 2020 and 2021.

Comparatively, all other renewable sources collectively provided 24.7% of the world's power output in 2023 – and this represents a 3% decrease from the 25.4% share in 2022. The reason for the renewables' reduced output can be found in climate change – hot temperatures and droughts troubled operations of hydropower plants, which saw their output decline. Non-renewable sources maintained their dominant position with a 69.8% share, going under the 70% mark for the first time, as their share decreased by 0.3 percentage points in 2023. This means growth in solar and wind was so strong that it could compensate for the decline in hydropower and force non-renewable sources' output to a new record low.

FIGURE 7 SOLAR AND RENEWABLE POWER AS A SHARE OF GLOBAL POWER 2018-2023



SOURCE: Ember (2024).

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1 Global solar market / continued

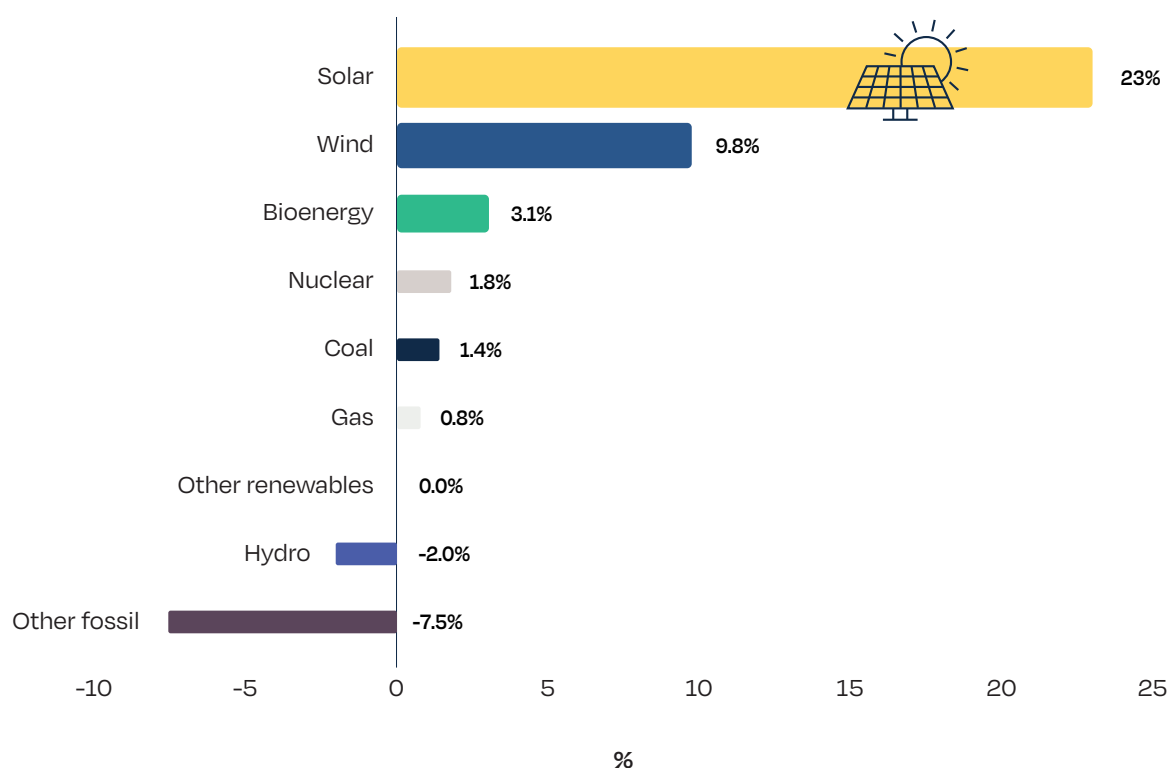
Solar's track record demonstrates the considerable scale of its market potential, and as it maintains strong cost leadership we can expect it to continue capturing a larger share than any other power generation technology. Spearheading the global transition to renewables, solar generated 1,631 TWh in 2023, up 23% from the 1,324 TWh generated in 2022 (Fig. 8). Wind is the second fastest growing technology in terms of global power generation, with a 9.8% growth rate, while hydropower saw a 2% decline. In the meantime, electricity generated by coal, the most carbon-intensive source of power generation, grew by 1.4%, and fossil gas-powered electricity grew slightly by 0.8%, while other fossil generation declined 7.5%, driven by a fall in oil generation. Nuclear electricity

increased by 1.8%. In absolute terms, solar added twice as much new electricity in 2023 as coal.

Solar's exceptional success over the last decade, driven by significant cost reductions, policy support and pressures from the energy crisis, is now driving the transition to a new clean energy economy. According to the International Energy Agency (IEA), clean energy added around 320 billion USD to the world economy in 2023, representing 10% of global GDP growth and corresponding to the GDP of a country like the Czech Republic.¹

Looking at the clean energy manufacturing sector in particular, supportive frameworks and policies across different regions have enabled significant investment

FIGURE 8 ELECTRICITY GENERATION GROWTH RATE FROM 2021 TO 2022, BY TECHNOLOGY



SOURCE: Ember (2024).

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1 IEA (2024): Clean energy is boosting economic growth.

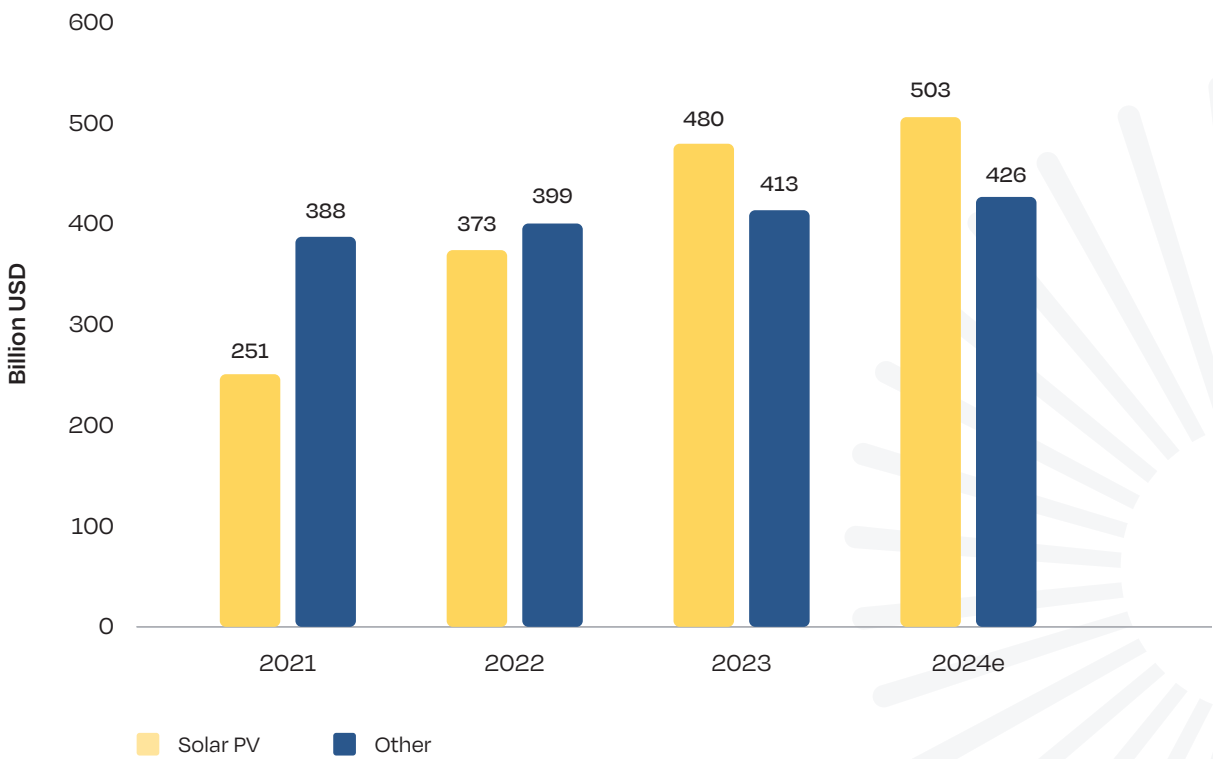
growth. In 2023, 200 billion USD was invested in the clean energy manufacturing sector globally, 40% of which (80 billion USD) went to solar PV manufacturing, and over 90% of this was invested in China. Also the Inflation Reduction Act (IRA) in the US, the Green Deal's Fit for 55 package and the Net Zero Industry Act (NZIA) in the EU, or the Production Linked Incentive (PLI) in India have enabled investment growth, or will do so when implemented.²

The strong investment activity in the clean tech sector can be understood in the context of the recovery from the COVID-19 pandemic, and the responses to the global energy crisis – immediately and regarding long-term geostrategic considerations – which provided a major boost to global investments in renewables, EVs,

and batteries. Out of an estimated 3 trillion USD invested in the energy sector in 2023, 63%, or 1.9 trillion USD, went to clean energy investments, according to the IEA.³

Specifically the power sector shows an increasing trend towards renewable power investments, led by the solar sector. In 2023, investments in solar PV have surpassed all other generation technologies combined, according to the IEA (see Fig 9). However, this level of investment needs to increase considerably to align with a 1.5°C trajectory. Annual investments in renewable power generation must reach 1.3 trillion USD by 2030, according to the Global Renewable Alliance (GRA), the International Renewable Energy Agency (IRENA), and COP28.⁴

FIGURE 9 GLOBAL ANNUAL INVESTMENT IN SOLAR PV AND OTHER GENERATION TECHNOLOGIES 2021-2024



NOTES: 2024e = estimated values for 2024. Other = electricity generation from all other technologies including coal, oil, natural gas, wind, hydro and nuclear.
SOURCE: IEA (2024).

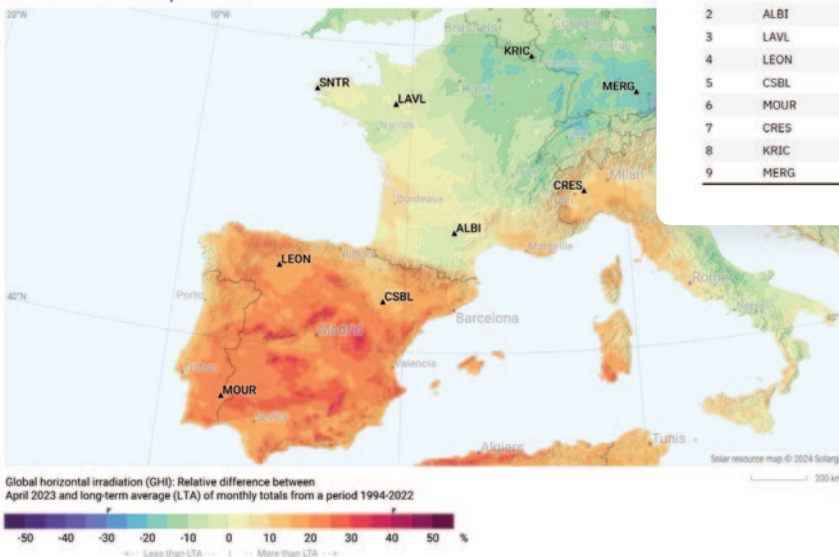
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2 IEA (2024): Advancing Clean Technology Manufacturing.
3 IEA (2023): World Energy Investments 2024.
4 COP28, IRENA and GRA (2023): Tripling renewable power and doubling energy efficiency by 2030: Crucial steps towards 1.5°C.

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2	ALBI	139.3	140.3	-1.0	-0.7
3	LAVL	128.2	128.8	-0.6	-0.5
4	LEON	188.1	161.1	27.0	16.8
5	CSBL	188.7	166.0	22.7	13.7
6	MOUR	204.9	167.5	37.4	22.3
7	CRES	160.4	145.5	14.9	10.2
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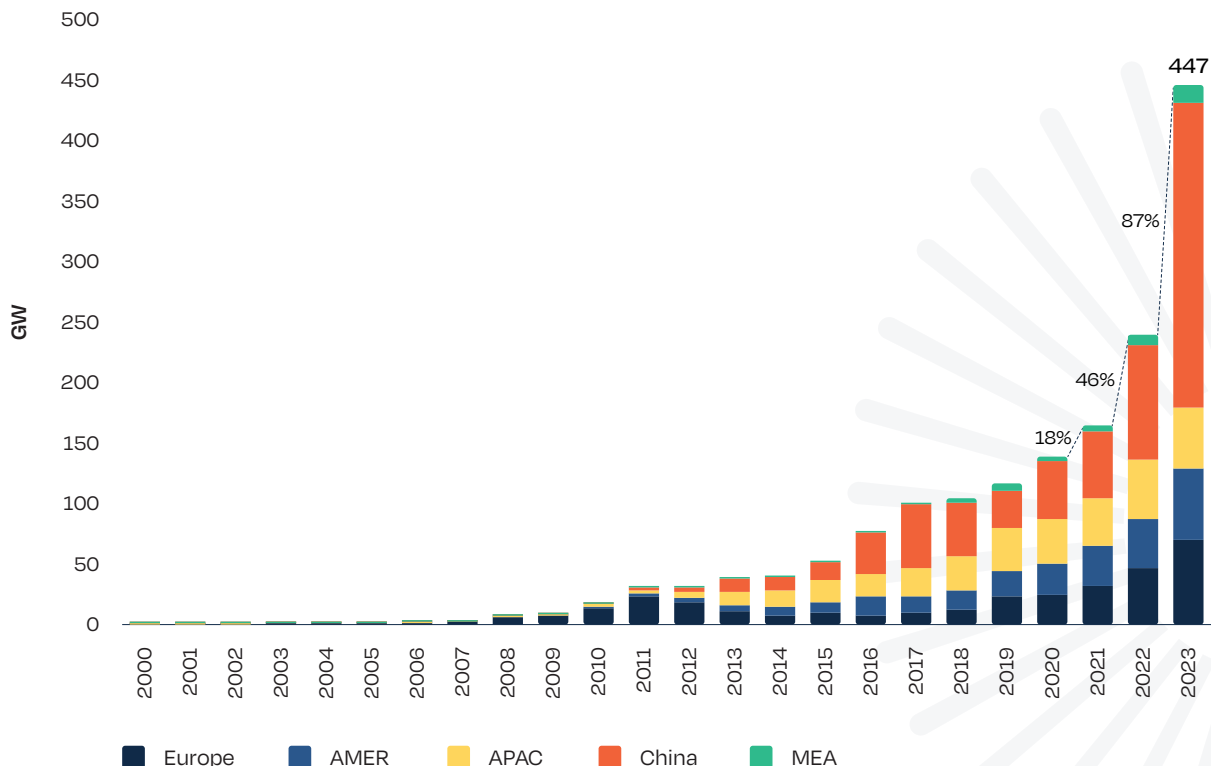
In 2023, the world grid-connected 447 GW of new solar capacity, far exceeding any solar analyst's expectations and marking an extraordinary 87% growth rate (see Fig. 10). This compares to previous year's addition of 239 GW and 46% year-on-year growth.

This unprecedented surge in installed capacity can be attributed to several key factors. Firstly, a very strong increase in global PV manufacturing capacities greatly improved availability of solar modules and balance-of-system products after the supply chain issues faced around the pandemic. Capacity expansions, which have been also part of technology shift towards TOPCon cell technology, have resulted in overcapacities that reached levels twice as large as demand for certain PV

products, triggering hefty competition among stakeholders. Prices for solar system components dropped sharply, and mostly for solar modules – by around 50% over the course of 2023.

In addition to demand reacting elastically to dropping prices, the effects of the 2022 global energy crisis continued to trigger demand in various regions of the world. Governments, individuals, businesses were looking to solar power as a reliable and cost-effective solution to high energy prices, while faster electrification of heat and transport sectors was increasingly on the radar. Pending order pipelines from 2021 and 2022 were often only installed in 2023, when products and installers were available and more affordable. All this comes on top of climate protection activities that have been continuing around the world.

FIGURE 10 ANNUAL SOLAR PV INSTALLED CAPACITY 2000-2023



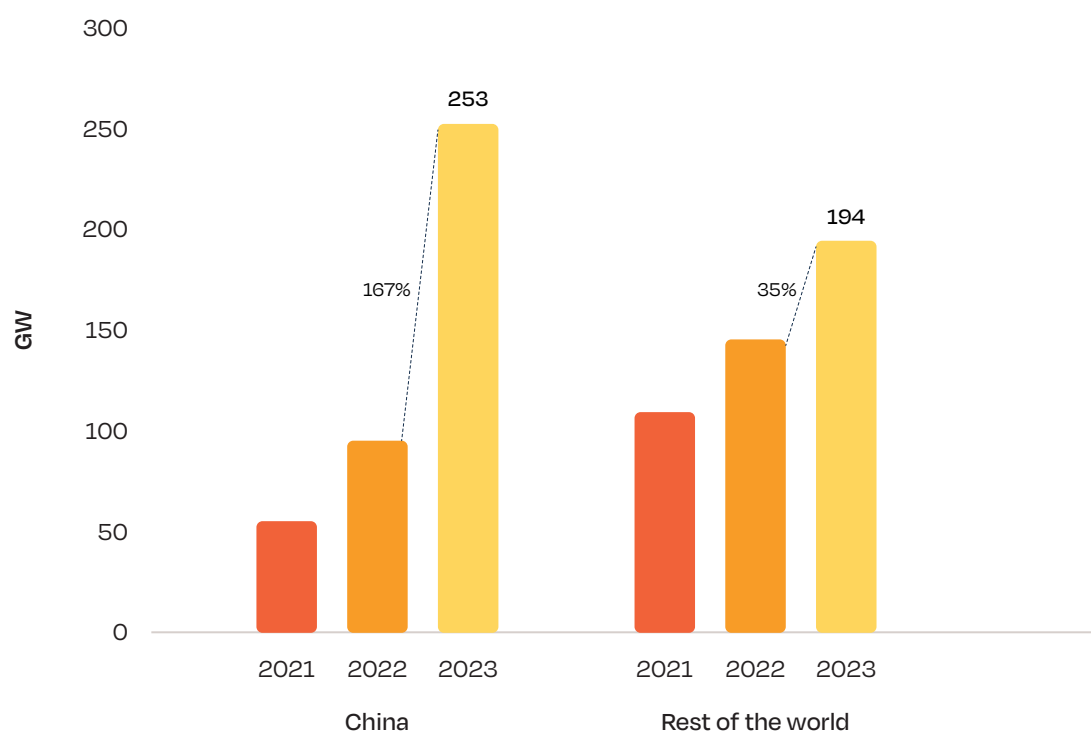
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1 Global solar market - Update 2000-2023 /continued

The combination of these factors resulted in the highest year-on-year growth since the beginning of the modern solar age. While the scale of 2023's growth was unprecedented indeed, it is important to highlight that most of this global market expansion was driven by China. The world's largest solar market for years, and also in 2022 with 95 GW annually installed, China broke any previous record by adding an incredible 253 GW of new solar PV capacity in 2023, marking a 167% year-on-year growth rate. Meanwhile, the rest of the world installed 'only' 194 GW of new solar PV, reflecting a 34% increase from the 145 GW deployed in 2022 (Fig. 11).

In our previous GMO edition, published in June 2023, we projected a global sector growth of 341 GW under our Medium Scenario, anticipating a 42% year-on-year market increase, one of the highest growth expectations among analysts at the time. While the estimates were almost right for the "rest of the world", which installed just 3% less than our 200 GW forecast, China far exceeded expectations, deploying 80% more solar capacity than our 141 GW forecast, and even 111% above the 120 GW the Chinese PV Industry Association (CPIA) forecasted in its optimistic scenario last year. In 2023, China installed 6% more solar PV capacity than the entire world deployed in 2022. In other words, without China's strong solar investments in deployment, the solar sector's growth would have been much more modest.

FIGURE 11 ANNUAL SOLAR GROWTH CHINA VS. REST OF WORLD 2021- 2023



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Top 10 global solar markets

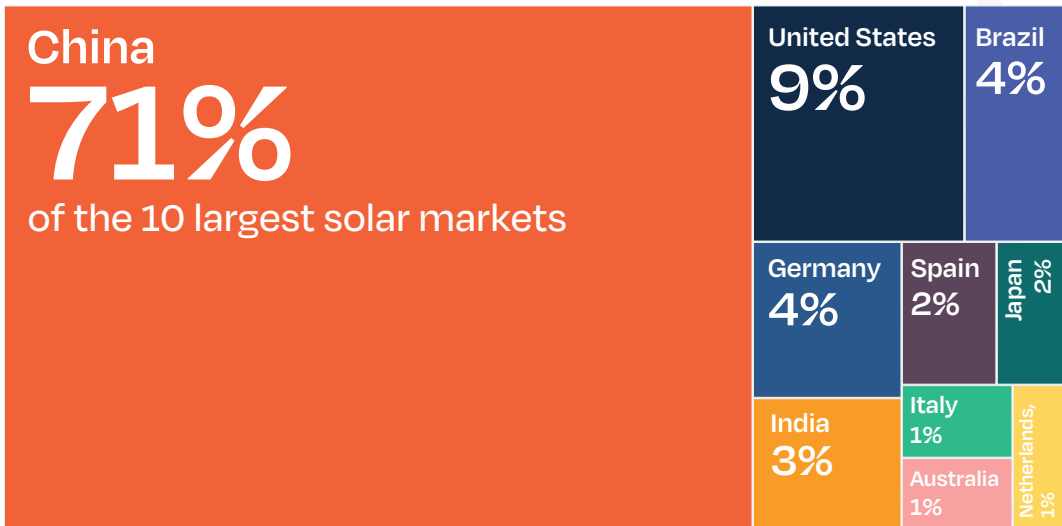
There are only few changes in the composition of the global top 10 markets. While 7 countries have moved up or down the ranks, the only new entrant is Italy, replacing another European country, Poland. Again, it's China that leaves its mark within the top 10: the largest global market installed more than twice the capacity of all the other nine combined (see Fig. 12).

Playing in a different league, also due to its sheer size, China has further cemented its position as the global No. 1 solar market (see Fig. 13). It installed almost eight times more PV capacity than the second largest market, and over half of the world's capacity additions. Over the past three years, the Chinese market experienced stellar growth, rising from 55 GW in 2021 to 95 GW in 2022 (+72%), and 253 GW in 2023 (+167%). Last year, around 62 GW out of the 253 GW new capacity was grid-connected in December alone.

Contrary to 2021 and 2022, the utility-scale sector left the rooftop segment behind again, following the higher impact of reduced product prices in this segment. China deployed 146 GW of utility-scale projects, representing 58% of new installations in 2023, while rooftops solar amounted to 107 GW, equal to 42% of the total. The cumulative solar PV capacity of China jumped to 656 GW by end of 2023, after having crossed the 400 GW mark in 2022, and the 300 GW mark in 2021. For more details on China, see the focus chapter at p. 61 and CPIA's market view at p. 84.

The **United States** experienced its best year for solar PV so far, installing 32.4 GW in 2023, a significant 48% boost from 2022. This jump marks a return to the growth path, which was interrupted in 2022 due to trade protection legislation impacts that limited imports. The delays caused by these trade actions were partly resolved in 2023, allowing for the realisation of postponed projects. The industry also

FIGURE 12 TOP 10 SOLAR PV MARKETS 2023



Note: Percentage indicates the share of aggregate top 10 installed capacity.

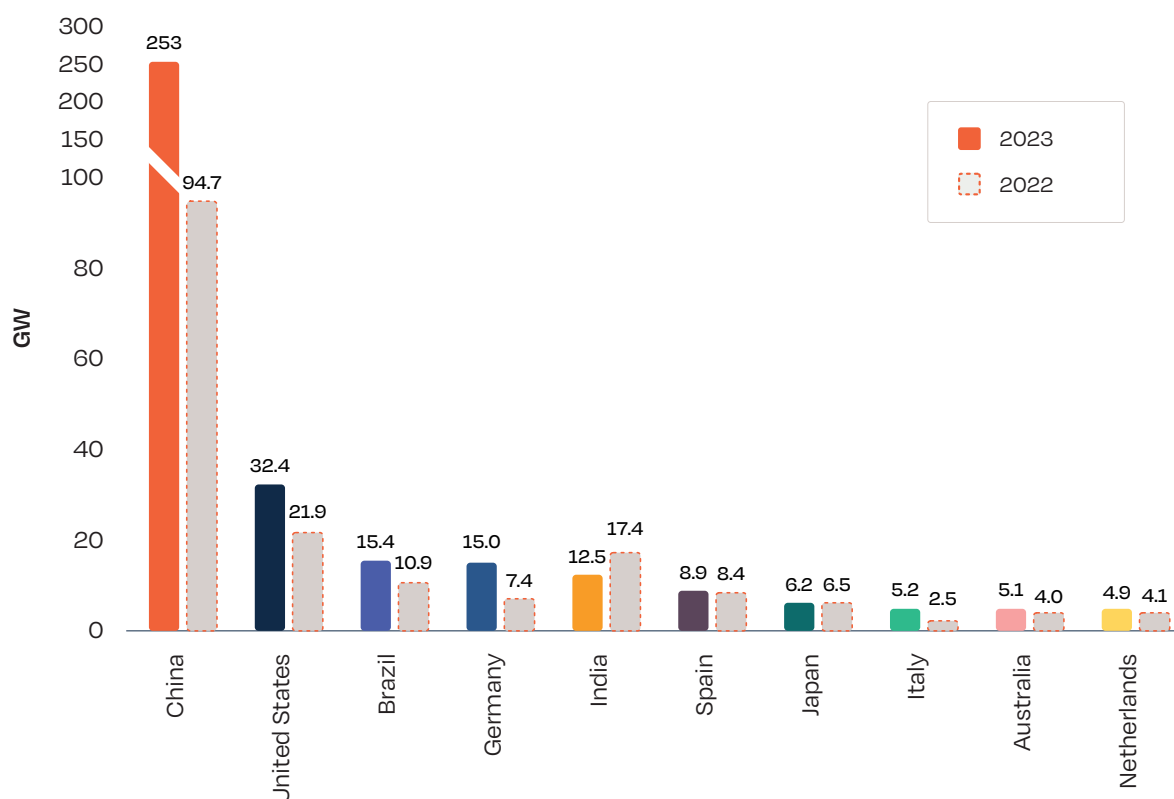
1 Global solar market - Update 2000-2023 / continued

benefited from the first full year of the IRA, which has largely improved the economic attractiveness of manufacturing projects but also solar deployment through an extension of the investment tax credit (ITC). In California, the US state with the largest operating PV capacity, new net-metering rules (NEM 3.0), taking effect in April 2023, led to a surge in residential installations in the first quarter, after which demand dropped drastically. Still, the residential segment grew by 13% year-on-year, and all new rooftop installations combined reached 9.8 GW in 2023, a 21% increase from 8.1 GW in 2022, while utility-scale installations soared to 22.6 GW, up 64% from 13.8 GW in 2022.

Brazil's solar PV sector grew significantly in 2023 and installed 15.4 GW, up 41% from 10.9 GW in 2022. After entering the top 10 for the first time on the 7th position

in 2021, the country progressed to rank 4 one year later, and now belongs to the three largest solar markets in the world. This strong growth was recorded despite macroeconomic challenges, grid-connection issues, and changes in the country's attractive net-metering law for systems up to 5 MW, which have been the fundament of Brazil's impressive solar boom. As of January 2023, PV system operators must pay fees for grid usage when feeding in solar power. Still, distributed solar added 10.3 GW, nearly the same level as the year before. On the other hand, the centralised solar PV segment augmented by nearly 50% to 4.1 GW, mostly from corporate Power Purchase Agreements (PPAs) in the unregulated market that offers more lucrative opportunities than the traditional government auctions. Brazil's total installed solar PV capacity surpassed 39.4 GW by the end of 2023.

FIGURE 13 TOP 10 SOLAR PV MARKETS 2022-2023



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Germany gained two positions, landing on the 4th spot of the world's largest solar markets after doubling its annual installed capacity to 15 GW in 2023, from 7.4 GW in 2022. By the end of August 2023, the country had already added 9 GW of solar capacity, meeting the government's target for the entire year. Germany's achievement marks the first time a European country has surpassed the 10 GW mark in annual installations, breaking the 12-year record previously held by Italy, which grid-connected 9.3 GW in 2011. The revision of the Renewable Energy Sources Act (EEG) in July 2022, which fully took effect at the beginning of 2023, has been instrumental for Germany to get on the roofs and ground its extensive project pipeline that arose from the 2022 energy price crisis. Key improvements included increased feed-in tariffs (FITs) for new rooftop systems, a halt to monthly reductions in feed-in rates for new systems until the end of 2024, and a cut of the national sales tax (VAT) on PV systems as of

January 2023. The current German government has acknowledged the critical role of solar energy, aiming for renewables to comprise 80% of electricity generation by 2030 and 100% by 2035, with a specific solar PV capacity target of 215 GW by 2030.

After strong growth in 2021 and 2022, it seemed India's solar PV sector was back on an upward trajectory. But the sector saw significant changes and challenges resulting in a 28% decline in new installations last year, with 12.5 GW added from January to December 2023, compared to 17.4 GW in 2022. This downturn costs the country two ranks, falling to 5th place in 2023. The market decline is attributed to fewer new tenders issued between 2020 and 2022, which led to a reduced pipeline of projects to be built in 2023. In addition, changes in policy and regulatory frameworks, such as revisions of tariffs and duties on imported solar modules, created uncertainty



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worldwide

20 

20 years
of solar
experience



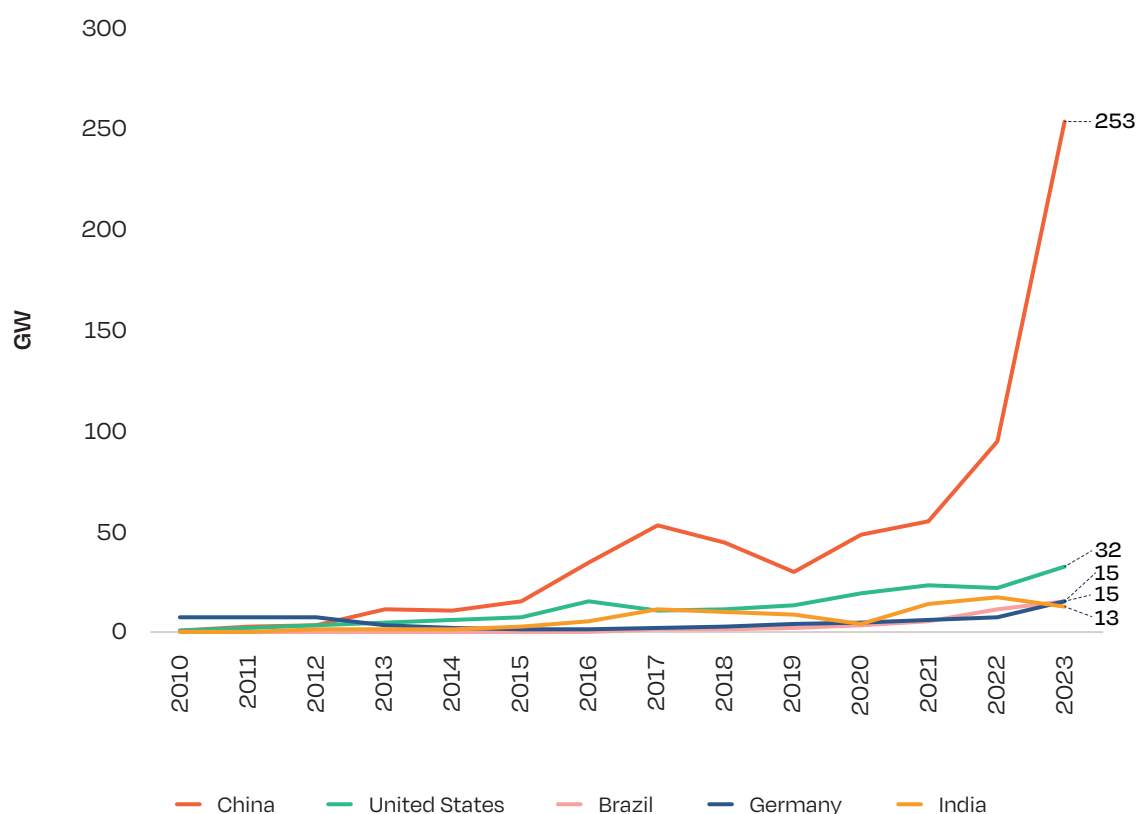
1 Global solar market - Update 2000-2023 / continued

that affected planning and financial viability of projects, resulting in delays and extension of project timelines. Also, the government's push for increased use of domestically made solar components did not help as very limited local production could not meet developers' needs. The issues are likely temporary, as challenges are being addressed, and over 50 GW of renewable tendered capacity was issued in 2023.

Within the top 5 markets, the spread between China and its peers has grown considerably (see Fig. 14). China, which has taken an outstanding role already as of 2016, plays in an entirely different league. In 2023, the #1 market was eight times larger than the #2, USA, and 20 times larger than the #5, India. That's significantly more than last year, when the market size difference had a factor of four and 11, respectively.

Spain, ranking as the second largest European market, secured the 6th position globally in newly installed capacity, reaching 8.9 GW. This represents a modest 5% year-on-year growth, contrasting sharply with the 76% surge shown in 2022. The country saw a 53% decline in the residential segment compared to the record performance in 2022, partially attributed to delayed subsidy payments tarnishing public perception on rooftop PV. Similarly, the commercial and industrial (C&I) segment experienced a downward trend, declining by 22%, despite falling module prices. On the contrary, ground-mounted PV confirmed its central role in Spain's solar landscape. The utility-solar segment was the only one to grow in 2023, with a notable uptick of 25%. Spanish solar parks are often developed through PPAs, underscoring the country's position as a global

FIGURE 14 TOP 5 SOLAR PV MARKETS 2010-2023



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leader in subsidy-free solar energy. With a huge PV project pipeline existing, growth could have been much higher if it were not for grid connection issues, along with high interest rates, declining wholesale electricity prices and decreasing solar capture rates that have made successful solar park developments much harder in Spain than in the past.

Japan maintained its 7th position, despite a marginal market decline. The country installed 6.2 GW of solar capacity in 2023, down 4% from 6.5 GW in 2022. Since its peak year in 2015, when it installed 10.8 GW, Japan has been on a downward trend in solar installations. The discontinuation of FITs for large-scale systems in 2022 marked a transition period, during which new business models reliant on PPAs and third-party ownership have started to gain traction. Anticipated growth in the residential and C&I segments is expected to reverse this trend and stimulate installations in the near future. Additionally, the country conducted four rounds of auctions for utility-scale projects throughout 2023, allocating a total capacity of 300 MW.

Italy emerged as one of the fastest-growing solar markets in the EU in 2023, installing 5.2 GW. This marks a 111% increase from the 2.5 GW installed in 2022, when the country re-entered the gigawatt scale for the first time since 2013. As a result, Italy rejoined the global top 10 on the 8th position. The energy crisis and the end of the government's Superbonus tax incentive scheme propelled residential solar, resulting in a notable 79% growth in this segment. After conditions were made much less attractive for the Superbonus, Italy's solar expansion has been primarily steered by the C&I segment, contributing approximately 46% of the installed capacity in 2023. Moreover, solar farms exceeding 1 MW experienced a peak month in December 2023, with installations reaching 362 MW, significantly surpassing the average monthly installation rate of 72 MW from January to November.

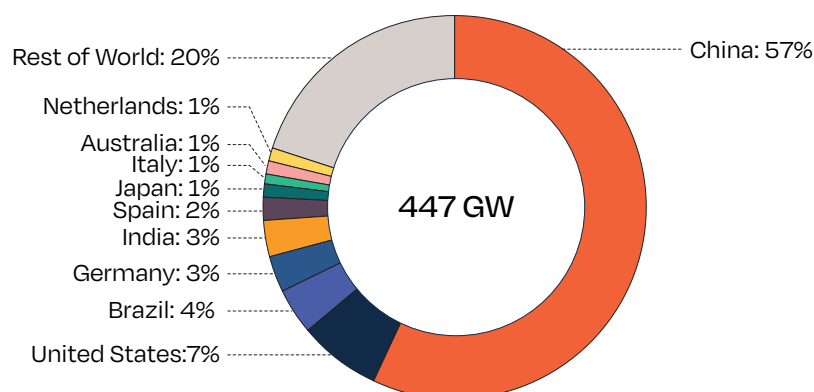
Australia has climbed one spot to 9th place after installing 5.1 GW in 2023, a 28% increase that was primarily attributed to the strong performance of the rooftop segment. Although it falls short of the country's best performance of 5.9 GW added in 2021, demand for rooftop systems has remained basically



100 MW, Guangdong Energy, China.

© Jinko Solar

FIGURE 15 TOP 10 COUNTRIES SOLAR SHARE 2023



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steady since 2022. Utility-scale projects accounted for 1.9 GW, a notable improvement from 1.2 GW in 2022. As of December 2023, out of 56 renewable energy projects under construction, 38 were large-scale solar projects, a decrease from 48 at the same point in 2022. But the amount of large-scale projects commissioned is expected to increase thanks to the Capacity Investment Scheme, which provides a national framework to foster new investments in renewable capacity.

The Netherlands switched ranks with Australia, dropping from 9th to 10th place despite 20% market growth to 4.9 GW. A robust rooftop market for years, the Netherlands' solar sector heavily relied on a net-metering policy for the residential sector and an auction programme for larger rooftop and ground-mounted systems. In 2023, the residential rooftop market surged by 39% to 2.6 GW, driven by record levels in response to the previous year's energy crisis and uncertainty about the future of the net-metering scheme. Although

both the C&I and ground-mounted solar sectors gained shares, they are increasingly encountering grid congestion issues. Additionally, large solar parks are facing severe space constraints, and have been leading the local industry to explore multifunctional PV applications such as Agri-PV, floating solar, and solar carports to address these challenges.

In 2023, more than ever, the market was dominated by China, which alone was responsible for 57% of the world's total additions, up 17 percentage points from the 40% it provided in 2022 (see Fig. 15). As a consequence of China's record performance, all the other top 10 markets lost global market share. The share installed in all other countries also decreased, by 5 percentage points to 20%. A positive development is a further increasing number of countries reaching annual GW market size: in 2023, 31 countries crossed the threshold, compared to 28 the year before. Details on these markets can be found in Chapter 4, where national industry associations active in the solar sector provide analysis on their home markets (see p. 81).

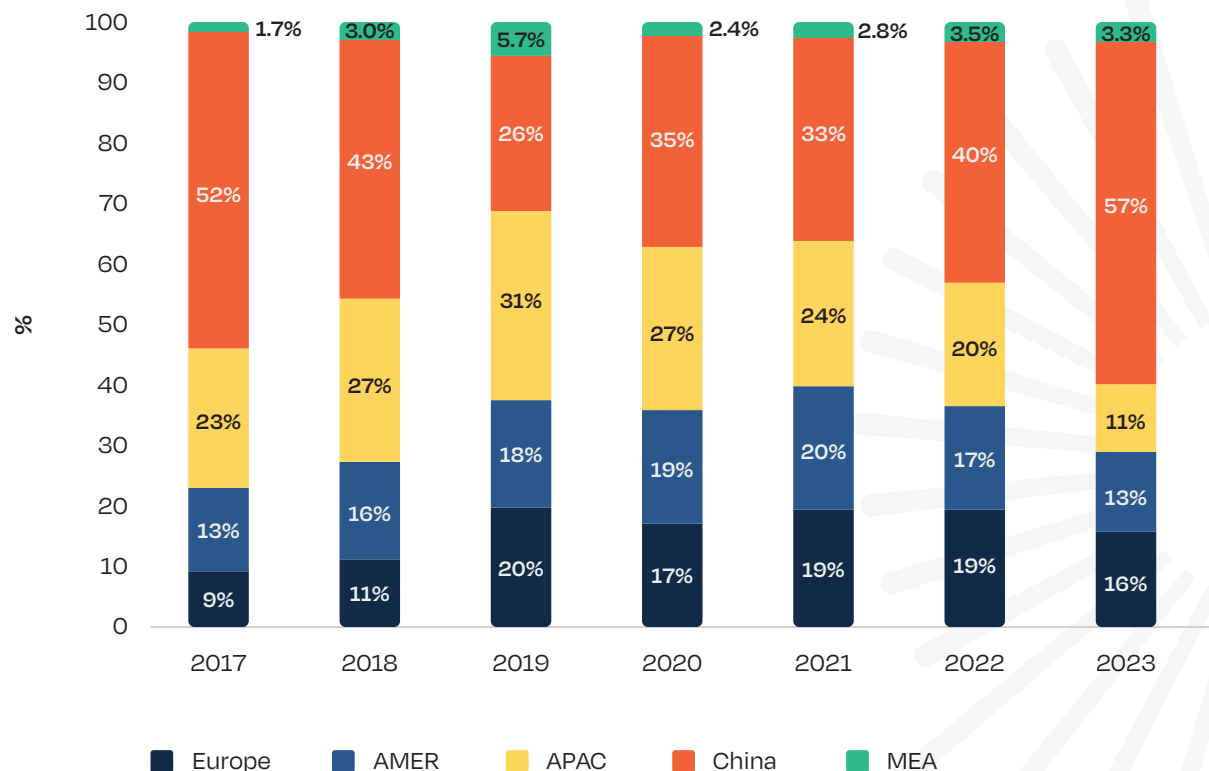
Regional Update

In 2023, no region other than Asia-Pacific (incl. China) managed to increase its market share. If China is looked at separately, all other regions lost market shares, showing how much the country dominates the solar world. China accounted for more than half of 2023 annual installations, a massive 57% of the global share unseen before, and up from 40% in 2022 and 33% in 2021. A similar though slightly lower level of dominance has been seen only once before, in 2017, when China surprisingly installed 54 GW equal to 52% of the global market share. Afterwards, it started a market transition project away from FITs to auctions and other policy frameworks, which resulted in new installations falling sharply for two years and recovered to new record levels only in 2021. However, in 2017, the global solar market was just over 100 GW, now it's more than four times larger, yet China holds an even greater share than ever before.

In Europe, the solar market continued its strong performance, increasing annual installed capacities substantially compared to the already exceptional growth years around the pandemic. After growth rates of 34% in 2021 and 45% in 2022, the region experienced a 51% year-on-year market boost to 70.1 GW in 2023, which meant a 16% contribution to the global market. Although it lost 3 percentage points in global market share, Europe has overtaken Asia-Pacific excl. China and the Americas, ending 2023 as the second largest solar region, one place up from the third rank in 2022, and the fourth position in 2021.

Within Europe, Germany has taken over the lead again, after Spain was not able to replicate its strong growth from 2022. Besides established solar markets Italy (5.2 GW), the Netherlands (4.9 GW), Poland (4.6 GW) and France (3.2 GW), several other European countries have passed the annual GW-market threshold. This includes

FIGURE 16 ANNUAL SOLAR PV INSTALLED CAPACITY SHARES 2017-2023



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1 Global solar market - Update 2000-2023 / continued

Türkiye (2.7 GW) and Austria (2.7 GW), which both exceeded the 2 GW mark for the first time with impressive growth rates of 69% and 166% respectively. Türkiye was the largest European non-EU market, while the United Kingdom and Switzerland also continued their upward trend, installing 1.8 GW and 1.5 GW respectively. The European Union registered 50% growth last year, with 26 out of 27 Member States installing more solar than the year before; the only exception was Denmark. Most of the European continent's solar deployment last year took place in the EU where 60.9 GW was deployed; that's equal to 87% market share, and 0.5 percentage point less than the year before. The 13% share of non-EU countries in 2023 is based on 9.3 GW of new PV capacity.

The Americas climbed one spot in the regional rankings to third place, capturing 13% of the global market share. The region installed 58.8 GW in 2023, a 43% increase from 41.0 GW added in 2022. But the continent remains heavily dependent on a few key markets, with the United States leading the solar way. In 2023, the United States increased its regional dominance, accounting for 55% of the continent's new installations, up by 2 percentage points from 2022. The only other market with a two-digit GW market is Brazil, whose share decreased by 1 percentage point to 26%. The Americas' third rank is shared by Chile and Mexico. Both markets added 1.6

GW but faced opposite developments last year. Unlike Chile, where solar activities slightly contracted from 1.8 GW to 1.7 GW due to congestion problems in the transmission lines, Mexico faced 40% demand growth despite government policies that have hindered investments in renewables while focusing on fossil fuels. Mexico's solar market is still reigned by utility-scale solar, but the rooftop segment is getting close – now at a 44% share, after it logged another record in 2023. Together, the US, Brazil, Chile and Mexico accounted for 86% of the Americas' 2023 solar additions, similar to the 87% share the four owned in 2022.

If Europe and the Americas both increased by one place in the ranking, it is partly due to the significant market share decline of the Asia-Pacific region (excl. China). In 2023, the region's share dropped from 20% in 2022 to 11% in 2023, the largest decrease among all regions, which meant a plunge from the second to the fourth position. Little growth of 2% to 50.2 GW in 2023, from 48.9 GW in 2022 was not enough to keep up with the other regions. Only Australia and Taiwan showed notable growth rates – the one by 28% to 5.1 GW, the other by 32% to 2.7 GW; but even these were significantly lower than the global average of 84%. All other major Asian solar nations disappointed. India's annual installations even plummeted by 28% to 12.5 GW in 2023, from 17.4 GW the year before. This



477 kW, Matosinhos, Portugal.

© Greenvolt

downturn is attributed to a lack of utility-scale development due to project delays, auctioning shortages, regulatory hurdles, and speculative behaviour on declining module prices. Also Japan's market slowed by 4% to 6.2 GW. Once a solar pioneer, the Japanese market has been stagnating for several years after policy makers slashed feed-in tariffs but failed to provide a framework that enables growth like in other former FIT markets. Although South Korea is home to one of the few successful non-Chinese cell/module makers, the current political elite seems to have abandoned any solar ambitions; political support has shifted from renewable energy towards nuclear. At 2.9 GW, the market was only 64% of the 4.7 GW that was installed in 2020. None of the other high-potential solar markets in that region, like Thailand, Malaysia, Philippines, and Indonesia, reached the GW-level in 2023.

The **Middle East and Africa (MEA)** region experienced significant solar growth of 78% in 2023, adding 14.8 GW. This broke the previous record set in 2022, when 8.3 GW of capacity was installed. Three MEA countries reached the GW-level in 2023, compared to two in 2022. South Africa maintained its top position after installing 3.2 GW, which meant a 142% growth rate. Regulatory reforms in 2022 have enabled unmatched growth in the private utility-scale solar segment, while energy security issues have driven demand for residential solar PV and battery storage systems.

Characterised mainly by a few very large PV projects, the United Arab Emirates (UAE) entered GW territory for the first time since 2019. It installed 3.1 GW, mainly based on the finalisation of the 2 GW Al Dhafra Solar Power Project near Abu Dhabi. The third GW-market in MEA was Saudi Arabia, which passed that threshold for the first time ever, after 1.9 GW was grid-connected, largely from the initial phase of the 1.5 GW Sudair Solar Project in Saudi Arabia. Israel remained the fourth-largest solar nation in the region, even though deployment decreased by 20% to 824 MW, down from 1 GW in 2022. Overall, MEA's global market share slightly fell to 3.3% in 2023, from 3.5% in 2022.

In summary, 2023 was a stellar year for solar energy, with a massive 87% market growth that no solar analyst predicted one year ago. A global growth rate at that scale was observed the last time in 2010 – then at 107%, but the market was only 17 GW large and 26 times smaller than today. Two primary factors contributed to this surge in installations. The sharp increase in manufacturing capacities in China has reached unprecedented levels, exceeding the TW-level from wafer to modules, and supplying plenty of low-cost and attractively priced product to the rest of the world. This significantly impacted module prices, which dropped by half, making solar energy more affordable than ever. Second, the lingering effects of the energy crisis drove orders in 2021 and 2022 to unmatched levels, with project realisations continuing in 2023.

While the manufacturing increase and the induced price effects will likely keep influencing the market in 2024, the immediate impact of the energy price crisis on installation levels is diminishing. However, the crisis undeniably left a mark amongst households, businesses, and policymakers, and highlighted the need for energy security and a faster development of renewables.

Nonetheless, the unprecedented market growth should not hide its skewness. The global solar PV market has never been as dependent on a single country as it was in 2023. More than every second solar module was installed in China. The country's annual market grew 72% in 2022 and a staggering 167% in 2023, while the rest of the world experienced a 32% growth in both years. At the same time, about eight PV modules out of ten are manufactured in China, which retains the strong majority of global PV production capacities. As the energy crisis dramatically highlighted, relying on a single supply source is not sustainable. A market contraction in China would inevitably lead to a global market contraction unless other regions accelerate their deployment of renewable energy. In the same vein, a downturn in China's PV manufacturing activities or global export accessibility would slow down the energy transition of other countries.

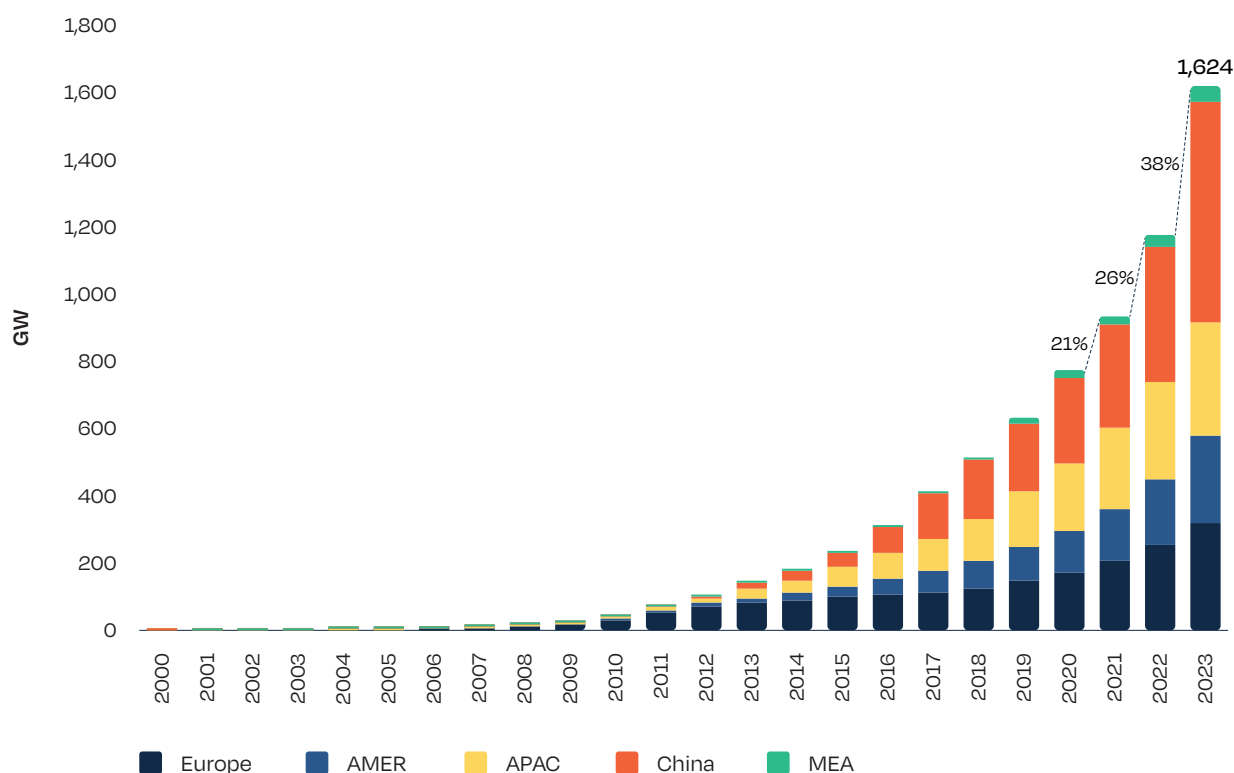
1 Global solar market - Update 2000-2023 / continued

While 2023 marked a historic year for solar PV growth, the future of the market hinges on continued global efforts to diversify manufacturing capacities and accelerate renewable energy deployment outside China. The lessons learned from the recent energy crisis emphasise the importance of a diversified energy supply. Similarly, supply chain constraints following the pandemic and energy crisis unveiled vulnerabilities in highly concentrated manufacturing. The year 2023 was the hottest on record, the world can simply not afford any disruption in renewable energy supply chain, as any slowdown in deployment of renewable energy is a slowdown in our global fight against climate change.

Cumulative solar installations until 2023

The global cumulative installed capacity of solar PV systems reached 1,624 GW by the end of 2023, up 38% from 1,177 GW a year earlier, when the solar fleet recorded 26% growth (Fig. 17). Global on-grid PV capacity is 38 times larger than the 41.4 GW operating at the end of 2010. Important installation milestones of the world's solar evolution include: reaching the first GW before 2000, surpassing the 10 GW level in 2008 and climbing up to 100 GW four years later in 2012. It took another six years until 2018 to exceed 500 GW, and just four more years for the market to double to the TW level in 2022. In 2023, solar PV also became the renewable energy technology with the largest capacity deployed worldwide, surpassing hydropower.

FIGURE 17 TOTAL SOLAR PV INSTALLED CAPACITY 2000-2023



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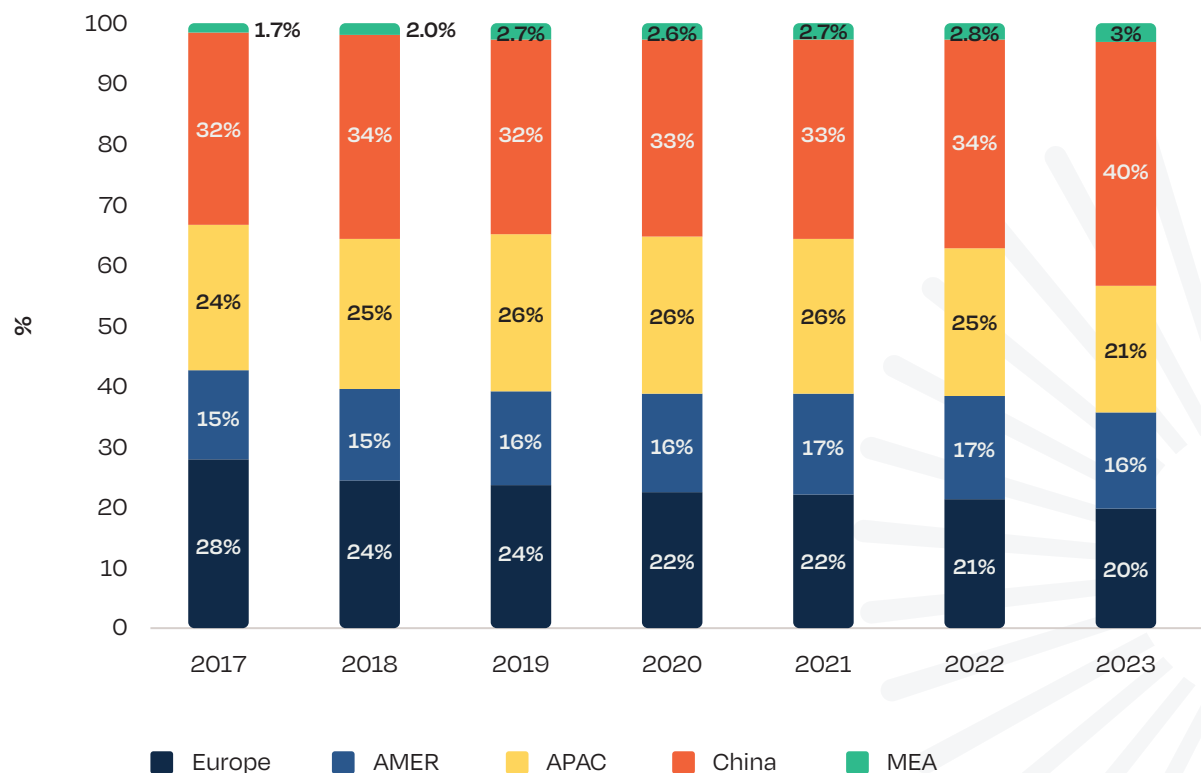
In the last few years, the regions' cumulative market shares remained relatively stable, as the total installed PV fleet had reached a level that makes it very difficult for annual installation fluctuations to have a notable impact. But the world's largest market **China's** explosive growth in 2023 was so outstanding that its total installed capacity share rose to 40%, from 34% in 2022 (see Fig. 18).

Combined, **APAC including China** now accounts for 61% of total solar PV installations, adding up to nearly 1 TW – to be exact, 995 GW –, a higher share than in any of the last six years. Without China, APAC lost more market shares than any other region – 4 percentage points down to 21%.

Europe's global market share slightly decreased by 1 percentage point to 20% despite significant solar growth in most of its markets. In terms of total operating solar capacity, Europe maintained its position as the second-largest region in 2023. The record addition of 70.2 GW strengthened Europe's second place with a cumulative PV capacity of 322 GW.

With a total installed PV capacity of 258 GW, the **Americas** kept its third place in the regional ranking in 2023, but lost 1% of global share to reach 16%. Although MEA experienced substantial growth of 45% after installing 48.1 GW, it did neither affect its overall solar positioning as the smallest global solar region nor its overall market share of 3%.

FIGURE 18 TOTAL SOLAR PV INSTALLED CAPACITY SHARES 2017-2023



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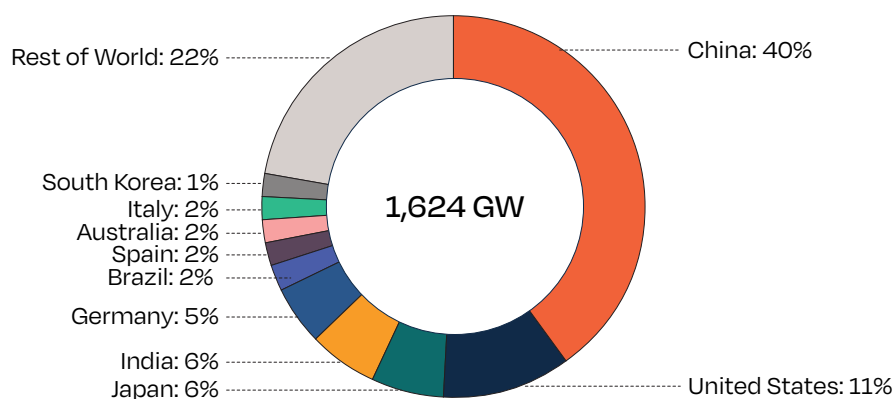
1 Global solar market - Update 2000-2023 / continued

The top 5 solar nations haven't seen any changes in their 2023 rankings compared to the previous year. Unsurprisingly, China is leading the global leader with a total solar power fleet of 656 GW, controlling 40% of the world's solar capacity at the end of 2023 (Fig. 19). The **United States** held steady its second place with 173 GW, although its market share fell by 1 percentage point to 11%. Japan maintained the third spot, probably for the last time, after its share has been dwindling for years, this time by 1 percentage point to 6%, based on an operational capacity of 90.4 GW. While India at 90.1 GW and Germany at 83.0 GW kept their fourth and fifth spots, despite losing 1% market share year-on-year, both might take over Japan in 2024. For the moment, India holds 6% and Germany 5% of the total installed global capacity respectively.

Altogether, the top 5 largest markets operated 67% of the global solar fleet in 2023, a slight increase from 66% in 2022. Their cumulative capacity surpassed the 1 TW level, reaching 1,092 GW by the end of 2023, compared to 774 GW one year prior.

There were no changes among the global top 5 solar power fleet operators – and it's very unlikely any other nation will enter soon. Brazil, in sixth place, owns less than half of Germany's total solar PV capacity. Within the lower half of the top 10, Brazil improved by four positions to #6 with 39.4 GW, while Australia moved down two spots to #8 with 36.1 GW installed. Spain is now ranked 7th with 36.3 GW, Italy on the 9th position with 29.8 GW, and South Korea closes the top 10, operating a solar fleet of 27.3 GW.

FIGURE 19 TOP 10 SOLAR PV MARKETS TOTAL INSTALLED SHARES 2023



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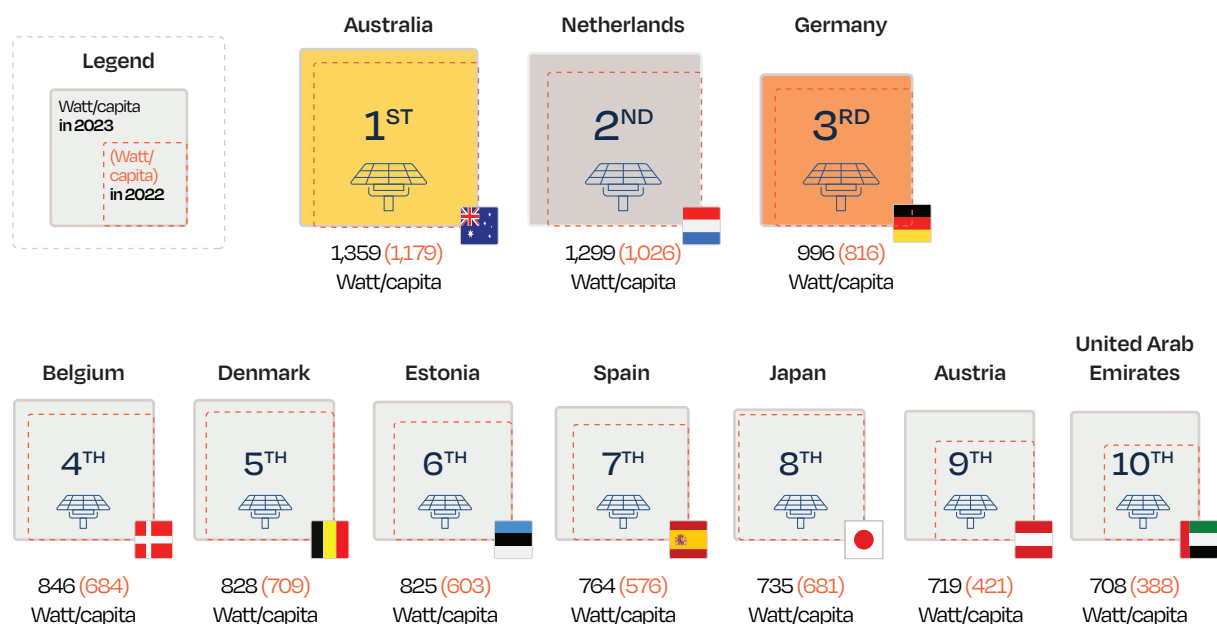
Solar watt per capita

Analysing the solar market on a watt-per-capita basis provides valuable insights on the development potential of countries beyond their absolute installed capacities.

Next to China, Germany stands out in the solar world – it's Europe's largest market, among the top 5 global solar power operators, and a top 3 market in terms of watt-per-capita installations. It's an example for countries where a growth environment has been created through supportive policy frameworks and ambitious targets. It is a likely contributing reason to why seven of the global top 10 markets with the highest solar per capita rates are European, and all but Australia, Japan and the United Arab Emirates are based in the European Union. In the EU, renewables have been high on the agenda for years, and during the energy crisis, it has embraced solar as a key technology to strengthen energy security.

The top 5 markets with the largest installed PV capacity per inhabitant saw few changes, the top 3 even remained the same (see Fig. 20). End of 2023, Australia led with 1,359 W/capita, followed closely by the Netherlands with 1,299 W/capita, and Germany with 996 W/capita. Like in 2022, only two countries in the world exceeded the 1 kW/capita threshold in 2023. Australia reached this milestone in 2021 and was joined by the Netherlands in 2022. Denmark shifted down one rank to the 5th spot with 828 W/capita because of a lacklustre solar performance in 2023. Belgium climbed one rank to enter the top 4 markets with 846 W/capita. Japan has fallen to the eighth place, being passed by both Estonia and Spain. It is telling that the world's largest market China is only ranked 12th, despite its global market dominance, and the global #2, United States on the 26th place when it comes to installed solar per capita. India, the world's most populated country, is only on the 74th position – clearly showing how much further growth potential there is for solar.

FIGURE 20 WORLD TOP 10 COUNTRIES SOLAR CAPACITY PER CAPITA 2023



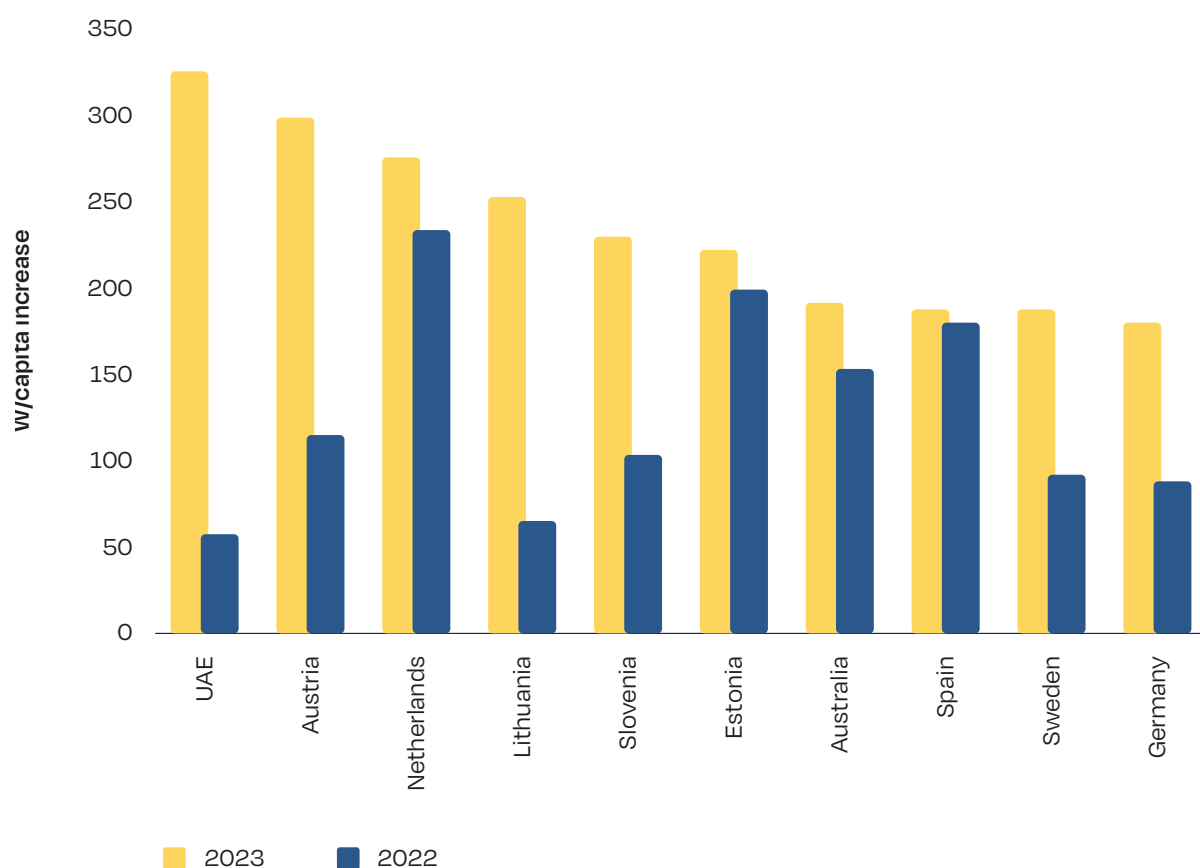
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1 Global solar market - Update 2000-2023 /continued

From an annual addition perspective, the watt-per-capita metric top ranking looks very different than the totals (see Fig. 21). This top 10 list contains six different countries and the top three do not appear in the 2023 Watt/capita ranking (China, US and Brazil). The European dominance is even more pronounced – with eight out of the top 10 coming from the EU. The United Arab Emirates secured the top spot with 323 W/capita added in a year. The country showed immense growth rising from a mere 54 W/capita in 2022 to the first place in one year, because of its large-

scale PV plant development in 2023. Austria and the Netherlands followed with 299 and 276 W/capita added. While Austria climbed three spots in the ranking, the Netherlands is no longer the top performer, shifting down two spots. Lithuania, Slovenia complete the top five and Estonia follows on the sixth spot, proving the case for strong development in Eastern European markets. Finally, Australia, Spain, Sweden and Germany complete the list, with all but Sweden being a top 10 market as well.

FIGURE 21 WORLD TOP 10 COUNTRIES IN ANNUAL SOLAR CAPACITY PER CAPITA ADDITIONS 2023



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FIGURE 22 COUNTRY RANKING 2023

	ANNUAL ADDITION		CUMULATIVE CAPACITY		ANNUAL WATT PER CAPITA		CUMULATIVE WATT PER CAPITA	
1.		China		China		United Arab Emirates		Australia
2.		United States		United States		Austria		Netherlands
3.		Brazil		Japan		Netherlands		Germany
4.		Germany		India		Lithuania		Belgium
5.		India		Germany		Slovenia		Denmark
6.		Spain		Brazil		Estonia		Estonia
7.		Japan		Spain		Australia		Spain
8.		Italy		Australia		Spain		Japan
9.		Australia		Italy		Sweden		Austria
10.		Netherlands		South Korea		Germany		United Arab Emirates

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A comparison of the different rankings of this chapter shows only Germany, Spain and Australia being listed in all four top 10 rankings (see Fig. 22). The strong difference between the United Arab Emirates in the annual and cumulative W/capita rankings, shows how relatively large the recent developments of utility-scale solar have been.

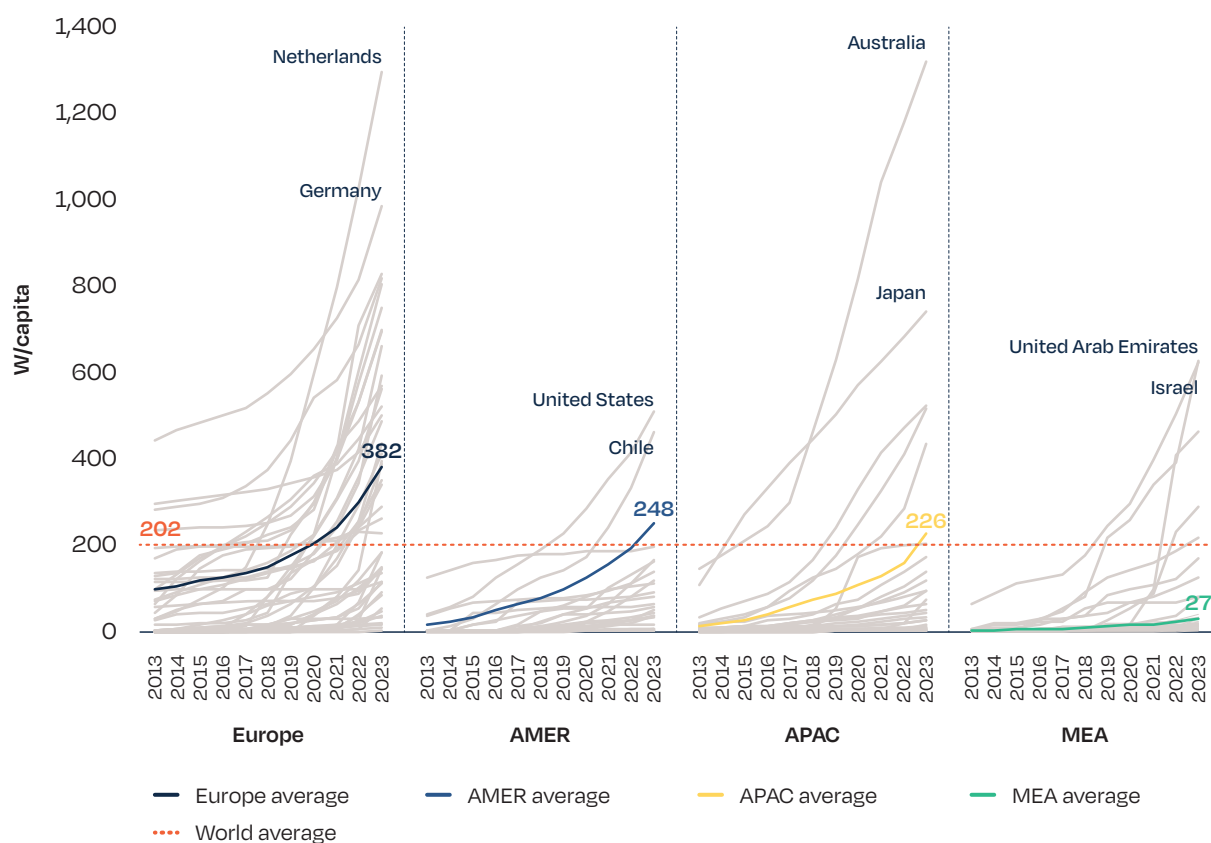
All three major solar regions exceeded the global average of 202 W/capita by end of 2023. Europe led with 382 W/capita, followed by the Americas with 248 W/capita, and APAC including China with 226 W/capita (see Fig. 23). In contrast, the Middle East and Africa region lagged dramatically behind, reaching only 27 W/capita. Considering the region's population of nearly 1.8 billion people – and on the one hand the economic power of some Arab countries and on the other the low electrification rates in many African nations, there is enormous potential for solar growth.

Looking at the speed of PV deployment in W/capita terms, APAC led with a 43% growth rate, increasing from 193 to 226 W/capita between 2022 and 2023. The Middle East and Africa region came second with a 41% growth rate, rising from 19 to 27 W/capita, while Americas and Europe show similar growth rates, with 29% and 28%, respectively.

The distribution of solar deployment on a watt-per-capita metric varies widely for the different regions. Europe demonstrates the most heterogenous distribution with 45% of the countries above the region's average, and the leaders significantly. In the Americas all but two countries – US and Chile – were below the regions average end of last year. In APAC, where 55% of the global population lives and which operates nearly two thirds of the world's solar fleet, the bulk of the countries remained below the region's average and only five were above, including Australia, a clear outlier where about every third household uses solar PV today.

1 Global solar market - Update 2000-2023 / continued

FIGURE 23 WATT PER CAPITA ACROSS REGIONS 2013-2023



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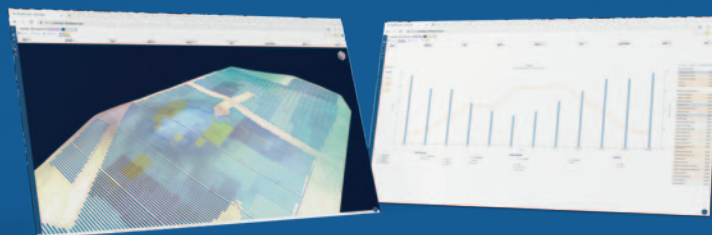
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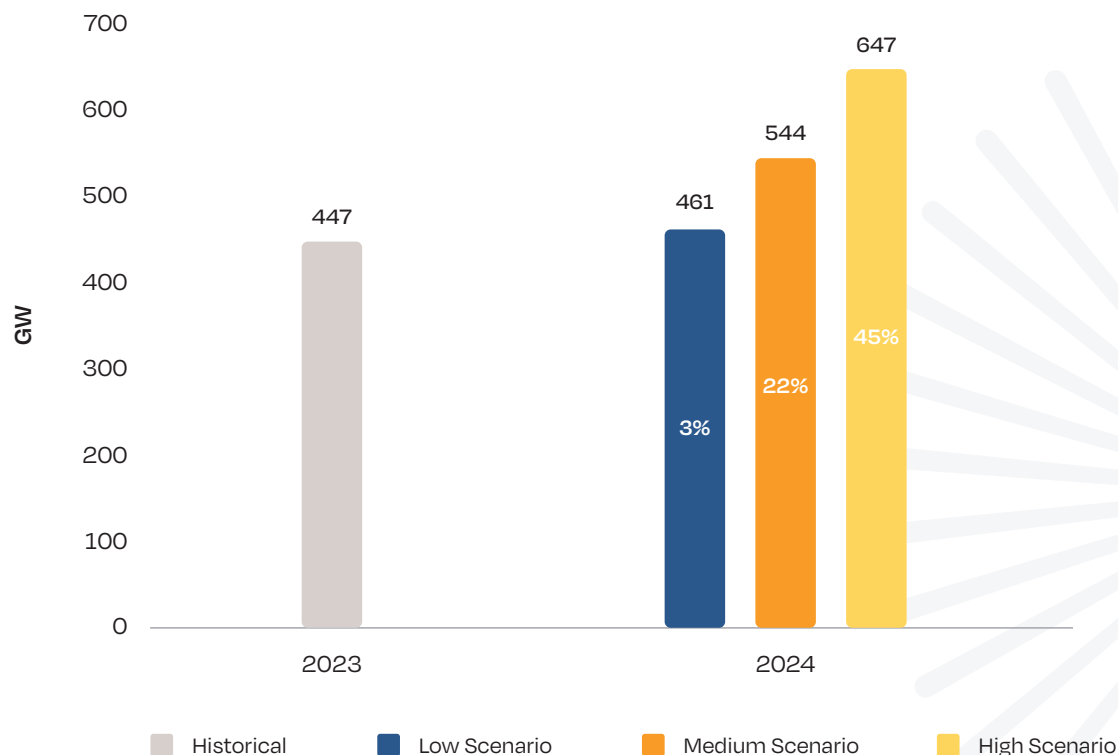
In the beginning of 2024, the solar PV sector found itself in a very different position than one year earlier. The energy crisis was over and power prices had rapidly advanced on a path towards the much lower levels seen in early 2021. Gigantic growth on the solar demand side met production capacities that expanded even faster, and to levels unseen before. The IEA estimates that China could have produced 800 GW, and not just 560 GW, if existing capacity were utilised at 85% instead of around 50%.⁵ No wonder PV products across the value chain and most system components have been under strong price pressure. At the start of 2024, Chinese solar silicon cost around 70% less than a year before, while factory-gate module prices dropped by 50% year-on-year; a

downward trend that have not come to halt in the first five months of 2024 – by our editorial deadline end of May, prices for silicon had tumbled another 40% and for modules by 10-15%, depending on technology.

What's bad for the manufacturing industry turns out to be a boon for the downstream sector – low product prices will stay a main driver in an environment of continuous high interest rates and very modest economic growth projections for major western solar markets, and deteriorating conditions in China, Brazil or India, according to the IMF's April World Economic Outlook.⁶

Amid these challenging circumstances, the global solar market is expected to continue growing in 2024, though at a much lower rate than in 2023. Our Medium Scenario forecasts a global growth rate of 22% to a market size of 544 GW, approximately 100 GW more

FIGURE 24 ANNUAL SOLAR PV MARKET SCENARIOS 2023-2024



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⁵ IEA (2024): *Advancing Clean Technology Manufacturing*.

⁶ IMF (2024): *World Economic Outlook*.

than in 2023 (see Fig. 24). As we've seen in recent years, current global market forecasting will depend largely on getting China's solar deployment right due to its outstanding position in the global PV landscape.

China reached new heights in 2023 due to a significant increase in PV module supply and a sharp decrease in module prices. In 2024, production capacity will still increase but not at the exceptional rate witnessed in the past two years. Similarly, module prices are not expected to decrease as steeply as they did from late 2022 to late 2023. China achieved new records in 2023 and raised the bar for solar; 2024 is expected to continue along similar lines, but without the same groundbreaking advancements.

In our Low Scenario for 2024, we project a market size of 461 GW, representing a modest 3% growth from 2023 levels. This very slow growth is based among others on assumptions that demand for solar PV in several regions may not increase as rapidly as anticipated following the energy crisis. Trade disputes and protective measures against cheap imported modules observed in the United States and India boil up and extend potentially to the EU. Interest rates will not decrease, local conflicts aggravate and depress economies. China could limit solar sector growth as its power networks and its energy sector increasingly face issues with the massive volumes of variable solar power systems feeding into the grid.

Our more optimistic High Scenario, on the other hand, projects 45% annual growth to a market volume of 647 GW in 2024. The scenario is based on the following assumptions: policy makers, businesses and individuals have learned their lessons from the energy crisis, with key markets continuing to implement strategic decisions in line with renewable energy commitments made in 2022-2023. A continuous oversupply will further provide very low-price modules, inverters and batteries further unlocking price-elastic demand, igniting strong growth in emerging and new markets. China will not limit growth of its solar market in a strategic move to support its domestic solar industry at a time the country faces low economic growth rates and Chinese manufacturers are confronted with trade barriers in various international markets.

Regional developments 2024

The Asia-Pacific region will continue to be dominated by its two largest markets, China and India. Together, both will account for 84% of the region's installed capacity in 2024, according to our Medium Scenario. Compared to 2023, India's market share is expected to increase slightly from 4% to 5%, while China's share will reach 79% of APAC's deployment, down from 83% in 2023.

After the impressive growth registered in 2023, China's market is expected to slow down in 2024, reaching a market size of 299 GW with an 18% annual increase. For the first time since 2021, this growth rate is lower than the expected global market growth of 22%, although the absolute volumes are so large that the forecasted level of installations still accounts for 55% of worldwide additions, only 2 percentage points less than in 2023. The High Scenario sees the Chinese market growing by 41% year-on-year to reach 356 GW in 2024. However, our Medium Scenario forecast is already optimistic compared to other analysts. CPIA estimates a market size between 190 and 220 GW_{AC}. Assuming a 1.3 DC/AC ratio, this places their highest estimate at 286 GW, which is below our Medium Scenario for 2024. Our upbeat forecast is driven by the significant pipeline of utility-scale projects, fuelled by declining costs of PV system components, and strong market developments in the first months of the year. From January to April 2024, China installed around 60 GW_{AC}, 25% more than in the same period 2023. 2024 will also be the year when China achieves its 2030 National Determined Contribution (NDC) target of 1.2 TW wind and solar capacity – 6 years in advance, and just another example of China overshooting its solar targets. For more insights on the Chinese market, see our focus chapter at p. 61 and the CPIA article at p. 84.

In APAC without China, India accounted for a quarter of all installed capacity in 2023 and is expected to sustain a similar 24% market share in 2024. After the market contraction occurred in 2023, we expect a rebound in 2024 with 19 GW and a 51% growth. This aligns with the government's ambitions since 2022 to reach 280 GW by 2030. Achieving this target would require the annual market to grow rapidly, averaging over 27 GW per year between 2024 and 2030. This growth is also meant to support local manufacturing of solar PV, funded under the government's PLI scheme. Considering India's vast and largely untapped

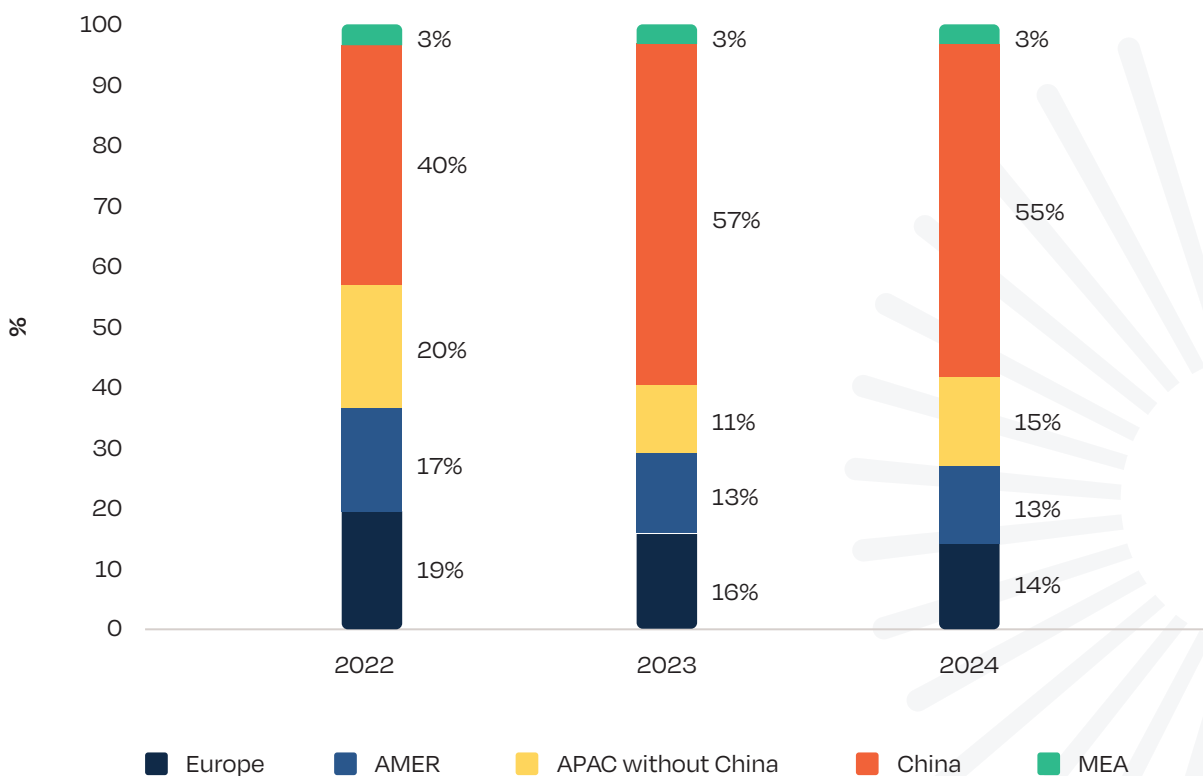
solar potential and the availability of stockpiled products ahead of the re-introduced local content measures for modules, bridging the gap until large volumes of domestically produced products become available, our High Scenario projects an increase to over 24 GW installed in 2024.

Different developments are expected across most Asia-Pacific GW markets in 2024. Japan is projected to grow by 7% to a 6.7 GW market, demonstrating remarkable resilience under less favourable market conditions. Australia is expected to increase by 22% to reach 6.2 GW, while Taiwan is set to reach 3 GW with a 10% growth rate. Pakistan will continue its growth trajectory with a 12% increase in 2024, reaching 1.4 GW. Malaysia is estimated to achieve the GW scale in 2024 with a 22% growth. Additionally, two countries are anticipated to surge from about 100 MW to the

GW-level in 2024: the Philippines, with a eight-fold growth to 1.5 GW, and Uzbekistan, with a nine-fold increase to 1.2 GW, highlighting the dynamism in the APAC market. On the contrary, South Korea's political shift towards inflexible nuclear energy and a massive reduction of its renewable target have negatively impacted its solar market. The market is expected to further progress on its downturn path, by 16% in 2024, reaching 2.5 GW.

APAC including China will dominate global demand even more in 2024, as we expect 2% growth to a global share of 70% (see Fig. 25). But this is likely not to be attributed to China's market share, which is expected to drop slightly from 57% to 55% in 2024. Without China, the region's share is projected to increase by 4 percentage points, from 11% to 15%.

FIGURE 25 REGIONAL SHARES OF GLOBAL SOLAR MARKET 2022-2024



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1 Global solar market - Prospects 2024-2028 / continued

Europe's market is projected to experience continuous growth in 2024. After it added 70.1 GW in 2023, Europe is expected to add 77 GW in 2024, a 10% annual market increase. However, due to China's dominance, the continent's market share will shrink by 2 percentage points to 14%. The EU is driving solar growth through its Green Deal and REPowerEU initiatives, aiming for carbon neutrality by 2050. The invasion of Ukraine by Russia has created momentum for several European countries to prioritise low-cost and versatile solar power as a means to reduce dependence on Russian gas and improve energy security. Today, power prices are getting close to pre-energy crisis levels, and the reduction of gas dependency is no longer the main driver. Instead, enabling regulatory frameworks and a profitable business case for solar PV need to sustain market growth, which, at 5% in the EU-27 for 2024, will be much lower than in the recent "emergency" times.

Within the EU-27, 19 out of 27 markets are projected to install more solar capacity than the year before, in contrast to the 26 markets that grew year-on-year in 2023. Germany's dominance is expected to continue with a 7% market growth to 16.1 GW, demonstrating its maturity after the 104% jump to 15 GW in 2023. Spain and Italy are estimated to remain in second and third positions respectively, with the Spanish market projected to decrease by 12% in 2024 and the Italian market to increase by 13%. The Italian market proves

resilient, with an increase expected despite the lowering of residential incentives, while Spain shows signs of decline in both utility- and residential scale, after solar reached a 17% share in the electricity mix in 2023. Further details on the EU-27 market as a whole are provided on p. 87.

Among non-EU countries, Türkiye is expected to maintain its significant market growth, installing 2.7 GW in 2023 (+69%) and projected to install 4.6 GW in 2024 (+68%). This unprecedented growth pace is also driven by the government's increased ambition to support its local solar manufacturing industry. The recent relaxation of rules surrounding unlicensed installations has created a new wave of installations in early 2024 by enabling a better business case for self-consumption installations. The United Kingdom is set to reach 2.4 GW in 2024 (+33%), cementing its position as the second-largest non-EU market. In third place, rooftop-driven Switzerland is anticipated to add 1.9 GW in 2024, a 23% annual growth. Overall, Europe's solar market is poised for growth, driven by legislative initiatives and energy transition plans that will drive further electrification, grid capacity, and flexibility measures. Despite facing challenges such as regulatory instability and grid congestion, Europe will remain an important market on the global solar map in 2024 and beyond. This is especially true for international manufacturers, who find it easier to export their solar products to this region compared to other large markets.



Residential PV in Germany.

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The Americas’ market share will remain stable at 13% in 2024, allowing the continent to maintain its position as the third largest region for solar installations, with a total of 70.3 GW. This market share closely aligns with Europe’s, as the Americas will see a strong increase of 11.5 GW in absolute installations (see Fig. 26). The primary driver behind solar deployment in the Americas is the United States, which is expected to hold a 59% share in 2024, up from 55% in 2023.

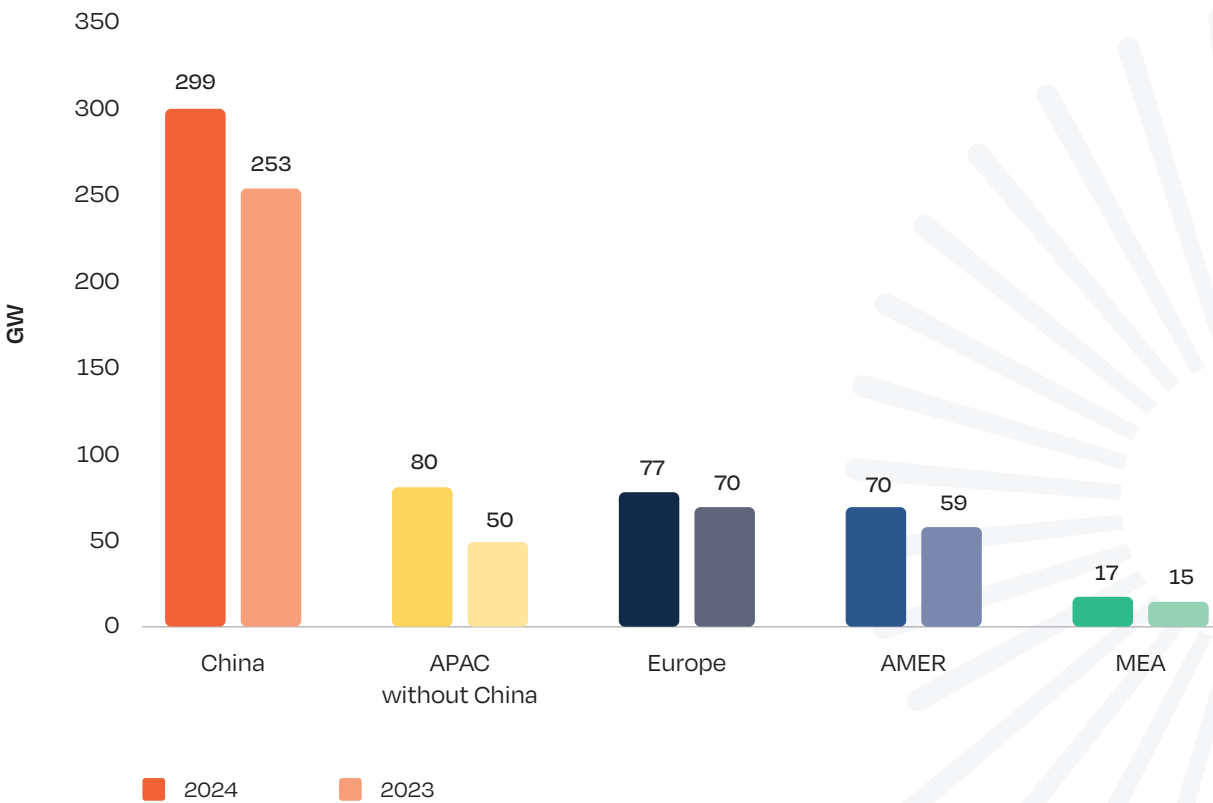
The August 2022 passed IRA demonstrates the Biden Administration’s strategy to transform the US into a global clean energy powerhouse. The ITC and recent interconnection reforms are major drivers of solar growth, both on the installation and manufacturing side. Bolstered by the IRA, significant investments in new factories have been announced by both domestic companies and solar technology leaders from abroad.

Despite a number of challenges, including increasing PV costs, trade tensions, the reform of net metering in California, and the upcoming presidential election, the United States remains one of the most attractive locations for solar investments today. Our Medium Scenario predicts a 27% growth in solar capacity to 41.3 GW this year.

In Brazil, the second most important market in the Americas, the outlook for this year appears somewhat less promising than last year, which showed exceptional growth. In 2024, the less favourable investment environment and the 2023 change in net-metering legislation are set to impact the market, which we expect to decrease 4% to 14.9 GW.

Chile and Mexico shared the fourth place in the Americas in 2023, with 1.7 GW installed each. While Chile’s market decreased by 10% from 1.8 GW, Mexico’s

FIGURE 26 REGIONAL PV DEVELOPMENTS 2023-2024



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1 Global solar market - Prospects 2024-2028 / continued

market increased by 40% from 1.2 GW, despite the challenging political environment for renewable energy deployment. However, the conditions of 2023 do not apply to 2024, and an 18% market decline to 1.4 GW can be expected in Mexico. On the other hand, Chile is set to regain pace, with an 8% growth reaching 1.8 GW in 2024, although the lack of transmission lines still pose a major challenge for mid-term solar market growth.

The increase in installations in 2023 resulted in a stable market share for the Middle East and Africa at 3%. The situation is expected to be similar this year. With the addition of 17.1 GW, the region is projected to experience a 15% annual growth. The main contributor is South Africa, which, for the second year in a row, may add more than one-fifth of all MEA installed capacity, reaching 3.5 GW. Unlike last year, this capacity is not matched by the United Arab Emirates, which will no longer be among the three GW-sized markets. In 2024, the UAE is expected to account for 5% of the installed capacity (0.8 GW), compared to 21% in 2023 (3.1 GW), simply because no major PV power plant projects are expected to be grid-connected this year.

Besides South Africa, Saudi Arabia is projected to be the other GW-sized market in 2024. After growing 400% in 2023 to a level of 1.9 GW, in 2024 the market is expected to decline by 20%, which means still GW-scale at 1.5 GW. Despite this, many smaller markets in the region, all with very favourable irradiation conditions, are increasingly recognising the cost advantages, business opportunities, and energy

security potential of solar power, leading to an upsurge in solar activities throughout the area. As such, 47 out of 63 analysed markets are forecasted to experience positive market growth, compared to 39 in 2023.

Global solar market developments 2025-2028

We anticipate substantial demand growth for solar PV power in the years 2025-2028, driven by further cost improvements, product availability, and the numerous benefits the technology provides. Climate emergency will continue to capture governments' attention, while energy security will remain a compelling argument to invest in solar power in a quickly fragmenting world order.

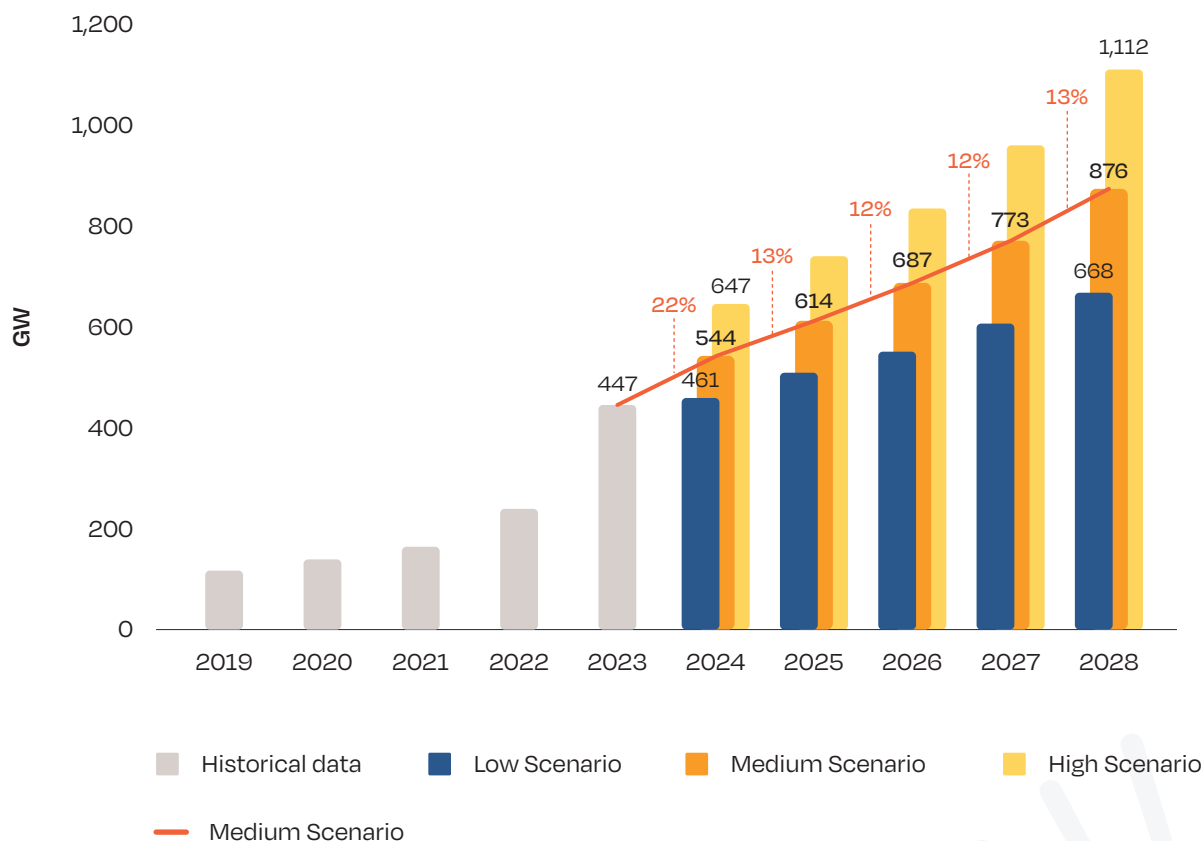
At COP28 in December 2023, policy leaders committed to tripling global renewable energy capacity to at least 11 TW by 2030 and doubling the annual rate of energy efficiency improvements from 2% to 4%. According to the IRENA, this target equals to around 5.5 TW solar capacity, and highlights substantial growth opportunities for solar PV, which is expected to contribute significantly (see Fig. 34 for an overview of PV cumulative capacity scenarios by 2030). As the pledge includes only few details about the required infrastructure, energy storage or solar targets, it remains hard to judge its actual impact, particularly in developing countries. Nonetheless, it creates certainty about the future global solar market demand, and a growing trend is all we can see in the near-term horizon. The real question is the inclination of that growth curve.



107 MW, Port Augusta, Australia.

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FIGURE 27 WORLD ANNUAL SOLAR PV MARKET SCENARIOS 2024-2028



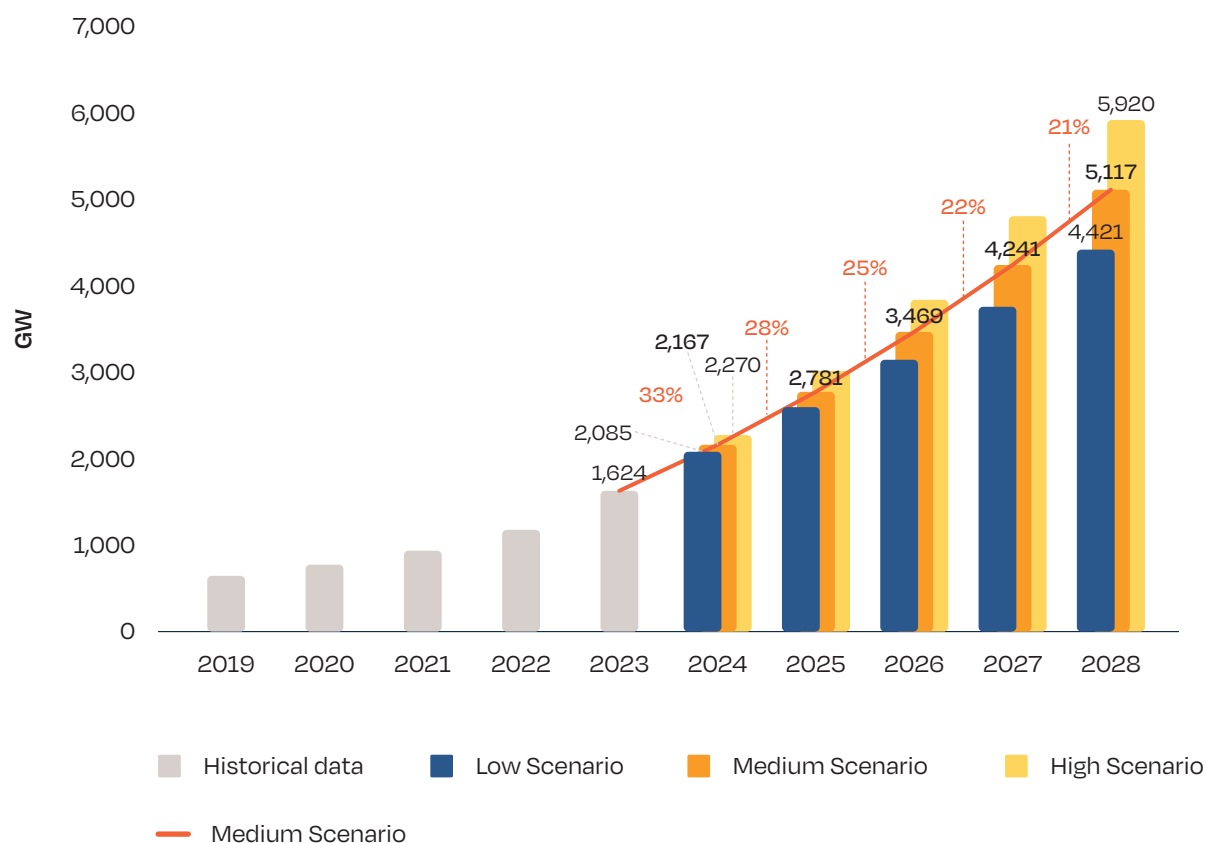
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Compared to our previous Global Market Outlook, aggregate annual solar additions between 2025 and 2027 have been increased from 1.6 TW to 2.1 TW. At the same time, annual growth rates have been lowered, reflecting our assessment that the surge in 2023 could have been the beginning of another market development phase after an extraordinary growth period fuelled by post-pandemic product availability and the energy crisis. We expect slightly lower growth rates in the range of 12-13% between 2025 and 2028, compared to previously forecasted 15-16% in the period 2025-2027. Our Medium Scenario projects the global market to reach 614 GW in 2025, a 13% increase from 544 GW in 2024 (Fig. 27). Looking forward, the Medium Scenario anticipates 12% increase to 687 GW in 2026, 12% to 773 GW in 2027, and 13% to 876 GW in 2028.

In the High Scenario, higher adoption of solar across most regions leads to annual growth rates of 15-16%, lifting annual additions beyond the 1 TW level already by 2028. This scenario is based on similar considerations as for 2024. We assume that energy market designs are quickly adapted to the needs of renewables, and grid infrastructure, storage and flexibility solutions will be deployed rapidly to limit curtailment and falling capture rates of solar power. On the contrary, our Low Scenario assumes that necessary policy frameworks and infrastructure upgrades continue to lag behind, while changes to less progressive governments, and culminating trade issues negatively impact solar expansion. Such a scenario translates into lower growth rates of 8-10% to reach annual installations of 668 GW in 2028, 40% less than in the High Scenario, and 24% less than our Medium Scenario.

1 Global solar market - Prospects 2024-2028 / continued

FIGURE 28 WORLD CUMULATIVE SOLAR PV MARKET SCENARIOS 2024-2028



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After surpassing the 1.6 TW level in 2023, the global solar power fleet is on track to exceed 2 TW by 2024 (see Fig. 28). Our Medium Scenario estimates the following milestones: 2.2 TW in 2024, 2.8 TW in 2025, 3.5 TW in 2026, 4.2 TW in 2027, and 5.1 TW in 2028 – numbers that are all significantly higher than forecasted in last year's GMO (see Box 1 at p. 78).

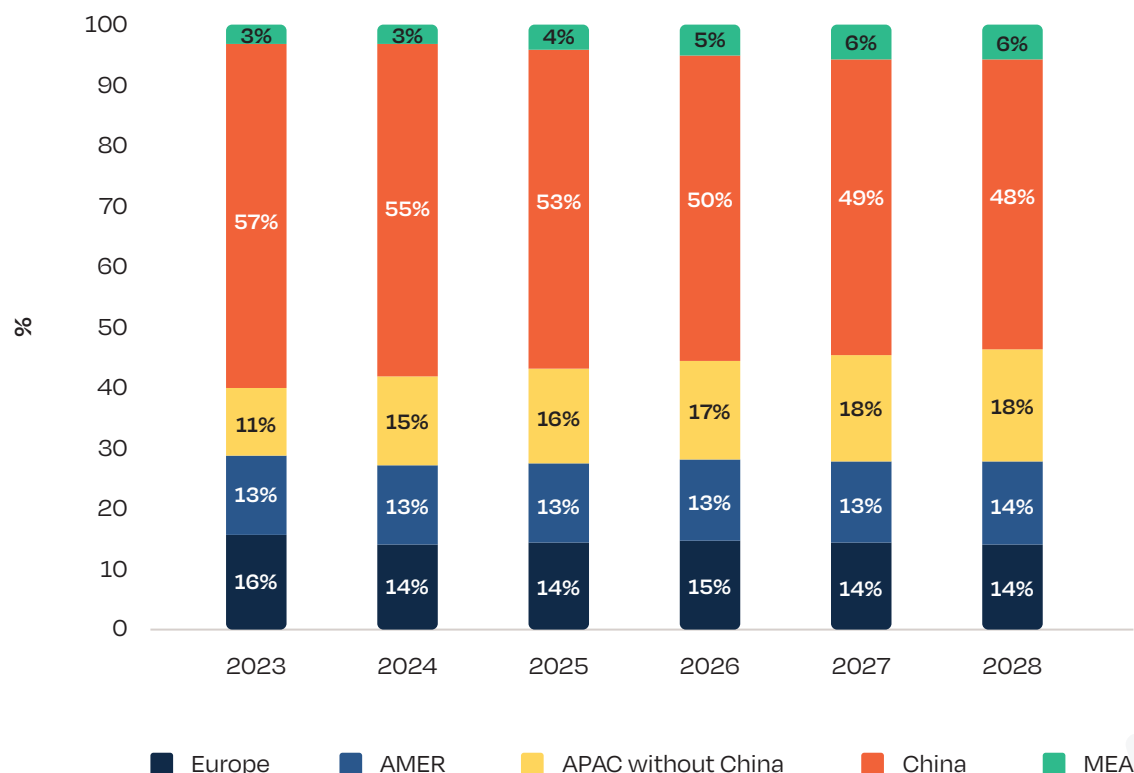
Under optimal conditions, total global solar capacity could even reach 3 TW in 2025 and potentially 5.9 TW by the end of 2028, while this year's Low Scenario in 2027 is more ambitious than our Medium Scenario in the previous GMO – 3.8 GW vs. 3.5 GW. In our Low Scenario, total solar capacity in 2028 reaches 4.4 GW, which is nearly three times the total capacity in 2023.

From 2024 to 2028, the global solar PV market is expected to see regional growth dynamics somewhat change compared to last year's GMO. After peaking at

57% in 2023, China's share is projected to progressively decline – to 55% in 2024, 50% in 2026, and 48% by 2028. We had forecasted that trend earlier – and although China proved us wrong several times, we believe that today's very low capex needs for systems paired with attractive LCOEs will create a strong momentum for emerging and new markets. Our latest market share expectations for China over the coming years are 14 to 15 percentage points higher than in last year's GMO.

Despite the European Union's 2030 solar targets and several policy framework improvements for the technology, solar in Europe is anticipated to grow slightly slower than in other regions. It will lose 2% market share, from 16% in 2023 to 14% by 2027 and 2028, significantly below the 19% we had assumed earlier for 2027.

FIGURE 29 EVOLUTION OF GLOBAL ANNUAL SOLAR PV MARKET SHARES UNTIL 2028



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Conversely, the Americas will defend its 13% share for the next four years before it gains 1 percentage point to 14% in 2028. The United States and Brazil will continue to be the foundation of the continent's growth, which will be flanked by more contributions of more Latin American countries in the coming years.

The APAC region without China is expected to experience the largest absolute regional growth. Due to its high population and largely untapped solar potential, we see its share increasing from 11% in 2023 to 18% in 2028. India will stand out, with its market growing nearly four-fold between 2023 and 2028.

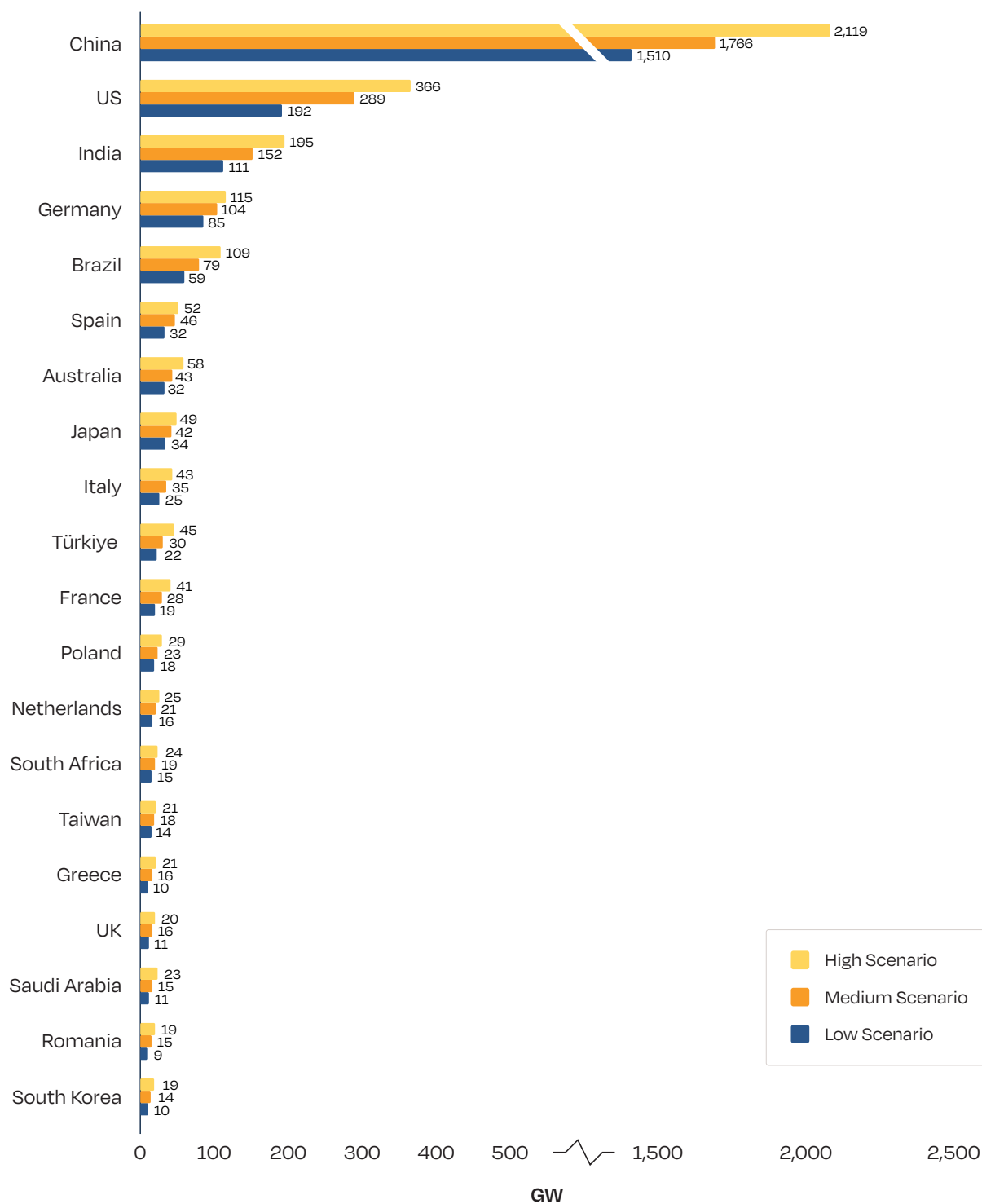
Though on a much smaller level, MEA is projected to see the largest relative gain, doubling its market share from 3% in 2023 to 6% in 2028, which will be driven by large projects in the Middle East.

The composition of the top 20 markets with the highest 5-year installation potential sees only minor changes from the previous edition (see Fig. 29). Among the top 10, Brazil surpassed Spain to enter the top 5, Italy moved up from 10th to 9th position, and Türkiye entered the top 10, rising from the 18th place last year. The second half of the ranking has two newcomers – South Africa and Romania replaced the UAE and Chile.

Again, this graph shows the outstanding position of China, which is expected to add almost 1.8 TW in the most likely scenario during the period 2024-2028. This is more than double last year's expectations of 873 GW and exceeds the 2023 global cumulative solar PV installations of 1.6 TW. It is also 75% higher than the combined additions of all other countries in the ranking and six times more than the United States, the second largest market, which is expected to add

1 Global solar market - Prospects 2024-2028 / continued

FIGURE 30 TOP 20 MARKETS SOLAR PV ADDITIONS 2024-2028



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1 Global solar market - Prospects 2024-2028 / continued

289 GW (+37 GW compared to last year's edition). Following China and the United States, India is estimated to install 152 GW (+7 GW) and Germany 104 GW (+16 GW). Each of these four leading countries is projected to add over 100 GW in the next five years, up from three countries last year.

Driven by China, the top 20 global markets are expected to install 2.8 TW over the next five years, which is 1 TW more than projected in our Global Market Outlook 2023, representing an 80% increase. Collectively, these markets will account for 88% of global new installations, up from 83% last year. Looking at the top 20 markets without China reveals a different growth trajectory. Our picture hasn't changed much as we expect these 19 countries to install just over 1 TW, an 11% increase from 903 GW projected in the last GMO.

The minimum 5-year capacity additions to enter the top 20 has increased again: countries need at least 14 GW, compared to 12 GW last year. Moreover, 13 countries are now expected to install at least 20 GW, up from 12 countries last year.

Under the Low Scenario, the top 20 markets are forecasted to add 2.2 TW until 2028, while in the High Scenario they are expected to install 3.4 TW, of which China would contribute 2.1 TW.

Our weather forecast for the top 20 countries anticipates a sunny policy and market outlook in most of the world's leading solar markets. Only four countries are facing clouds or even rain on the horizon, one more than last year, while two countries are now expected to experience single-digit annual growth rates, compared to one last year (Fig. 31).

For most of the listed countries, we expect compound annual growth rates (CAGRs) around 20% or above and, as last year, the highest five-year growth rate is expected in Saudi Arabia, now at 42% CAGR. Long delayed, the sunny oil nation reached the GW-scale for newly installed capacity for the first time in 2023, and the completion of numerous GW-scale projects might mark the turnaround point towards reaching its 40 GW target by 2030. At the other end of the spectrum, Japan stands out again with a relatively low CAGR of only 8%, the same as last year. Still, Japan is a mature solar market with a large installation base, and our growth projections for the next five years see additions of nearly 42 GW, more than most other countries in the top 20 list.

Our weather radar shows rain only for South Korea, which, along with Japan, is the only country with a one-digit CAGR. But South Korea has installed much less solar capacity than Japan. The government's decision to prioritise expansion of its nuclear fleet over renewables in 2023 was accompanied by a strong reduction of its overall renewable energy target from 30.2% to 21.6% by 2030. Ranked 14th in last year's report, we see South Korea now as the least promising of the top 20 solar markets. Its CAGR decreased from 11% to 9% after our installation estimate was cut to 13.8 GW, nearly half of what we anticipated one year ago.

France's solar story has been mostly disappointing considering its huge potential – being one of Europe's leading economies with vast space and plenty of sunshine. While the majority of upcoming installations are anticipated to be ground-mounted, they are meant to take place in designated go-to areas, which are not yet fully defined. Agri-PV also holds significant potential but remains a subject of heated debate in the country, currently hindering the emergence of very large solar power plants. Considering the slow pace of legislative progression, we have a cloudy outlook for the country, which, by 2028, is expected to have only less than two-thirds of the installed capacity of Spain, and less than a third of the German installed power fleet.

Despite a 17% CAGR and higher expected installed capacity, some clouds are appearing on the Italian skies. In May 2024, the government enacted a decree banning large-scale solar installations on productive agricultural land, although the ban is not applied retroactively, protecting some of the projects in the pipeline, and Agri-PV projects are not affected. Still, the government intervention poses a significant barrier on the way to meet the country's 2030 solar target.

New uncertainties also loom over the Netherlands as the new government plans to terminate the country's successful net-metering scheme for residential customers. Now set to end in 2026, the country is expected to face a boom-and-bust period in the next few years, something the industry would have liked to avoid. It rather preferred a gradual phase-out enabling the market to adapt and provide time to build up the much needed residential storage market, which has been incompatible with the net-metering scheme.

The cloudy skies we saw last year over Brazil and Türkiye have now cleared. Changes to the net-metering scheme for distributed systems in early

FIGURE 31 TOP SOLAR PV MARKETS' PROSPECTS

Country	2023 Total capacity (MW)	By 2028 Total capacity Medium Scenario (MW)	2024-2028 New capacity (MW)	2024-2028 Compound annual growth rate (%)	Political support prospects
China	656,045	2,422,149	1,766,104	30%	
United States	173,185	462,443	289,258	22%	
India	90,069	241,738	151,669	22%	
Germany	82,979	186,498	103,519	18%	
Brazil	39,443	118,356	78,913	25%	
Spain	36,273	82,251	45,978	18%	
Australia	36,109	79,043	42,934	17%	
Japan	90,357	131,889	41,532	8%	
Italy	29,844	64,388	34,544	17%	
Türkiye	12,239	42,553	30,314	28%	
France	18,912	47,401	28,489	20%	
Poland	16,832	39,442	22,610	19%	
Netherlands	22,916	44,061	21,145	14%	
South Africa	9,291	28,634	19,343	25%	
Taiwan	12,443	30,591	18,148	20%	
Greece	7,149	23,436	16,287	27%	
United Kingdom	17,697	33,419	15,722	14%	
Saudi Arabia	3,181	18,649	15,468	42%	
Romania	3,350	18,130	14,780	40%	
South Korea	27,252	41,100	13,848	9%	

2023 initially raised concerns about Brazil's continued market growth. But the 41% growth in 2023 and the installation of 7.5 GW in the first quarter of 2024 have eliminated our doubts for the moment. Regarding Türkiye, important regulatory amendments have

effectively simplified bureaucratic procedures for consumers to harness solar energy for self-consumption. Mainly thanks to its growing rooftop PV segment, the country is poised to almost double its 2023 annual market by adding 4.5 GW this year.

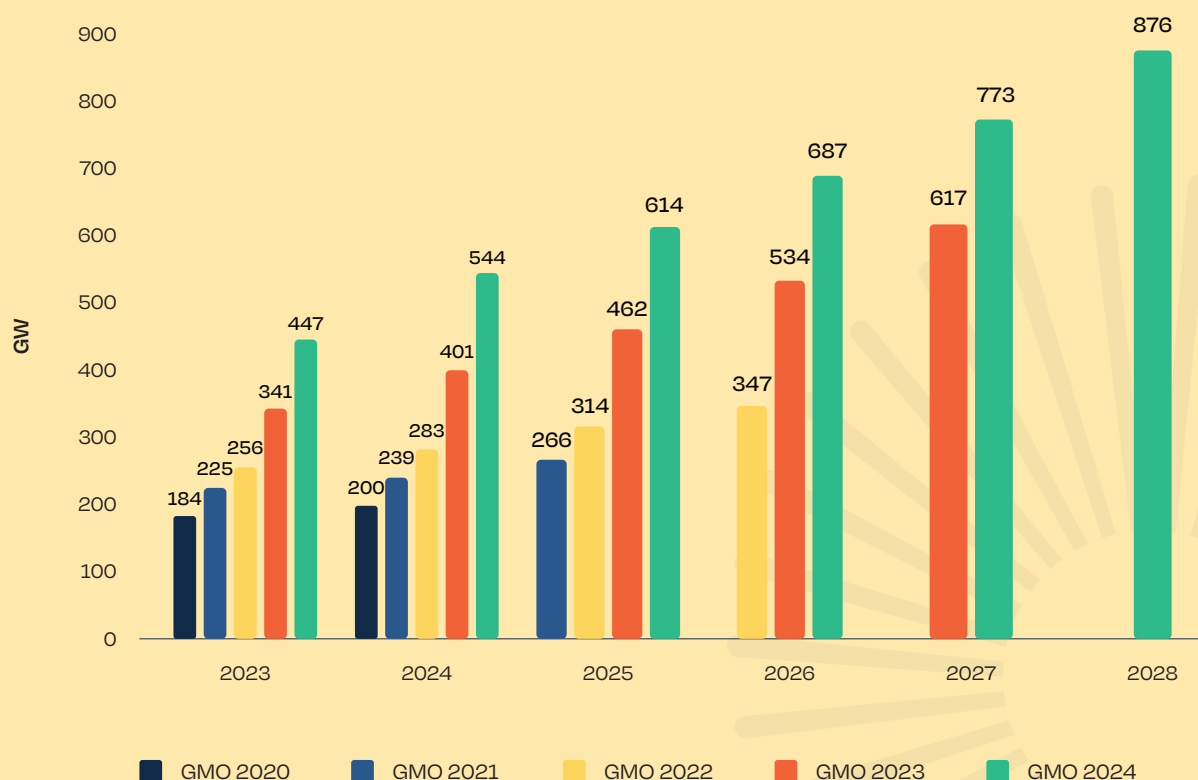
Box 1: Looking back and forth

This year's Global Market Outlook offers a significantly more upbeat outlook than the GMO 2023 projections, which underestimated the acceleration in the solar industry, despite our previous forecast was more optimistic than most solar analysts'. As shown in Fig. 32, our annual market forecast of 341 GW for 2023 in last year's GMO turned out to be 106 GW lower than actual installed solar capacities. This strong underestimation comes mainly from getting one market's growth wrong – the Chinese solar market grew 112 GW more than expected. Without China, our estimates were almost right for the “rest of the world”,

which installed just 3% less than our 200 GW forecast. However, the graph shows a history of underestimating solar market growth, and upward revisions for each new edition.

The forecast for newly installed capacity in 2024 has been revised to 544 GW, a 143 GW increase from the previous year's projection of 401 GW. In our GMO 2021, our 2024 outlook was less than half the new capacity we expect today. Just one edition ago, surpassing 500 GW was estimated to occur only by 2026. Our latest Medium Scenario outlook has increased by 33% to 614 GW for 2025, by 29% to 687 GW for 2026, and by 25% to 773 GW for 2027. Our current Medium Scenario forecasts are roughly twice as much as we expected only 2 years ago, in the GMO 2022.

FIGURE 32 COMPARISON MEDIUM SCENARIO GMO 2024 VS PREVIOUS GMO EDITIONS



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Segments 2023-2028

The key drivers behind the rooftop and large-scale segments have shifted dynamics in recent years. In 2022, high module prices forced developers to postpone installations of large-scale projects, while high retail electricity prices boosted rooftop deployment. A year later, in 2023, a sharp decrease in module prices spurred large-scale installations, while a 'normalisation' trend of energy prices towards pre-energy crisis levels slowed demand for rooftop systems toward the end of the year.

While both the utility-scale and rooftop segments experienced strong growth in 2023, utility-scale solar stood out as the largest and most dynamic segment. The decrease in module prices was a significant factor, backing a 110% increase to 251 GW in the segment, up from 120 GW the previous year. Many projects that had been delayed due to high module prices, supply chain and EPC bottlenecks in 2021-2022 were realised in 2023. Looking at markets, the key factor was the surge of large-scale projects that were connected at the end of 2023 in China, growing by 236% year-on-year. Large-scale solar in China accounted for 58% of all utility-scale projects connected to the grid worldwide. The year also marked a turning point for the world leading solar market, as the large-scale segment surpassed rooftop capacity additions, with 146 GW in 2023 (+103 GW compared to 2022), compared to 107 GW for rooftop installations (+56

GW). Globally, the utility-scale segment represented 56% of all new installations in 2023, up from 50.1% in 2022.

Numerous other countries made significant progress in their utility-scale solar installations in 2023. The United Arab Emirates and Saudi Arabia both connected several utility-scale plants, achieving 2.9 GW (+544%) and 1.7 GW (+383%) respectively. Both countries are almost fully relying on large-scale solar projects to meet their climate goals. Several other countries also managed to double their utility-scale market: Sweden (+403%), South Africa (+136%), Italy and Poland (+103% each). While the United States experienced an 18% decrease in annual installations in 2022 due to various trade and supply challenges, the support from the IRA helped the segment to recover. The United States large-scale segment grew to 22.6 GW in 2023, up from 13.8 GW in 2022 (+64%). This growth is expected to continue, reaching 55.9 GW by 2028, maintaining its position as the second-largest utility-scale market in the world after China.

In 2024, we anticipate the worldwide solar utility-scale segment to achieve 300 GW, reflecting a 20% annual growth rate, and market share of 55%. Over the coming years the ground-mounted solar segment will continue its ascent – reaching 346 GW (56% share) in 2025; 391 GW (57%) in 2026; 446 GW (58%) in 2027; and 513 GW (59% share) in 2028.



100 MW, Puertollano, Spain.

© Iberdrola

1 Global solar market - Segments 2023-2028 / continued

While there is increasing resistance to large-scale power plants in densely populated countries, stakeholders have been working on manifold solutions – from involving locals in the planning process, hiring residents for construction and even sharing some profits. Industry associations have created best practice guidelines to enable solar plant building in harmony with nature to support biodiversity and optimise land-use. Though still in their infancy, multiple-use applications like Agri-PV and Floating Solar or hybrid renewable plants are further diversification trends for large-scale solar. However, where space is no issue, we are increasingly seeing very large-scale ground-mounted power plant large-scale ground-mounted power plants, that will be increasingly linked to green hydrogen production are likely to contribute to this trend in the final years of the decade.

Rooftop Solar

In parallel, the growing rooftop market continued to demonstrate strong interest from the residential and commercial sectors to generate and self-consume electricity. Despite the lingering effects of the energy crisis through 2023, the huge unbuilt project pipeline and quickly falling product prices spurred installations for a large part of the year. This resulted in 196 GW being connected in 2023, up from 120 GW the previous year. With the biggest driver – very high energy prices – continuously decreasing, rooftop PV systems lost

market share, representing 44% of all new PV installations in 2023, down from 49.1% in 2022.

The rooftop segment showed varying trends across different countries, with several among the top 20 solar markets installing fewer rooftop systems than in the previous year. The most significant declines were observed for Spain (-43%), India (-32%), and Poland (-25%). While the first two have been traditionally ground-mount solar territory, Poland's solar market has been built on rooftop systems. In both European countries, issues with incentive schemes were the main cause for the hefty market share losses. Though the share of China's rooftop segment contracted in 2023, it was still 109% larger than the year before, and leading the world by far with 107 GW of newly building-attached capacity.

Most of the major solar markets experienced strong growth in their rooftop segment, in particular in Europe, that was most heavily impacted by the energy crisis. Many countries had improved policy frameworks to accelerate solar rooftop installations – and these were often still available in 2023. The largest growth in Europe's rooftop segment took place in Austria (+175%), Germany (+138%), Italy (+113%), and Sweden (+93%). The South African rooftop market also surged with three-digit rates (+164%). Triggered by numerous load-shedding events throughout the year, households and businesses increasingly invested in solar PV and battery storage to protect themselves from very frequent blackouts.



O&M at Adelschlag Solar park, 3.3 MW. Adelschlag, Germany.

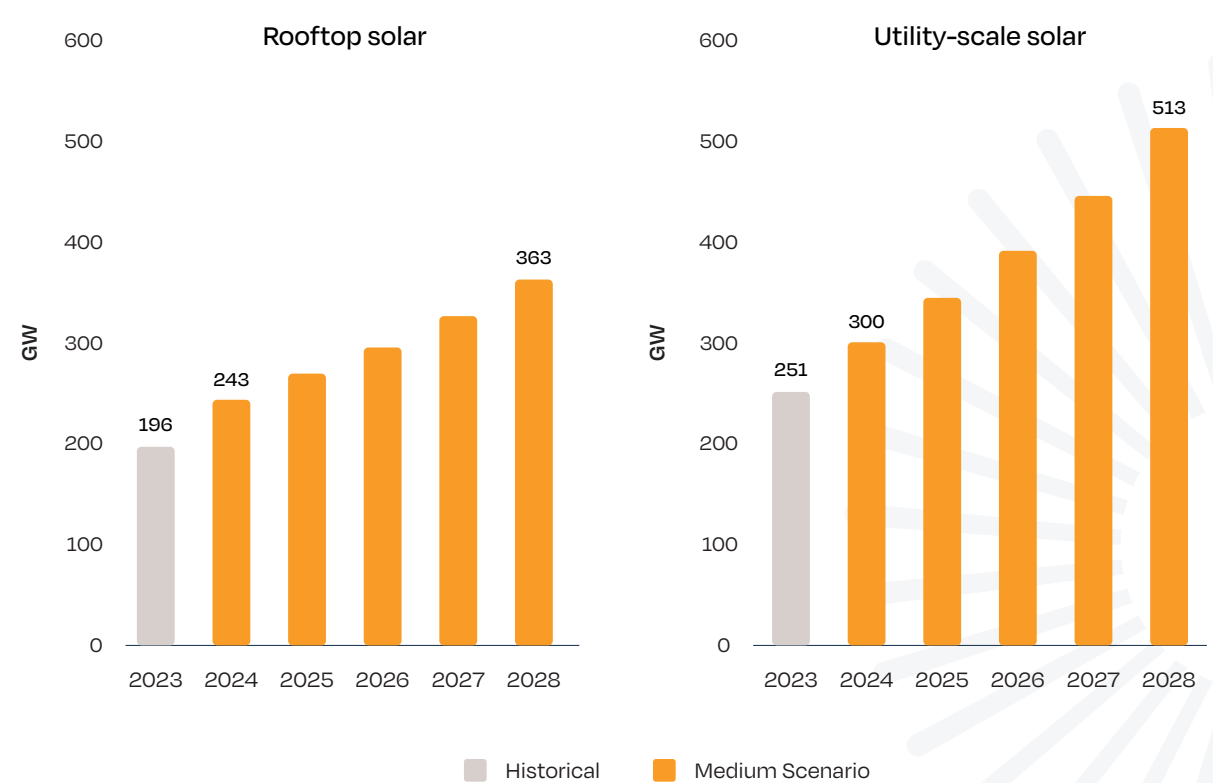
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The introduction of solar mandates to put solar on buildings in several countries could further boost the rooftop market, as could the electrification of the transport and heating sector, with many countries promoting heat pumps to decarbonise heating. Additionally, residential and commercial power consumers are evolving into prosumers, PV modules are starting to become building materials, and smart cities are embracing small-scale distributed solar combined with storage and digital solutions. Several sub-national actors have followed California's example of mandating solar installations in new-build homes. In March 2024, the European Parliament adopted the solar mandates as part of the Energy Performance of Buildings Directive. As of 2026, Member States will have to gradually implement requirements for solar installations in

building renovations, and as of 2030 public bodies must retroactively install PV on their buildings.

Consequently, the forecast for rooftop solar development looks bright. In 2024, the rooftop segment is expected to increase to 243 GW, representing a 24% growth from 2023. In the following years, we anticipate a slight slowdown in the growth rate, with annual increases ranging from 10% to 11%. As energy prices return to lower levels in Europe, we are already observing a slowdown in demand in the first months of 2024. Nevertheless, the annual installed rooftop capacity is forecasted to increase to 268 GW in 2025, and up to 363 GW in 2028 in our Medium Scenario, representing an 85% growth from 2023 levels.

FIGURE 33 SOLAR PV ROOFTOP AND UTILITY-SCALE SEGMENTS SCENARIOS 2024-2028



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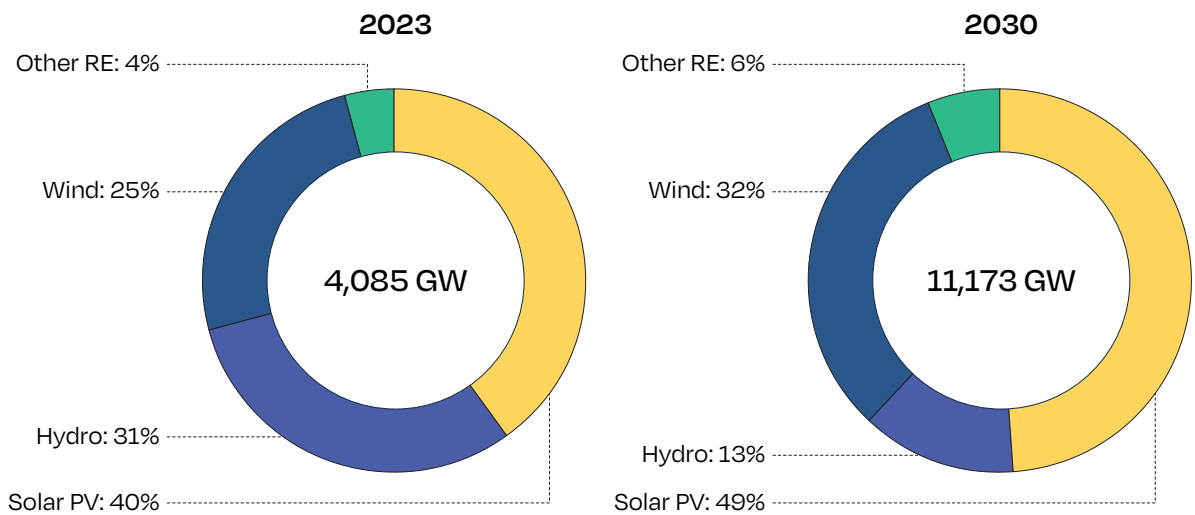
1 Global solar market - Solar outlook to 2030

Solar outlook to 2030

Our GMO forecasts traditionally have a 5-year horizon, which ends in 2028 for this edition. As 2030 is considered an important milestone on the way to climate neutrality later this century, several national and multi-national government bodies have set renewables and climate targets for that year; a good reason for us to look into the solar goals and market estimates of major stakeholders.

A key achievement of COP28 in Dec. 2023 was an agreement of over 130 countries, including the G7, on a first concrete target for renewables by 2030 – tripling capacity to 11 TW. Though no breakdown for the different technologies was given, solar will play the major role, as it has been adding more capacities than any other power generation technology for the past years. IRENA estimates that solar PV will contribute 5.5 TW or half to the 11 TW in 2030, up from 1.6 GW at the end of 2023 (see Fig. 34).⁷

FIGURE 34 GLOBAL INSTALLED RENEWABLE ELECTRICITY GENERATION CAPACITY IN THE 1.5 C° SCENARIO, 2023 AND 2030



Source: SolarPower Europe (2024), IRENA (2024).

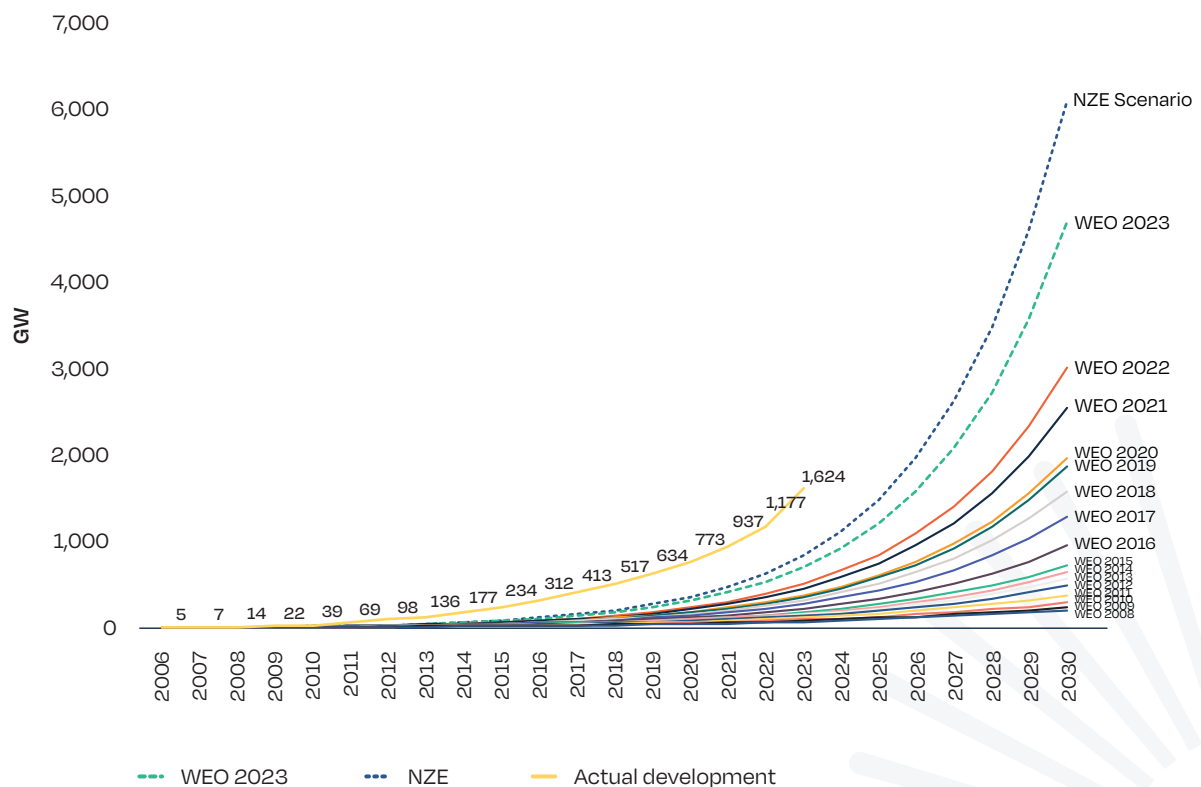
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7 IRENA (2023): [World Energy Transitions Outlook 2023](#).

But is this estimate for 2030 realistic? What seems ambitious at first sight, would mean average solar additions of 558 GW per year to install 3.9 TW by 2030, an annual level we expect to be almost met in 2024. Would it be realistic that there will be no growth for any of the coming years until 2030?

To put things into perspective: any global solar market analysis has been severely underestimating growth, that included us (see Box 1 p. 48), though the best-known example is the IEA annual flagship report, the World Energy Outlook (WEO) (see Fig. 35).

FIGURE 35 IEA WORLD ENERGY OUTLOOK FORECASTS VERSUS ACTUAL HISTORICAL DEVELOPMENT OF SOLAR PV



Source: IEA (2008-2023). Note on scenarios considered: for WEO 2008 and 2009, the Reference Scenario; for WEO 2010 to 2018, the New Policies Scenario; for WEO 2019 to 2023, the Stated Policies Scenario; NZE is the Net Zero Emissions by 2050 Scenario, updated in 2023.

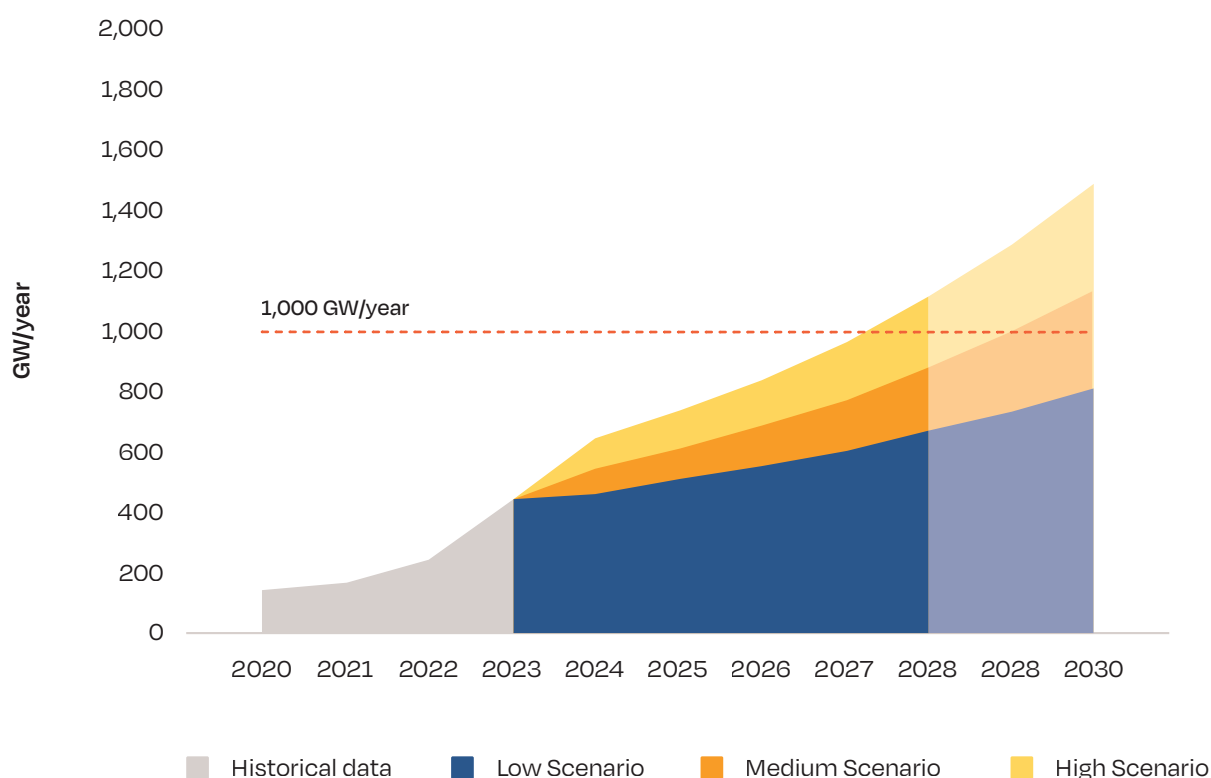
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1 Solar outlook to 2030 / continued

When analysing historical market data between 2010 and 2023, PV additions almost doubled every 3 years. The deployment pace even accelerated in the 2 years after the pandemic to annual growth rates of 46% in 2022 and almost 90% in 2023. Simply extrapolating our 2024-2028 scenarios to 2030 shows annual TW

market levels before 2030 for both our Medium Scenario and High Scenario. Our Medium Scenario would reach the TW mark in 2029, the optimistic one already in 2028; only the Low Scenario would remain below the TW level by 2030, reaching 810 GW at that time (see Fig. 36).

FIGURE 36 SOLAR GROWTH SCENARIOS – TOWARDS 1 TW ANNUAL GLOBAL MARKET BY 2030



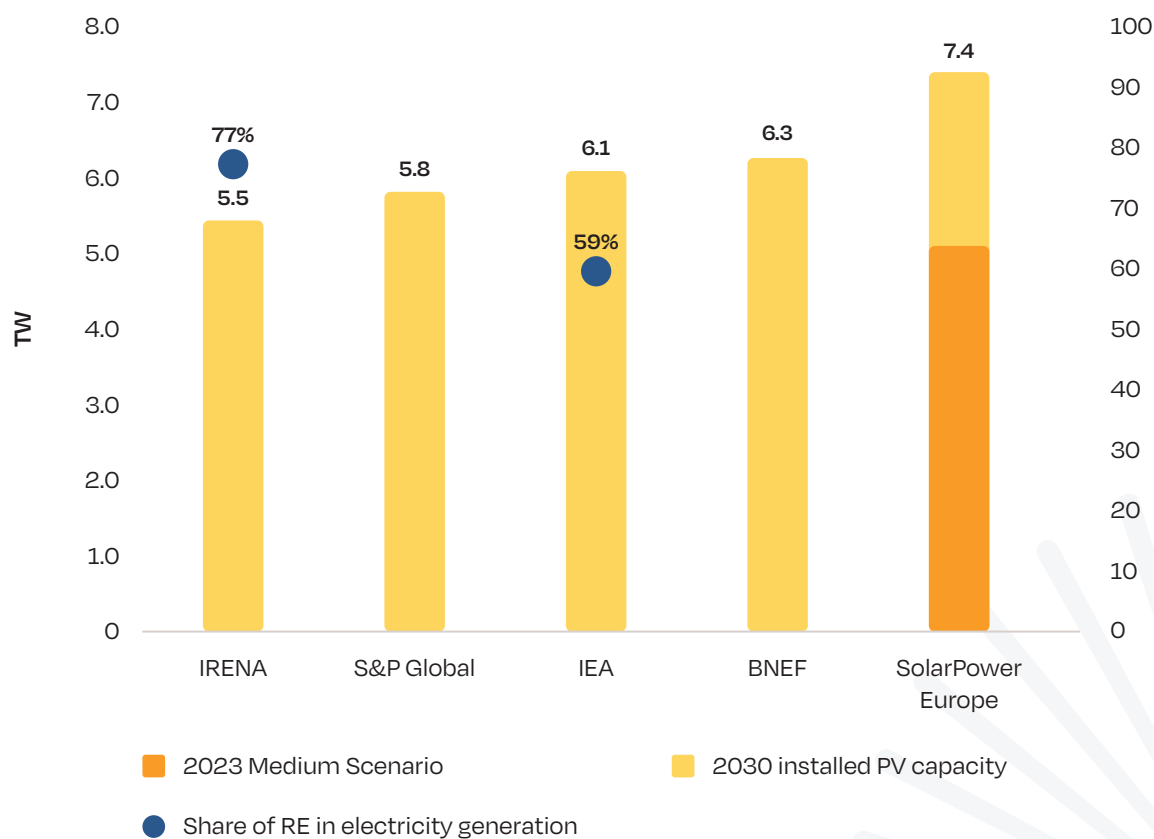
Note: Installation volumes beyond 2028 are based on a simple extrapolation of each 2024-2028 scenario to 2030.

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However, any of the most recent 2030 estimates of leading solar analysts has been somewhat higher than the numbers in IRENA's COP28 report, with

BloombergNEF presenting the most optimistic assumptions of 6.3 GW, when excluding our "simple" extrapolation of our Medium Scenario that stands out at 7.4 GW.

FIGURE 37 GLOBAL SOLAR PV CUMULATIVE INSTALLED CAPACITY SCENARIOS 2030

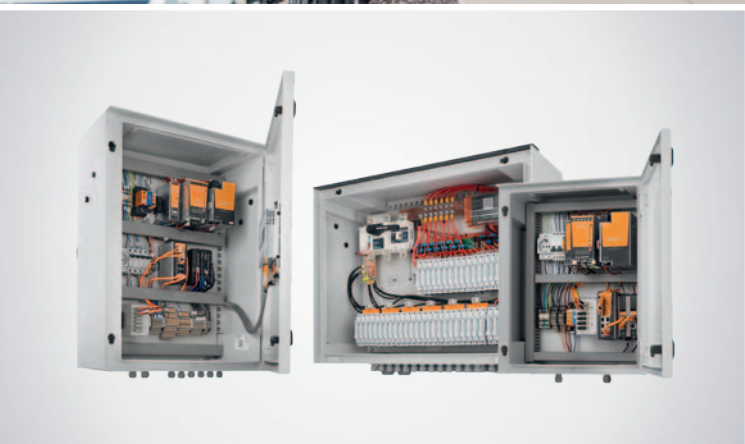


Source: IRENA (2023), S&P Global (2024), IEA (2023), BNEF (2024) and SolarPower Europe (2024).
Note: SolarPower Europe value reflects a simple extrapolation of the GMO Medium Scenario 2024-2028 to 2030.

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2

GSC Policy recommendations

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Policy Recommendations for Global Solar PV Deployment

Solar PV's scalability and cost-effectiveness make it an ideal solution to keep pace with growth in power demand and address the needs of the urgent and just global energy transition to clean energy. Despite the exponential growth of solar PV in the last decade, several challenges remain that hinder deployment to its full potential. The Global Solar Council encourages policymakers to urgently adopt the recommendations below in order to foster a supportive policy environment for solar at the global level and address the barriers slowing down its deployment:

- **Encourage solar PV deployment across all market segments, and set national targets that match its full potential**

Realising the deployment potential of solar PV will require clear, ambitious, and binding targets, supported by policies, regulations and financing mechanisms. Clear and stable policies are essential for attracting investors and low-cost financing. Retroactive changes should, by all means, be avoided as this will spoil investor appetite and stall market growth. Whilst support schemes such as feed-in tariffs, or green certificates can help boost solar deployment in the short term, they must be complemented with structural regulatory improvements in auction schemes and electricity market reform. In addition, governments should support solar innovations, and new applications such as off-grid solar, Agri-PV, floating PV, and Building Integrated PV to fully unleash the solar potential in their countries..

- **Ensure finance flows align with investment needs**

Investments are not flowing at the pace, scale, or to the geographies, needed for global energy access or achieve the energy transition. 12 trillion USD are needed to deliver the global target of tripling renewable energy by 2030. It is crucial to increase public funding for solar PV projects and encourage private investment to achieve low-cost solar finance. Multilateral Development Banks and bilateral Development Finance Institutions should be supported to scale up their project financing and de-risking instruments, especially to enable solar in regions where energy access remains a challenge. We also need to engage as a sector with the major private investors in order to resolve any challenges they have when investing in PV e.g. small project sizes, standardisation, ESG.

- **Unleash a flexibility revolution by scaling-up investment in battery storage, grids and electrification**

The unprecedented pace of newly added solar and renewables is rapidly and fundamentally changing energy systems, and requires next level investment in energy system flexibility. Failing to plan for these changes equals planning for failure. As renewables grow at all voltage levels and electrification accelerates – offering new and high potential for flexibility – grid planning is more important than ever.

Coordinating grid upgrades with capacity addition schedules is key to avoiding the curtailment of solar projects and creating continued market certainty for investors. System operators should develop

2 GSC Policy recommendations / continued

tools to assess and plan for future flexibility needs, both on transmission and distribution levels, including assessing how distributed solar PV and storage reduce the need for high-voltage transmission lines by bringing generation closer to consumption. It will also allow the integration of many flexibility measures, like battery storage and demand response, that can accommodate and manage new capacities of solar energy.

Establishing a global storage target is a critical step, as it provides a clear roadmap for investment and development. Targets like the one announced by the G7 to install 1,500 GW of global energy storage by 2030 – a sixfold increase from today's levels – is a step in the right direction and should be adopted at the UNFCCC level to complement the COP triple renewables goal. In addition, interconnections with other countries can be a mutually beneficial way for one party to balance their grid, and the other to secure access to cost-competitive solar electricity, making regional integration of electricity systems and markets a viable way to increase solar penetration.

- **Scale up and diversify low-cost and resilient solar PV supply chains**

Significant efforts should be made to diversify supply chains, aligning national and regional initiatives to reach free, fair, open, and resilient global supply chains based on international standards. It should be robust, cost-effective and can adequately support the growing demand for solar globally. Countries should increase investor confidence by combining energy transition strategies with policies that reward sustainable and resilient supply chains, while promoting open trade, international cooperation, cost reductions and sustainable growth of the global solar PV market. Additionally, countries need to invest in research and development and foster innovation through government funding, private sector incentives, and international collaboration. It paves the way for technological breakthroughs that result in advanced manufacturing processes, resource efficiency improvements, and increased environmental sustainability, all of which contribute to the affordability and widespread adoption of solar PV.



3.3 MW, Adelschlag, Germany.

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- **Streamline permitting schemes to accelerate solar PV deployment**

It's crucial to harmonise permitting procedures across the country for all grid operators and for all regions. The permitting process can be further simplified and expedited by establishing "one-stop shops" and include maximum waiting times after which permits are automatically granted. Permitting for solar PV projects in many countries can take any time between two weeks and several years, so it is important to build trust with consumers and developers by showing transparency in the review methodology and decision-making process, and by providing guarantees of connection to the grid. In addition, the administrative burden of permitting procedures should be flexible and dependent on multiple variables like location and available infrastructure, or on the size of the installation, ensuring that processes for rooftop solar and mini-grids are quicker than those for utility-scale.

- **Adopt sustainable Environmental, Social and Governance (ESG) practices based on global standards and multi-stakeholder initiatives**

Governments should set clear regulations and standards on ESG practices, complemented by due diligence processes and a clear route-to-compliance. Governments should embrace multi-stakeholder initiatives tailored to solar PV that support achieving the ESG goals. Regulators and the solar industry must also work together to put in place extended producer responsibility regulations for the proper disposal, reuse, and recycling of end-of-life solar system components, such as modules, inverters, and batteries, which can help guarantee an environmentally sound level of waste management. Establishing designated processing

facilities is a key part of the solution, and ideally it would feed the output material into local supply chains to service new raw materials, products and renewable energy equipment.

- **Nurture the development of solar skills to match the needs of sector growth**

The solar PV sector is a significant source of job creation. IRENA estimates solar PV employment at 4.9 million in 2022 globally, which is the highest in relation to any other power generation technology. As the sector grows, so does the need for skilled workers to manufacture, install, and maintain solar PV systems. By investing in training programmes and educational initiatives focused on solar technology, governments and organisations can equip individuals with the necessary skills to fill these in-demand positions. This not only fuels the growth of the solar PV sector but also empowers individuals to participate in the clean energy revolution, creating a more sustainable and equitable future for all.

The global nature of the solar sector requires harmonisation and consistency in skills development and certification. Solar energy projects are implemented across borders, with companies and workers often crossing national and regional boundaries to contribute to the growth of the industry. It is important to adopt, recognise, and support global certification schemes like the Solar Training Standards Initiative implemented by the Global Solar Council in partnership with the Global Wind Organisation. By harmonising skills development and certification at a global level, the industry can ensure that workers possess universally recognised and transferable skills, enabling seamless mobility and enhancing workforce efficiency.

Author: Global Solar Council (GSC).



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3

Focus: China solar PV market

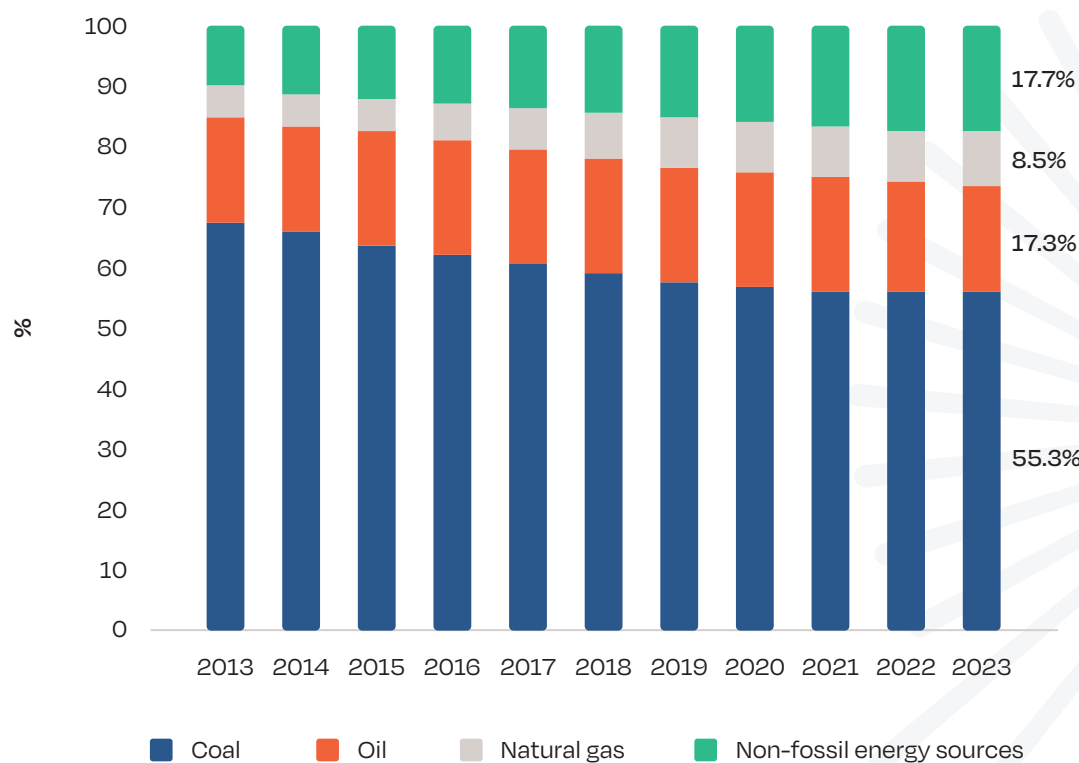
50 MW floating PV park in Jining, Shandong, China. © Sungrow

1. Introduction

China has been optimising its energy consumption structure in recent years, with a growing share of non-fossil energy sources. By the end of 2023, non-fossil energy sources accounted for 17.7% of its total energy

consumption, while coal's share had decreased to 55.3% (see Fig. 38). Over the past decade, coal consumption has dropped by about 12%, while non-fossil energy sources have increased by more than 7%. This shift has played a crucial role in China's transition to low-carbon energy.

FIGURE 38 STRUCTURE OF ENERGY CONSUMPTION IN CHINA, 2013-2023



SOURCE: National Bureau of Statistics.

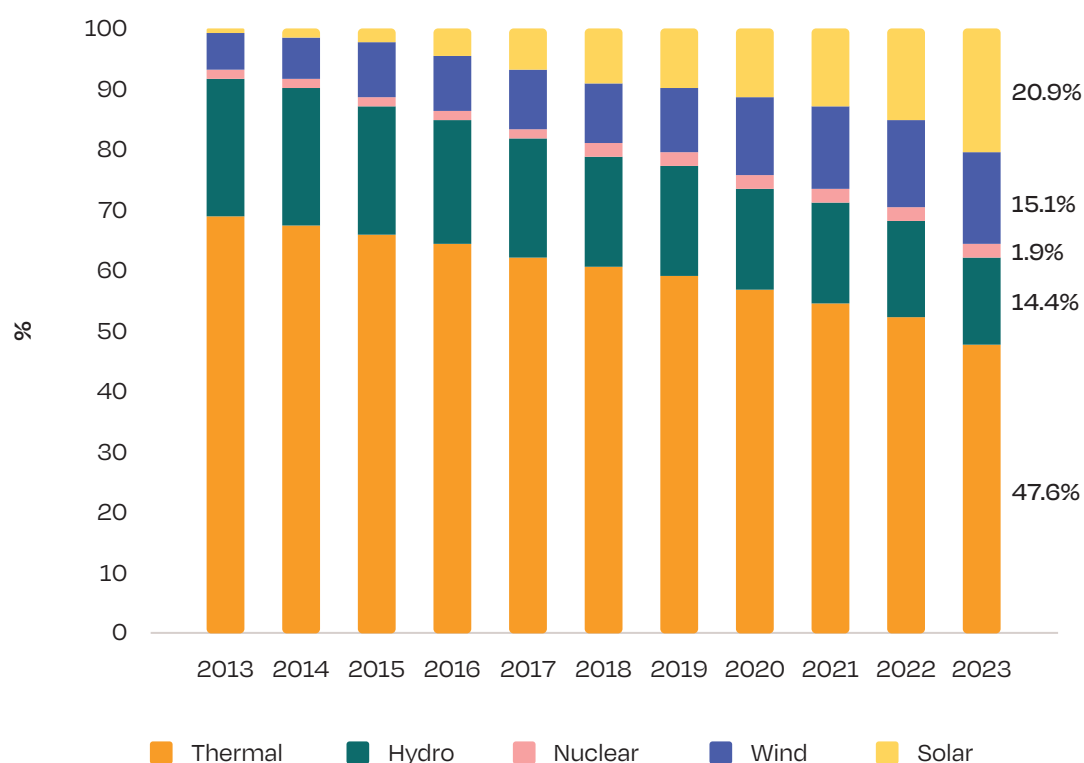
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3 Focus: China solar PV market / continued

In 2023, China's total installed electricity generation capacity reached 2.92 TW, up by 13.9% year over year. More specifically, the installed capacity of coal power was about 1.165 TW, accounting for less than 40% of total capacity; while non-fossil energy sources was at

1.57 TW, accounting for 53.9%, exceeding 50% for the first time ever. In addition, China's combined installed capacity of wind and solar amounted to 1.05 TW, accounting for 36%, marking a new breakthrough (see Fig. 39).

FIGURE 39 SHARES OF INSTALLED CAPACITY OF DIFFERENT ELECTRICITY SOURCES IN CHINA, 2013-2023



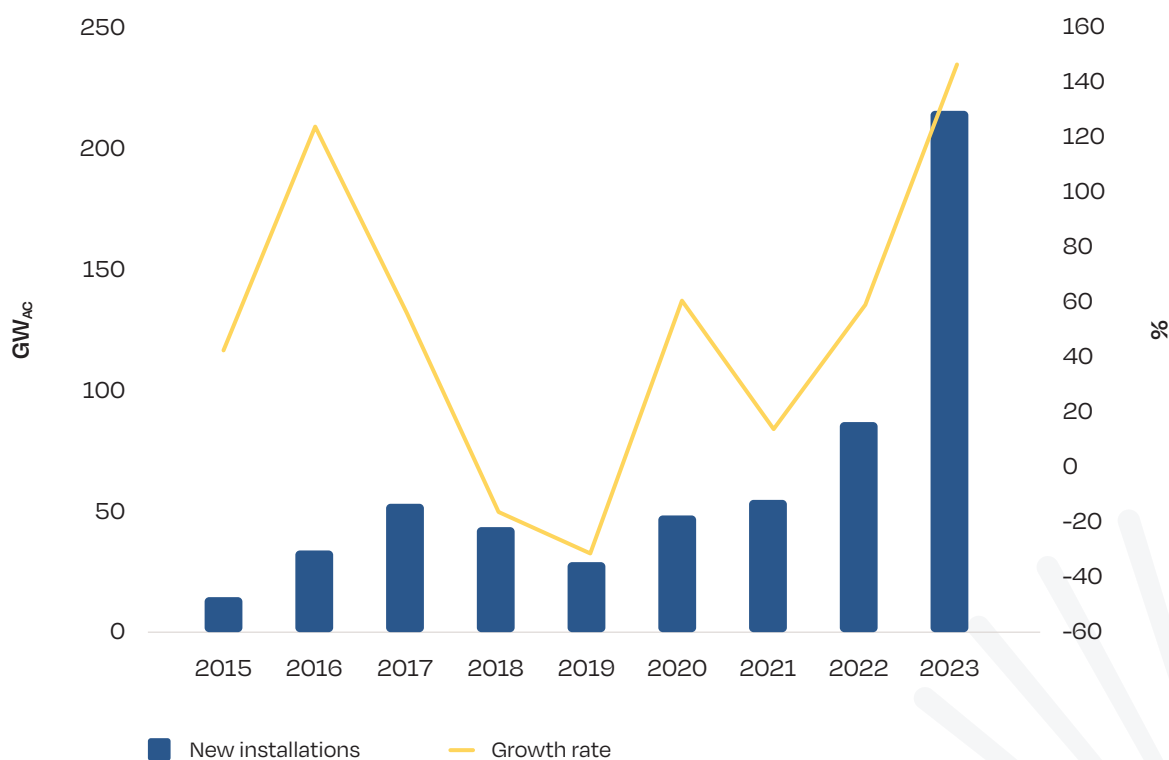
SOURCE: China Electricity Council.

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China's solar PV market grew much better than expected in 2023; 216.3 GW⁸ of capacity was newly installed, a 148.1% increase year over year, hitting a record high and accounting for approximately 60% of

the world's total. By 2023, China's accumulative installed solar PV capacity amounted to 608.9 GW, surpassing hydropower and becoming the country's second-largest electricity source after coal power capacity.

FIGURE 40 COMPARISON OF NEWLY INSTALLED SOLAR PV CAPACITY 2015-2023



SOURCE: National Energy Administration (NEA).

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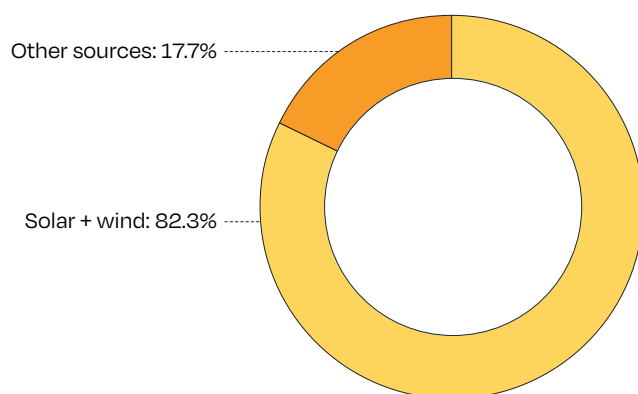
8 All PV capacity in this chapter is expressed in AC values.

3 Focus: China solar PV market / continued

In general, the installed capacity of renewable energy sources, namely wind and solar PV, has been increasing rapidly, and for the first time, the share of installed capacity from non-fossil energy sources has exceeded 50% (see Fig. 41). This highlights China's

growth momentum in the renewable energy sector. The solar PV industry in particular, has experienced better-than-expected development, with new installations hitting a record high. This demonstrates the huge development potential of the industry.

FIGURE 41 NEWLY INSTALLED SOLAR PV AND WIND VS OTHER SOURCES IN 2023



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17.9 MW, Datang Sanmenxia thermal power plant, China.

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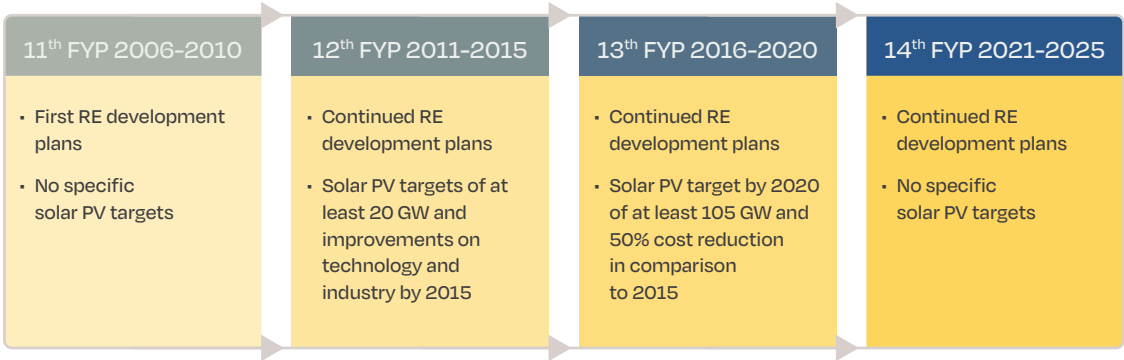
2. Policy objectives

Solar PV and renewable energy targets in the Five-Year Plan (FYP)

Renewable energy targets, including solar PV targets, were medium- and long-term development goals established by the central government at the national

level. These targets are primarily initiated and will be implemented through the FYPs. Attention to renewable energy began during the 10th FYP period of 2001-2005. Renewable energy development plans were formulated during the 11th, 12th, and 13th FYP periods. Special plans for wind and solar power were only introduced during the 12th and 13th FYP periods.

FIGURE 42 TIMELINE OF SOLAR PV PLANS MENTIONS IN THE FIVE-YEAR PLANS



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3 Focus: China solar PV market / continued

In the ongoing 14th FYP period (2020–2025), China introduced the *14th Five-Year Plan for Renewable Energy Development*. This plan, which is built on carbon peaking and carbon neutrality targets, outlines key targets for renewable energy development and utilisation by 2025 (see Table 3). These targets include renewable energy total installed capacities, electricity generation, electricity consumption, and renewable energy non-electricity utilisation targets in order to achieve a 20% share of non-fossil energy consumption by 2025. 2024 is a key year for achieving the objectives and tasks of the 14th FYP. China will make sustained efforts to promote high-quality development in the renewable energy sector by establishing a sound policy mechanism for guaranteed new energy consumption, advancing the construction of large-scale wind and solar power bases, and steadily facilitating new energy access to the electricity market.

The Chinese government has issued guiding documents to direct energy development at different stages. For example, the *Strategy for Energy Production and Consumption Revolution (2016–2030)* was released in 2016. It specified that the share of non-fossil energy sources in primary energy consumption should reach 15% by 2020, 20% by 2030, and 50% by 2050. In 2021, the State Council issued *The Opinions on Facilitating Carbon Peaking and Carbon Neutrality by Implementing the New Development Concept in a Complete, Accurate, and Comprehensive Manner*, proposing that by 2030, the total installed capacity of wind and solar should reach more than 1.2 TW. According to the latest data released by NEA, as of the end of March 2024, the combined installed capacity of wind and solar PV in China has exceeded 1.1 TW, indicating a high probability of meeting the target ahead of schedule.

TABLE 3 MAJOR TARGETS FOR RE DEVELOPMENT AND UTILISATION IN THE 14th FYP

CATEGORY	UNIT	2020	2025	ATTRIBUTES
1. Renewable energy utilisation for electricity generation				
1.1. Share of total renewable energy in electricity consumption	%	28.8	33	Expected
1.2. Share of total non-hydro renewable energy in electricity consumption	%	11.4	18	Expected
1.3. Renewable energy generation	TWh	2210	3,300	Expected
2. Renewable energy non-electricity utilisation	10,000 tons	—	6,000	Expected
3. Total renewable energy utilisation	100M tce	6.8	10	Expected

"Dual carbon" targets

On September 22, 2020, President Xi Jinping announced at the 75th session of the UN General Assembly that China aims to peak carbon dioxide (CO₂) emissions before 2030 and achieve carbon neutrality before 2060, which are referred to as the "dual carbon" targets. The specific concepts and implementation paths have been further refined and deepened since then. The near-term

goals include lowering CO₂ emissions per unit of GDP by 18% from the 2020 level by 2025, peaking CO₂ emissions around 2030, lowering CO₂ emissions per unit of GDP by more than 65% from the 2005 level by 2030, reaching around 20% non-fossil energy sources in primary energy consumption, and increasing the forest stock volume by about 4.5 billion m³ from the 2005 level. The long-term goal is to achieve carbon neutrality, or "zero" CO₂ emissions, by 2060. (see Table 4).

TABLE 4 DEVELOPMENT TARGETS FOR ENERGY INTENSITY, CARBON INTENSITY, SHARE OF NON-FOSSIL ENERGY SOURCES, ETC.

SEGMENT	2020	2025	2030	2060
Decline in energy consumption per unit GDP	15.3% (from the 2015 level)	13.5% (from the 2020 level)		
Decline in carbon emissions per unit of GDP	18.8% (from the 2015 level)	18% (from the 2020 level)	Over 65% (from the 2005 level)	
Share of non-fossil energy consumption	15.9%	20%	25%	Over 80%
Installed capacity of wind and solar power	530 GW		Over 1.2 TW	



700 MW, China.

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3. Development of China's solar PV industry and major policy evolution

Development of China's solar sector

The history of the development of China's solar PV industry can be divided into several key stages: infancy, private sector as a driver of development, and large-scale expansion.

Prior to 2000, the solar PV industry was in its infancy. At this stage, deficiencies were found in domestic market size, manufacturing capacity, systematic R&D framework and investment, and vertical integration in the supply chain.

Between 2001 and 2010, the private sector played a significant role in the rapid growth of the industry. Overseas Chinese entrepreneurs kick-started the industry's development, and China's "investment promotion" policy attracted more overseas talent to return home and launch their own businesses. The limited size of the domestic market was mitigated by accessing foreign markets through improved quality and reliability. The involvement of private companies and joint ventures helped address the lack of technical knowledge, while foreign investment met the financing needs of start-ups. Additionally, the use of vertically integrated supply chains led to a dramatic reduction in costs.

Between 2011 and 2017, the industry experienced a significant growth. By the end of 2013, China's cumulative installed solar PV capacity had reached 125.7 GW, accounting for 7.5% of the total installed electricity capacity. However, there were some challenges, such as limited domestic market size, higher electricity generation costs than traditional thermal power, and a lack of relevant technical standards. To address these issues, the government adopted several policies to stimulate the growth of the domestic market. These included feed-in tariff (FIT) policies, demonstration projects, and improved regulation and management of the market by introducing the *Specification Conditions for the Photovoltaic Manufacturing Industry* and the "PV Industry Standardised Technical System." Additionally, efforts were made to build up research and development capacity and promote innovation.

Since 2017, technological advances and efficiency improvements in smart control systems, among others, have reduced the cost of solar PV electricity,

paving the way for the grid-parity era. Concerns over subsidy delays or cancellations, particularly the change in FIT policies, led to a sharp decline in newly installed capacities in 2018 and 2019. However, new installations grew rapidly after the full subsidy policy was lifted in 2021 and exceeded the 2017 level due to a rush to install. Later, the industry entered a period of rapid growth in the grid-parity era.

Subsidies for solar PV electricity generation

Starting in 2009, China has implemented several key projects and initiatives to promote the domestic development of the Solar PV market. These include the solar PV Building Application & Demonstration Projects, the Golden Sun Demonstration Project, and power plant concession bids. The 2009 Golden Sun programme offered subsidies to on-grid and off-grid solar PV generation projects, targeting 500 MW of installed capacity nationwide by 2012.

In 2013, the State Council issued the *Several Opinions of the State Council on Promoting the Healthy Development of the Photovoltaic Industry*, which for the first time clarified the implementation period of FIT and subsidies to be 20 years and identified the Renewable Energy Development Fund (REDF) as the funding source. Subsequently, the National Development and Reform Commission (NDRC) issued various documents outlining subsidy methods for different types of projects that gradually established subsidy standards and policies.

In June 2021, NDRC issued the *Notice on Matters Relating to the New Energy Feed-in Tariff Policy for 2021*. It explicitly stated that starting in 2021, new centralised solar PV plants and industrial and commercial distributed solar PV projects would no longer receive subsidies from the central government but would instead be subject to grid parity. In April 2022, the NDRC issued a document to eliminate the national tariff subsidy for residential solar PV and implement full grid parity for solar PV electricity generation, marking the official end of the solar PV subsidy policy.

Guarantee Mechanism for Renewable Energy Consumption

The volatility and intermittency of renewable energy-based electricity generation, along with the rapid growth

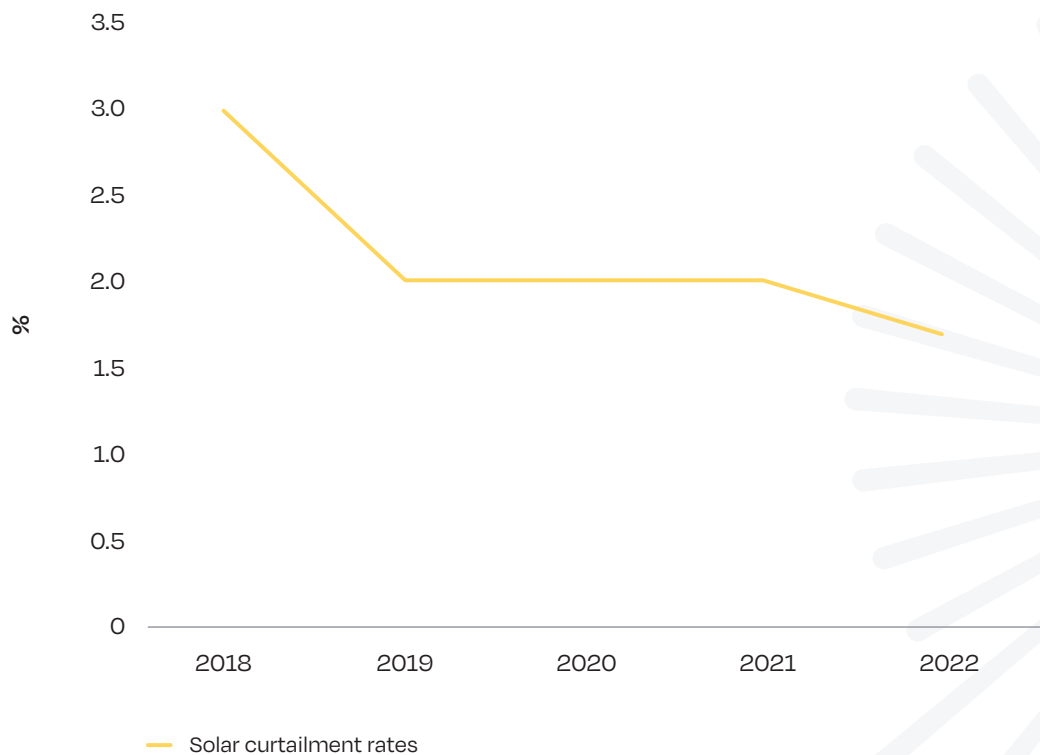
in the scale of development, has resulted in wind and solar energy curtailment. To address this, the Chinese government has introduced a number of policies. Important mechanisms, like feed-in tariffs, were established in the *Renewable Energy Law of 2005*, which was amended in 2009 to further define prioritised grid access and guaranteed purchase of renewable energy, providing the base for China's solar PV growth. Between 2013 and 2017, the relevant departments introduced measures to explore “wind energy for heating” and issued a notice on grid-connected wind energy consumption. As a result of these measures, China's annual average wind curtailment and solar curtailment rates showed a downward trend.

Subsequently, there were improvements in the relevant policy mechanisms on the consumption side. In 2018, China issued the Clean Energy Consumption Action Plan (2018-2020), which clarified the guarantee mechanism for renewable energy consumption as a key long-lasting mechanism on the

matter. In 2019, the Notice on Establishing and Perfecting the Guarantee Mechanism for Renewable Energy Electricity Consumption formally established this mechanism. It set the share for renewable energy consumption for provincial administrative regions and stipulated relevant supporting systems such as assumption of responsibility by the relevant main entities, consumption accounting, and trading methods. China's annual average wind curtailment and solar curtailment rates dropped significantly after the implementation of the guarantee mechanism.

According to the “Long-Range Objectives Through the Year 2035”, China plans to expedite the development of renewable energy. It is expected that by 2035, renewable energy will become the primary source of energy consumption in China. Therefore, further developing and improving the guarantee mechanism for renewable energy consumption will become a long-lasting mechanism for promoting the widespread utilisation of renewable energy.

FIGURE 43 SOLAR CURTAILMENT RATES IN CHINA 2018-2022



SOURCE: National Energy Administration (NEA).

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3 Focus: China solar PV market / continued

County-wide distributed solar PV development

On June 20, 2021, NEA issued the *Notice on the Submission of Pilot Programmes for the Development of Distributed Rooftop Photovoltaics in Entire Counties (Cities and Districts)*. The Pilot Programme aimed to realise the economies of scale for rooftop solar PV through the intensification of integrated resources while simultaneously encouraging its installations in a county (city or district). Thus, it reduced the customer acquisition costs, the soft cost of the development of residential installations, and the development costs of the solar PV power plants. In September 2021, NEA published a list of 676 pilot counties, cities, and districts in 31 provinces, autonomous regions, and municipalities. The construction scale of the pilot counties, cities, and districts ranged from 150 MW to 300 MW, and the cumulative capacity filed by pilot counties reached approximately 165 GW. The launch of the Pilot Programme set off a distributed solar PV development boom, which promoted its development and offered opportunities for urban and rural residents to utilise their rooftops to increase their incomes. In 2021, the installed capacity of distributed projects accounted for about 53% of the national newly installed solar PV capacity, surpassing that of centralised power plants for the first time. In 2022, its share increased to 58%. However, the progress of the Pilot Programme fell below expectations. By the end of 2022, the cumulative grid-connected capacity in the pilot areas nationwide was 29.76 GW.

Integrated energy planning

On January 29, 2022, NEA released the *14th Five-Year Plan for Modern Energy System*, listing “secure, green, and low-carbon” as the first of the four basic principles. According to the five main objectives of the Plan, CO₂ emissions per unit of GDP will fall by 18% in five years. By 2025, the plans aim to increase the share of non-fossil energy consumption to around 20%, non-fossil power generation to 39%, and electricity in end-use energy to around 30%. In addition, the Plan put forward the goal of “significantly enhancing the capacity for innovation and development.” To achieve these aims, efforts will be made to integrate new energy technologies, reach milestones in constructing new grids, and significantly improve the innovation capacity for safe and efficient energy storage, including hydrogen energy technology. In addition, actions will be taken to accelerate the scale-up and application of technologies for reducing pollution and cutting carbon emissions, and ensure that investment in energy research and development during the 14th FYP period will increase by 7% annually, and that breakthroughs will be made in about 50 key technology areas.

By 2035, significant progress is expected in the development of the energy sector and in establishing a modern energy system. Energy security and guarantee capacity should be substantially improved with major progress in upgrading the grid. Green production and consumption patterns should be



Suqian, Jiangsu, China.

© Trina Solar

widely formed, with a substantial increase in the share of non-fossil energy consumption, making renewable energy electricity generation the main energy source in the mix. Additionally, total carbon emissions will steadily decrease after reaching the peak.

4. Market developments in 2023-2024

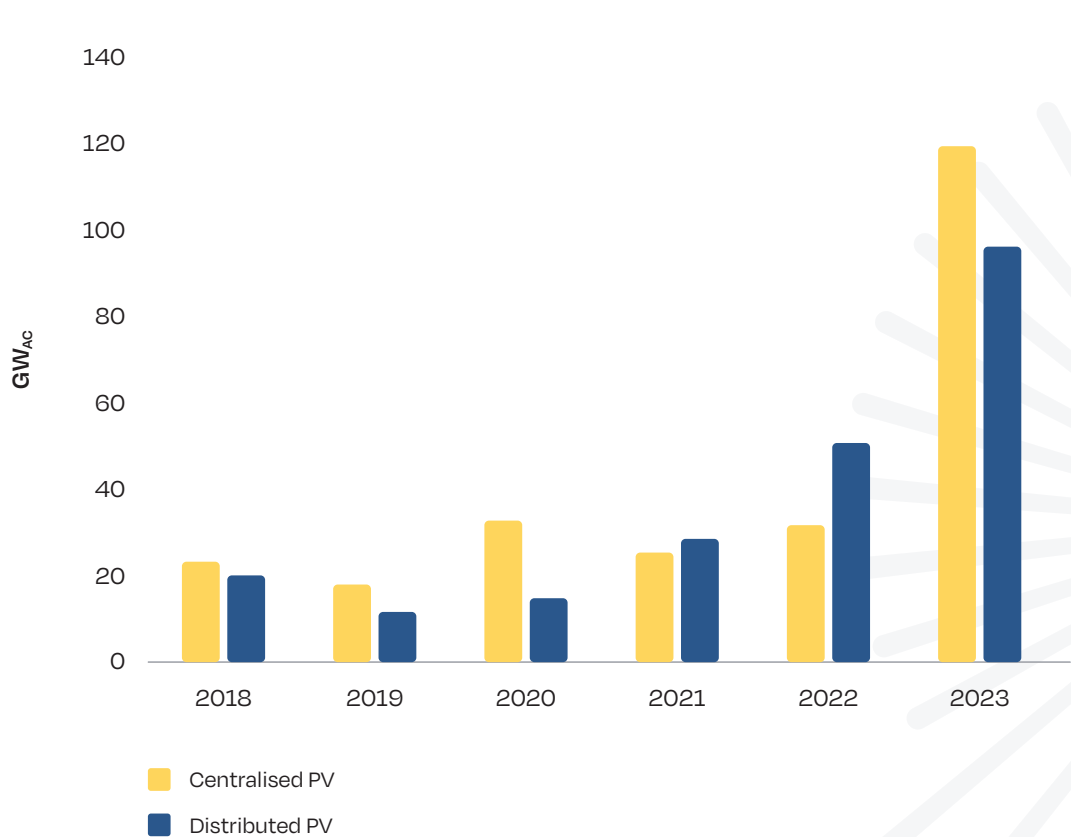
Current status of China's solar PV market

In 2023, China's total installed solar PV capacity reached 608.9 GW, with distributed projects accounting for 254.4 GW and centralised projects for 354.5 GW. In 2023 alone, China added 216.3 GW of solar PV capacity, showing an increase of 148.1% year over year. Specifically, distributed projects had a capacity of 96.3 GW, accounting for 44.5% of the installations, while centralised projects had a capacity of 120 GW,

accounting for 55.5%. In comparison to the period from 2018 to 2022, both centralised and distributed new installations showed remarkable growth, and reached the 100 GW scale. Additionally, the new installations of centralised projects surpassed that of distributed projects, reversing the trend in the past two years. This is mainly due to the accelerated construction of large-scale wind and solar complementary bases in the northwest region. The first batch of projects was required to be connected to the grid before 2023, while the second batch of projects was required to have an operating rate of more than half.

In 2023, there were 20 provinces with cumulative installed capacity of more than 10 GW and 16 provinces with a capacity of more than 20 GW. Shandong Province had the highest installed capacity (56.9 GW), followed by Hebei (54.2 GW), Jiangsu (39.3

FIGURE 44 DISTRIBUTION OF NEW INSTALLATIONS BY TYPE 2018-2023



SOURCE: National Energy Administration (NEA).

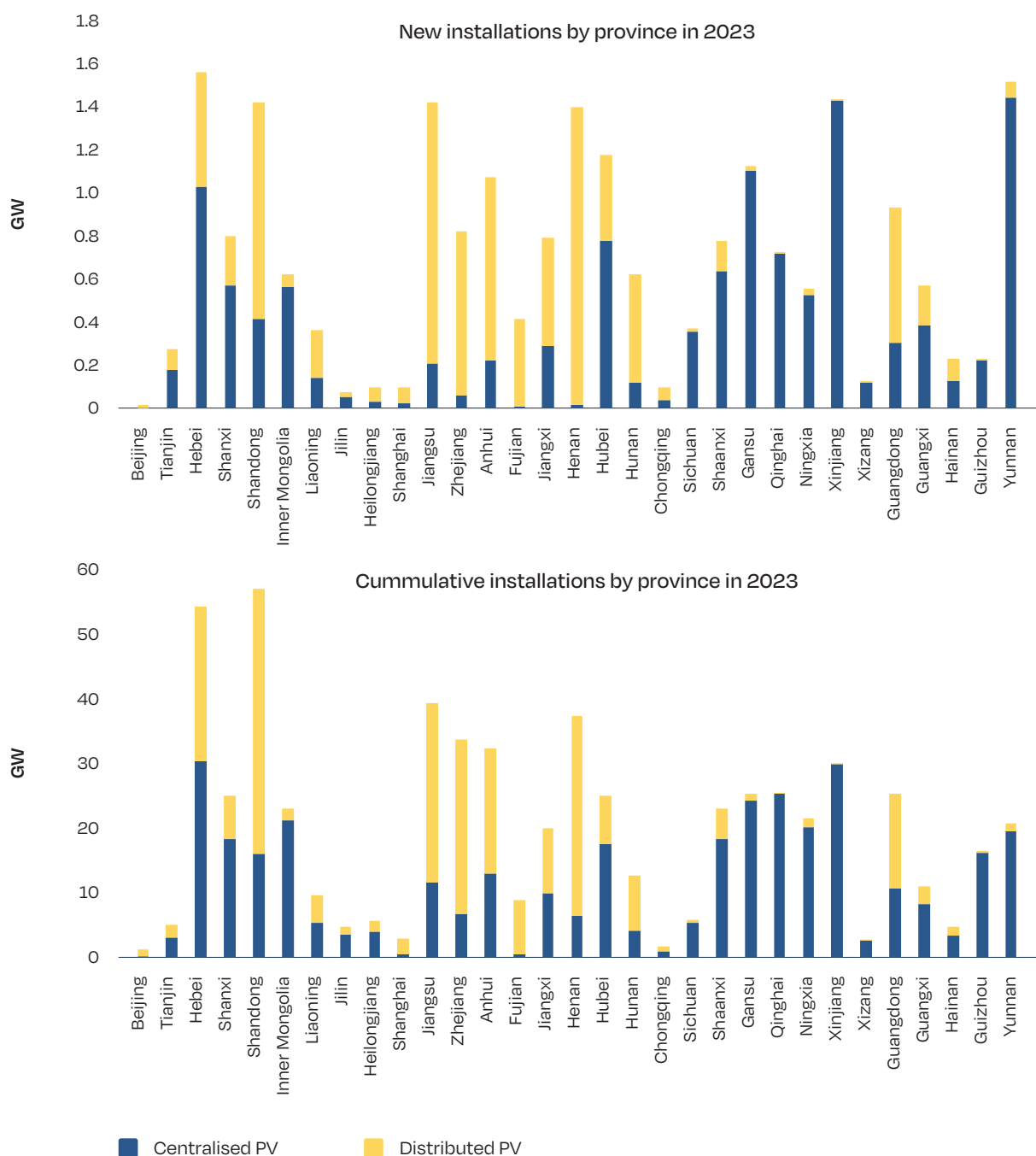
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3 Focus: China solar PV market / continued

GW), Henan (37.3 GW), Zhejiang (33.6 GW), and Anhui (32.2 GW). Additionally, 15 provinces had accumulated installations of centralised plants of more than 10 GW, the top five being Hebei, Xinjiang, Qinghai, Gansu, and

Inner Mongolia. Furthermore, eight provinces had total installations of distributed solar PV of more than 10 GW, the top five being Shandong, Henan, Jiangsu, Zhejiang, and Hebei.

FIGURE 45 ANNUAL AND CUMULATIVE INSTALLATIONS BY PROVINCE IN 2023



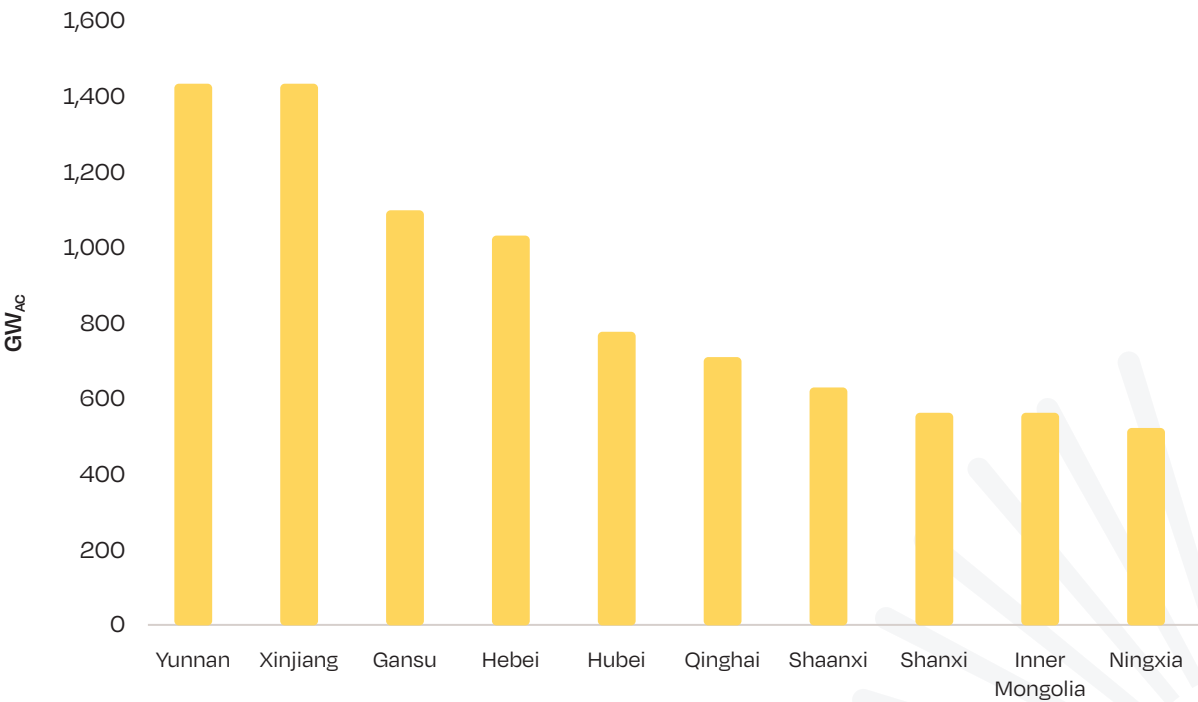
SOURCE: National Energy Administration (NEA).

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In 2023, the new installations of centralised solar PV projects exceeded 120 GW, accounting for over 55% of total installations. The first phase of base projects was required to be connected to the grid by the end of 2023. More than half of the second phase of base projects have started, and the third phase of base projects has completed nearly 25% of the preliminary work.

The top 10 provinces and autonomous regions with the highest centralised installations were Yunnan and Xinjiang, ranking first and second, with new installations close to 15 GW. They were closely followed closely by Gansu and Hebei, both exceeding 10 GW. The remaining six provinces were Hubei, Qinghai, Shaanxi, Shanxi, Inner Mongolia, and Ningxia, with new installations in the range of 5-10 GW.

FIGURE 46 TOP 10 PROVINCES AND AUTONOMOUS REGIONS IN TERMS OF CENTRALISED SOLAR PV INSTALLATIONS IN 2023



SOURCE: National Energy Administration (NEA).

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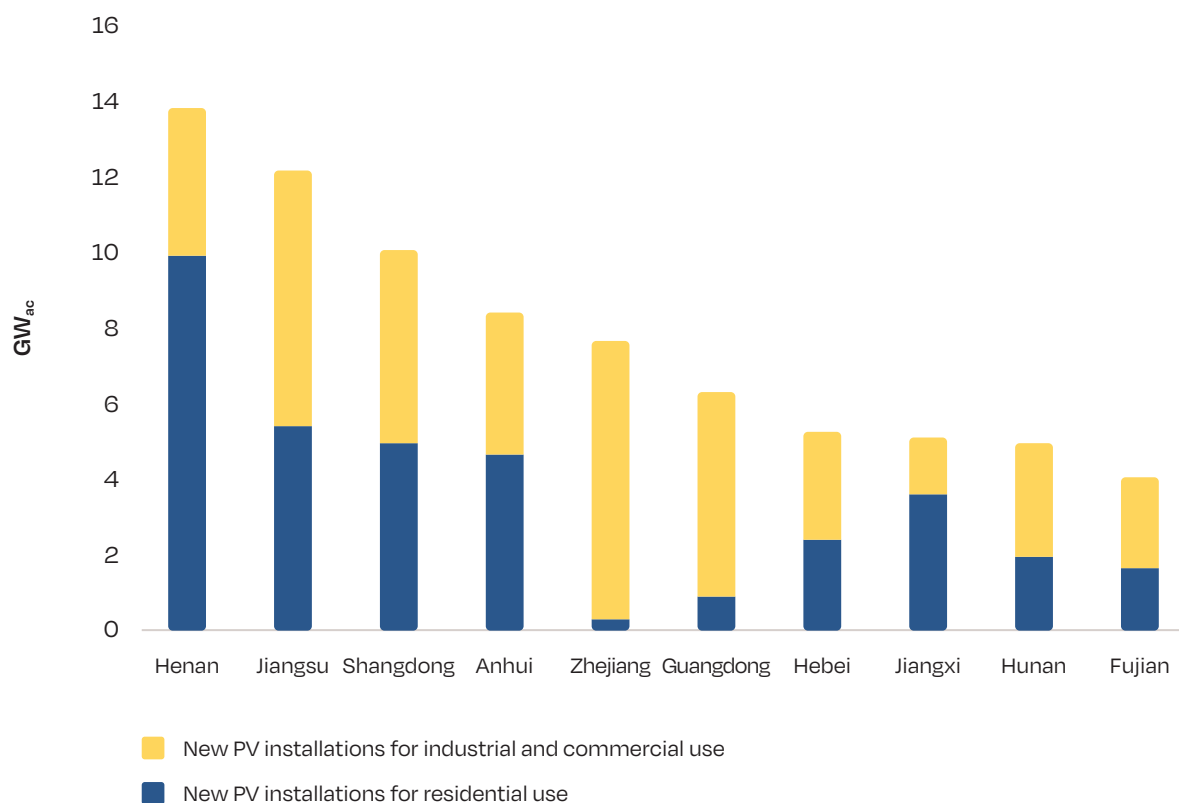
3 Focus: China solar PV market / continued

Newly installed distributed solar PV nearly reached 100 GW in 2023, representing around 45% of annual deployed capacity. Of these, 43.5 GW were for residential use, accounting for 45.2% of the total for distributed installations, while commercial and industrial projects amounted to 52.8 GW, accounting for 54.8% of the total. China's distributed PV market last year showed faster growth in C&I projects, and a shift in focus to the residential distributed market. This is clearer in Zhejiang and Guangdong, where local economic development in these two provinces mainly relies on export-oriented enterprises that lack land space for large-scale construction.

Existing Supporting Policies

In China, the sources of green electricity (referring to electricity produced with zero or near-zero CO₂ emissions) have recently been dominated by wind and solar PV. Green electricity products have both energy and environmental value, and the price of transactions in the green electricity market reflects a green premium, whereas electricity users who buy green electricity products will be provided with certificates. Green Electricity Certificates (GECs), also known as Renewable Energy Certificates (RECs), are a policy tool based on the Renewable Portfolio Standard (RPS). In order to increase flexibility of the obligated parties of RPS to fulfil the quota targets, RPS will often be adopted to provide more flexible market-based mechanisms. The purchase of RECs is the means and proof of meeting the RPS.

FIGURE 47 TOP 10 PROVINCES IN TERMS OF DISTRIBUTED SOLAR PV INSTALLATIONS IN 2023



SOURCE: China Photovoltaic Industry Association (CPIA).

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China has been exploring market-based mechanisms for green electricity on the basis of the traditional electricity market, with green electricity and GEC trading as the main varieties, forming a system of market-based mechanisms that comprehensively reflect the value of electric energy and the environmental value of green electricity. With the elimination of solar PV tariff support schemes and the implementation of the grid parity policy, there has become an urgent need to develop sound green electricity and GECs trading markets. In February 2023, NDRC, the Ministry of Finance (MOF), and NEA issued the *Notice on Matters Relating to the Participation in Green Electricity Trading of Green Electricity Projects Enjoying Subsidies from the Central Government* to steadily encourage the projects under the national FIT subsidies to participate in green electricity trading. Under these new rules, the revenue generated by green

premiums of grid parity projects will be given to the electricity generation enterprises. For subsidised projects, the green premium will be given to the government to be used for subsidies, so that the subsidy the project receives stays the same in the end. This means that green electricity premium income and national renewable energy subsidies cannot be achieved at the same time.

In July 2023, NDRC, MOF, and NEA jointly issued the Notice on Promoting Renewable Energy Electricity Consumption through *Full Coverage of Renewable Energy Green Electricity Certificates*, comprehensively restructuring the GEC system and achieving full coverage of GECs. At present, China's green electricity and GEC trading markets have gradually established corresponding mechanisms (see Table 5).

TABLE 5 CHARACTERISTICS OF GECS AND GREEN ELECTRICITY TRADING MARKETS

TRADING MECHANISMS	GECS	GREEN ELECTRICITY
Subject matter of the transaction	Environmental rights and benefits	Physical quantities + environmental rights and interests
Transaction characteristics	Separation of certificates and electricity	Unification of certificate and electricity
Transaction price	Environmental value	Electrical energy value + environmental value
Transaction scope	Not limited by geography	Inter-provincial + Intra-provincial
Transaction priority	/	Prioritised organisation, implementation, settlement and guarantee

3 Focus: China solar PV market / continued

The guaranteed purchase mechanism of renewable electricity is an important tool for fostering the renewable energy market and the sector. It is also an important legal system for ensuring the smooth development of the renewable energy electricity sector during its start-up and growth phases. The mechanism can be traced back to the *Measures for the Supervision of Full-Amount Purchase of Renewable Energy Electricity by Grid Enterprises*, implemented on a trial basis since September 1, 2007. It stipulates that grid operators should purchase grid-connected electricity in full amount from renewable energy projects covered by their electricity grids. The policy was adjusted several times in response to changes in the policy and market environment. Since the 13th FYP period, additional policy documents were released in order to implement the full-amount guaranteed purchase system, including the document that NDRC and NEA have issued on the *Measures for the Administration of the Full-Amount Guaranteed Purchase of Renewable energy Electricity Generation, the Action Plan for Clean Energy Consumption, and the Notice on the Establishment of a Sound Guarantee Mechanism for the Consumption of Renewable Energy Electricity*. Under the guidance of these documents, the State has approved the minimum guaranteed annual utilisation hours for wind and solar PV electricity generation in key regions. In addition, they have set the share of renewable energy consumption in provincial administrative regions. A

national monitoring and evaluation system has been established for the development of wind and solar PV generation to monitor and release market and industrial developments on an annual basis and provide information and early warning for the industry. The state has also coordinated and promoted the construction of relevant electricity grid infrastructure, stimulated thermal electricity flexibility retrofits, and intensified the assessment of electricity grid operations.

The implementation of the National Full-Amount Guaranteed Purchase Mechanism for renewable energy has achieved positive results. During the 13th FYP period, China's wind electricity utilisation rate increased from 82.7% to 96.5%, while the solar PV electricity utilisation rate increased from 89.5% to 98%. This led to a significant improvement in the level of renewable energy consumption. Additionally, during the current 14th FYP period, there are continuous improvements in the rate of renewable energy utilisation and energy curtailment mitigation.

The *Measures for the Supervision of Full-Amount Guaranteed Purchase of Renewable Energy Electricity* came into force on April 1, 2024. The major change in this new policy is that grid operators will no longer be obligated to purchase all renewable energy electricity, and the model has shifted to a guaranteed purchase. This means that part of the renewable energy generation will participate in market-based trading.



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5. Opportunities and Challenges Ahead

Opportunities

The development of the solar PV sector has been already constrained by problems on the consumption side. As the large-scale solar industry continues to grow, it is crucial to create application scenarios that can address the demand side and gradually replace traditional energy sources. By coupling solar PV with battery storage and hydrogen production, solar can reduce curtailment, and ease the bottleneck to scaled-up grid access. Adding flexibility tools, including green hydrogen, offers an effective way to better match solar growth with demand.

In 2023, many wind and solar PV hydrogen production projects were launched, making it the first year of rapid development of green hydrogen. The Xinjiang Kuqa Green Hydrogen Demonstration Project, fully constructed and operational by August, holds great significance as China's first 10 kt solar PV direct green hydrogen production project. The project provided the first feasible sample of large-scale production, storage, transportation, and utilisation of green hydrogen. At full capacity, it can produce 20 kt of green hydrogen annually, which can replace natural gas hydrogen production in oil refinery processing, and lower carbon emissions.

In the short term, the three northern regions will become the main concentration of China's electrolyser-based hydrogen production capacity due to manufacturing costs and consumption potential. Xinjiang, Ningxia, and Inner Mongolia have abundant solar and wind resources, with a high rate of utilisation hours. This plays a key role in rapidly reducing combined hydrogen costs. Additionally, these regions are home to China's oil and gas, coal, and chemical industry bases, with a number of use cases for hydrogen application for oil and gas processing, methanol production, and more. This means it can be used for technical verification for large-scale industrial applications of green hydrogen and to create a demand for its consumption.

As the renewable energy subsidy policy became history, the guaranteed purchase system for all power generated has also begun to be phased out. The Chinese government will step up efforts in building renewable electricity market-based mechanisms in the future, bringing new opportunities for the solar PV industry. Market based solutions will increase the risk for renewable energy project owners compared to the previously guaranteed electricity offtake with guaranteed quantity and price. For instance, the risk of gains from competing in the market through quotation has increased significantly.



21 MW, Ningbo Yuyao, China.

© Sungrow

3 Focus: China solar PV market / continued

Renewables' characteristics, such as volatility, intermittency, and low marginal cost, can lead to lower electricity prices in electricity market competition. As a result, the value of electricity generated by solar PV remains consistently lower than that of thermal-based generation and other sources of non-intermittent electricity. While the price risk solar PV faces is even more pronounced, a more market-driven approach could significantly boost the installation of new renewable energy capacities.

In March 2024, NDRC issued the Measures for the Supervision of Full-Amount Guaranteed Purchase of Renewable Energy Electricity. It further incentivised renewable energy projects to participate in electricity market transactions. While the environmental value of green electricity is captured through GECs, supporting measures should be implemented to maximise the benefit of solar PV generation. Such measures include expansion of the full coverage and application of GECs, strengthening the connection between GECs and the domestic carbon market as well as international recognition, and further enhancing the visibility of GECs. The decreasing cost of solar PV allows for market-based approaches to foster affordable renewable energy consumption or even manage the curtailment of generation. This can help to better allocate grid-regulated resources, providing more

opportunities for renewable energy consumption and increasing renewable energy capacities.

The renewable energy sector worldwide employed 13.7 million people in 2022, almost doubling the 7.3 million employed in 2012. China is a global leader in renewable energy employment, accounting for 41% of the total jobs created globally, at the same time China accounted for about 56% of solar PV employment worldwide.⁹ China's solar PV industry offers a large number of jobs, especially in the manufacturing and construction sectors. While labour demand in some industries has been decreasing due to increasing automation, the deployment side of solar is much more job-intensive.

Challenges

Problems of grid access and electricity consumption for centralised projects persist; however, distributed solar PV has begun to face similar problems in recent years. The "self-generation and self-consumption" mode affects the net load, exacerbating electricity generation-consumption balance problems and causing sharp changes in electricity prices in provinces where the spot market operates. Distributed solar PV in Shandong and Henan, among others, has been developing rapidly, and because of that the electricity load profile is shifting from a "duck curve" into a "canyon curve".



1 GW solar PV park producing green hydrogen in Delingha, Qinghai, China.

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9 IRENA and ILO (2023), *Renewable energy and jobs: Annual review 2023*

Since last year, distributed projects, which originally had convenient grid access conditions, are now encountering opposition from different energy actors. There is an increasing number of cities and counties that have limited distributed capacity on the distribution grid and have restricted its rapid deployment. Additionally, Shandong, Henan, and Hebei, along with other major solar PV provinces, have adjusted the peak-valley tariff range for users, and are using price mechanisms to optimise the development of distributed installations.

The *Energy Work Guidance* released by NEA in March 2024 mentions an upcoming revision and publication of the *Measures for the Administration of Distributed PV Electricity Generation Projects*. In addition, it described further advancements of pilot programmes for enhancing grid capacity for distributed projects. This indicates that grid capacity has become a bottleneck for the development of distributed projects, and efforts should be made to unlock the full potential of grid access and foster the healthy growth of distributed solar PV on the grid. In that direction, energy storage has increasingly become a key technology for China to support the growth of renewable energy capacities. By serving as a regulator for the energy system, energy storage can enhance the grid's capability to integrate, distribute, and manage clean energy. Yet, the development of energy storage also faces multiple challenges, including at the supply chain and regulatory framework level.

Based on market data, the solar PV industry has been experiencing challenges in supply and demand since 2022. Installed capacity has gained a relatively large boost in recent years, with increased supply, superimposed on panic price cuts by some companies. However, due to declining profitability, it is worth pondering how much of the company's originally planned capacity can still be achieved from a practical point of view. Furthermore, price fluctuations are a natural feature of any market. According to economic principles, the current low prices are unlikely to persist in the long term. In fact, signs of stabilisation in the prices of some products are already being seen. In recent years, the "New Trio"

of industries (solar panels, electric vehicles, and lithium batteries) from the solar PV industry side has grown close to 70% globally, while domestic growth in China has approximately reached 150%. From the perspective of industry and economic development, only maintaining an appropriate "surplus" status can trigger competition in the industry, address the price monopoly, foster technological progress, and ultimately emancipate productive forces.

6. Conclusions

China's solar PV market is expected to remain strong in the coming years with an anticipated significant growth in the future. According to CPIA, China's new solar PV installations will reach between 252 GW and 317 GW in 2030, based on conservative estimates and optimistic estimates, respectively (see CPIA's market update on China at p. 84). However, in 2024, due to a combination of factors, a small decline in new installations, the same as that in 2023 can be expected for both conservative and optimistic forecasts. Despite these challenges, China's PV industry is well-positioned for continued growth.

China's solar PV industry has transitioned from a period of rapid growth to a market-driven phase. As a result, several significant challenges are presented and need to be urgently addressed. Both centralised and distributed projects are under pressure because of the uncertainty of consumption patterns. Additionally, the traditional FIT policy has been undergoing significant changes, prompting the need for a new model as solar PV increasingly participates in electricity market transactions. For example, a number of provinces have begun to explore time-of-use tariff mechanisms, which will become more common in the foreseeable future. The introduction of this type of market-driven tariff policy will inevitably have an impact on the economic efficiency and investment decisions of solar PV projects. The industry also faces challenges related to land, water, and sea use restrictions, which require supporting measures on policy and project levels. Furthermore, there are challenges related to grids, storage, land costs, and the need for additional investment in development.

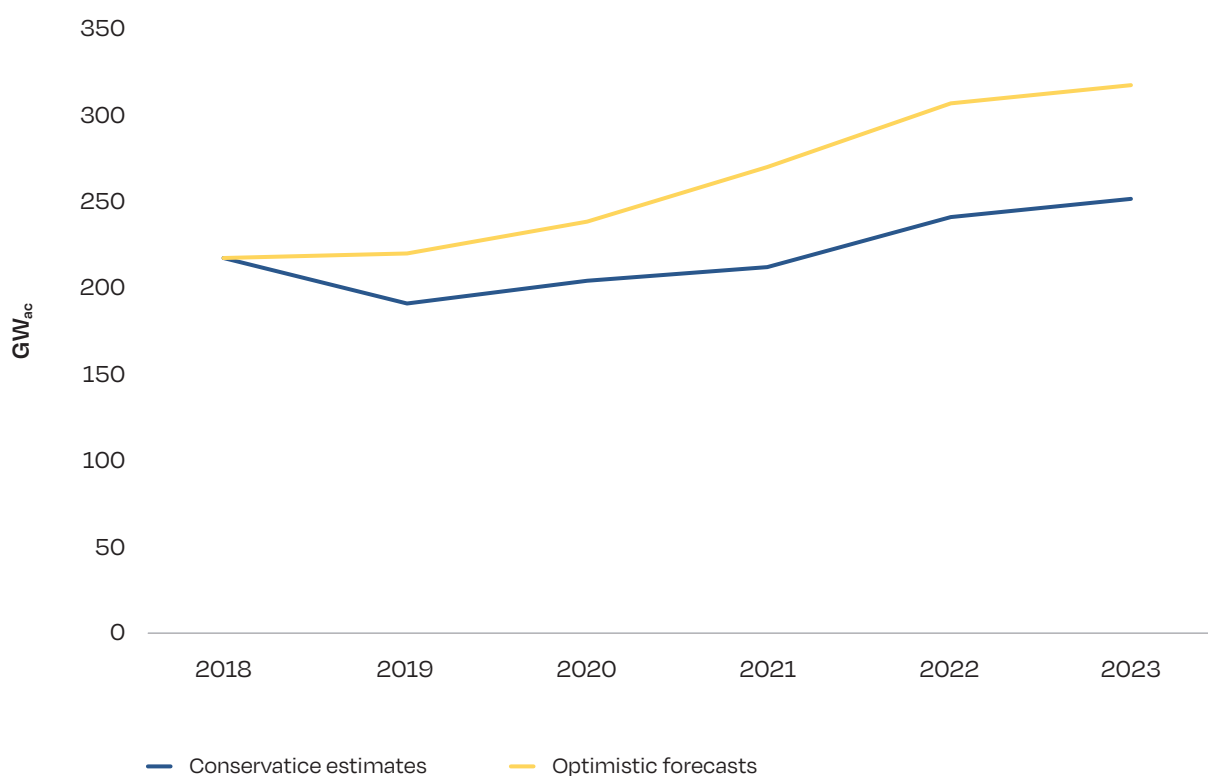
3 Focus: China solar PV market / continued

Increased solar PV production capacity has intensified price competition in China's industry, posing a survival test in the market for many businesses. However, internal competition is necessary for the industry to achieve high-quality development. It will drive efficiency improvements, enhance individual

company competitiveness, and ultimately lead to industry-wide upgrades. Despite these challenges, China's solar PV industry is well-positioned for continued growth and ultimately realise great development as a whole.

Author: Chinese Renewable Energy Industries Association (CREIA) & Global Solar Council (GSC).

FIGURE 48 CHINA ANNUAL SOLAR PV INSTALLED CAPACITY FORECAST 2023-2030, BY CPIA



SOURCE: CPIA.

© SOLARPOWER EUROPE 2024



In 2023, 31 countries across the world installed more than 1 GW of new solar capacity. That's three more than in 2022, when 28 countries reached this level (see Fig. 49). The new entrants are the UAE, Sweden, Saudi Arabia, Romania and Bulgaria, while Denmark and Israel, which were GW markets in 2022, did not reach that scale in 2023.

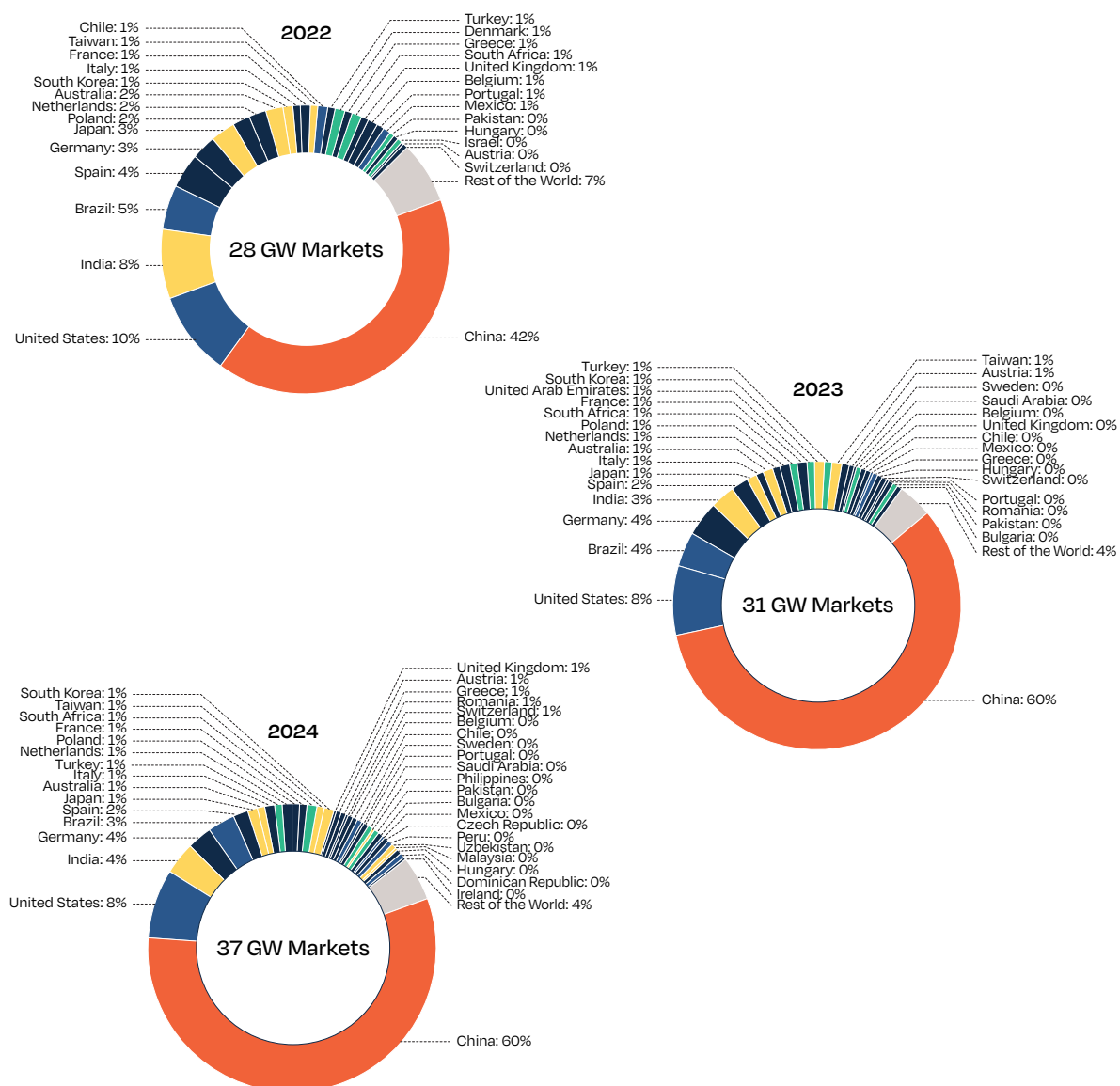
The increasing number of GW-scale markets illustrates the continued solar momentum, as more countries find in solar PV a low-cost and versatile solution to their energy needs and decarbonisation strategies. According to our Medium Scenario forecast, the upward trajectory will continue in 2024, as we anticipate six more countries will join the group to a total of 37 GW-scale markets. In 2025, we expect to reach the milestone of 50 GW-scale markets worldwide.

As in previous Global Market Outlooks, national solar associations from markets that have added more than 1 GW in the previous year have been invited to present their local expert views on their home markets (which sometimes differ from our estimates that are based on several sources). Many of these associations, like SolarPower Europe, are members of the Global Solar Council (GSC), which is a long-time supporter of the Global Market Outlook.

The number of GW markets in the European Union keeps increasing, growing from 12 GW markets in 2022 to 14 GW markets in 2023. This year, the EU-27 is assessed as a whole, ranking #2 in terms of combined annual installed capacity; only European markets outside the EU are featured individually. For a detailed analysis of each EU GW market from the point of view of our national association members, please read our latest EU Market Outlook for Solar Power, published in December 2023.

4 GW-scale markets / continued

FIGURE 49 GW-SCALE SOLAR PV MARKETS 2022-2024



© SOLARPPOWER EUROPE 2024

1.	CHINA China Photovoltaic Industry Association (CPIA)
2.	EU-27 SolarPower Europe
3.	UNITED STATES The Solar Energy Industries Association (SEIA)
4.	BRAZIL Brazilian Photovoltaic Solar Energy Association (ABSOLAR)
5.	INDIA National Solar Energy Federation of India (NSEFI)
6.	JAPAN Japan Photovoltaic Energy Association (JPEA)
7.	AUSTRALIA Smart Energy Council (SEC)
8.	SOUTH AFRICA South African PV Industry Association (SAPVIA)
9.	UNITED ARAB EMIRATES Middle East Solar Industry Association (MESIA)
10.	SOUTH KOREA Korea National University of Transportation
11.	TÜRKIYE Turkish Solar Energy Association (GÜNDER)
12.	TAIWAN Taiwan Photovoltaic Industry Association (TPVIA)
13.	SAUDI ARABIA Middle East Solar Industry Association (MESIA)
14.	UNITED KINGDOM Solar Energy UK (SEUK)
15.	CHILE Chilean Solar Association (ACESOL)
16.	MEXICO Mexican Association of Solar Energy (ASOLMEX)
17.	SWITZERLAND Swissolar
18.	PAKISTAN Pakistan Solar Association (PSA)

1. China

Overview of PV developments

In 2023, China's new installed PV capacity exceeded 216 GW_{AC}, an increase of 148% year-on-year. New installations reached a new record high, and the speed of new PV installation in China is continuing to accelerate. By the end of 2023, the cumulative PV capacity reached 609 GW_{AC}.

The fast development of China's PV installed capacity in 2023 was mainly due to the combined effect of connecting the first set of large utility-scale projects to the grid, and the rapid decline in the price of PV system cost components. By the end of the year, China's polysilicon production stood at about 1.43 million tonnes, up 67% from 2022; wafer production reached 622 GW, an increase of 68%; cell production totalled 545 GW, up 65%; while module production reached 499 GW, up 69% over the previous year.

National targets for solar PV

In April 2024, the National Energy Administration issued the "Guidance on Energy Work in 2024", providing guidance on the deployment of energy work throughout the year, as well as development targets. The document highlights the need to keep optimising energy infrastructure, and increased targets for the proportion of installed power generation from non-

fossil energy sources to 55%. Specifically, it increased the share of power generated by wind and solar to more than 17% of the country's national generation capacity.

The year 2024 marks the 75th anniversary of the founding of the People's Republic of China. It will also be a crucial year to get on track to reach the 2025 targets of China's 14th Five-Year Plan. The country will continue to consolidate the advantages of the development of the PV industry, expand the supply of clean, low-carbon energy, and actively promote the decarbonisation and low-carbon transformation of China, both at the local and national levels.

Drivers for solar growth

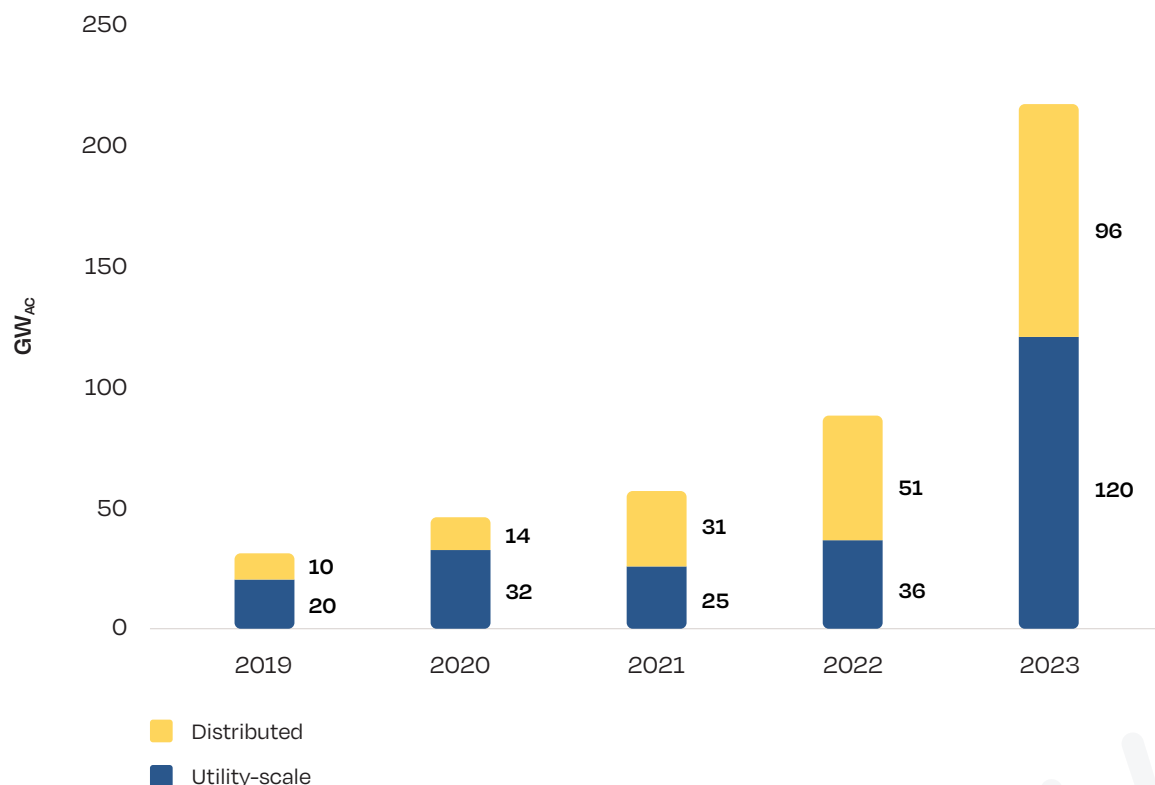
In the future, it is anticipated that China's PV industry will evolve towards greater efficiency, greener practices, and enhanced smart capabilities. The PV industry is expected to align with the construction of new electric power systems, enhancing efficiency and the use of green, intelligent processes in value chains for the manufacturing sector. In terms of applications, China will promote the development of key technologies for integrated 'solar-storage-grid' solutions.

In order to actively advance carbon-peaking and carbon-neutrality, China is accelerating the deployment of a new energy system. The rapid development of PV and other forms of renewable energy is now an inevitable trend, and the expansion of green electricity lays a solid foundation in this regard.



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FIGURE GW 1.1 CHINA ANNUAL SOLAR PV MARKET 2019-2023, BY CPIA



SOURCE: CPIA.

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Another driver for solar growth in China comes from a growing electricity demand. Electricity consumption is expected to achieve an average annual growth of 6% over the 14th Five-Year Plan period, which will require additional renewable energy capacity.

Solar power can be integrated seamlessly with various sectors, including construction, transportation, and agriculture. These dual land-use projects allow for the exploration of new PV applications and business models, further broadening the scope for solar development.

Utility-scale vs. distributed and rooftop developments

In 2023, utility-scale new installed capacity reached 120 GW_{AC}, accounting for 56% of all installations. Distributed PV annual installed capacity reached 96 GW_{AC}, providing the remaining 44% share. Having overtaken new distributed installations, new utility-scale installations have become the main driver for new

PV installations in 2023. The residential sector installed 43.5 GW_{AC}, accounting for 45% of the newly distributed PV additions, while the commercial and industrial (C&I) segment added 52.8 GW_{AC}, accounting for 55% of the newly distributed PV additions.

Challenges for the market

Although the participation of solar PV in the electricity spot market trading has become a trend in most provinces, introducing new energy sources on the spot market can cause fluctuations in electricity price. Unlike medium and long-term trading, spot power market trading altered the previous solar PV business model. Additionally, different provinces have different rules for spot power trading, making it difficult to predict PV revenues, which is becoming a new challenge. In the future, under the spot market system, electricity prices will change with time and location.

4 GW-scale markets / continued

Therefore, PV enterprises should break away from the fixed-price and explore ways to participate in spot market-based trading of PV power generation.

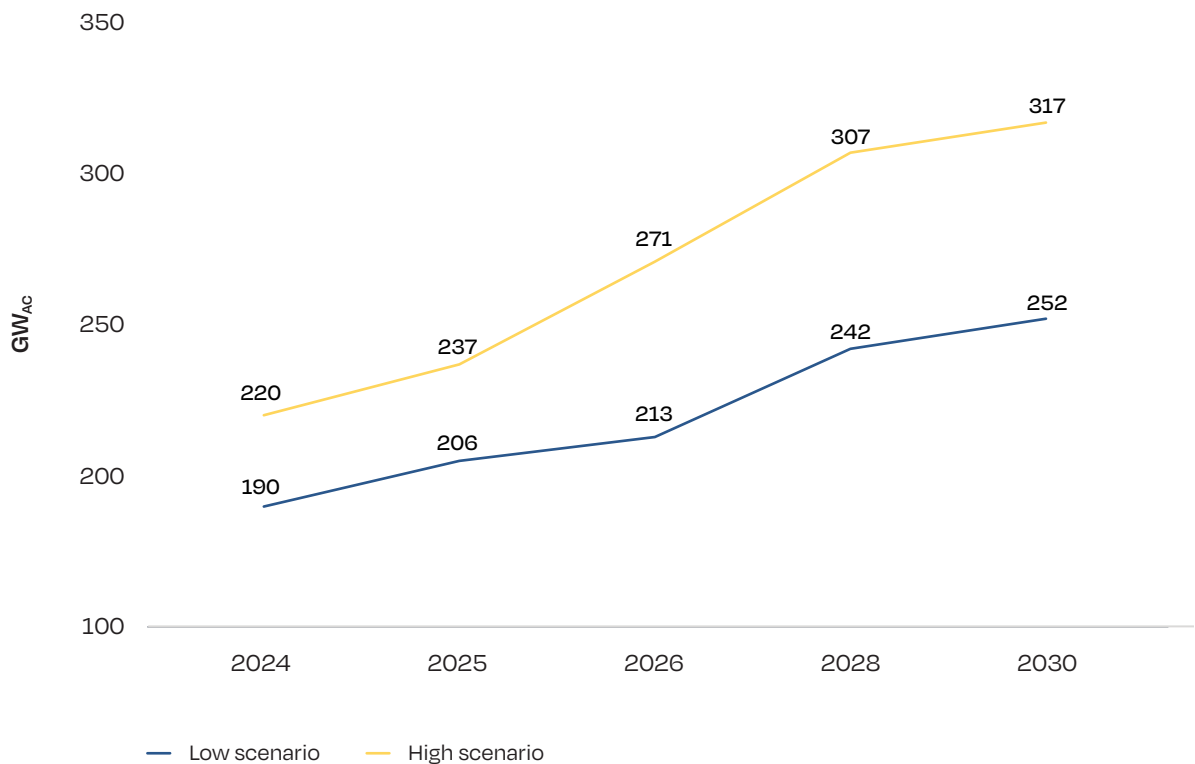
Outlook for the years 2024-2030

In December 2023, at COP28, 118 countries signed the Global Renewable Energy and Energy Efficiency Commitment to triple global installed renewable energy generating capacity by 2030, and double the average annual rate of improvement in global energy efficiency. This will contribute to advance the deployment of renewable energy, especially solar, which is the fastest growing renewable source.

In 2023, China's PV industry has seen substantial growth both in terms of manufacturing and installations. For 2024, the Chinese Photovoltaic Industry Association (CPIA) expects China's annual PV market to reach 190-220 GW_{AC}, keeping the record-breaking order of magnitude reached in 2023, while global installed capacity may reach 390-430 GW_{AC}. By 2030, we can expect China's average annual PV installed capacity to stay above the 250 GW level (see Fig. GW 1.2).

Author: China Photovoltaic Industry Association (CPIA).

FIGURE GW 1.2 CHINA PV MARKET SCENARIOS 2024-2030, BY CPIA



SOURCE: CPIA.

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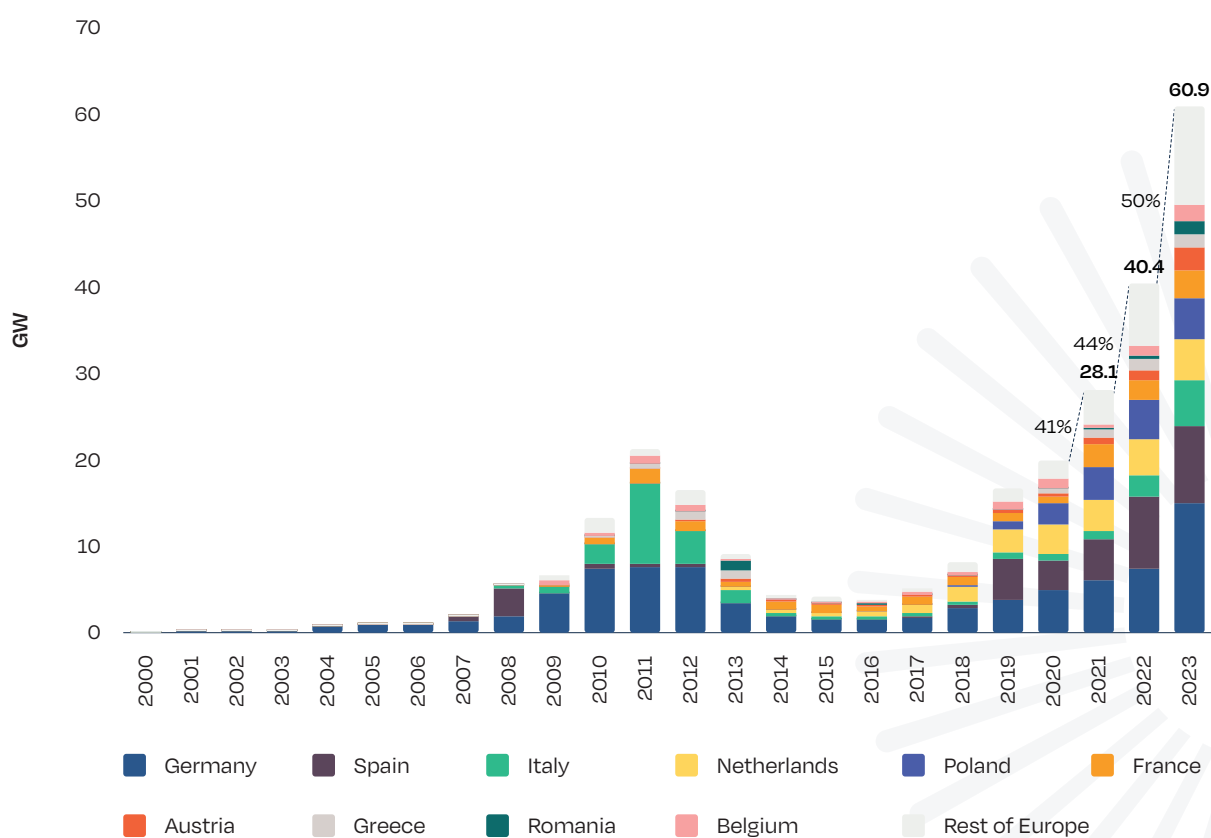
2. EU-27

Overview of PV developments in 2023

2023 was a remarkable year for solar PV deployment in the EU, which added 61 GW and reached 264 GW of cumulative capacity, equal to 50% year-on-year growth, up from the 40.4 GW installed in 2022 (see Fig. GW 2.1.). This is significantly higher than the 55.9 GW SolarPower Europe estimated in its Dec. 2023 published [EU Market Outlook](#), a number that was gradually adjusted upwards over the last few months as official numbers from Member States became available. A combination of high energy prices, political

support, improved permitting and administrative framework, a higher than expected order pipeline from 2022 that was finally met with wide product and installer availability, created the conditions for the record growth in 2023. The year 2023 showed the best annual growth rate since 2018, and was the year with the highest absolute market growth ever, with over 20 GW more than the market size in 2022. Germany led this surge, installing 15 GW or a quarter of the total EU installations, while Spain and Italy completed the podium, with over 14 GW combined. Market diversification is also increasing, as 14 out of 27 Member States crossed the GW scale in annual capacity additions.

FIGURE GW 2.1 EU-27 ANNUAL SOLAR PV INSTALLED CAPACITY 2000-2023



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4 GW-scale markets / continued

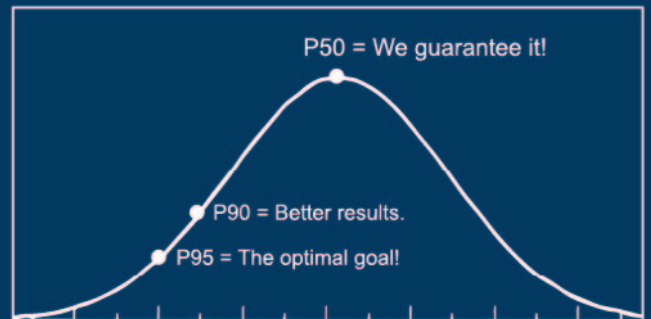
Germany emerged as the top performer in 2023, with a 104% annual growth and exceeding its 9 GW national target for 2023 by a large margin. The market is showing maturity, with regulatory stability and high public acceptance of PV, as well as a balanced contribution to capacity additions from all PV segments. Contrary to earlier expectations, Spain slightly rose (5%) in 2023, bringing the market to 8.9 GW, while the Italian market more than doubled (111%) and reached 5.2 GW. Other EU GW markets were the Netherlands (4.9 GW), Poland (4.6 GW), France (3.2 GW), Austria (2.7 GW), Sweden (2 GW), Belgium (1.9 GW), Greece (1.6 GW), Hungary (1.6 GW), Portugal (1.5 GW), Romania (1.5 GW), and Bulgaria (1.2 GW). Czech Republic missed the GW mark by around 20 MW. Among the 27 Member States, 26 installed more solar capacity than the year before.

Drivers & Challenges

Government policies and regulations play a critical role in shaping growth for renewable energy in the EU, particularly as the sense of urgency following the energy price crisis diminishes. Energy security concerns and high energy prices catalysed rapid growth in EU solar PV installations during 2022-2023. Additionally, as countries sought to mitigate volatile fossil fuels, EU policymakers adopted supportive measures for solar deployment, among which the setting of the 750 GW (600 GW_{ac}) REPowerEU target by 2030, improved permitting conditions, and a EU rooftop solar strategy. Nevertheless, high energy prices have been the key driver of growth in 2022 and 2023, making many projects economically viable, despite several bottlenecks and a high interest rate environment. Today, with energy prices receding towards pre-crisis levels, the role of policymakers has become even more critical to maintain high installation rates and prevent future dependency on fossil fuels.

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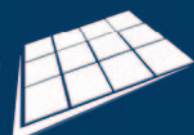
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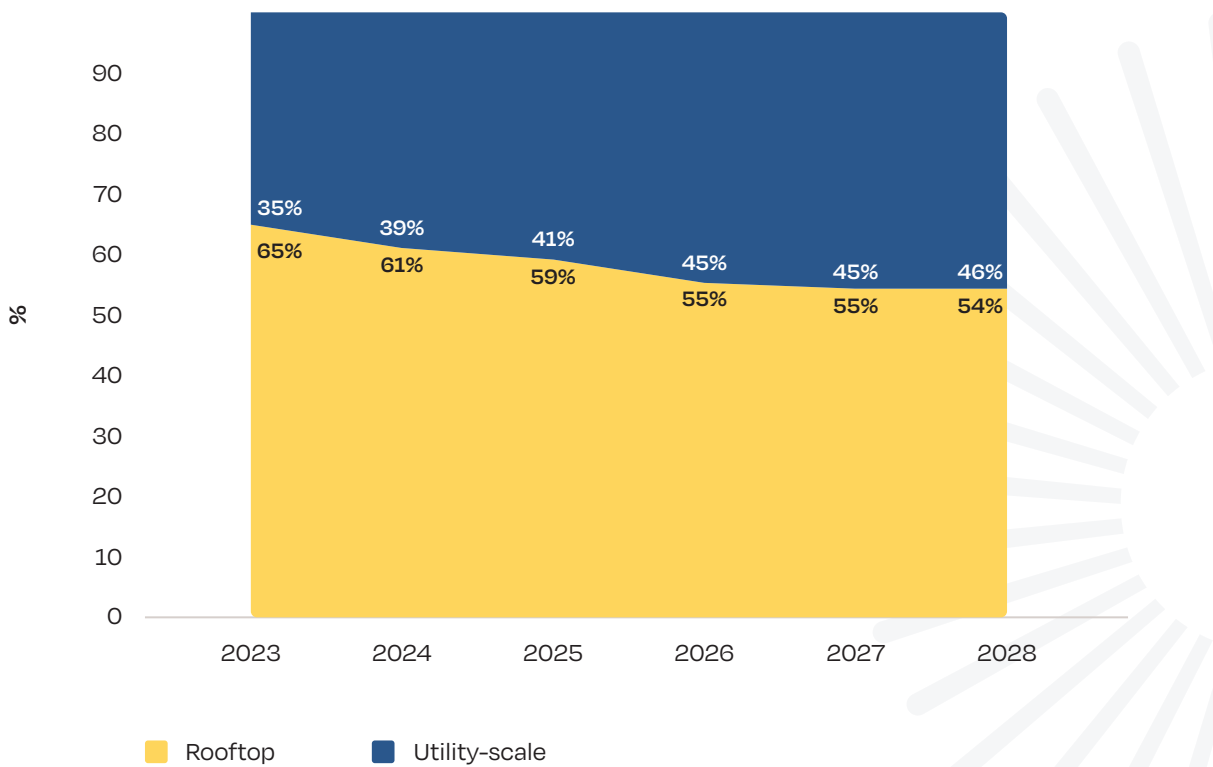
A promising development is the recent increase in solar ambition under the National Energy and Climate Plans (NECP) targets. However, despite this progress, the targets are still 30% below SolarPower Europe’s Medium Scenario trajectory projected for 2030, and 17% below the REPowerEU target.

Compounding this uncertainty are the recent and upcoming Member State and EU elections, raising concerns about policy stability and adherence to deployment targets. For example, the newly elected government coalition in the Netherlands has announced an abrupt halt of policy support for residential solar (compared to a gradual phaseout) and reduced future support for electrification. Such sudden policy changes, particularly from governmental shifts, significantly harm investor

certainty and public perception surrounding renewable energy, and pose a threat to continued solar market growth. Fortunately, EU and national policies adopted during the period of heightened urgency could help mitigate the impact of less solar-oriented government changes, and are expected to continue influencing solar deployment in the medium term.

In this regard, a significant driver to solar growth in the EU is the strong increase in auctioned solar capacity, particularly in the utility-scale segment. The momentum created by larger auctioned capacities will lead to an increase in utility-scale contributions, growing from 35% in 2023 to 45% and beyond from 2026 onwards (see Fig. GW 2.2.). Still, rooftop solar is expected to remain the largest market segment until the end of our forecast period in 2028.

FIGURE GW 2.2 EU-27 ANNUAL ROOFTOP VS. UTILITY-SCALE SEGMENTATION 2023-2028



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4 GW-scale markets / continued

PV technology advancements, economies of scale at production level, and large overcapacities in global PV manufacturing have contributed to a drop in product prices, which decreased overall significantly, and for solar modules by 50% over the course of 2023. These price cuts have partially masked the worsening in the business case for large-scale solar. Within a high interest rate environment and rising PV penetration shares, large-scale solar projects increasingly experience high curtailment and price cannibalisation. This is particularly visible in Southern European markets, which exemplify the challenges that await the rest of Europe if grid and flexibility constraints are insufficiently addressed. Without adequate grid development, storage deployment and demand response, the growth of the solar market is capped. After a year of record growth, these issues only magnify, revealing further vulnerabilities in the grid during peak periods. As such, the future of solar is tightly bound to policymakers' focus on grid reinforcement and flexibility-boosting policies. This includes clear policy actions supporting battery storage and the electrification of heating, cooling and transportation.

Until recently, product component availability and skills were identified as crucial bottlenecks to solar growth in the EU. Today, access to components does not pose a major hurdle anymore. However, exceptions exist, such as a shortage of power transformers that continues to influence large-scale deployment timelines. On skills, solar companies have made considerable strides in

enhancing the efficiency of their teams and attracting additional workforce, also from less flourishing sectors, like construction. As such, the skill shortage in the EU solar industry has become less of a barrier to growth.

Finally, the change in the economic and financial landscape has been detrimental to the solar business case. The lifting of several bottlenecks and the availability of low-cost components have somewhat masked the damaging effects of a more challenging financial environment for solar. Persisting high interest rates are no longer accompanied by high energy prices and low grid congestion. Additionally, the continuation of administrative hurdles, regular court cases, and unstable policy frameworks has undermined investor certainty by making solar a comparatively riskier asset to invest in. Consequently, the cost of capital has increased, adding to the financial burden on solar projects. In response, de-risking contracts such as PPAs have risen in popularity. This is also notably linked to the higher cost-bearing capacity of large corporations with sustainability targets, which will continue to be a key driver. On small-scale deployment, the high inflationary environment is hurting purchasing power, postponing many households' decisions to invest in solar PV. This is especially true for lower-income households who are unable to access affordable green loans or other policy measures to help bridge the initial investment barrier. For continued market growth, it is essential that both the business case of large-scale solar and the accessibility of small-scale solar is improved.



48 MW, Tábua, Portugal.

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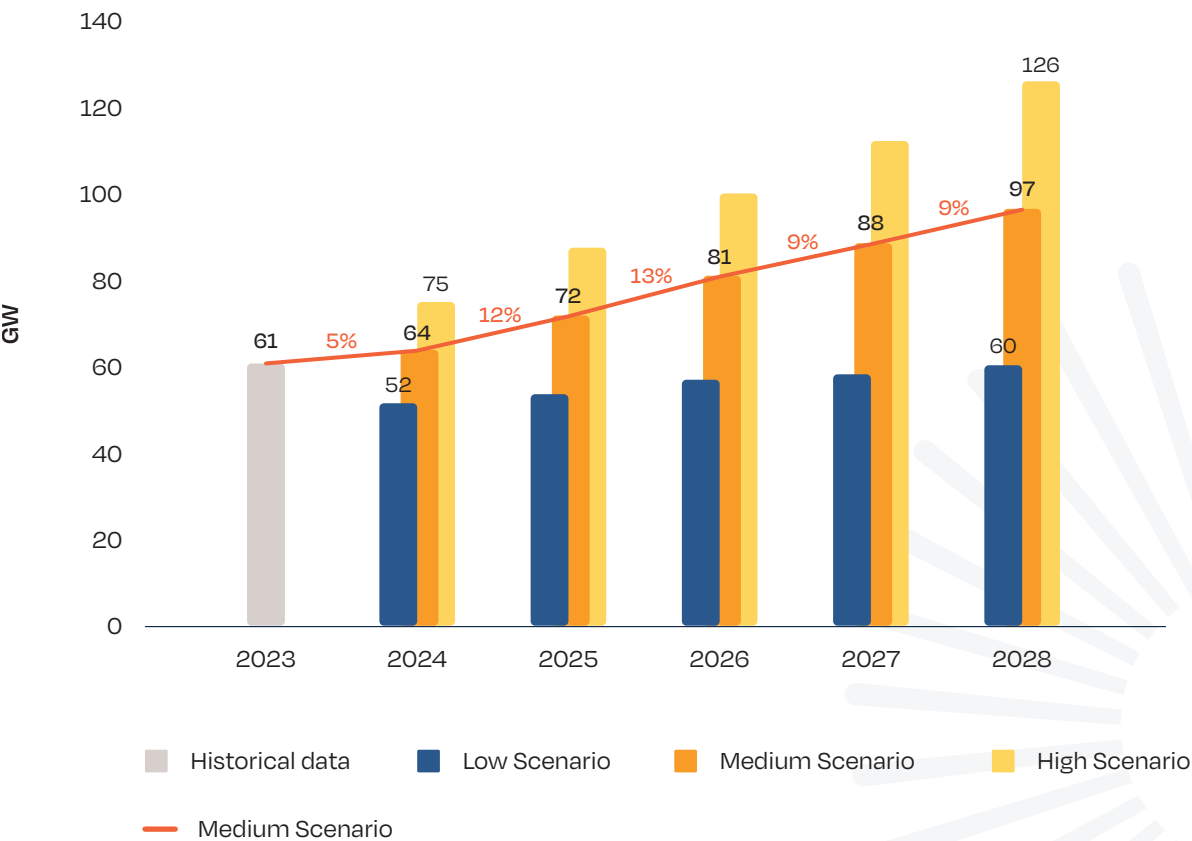
Outlook

After the record year registered in 2023, growth is expected to slow significantly to single digits in 2024 – the first time since the lift of trade measures in 2018. This reduction can be attributed to the high baseline set by 2023 and a changing market environment. While more mature and well-established markets are now stabilising their growth path, cyclical markets face downturns because of unadjusted policy support and regulatory

instability. Consequently, 19 out of 27 markets are expected to grow year-on-year, 7 less than in 2023.

In our most likely scenario, we expect the EU market to grow 5% to 63.9 GW in 2024, followed by a partial recovery in the next two years, up 12% to 71.8 GW in 2025 and 13% to 80.9 GW in 2026 (see Fig. GW 2.3.). By 2028, our Medium Scenario anticipates 96.7 GW of annual capacity additions, although the High Scenario reaches the 100 GW mark already in 2026, and 126 GW in 2028.

FIGURE GW 2.3 EU-27 ANNUAL SOLAR PV MARKET SCENARIOS 2023-2028



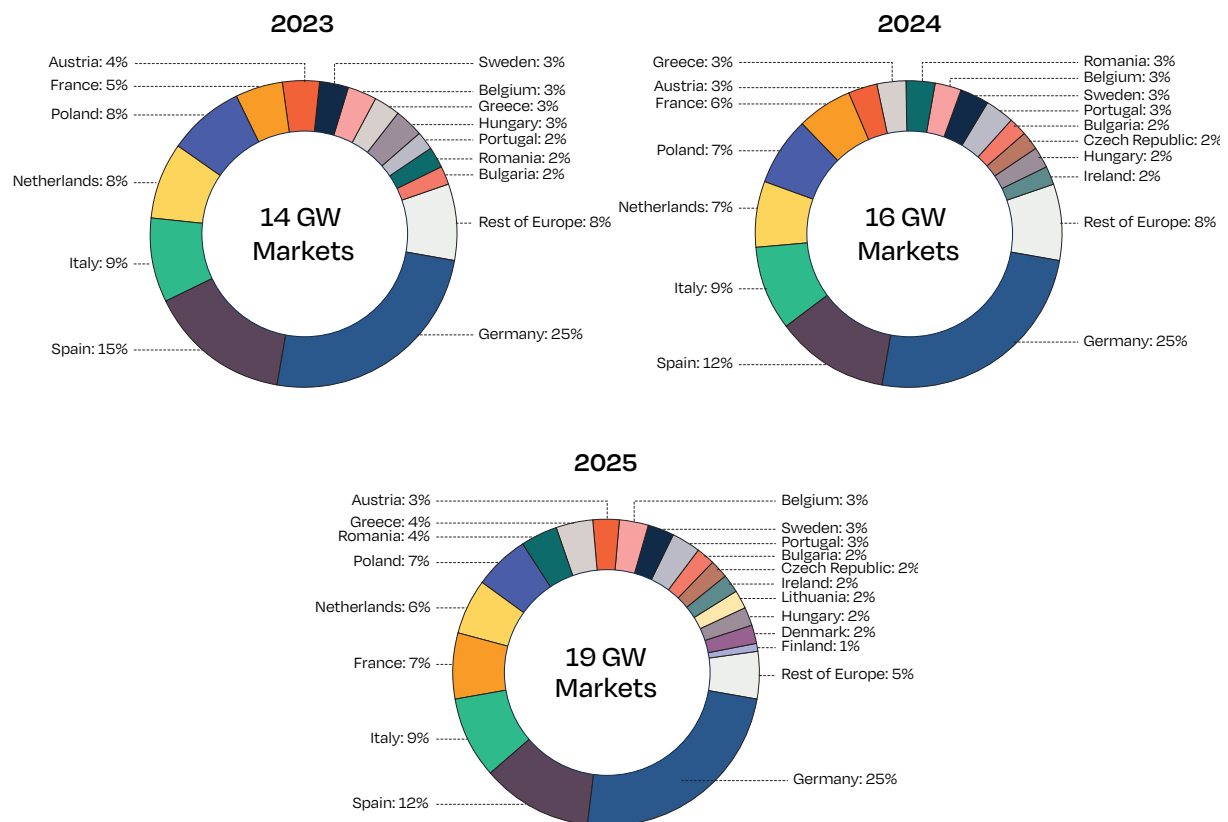
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4 GW-scale markets / continued

Market concentration is decreasing rapidly. In 2019, the top 10 markets installed 95% of solar capacity in the EU, and the top 5 accounted for 78%. Since then, that share has been steadily decreasing to 83% for the top 10 in 2023, and 65% for the top 5. By the end of

2024, the top 10 are estimated to account for 80% of the installed capacity while the top 5 will only cover 60%. This trend is expected to continue, with 16 GW-scale markets by the end of 2024 compared to 14 GW markets in 2023. By 2025, 19 out of 27 EU markets will reach the GW scale (see Fig. GW 2.4).

FIGURE GW 2.4 EU-27 GW MARKETS



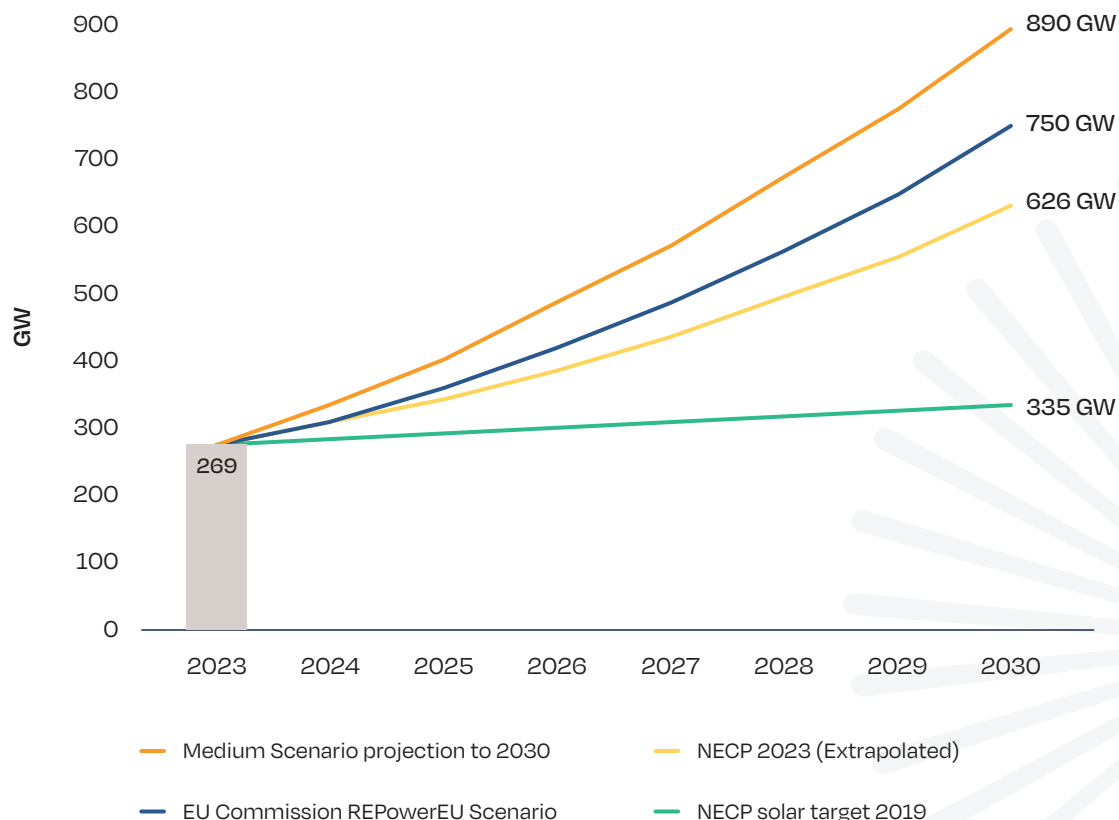
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Our cumulative installation forecast indicates that NECP targets set by Member States are still well below expected solar market developments. Even though the aggregated NECP targets are 87% higher than the original targets measured in 2019, they remain significantly below EU ambition and projected market developments (see Fig. GW 2.5). Our extrapolation of the Medium Scenario to 2030 results in 890 GW of operating solar PV capacity by the end of the decade, which is 42% higher than the aggregate NECP targets. This means that the EU is expected to reach the aggregated NECP target by 2028 already. The

REPowerEU target, which is more ambitious, is still 19% below the Medium Scenario projections and is expected to be reached by 2029 in the Medium Scenario and 2028 in the High Scenario. It is important to note that these under-ambitious targets can have negative impacts on renewable energy deployment. For instance, insufficiently ambitious targets can lead to an underinvestment in grid infrastructure and demand response boosting policies. Realising true growth potential too late can cause significant hurdles in the future.

Authors: Jonathan Gorremans, Raffaele Rossi & Michael Schmela, SolarPower Europe.

FIGURE GW 2.5 EU-27 INSTALLED SOLAR PV CAPACITY SCENARIOS 2023-2030



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3. United States

Overview of 2023 PV development

The US solar industry had a record-breaking 2023.

In the first full year of the Inflation Reduction Act's clean energy incentives, US solar businesses dramatically expanded deployment across market segments, and domestic solar manufacturers moved forward major investments across the solar value chain.

While the US solar market is growing rapidly, the siting, permitting, and interconnection of solar projects remains a key challenge for developers, and global market conditions continue to put pressure on US producers.

According to the [Solar Energy Industries Association's \(SEIA\) Solar Market Insight Year in Review](#), the US added a record 32.4 GW of new solar capacity in 2023. This represents a 51% increase from 2022. In terms of total capacity, the US now operates a solar fleet of more than 180 GW, up 22% from 148 GW in 2022 (see Fig. GW 3.1).

Solar deployment increased across all market segments. The utility-scale segment captured the largest share of new capacity and the largest growth rate, with 22.5 GW and a 77% increase from 2022. The residential sector installed 6.8 GW, growing 13% from 2022, while the C&I segment installed 1.8 GW (+19%) and the community solar sector added 1.1 GW (+3%).

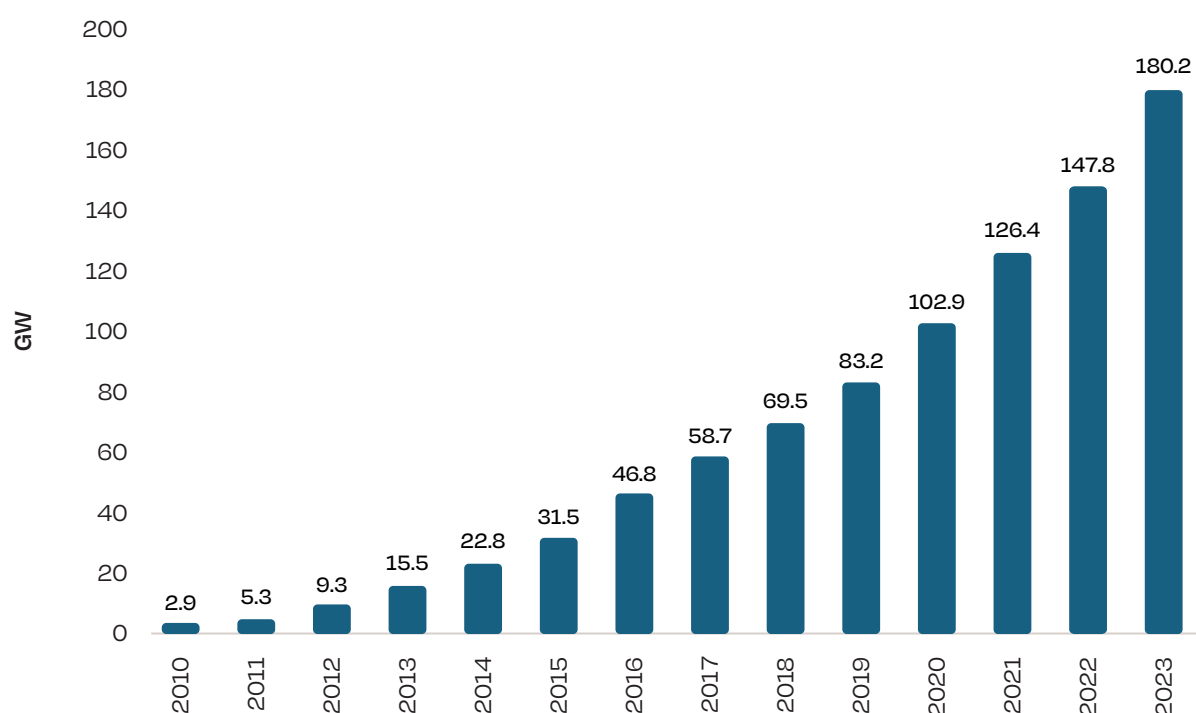
Drivers for solar growth

In 2021, the United States set a goal to reach 100% carbon-free electricity by 2035 and, since then, the US solar industry has raced to meet this goal.

Solar was the top technology for new electricity generation capacity in 2023, accounting for 53% of new generating capacity. This was the fifth consecutive year that solar led new capacity additions and the first time solar accounted for over half of new electricity generating capacity.

In fact, it was also the first time in 80 years that a renewable electricity source accounted for over 50% of new annual capacity additions to the US grid (see Fig GW 3.2).

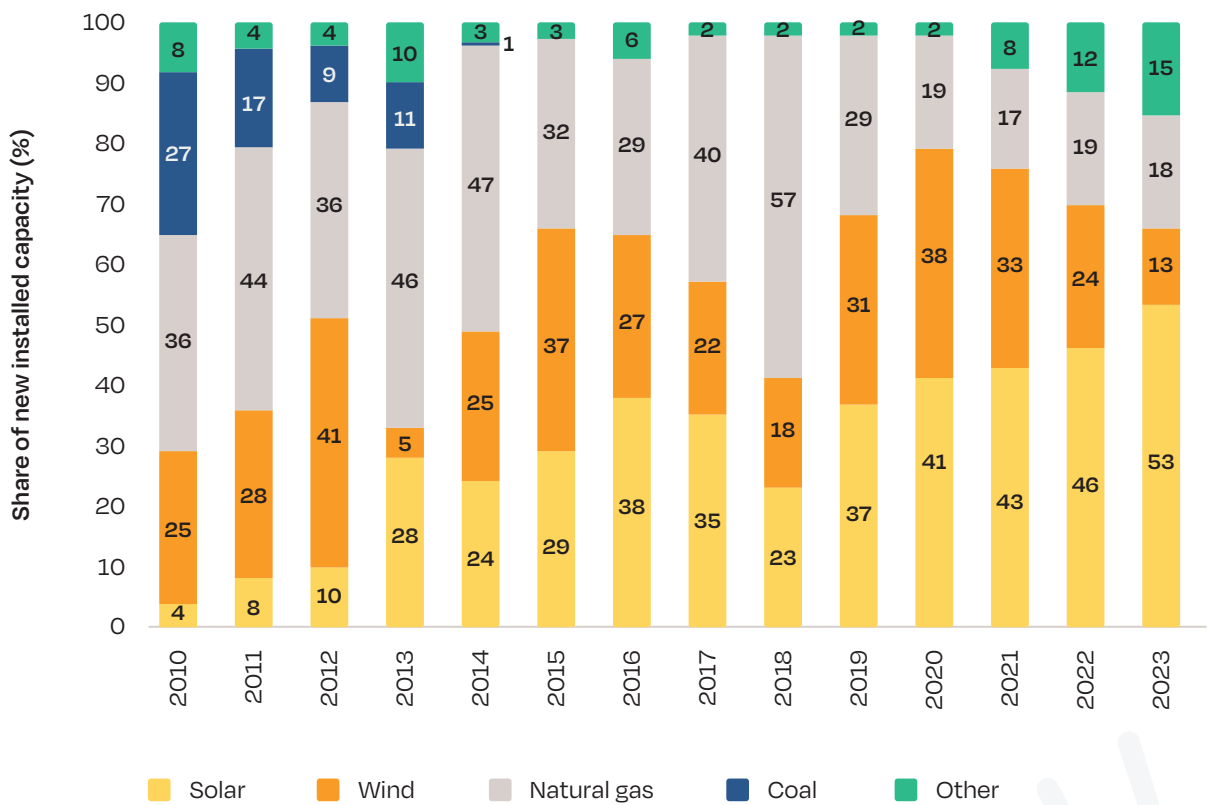
FIGURE GW 3.1 UNITED STATES CUMULATIVE SOLAR PV CAPACITY 2010-2023, BY SEIA



SOURCE: SEIA/Wood Mackenzie Solar Market Insight Report 2023 Year in Review.

© SOLARPOWER EUROPE 2024

FIGURE GW 3.2 NEW US ELECTRICITY GENERATION CAPACITY ADDITION 2010-2023, BY SEIA



SOURCE: SEIA/Wood Mackenzie Solar Market Insight Report 2023 Year in Review.

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Elm Branch solar park, 163 MW, Texas, United States.

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4 GW-scale markets / continued

By 2040, solar is expected to be the largest source of total US electric generating capacity. Long-term deployment and manufacturing incentives in the Inflation Reduction Act (IRA) are foundational to meeting these milestones.

The IRA includes a 10-year extension of the investment tax credit (ITC), which is critical to financing projects and driving deployment. Growth may also be boosted by recent interconnection reforms. In 2023, the Federal Energy Regulatory Commission (FERC) issued an order which attempts to reduce solar project development timelines by penalising transmission providers for not conducting interconnection studies in accordance with defined deadlines.

The IRA also included new incentives for domestic solar manufacturing across the value chain. Some of these facilities have already started production, including new inverter, racking, and module manufacturing. More facilities are expected to come online in the next few years, including missing parts of the US solar value chain such as wafers and cells. In 2023 alone, solar module manufacturing capacity grew from 8.5 GW to over 16 GW.

Challenges

While the US solar industry is larger than ever, many technical, policy, and regulatory issues continue to restrict the long-term pace of solar growth.

The most pressing challenges are related to siting, permitting, and interconnecting solar projects. Long interconnection queues with US regulators and a variety of local permitting challenges can alter the timeline and cost of solar projects. The move by FERC to reform interconnection regulations is positive, but success will ultimately depend on effective implementation. Addressing these barriers will be key to meeting US clean energy goals and the country's growing power demand.

The value of solar in the US is also impacted by federal and state policy changes. The Federal Reserve's decision to maintain high interest rates will limit solar growth across the rooftop and utility-scale sectors. In addition, the outlook for rooftop solar could be impacted if more states follow California and transition away from net metering to programmes that lower export compensation rates for solar customers.

The US manufacturing buildout is expected to reduce supply chain risk and encourage more deployment. While domestic manufacturing is rapidly expanding, it takes time to build domestic supply chains, and falling global solar prices are putting pressure on domestic producers. Meanwhile, the risk of potential trade action continues to inject uncertainty into the US deployment outlook.

Finally, there is policy uncertainty tied to the upcoming 2024 presidential election. This could potentially shift US clean energy goals, the impact of recently passed incentives, and the urgency to fix the regulatory challenges impeding more exponential solar deployment.

Author: The Solar Energy Industries Association (SEIA).



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4. Brazil

Overview

In 2023, the Brazilian solar PV market has shown its resilience despite an apparently unfavourable macroeconomic environment in the country. The basic interest rate (SELIC rate) was 13.75% during the first semester of 2023, then slowly decreased in the second semester to reach 11.75% by the end of the year. In Brazil, this basic interest rate has a substantial impact on the local economy, as well as the access to financing. For the solar PV sector, a high cost of capital can significantly impact the availability and feasibility of loans for new solar PV systems.

Adding to this challenging context, several electricity distribution utilities started to block the connection of new distributed solar PV projects, under alleged claims of reverse power flow in different points of the grid. This negatively impacted small and medium solar PV systems' market growth in some areas of the country, resulting in hundreds of millions of US dollars in postponed investments.

Despite these challenges, Brazil's solar PV market remains on a steady upward trajectory, solidifying its relevance in the renewable energy sector. At the end of 2023, the nation achieved a total installed solar PV capacity of 37.9 GW_{AC}, surpassing official government forecasts yet again. Notably, 2023 witnessed the

addition of 12.3 GW_{AC} of new solar PV capacity, with about 8.3 GW_{AC} attributed to distributed generation and nearly 4.1 GW_{AC} to centralised generation.

As a result of this impressive growth, Brazil was ranked as the third largest solar PV market in 2023 in terms of added capacity, and sixth in terms of cumulative installed capacity.

Distributed and centralised PV system developments

In Brazil, solar PV systems are categorised into two main types:

1. **Distributed generation:** this encompasses small- and medium-sized solar PV systems, each equal to or below 5 MW in capacity. These systems participate in the Brazilian national net-metering program, allowing for the exchange of surplus electricity with the grid;
2. **Centralised generation:** large-scale solar PV power plants with capacities exceeding 5 MW. These plants commercialise their electricity through regulated market auctions administered by the Federal Government or through bilateral PPAs within the free electricity market.

As established by the legal framework for distributed renewable generation in the country (Law nº 14,300/2022), the beginning of 2023 saw a change in the net-metering mechanism for new solar PV distributed generation projects. The law introduced a



Parnaíba - MS, Brazil.

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4 GW-scale markets / continued

gradual fee for the use of the grid when electricity is injected into it. Since this change only applied to new systems requested from 2023 onwards, there was a peak of applications for new distributed generation solar PV in 2022. As a result, the distributed generation market boomed to record numbers in 2022. In light of this, there were concerns as to how these new conditions would affect the market in 2023. The good news is that the solar PV distributed generation market remained strong in 2023, decreasing by less than 0.2 GW_{AC} (-2%) when compared to 2022.

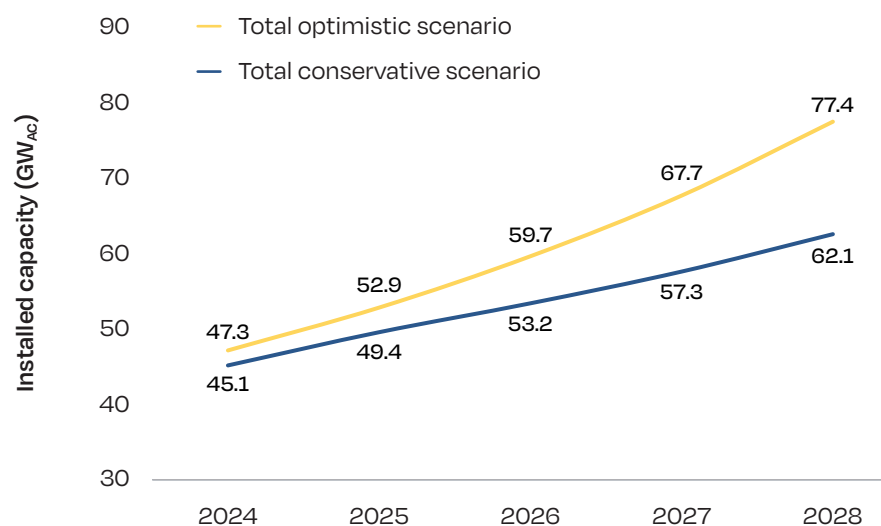
On the other hand, Brazil has made significant progress in large-scale solar PV power plants. In 2023, the annual installed capacity of new centralised generation solar PV systems increased by 47.5%, from

2.8 GW_{AC} in 2022 to 4.1 GW_{AC} in 2023, mostly selling its electricity through bilateral PPAs on the wholesale electricity market.

Solar PV Forecast

The Brazilian Solar Photovoltaic Energy Association (ABSOLAR) forecasts a lower but still robust market performance for solar in 2024. It is expected an annual addition of new 9.4 GW_{AC} of new solar PV capacity, considering both centralised and distributed generation. The projected slowdown is driven by the lack of transmission infrastructure that limits the dispatch of new centralised generation in some regions and limitation for new connection of distributed generation projects.

FIGURE GW 4.1 FORECAST OF SOLAR PV INSTALLED CAPACITY IN BRAZIL, BY ABSOLAR



Distributed Solar Conservative Scenario (MW _{AC})	30,940	32,851	33,699	34,309	34,916
Distributed Solar Optimistic Scenario (MW _{AC})	32,375	34,979	37,940	41,181	44,981
Centralised Solar Conservative Scenario (MW _{AC})	14,179	16,570	19,462	22,962	27,197
Centralised Solar Optimistic Scenario (MW _{AC})	14,903	17,940	21,736	26,482	32,413
— Total Conservative Scenario (MW _{AC})	45,119	49,421	53,161	57,271	107,094
— Total Optimistic Scenario (MW _{AC})	47,278	52,919	59,676	67,663	77,394

SOURCE: ABSOLAR (2023) and official projections from EPE (2022).

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According to ABSOLAR and based on official projections from the Brazilian Energy Research Office (EPE), solar PV may reach between 62.1 GW_{AC} (Conservative Scenario) and 77.4 GW_{AC} (Optimistic Scenario) of cumulative installed capacity by 2028. For distributed generation, the cumulative installed capacity until 2028 may reach between 34.9 GW_{AC} and 45 GW_{AC}. For centralised generation, ABSOLAR estimates average annual capacity installations between 3.3 GW_{AC} and 4.2 GW_{AC} until 2028. The total yearly forecast for both centralised and distributed generation is between 4.8 GW_{AC} and 7.9 GW_{AC} of solar PV average capacity additions between 2024 and 2028.

Challenges and opportunities

Despite the promising future of solar PV in Brazil, there are regulatory and infrastructure bottlenecks that must be considered. Distributed generation is facing challenges for new connections to the grid. This problem is based more on legal and regulatory grounds than on technical difficulty, as electricity distribution utilities are using misinterpretations in the Law nº 14,300/2022 as a justification to deny new connection applications. Furthermore, the lack of transmission infrastructure in certain regions could continue to limit the development of new large-scale

projects. ABSOLAR has been working to defend and represent the solar PV sector towards the Federal Government, the regulatory agency, other technical authorities and the Brazilian National Congress to develop solutions for these problems. To further unlock the immense Brazilian solar PV potential, it is essential to structure good public policies, programmes, and incentives, at the federal, state, and municipal levels.

The current Federal Government has a special focus on social and environmental topics. This is a positive environment for solar PV use in social programmes developed by the public authorities, as an alternative to the country's high electricity price. Taking this into consideration, ABSOLAR is recommending the implementation of solar PV systems in low-income households through the "[Minha Casa Minha Vida](#)", the "[Luz para Todos](#)" and the "[Mais Luz para a Amazônia](#)" government programmes. Additionally, the association is motivating the authorities to expand the use of solar PV in government-owned buildings, assisting the government in reducing their electricity costs and carbon footprint.

In addition to solar, Brazil has an immense potential in energy storage and green hydrogen technologies. According to McKinsey's 2021 article "[Green](#)



817.9 kW Ana Nery Hospital, Santa Cruz do Sul-RS, Brazil.

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4 GW-scale markets / continued

Hydrogen: an opportunity to create sustainable wealth in Brazil and the world", green hydrogen alone may bring more than 200 billion USD in new investments to the country by 2040.

It is important to highlight that, since December 2023, Brazil holds the G20 presidency, representing the most industrialised economies in the world. Additionally, in November 2025 Brazil will host the 30th Conference of the Parties to the United Nations Framework Convention on Climate Change (UNFCCC COP30), in the city of Belém, the capital of the state of Pará. In light of this, the country currently has a

strategic role on the international stage, and has a unique opportunity to showcase its economy and market, both to the international community and renewable energy investors.

ABSOLAR will continue to defend solar PV in Brazil, energy storage, and green hydrogen sectors, and will continue to develop and implement strategic recommendations to increase the widespread adoption of solar PV and sustainable technologies in the country.

Authors: *Dr. Rodrigo Lopes Sauaia*, CEO; *Dr. Vinicius Suppion*, Technical and Regulatory Specialist; ABSOLAR.



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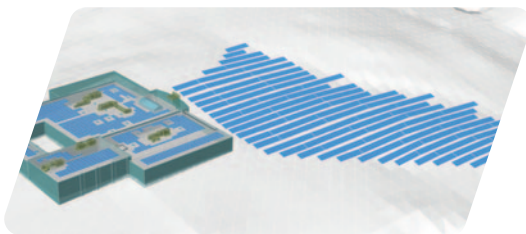
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5. India

Overview of PV developments

2023 has been one of the best years in India's solar industry. The country installed around 15.5 GW of solar in India's Financial Year (FY) April 2023 to March 2024, reaching 81.8 GW of cumulative capacity (see Fig GW 5.1). India's renewable energy capacity as of 31st March 2024 stands at 143 GW without including large hydropower installations, positioning India at 4th place globally in terms of installed RE capacity, and 4th place in terms of solar energy installations, which constitute 57% of the total RE capacity.

Out of the total 81.8 GW cumulative solar capacity, 64.4 GW come from utility-scale installations, while grid-connected solar rooftop installations contribute 11.9 GW. Additionally, hybrid projects integrating solar components have added 2.6 GW to the nation's solar capacity, while off-grid solar systems contribute 3 GW. Furthermore, it is predicted that in the next financial year 2024-2025, India will add around 26-30 GW of solar energy installations. Along with accelerating solar deployment, India's solar manufacturing saw a 100% growth, increasing module manufacturing capacity from 21 GW to 44 GW.

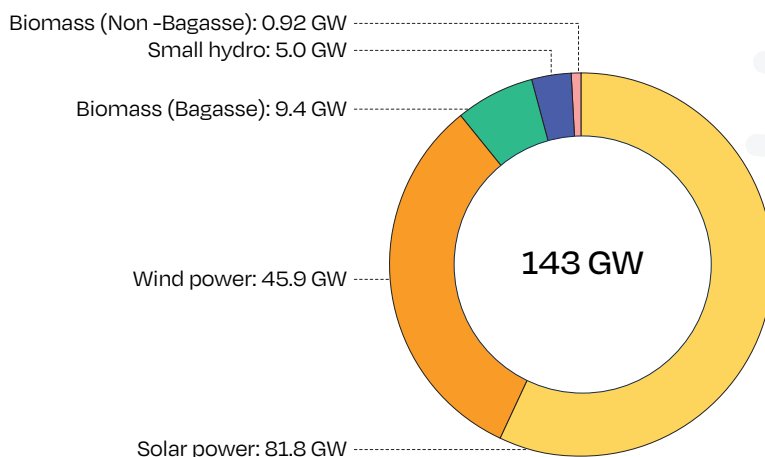
Utility-scale and rooftop developments in 2023-24

In the objective of escalating residential rooftop installations in India, in February 2024, the government launched the PM Surya Ghar: Muft Bijli Yojana scheme, aiming to provide free electricity to households in India. Under the programme, households will be provided with a subsidy to install solar panels on their roofs. The subsidy will cover up to 40% of the cost of the PV system. The scheme is expected to benefit 10 million households across India and add 30 GW of capacity in the next 3 years. It is also estimated that the scheme will save the Indian government 9 billion USD (8.3 billion EUR) per year in electricity costs.

On the utility-scale front, the development of Solar Parks and Ultra Mega Solar Power Projects, launched in December 2014 with an initial capacity of 20 GW, has seen a significant increase to 40 GW by 2025-26. As of November 30, 2023, the Ministry has awarded 50 solar parks across 12 states, totalling 37.5 GW in capacity, with 10.4 GW already commissioned.

In FY 2023-2024, India's renewable tenders reached 70 GW, far surpassing the annual bidding target of 50 GW. This highlights the importance of utility-scale solar and hybrid installations in the coming two years.

FIGURE GW 5.1 RE SECTOR-WISE CUMULATIVE CAPACITY, BY NSEFI



SOURCE: NSEFI

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4 GW-scale markets / continued

India's government has also announced an additional 13 GW of renewable energy capacity alongside a 12 GWh BESS in Ladakh, with the construction of an Inter-State Transmission System to facilitate power evacuation and grid integration for these projects.

In 2023, the government has also set Renewable Purchase Obligation (RPO) targets, mandating an increasing share of renewable energy in the total energy mix until 2030, indicating a significant leap towards sustainability.

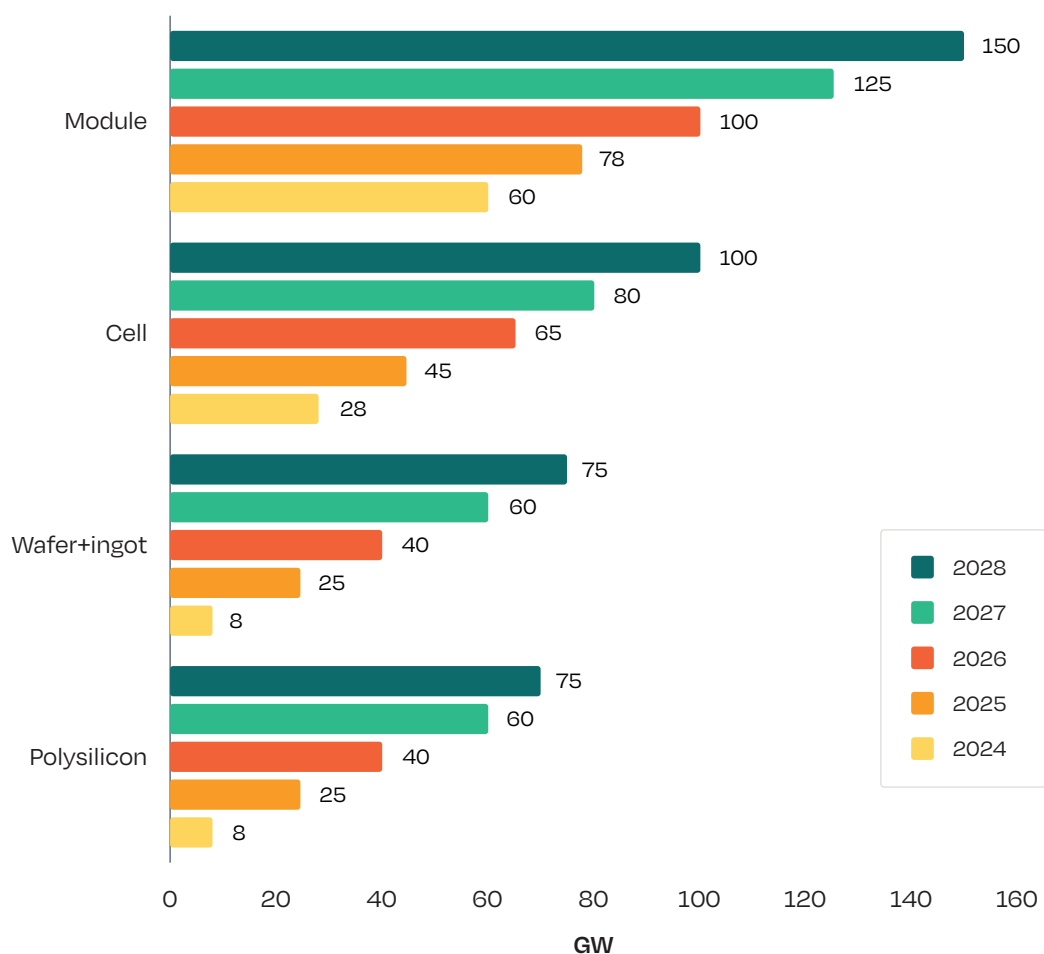
With the implementation of the Production Linked Incentive (PLI) in 2022, aiming to escalate and promote domestic manufacturing in India, production

has started to meet the set targets to increase solar manufacturing to 100 GW for 2026 (see Fig. GW 5.2). The targets, which have been revised upwards from last year, will bring India to become the world's second-largest solar manufacturing hub by 2025.

Challenges

Transmission Infrastructure. With growing PV capacity additions and ambitious targets, timely deployment of grids will be a key issue to address in the coming years to achieve India's transmission targets, especially beyond 2025. While the Indian government has already pledged investments in creating a dedicated Inter State

FIGURE GW 5.2 INDIA'S MANUFACTURING OUTLOOK FOR 2024-2028, BY NSEFI



SOURCE: NSEFI

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Transmission System (ISTS) along with a Green Energy Corridor, there are also requirements to establish transmission systems in states with moderate availability of solar resources.

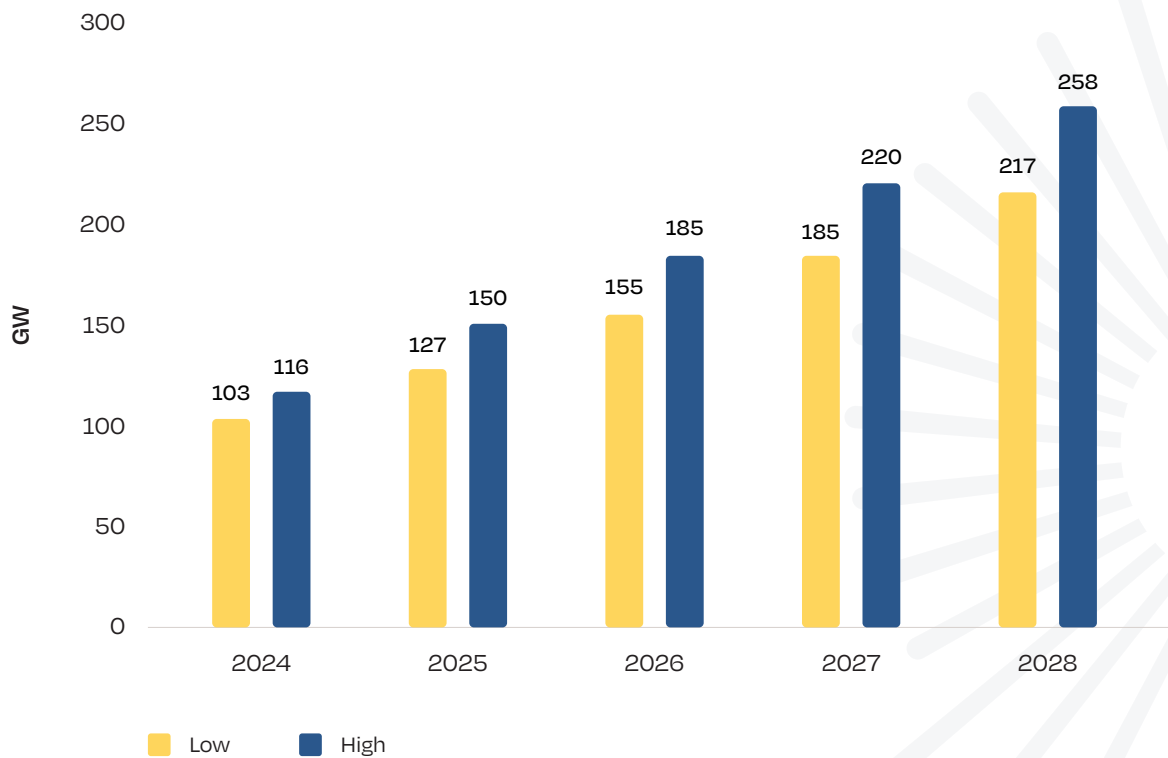
Land Availability. In the coming years, due to the sheer volume of solar generation that will be integrated to the grid, there are a number of challenges that India needs to address, including land availability. It will be important to mobilise two possible solutions to address this challenge. First, the government, taking cue from the success of Ultra Mega Solar Parks, can look at identifying hubs where integrated renewable energy generation, including storage, can be stationed. A second solution to land-use constraints can be dual use of land, whereby solar generation is co-located with agricultural activities, for example low risk crops including horticulture. This can also increase land use efficiency while helping the nation achieve its energy targets.

Outlook

India already exceeded the target of tendering 50 GW a year in 2023, and obtained bids for 70 GW of RE projects with a major share coming from solar. The majority of these projects will be commissioned in the next three years. This indicates that India will add as much solar in the next three years as it has added in the last 10 years. The country is also poised to reach the 100 GW mark before April 2025, and will cross the 150 GW mark by February 2027.

Authors: *Subrahmanyam Pulipaka*, CEO; *Shubhang Parekh*, Manager - International Relations and New Projects; National Solar Energy Federation of India (NSEFI).

FIGURE GW 5.3 INDIA 5-YEAR SOLAR OUTLOOK



SOURCE: NSEFI

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4 GW-scale markets / continued



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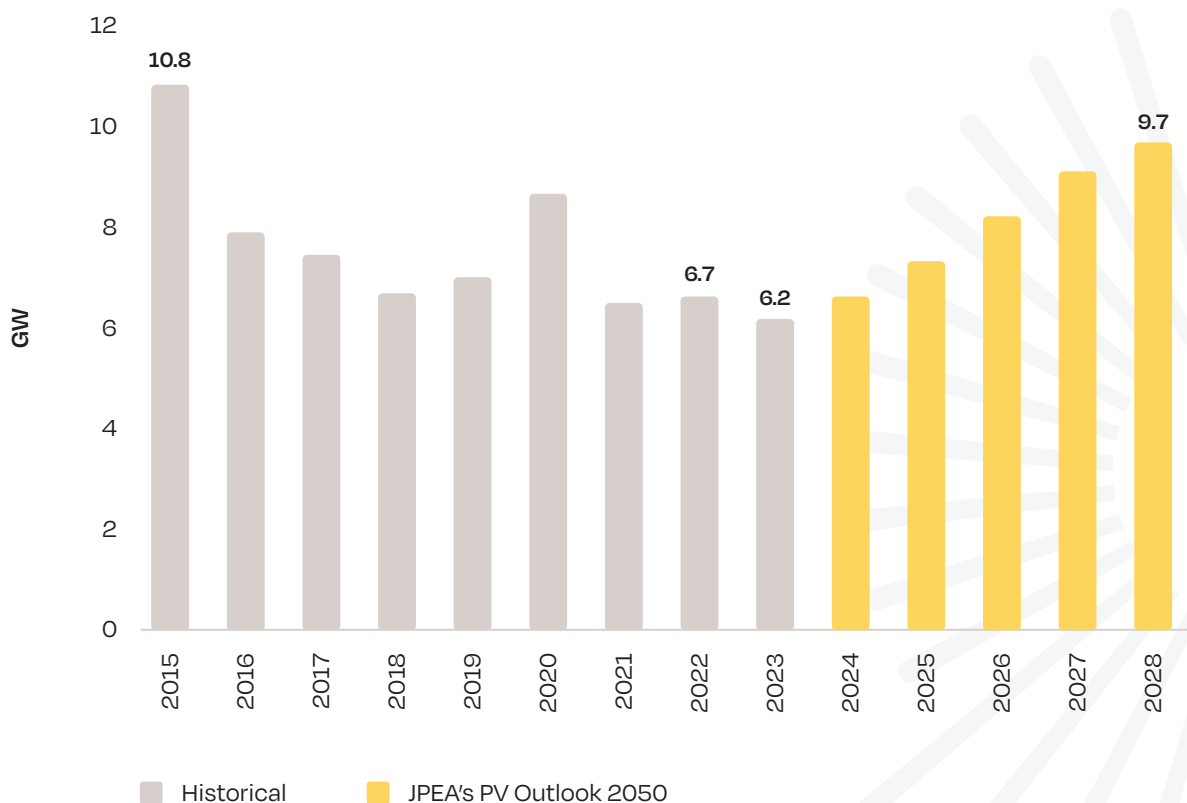
Overview of PV developments

Having achieved record capacity addition of 10.8 GW in 2015, the Japanese PV market has been on a downtrend following the reduced FIT support for solar PV. In 2023, Japan installed around 6.2 GW of new solar PV capacity, down some 0.5 GW from 2022. The cumulative installed capacity at the end of 2023 is expected to reach 91.3 GW. Even with the reduced FIT support, Japan's PV market is expected to start trending upward again from 2024, due to the growth in residential and C&I rooftop markets, and new corporate renewable PPA models. Japan's emissions reduction target of 46–50% by 2030 will require a large increase in the share of renewable energy, in particular solar PV.

Japanese solar and renewable energy targets

- **The government's PV target:** According to the Long-term Energy Supply and Demand Outlook (Energy Outlook) published by the Ministry of Economy, Trade and Industry (METI), the cumulative installed PV capacity target for 2030 is 130–147 GW (104–118 GW_{AC}) to meet the new carbon reduction target of up to 50% by 2030.
- **Renewable Energy Target:** According to the METI's Energy Outlook, the renewable part of its energy generation mix is 36–38% by 2030.
- **Japan Photovoltaic Energy Association's (JPEA) vision (PV OUTLOOK 2050):** In JPEA's PV OUTLOOK 2050, the cumulative installed PV capacity is expected to be around 154 GW (125 GW_{AC}) in 2030. This ambitious target – which is 7 GW higher than

FIGURE GW 6.1 JAPANESE SOLAR PV MARKET SCENARIOS 2024-2028, BY JPEA



SOURCE: JPEA.

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4 GW-scale markets / continued

METI's target – means, on average, that around 9 GW of solar PV will have to be installed every year from 2024 until 2030.

Drivers for solar growth in Japan

The FIT scheme has been the strongest driver of solar growth in Japan since its introduction in July 2012. However, the relevance of this FIT scheme has decreased, and a more market-oriented Feed in Premium (FIP) has been introduced in April 2022. Instead of setting a fixed feed-in tariff rate, the FIP scheme allocates a certain amount of premium in addition to the wholesale electricity price. This way, the remuneration level is connected to the current electricity prices. Under the new framework, larger projects will be subject to the FIP remuneration, while the FIT is maintained for smaller systems. The new FIT/FIP scheme is expected to be a new driver for solar demand.

The 'self-consumption business model' for commercial and industrial users is growing rapidly in Japan. On-site, self-consumption PV systems are becoming more attractive to business users, as the LCOE of PV power generation is already competitive with the retail electricity prices of commercial and industrial users.

An additional driver to solar growth stems from policies setting **PV mandates for new buildings**. The Tokyo Metropolitan Government, and Kawasaki City, will make it mandatory to install PV systems on new buildings, including detached houses, starting in 2025. If similar installation mandates spread to municipalities across the country, the PV market, especially residential rooftop, will expand significantly.

In addition to the needs of renewable energy users, the **PPA model** is beginning to gain traction in the Japanese PV market, driven by government subsidies and rising electricity prices. According to METI, the installed capacity of the PPA model outside the FIT Scheme reached some 0.6 GW in 2022.

Utility-scale vs. distributed & rooftop solar developments

In 2023, residential PV (below 10 kW) capacity additions totalled some 1.0 GW_{AC}, same as in 2021. We expect this segment to grow further, supported by the FIT and various subsidies for net-zero energy houses (ZEH), battery systems, etc. Beyond 2025, municipal PV installation mandates for new buildings, including those from the Tokyo Metropolitan Government and Kawasaki City, could be a strong driver of residential rooftop market growth.



102.3 MW Terras Energy Yakumo Solar Park, Hokkaido, Japan.

© Terras Energy

Distributed solar PV under 1 MW, mostly ground-mounted, is on a downward trend since 2016, mainly due to reduced FIT support. This segment requires a business transformation, for example, from a simple ground-mounted system to a self-consumption system integrated with renewable energy users' and/or local community's energy demand. The segment is also expected to grow again with the growth of corporate PPAs outside the FIT scheme.

Large solar PV systems of 1 MW and above, including utility-scale systems, are also trending downwards. In addition to the FIT termination, power grid constraints and land availability have also contributed to reduced demand. This segment is likely to start growing again in the medium term, as soon as these constraints are overcome, and with improved cost competitiveness. Following the introduction of the FIP regime, many investors and developers are preferring to wait due to the significant uncertainties about future electricity prices. At the same time, they are turning more and more towards on-site PPAs.

Challenges

- **Smooth transition from FIT to FIP:** FIPs were introduced in 2022 as a mandatory incentive mechanism for large-scale solar PV (250 kW and above from 2024), and optional for distributed solar PV (50 kW – 250 kW from 2024). One of the biggest challenges for the industry and for policymakers is the smooth transition from FITs to more market oriented FIPs.
- **Business model transformation:** The role of FIT/FIP will gradually shrink in the coming years. With the

emergence of PPA type business models, this decade will see the transition towards a market growth with little reliance on the FIT/FIP regime.

- **Grid constraints:** Limited grid capacity and curtailment risks are the primary causes for the downward market trend in Japan. In 2023, the curtailment of solar PV in the Kyushu area increased to around 8%, and the risk became apparent. METI is working on additional mitigation measures to minimise curtailment risk, including demand-side management and flexible supply-side operations. Moreover, METI is developing the long-term grid expansion programme to accommodate large amounts of renewable energy.
- **Land availability:** New business models without dedicated land space (e.g., on-site self-consumption models), and utilisation of unused/abandoned farmland are a solution to the limited land availability problem. To date, conversion of unused/abandoned farmland to solar farms is very limited as it requires strict legal procedures, and the local authorities' permission. The government is now tackling those constraints by reforming existing laws and regulations.
- **Cost competitiveness:** The cost of solar PV in Japan still needs to be reduced to compete with the conventional gas and/or coal fired power plants. The average cost of the ground-mounted Solar PV of 250 kW is estimated at around 12 JPY/kWh (0.073 EUR/kWh). The FIT for a ground-mounted PV system between 10 and 50 kW is set at 10 JPY/kWh (0.061 EUR/kWh) and 8.9 JPY/kWh (0.054 EUR/kWh), for systems from 50 kW to 250 kW for 2025.

Author: *Takeaki Masukawa*, Secretary General, Japan Photovoltaic Energy Association (JPEA).

7. Australia

Overview

In 2023, Australia saw the first year of reduced growth since 2013, with 4.6 GW of solar PV capacity installed. Rooftop solar made up the majority of this with 3.17 GW of new installations, as Australia continues to lead the world for the share of households with rooftop solar, with over one-third of all residential homes using rooftop solar.

The percentage of overall electricity generation from renewables continues to rise, reaching 39% of total generation in 2023, compared to 36% in 2022. However, there has been a noticeable stall in utility-scale solar developments.

The Australian Labor government continues to advocate for Australia's role in the renewable energy transition, committing significant resources to solar PV manufacturing and capacity generation. Notably, the announcement of a Net Zero Economy Authority highlights the current government's role in guiding Australia to Net Zero.

National Targets

The Albanese government remains committed to achieving 82% renewable energy by 2030. The focus will remain on the development of utility-scale projects, and integration of the grid across the country. Transmission needs the most development in this space, with various social licence issues arising with landholders.

Recent slowdowns in renewable energy capacity growth, especially in utility-scale projects, are due in a major part to the planning and approvals process. Bottlenecks in government mean that some projects face months of delays before a site can be developed. This also impacts investment in utility-scale solar PV projects.

Drivers

Announced in 2023, the Capacity Investment Scheme aims to achieve 23 GW of new renewable electricity generation, with another 9 GW of dispatchable capacity, between now and 2027. It focuses on utility-scale development, which has seen a significant slowdown in the last year. The Scheme aims to increase utility-scale capacity mostly by encouraging investments, following a major drop in 2023.



© Tindo Solar

Announced in early 2024, the Solar Sunshot Programme aims to use 1 billion AUD to boost Australia's solar PV manufacturing capacity. The government believes developing domestic capability in this area of the supply chain is essential for maintaining Australia's prominence as a renewable energy powerhouse. This announcement follows more than 830 million AUD in previous funding for large-scale projects.

Utility-scale vs distributed

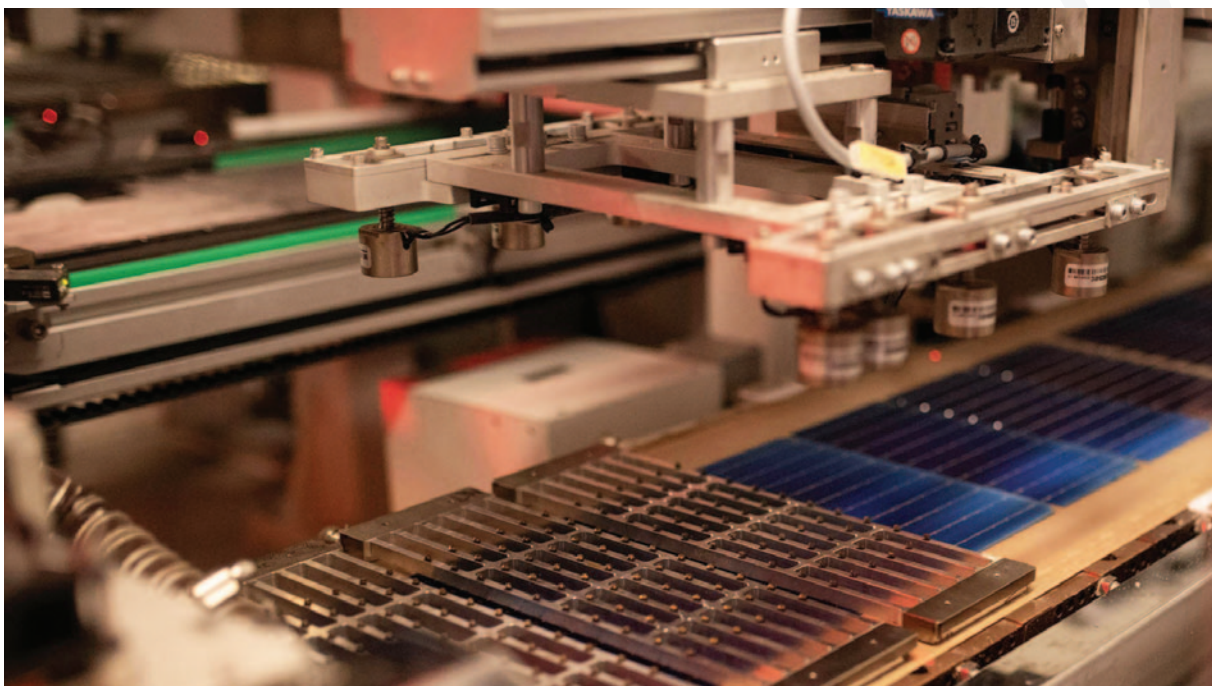
Out of the 56 renewable energy projects under construction as of December 2023, 38 were utility-scale PV projects. This share is down from 2022, due to an abnormally large number of projects being approved in previous years. However, the trend of recent slowdowns in utility-scale capacity growth is evident from a slow pipeline of approvals.

Only 1.5 billion AUD was secured for utility-scale developments in 2023, as opposed to 6.5 billion AUD secured in 2022. This is likely due to a slowing approval process and a backlog of projects undertaken in previous years.

Market Challenges

The major challenge is network connection & transmission for utility-scale solar PV, and market design and operation that continue to penalise PV (& wind). There have been some major policy initiatives by the Federal Government, now a strong supporter of renewables, and in States and Territories. These have been accompanied by financial commitments which should see growth in our sector. Policies and funding now mandate firmed supply, so we are seeing a massive increase in hybrid plants with batteries and addition of batteries to existing facilities.

The national opposition party coalition's obsession with nuclear power continues to grow. With Australia's current position and history, nuclear power is the most expensive and slowest energy source to develop as we continue to transition away from coal and gas power stations. The possibility of nuclear power is being flaunted as a delay tactic to extend the life of our coal-fired power stations, going against the booming solar PV industry, which already provides the cheapest electricity.



Tindo Solar Pv manufacturing plant in Adelaide, Australia.

© Tindo Solar

4 GW-scale markets / continued

Outlook for coming years

Australia still has enormous untapped potential for solar PV in the residential, commercial and large-scale solar fields.

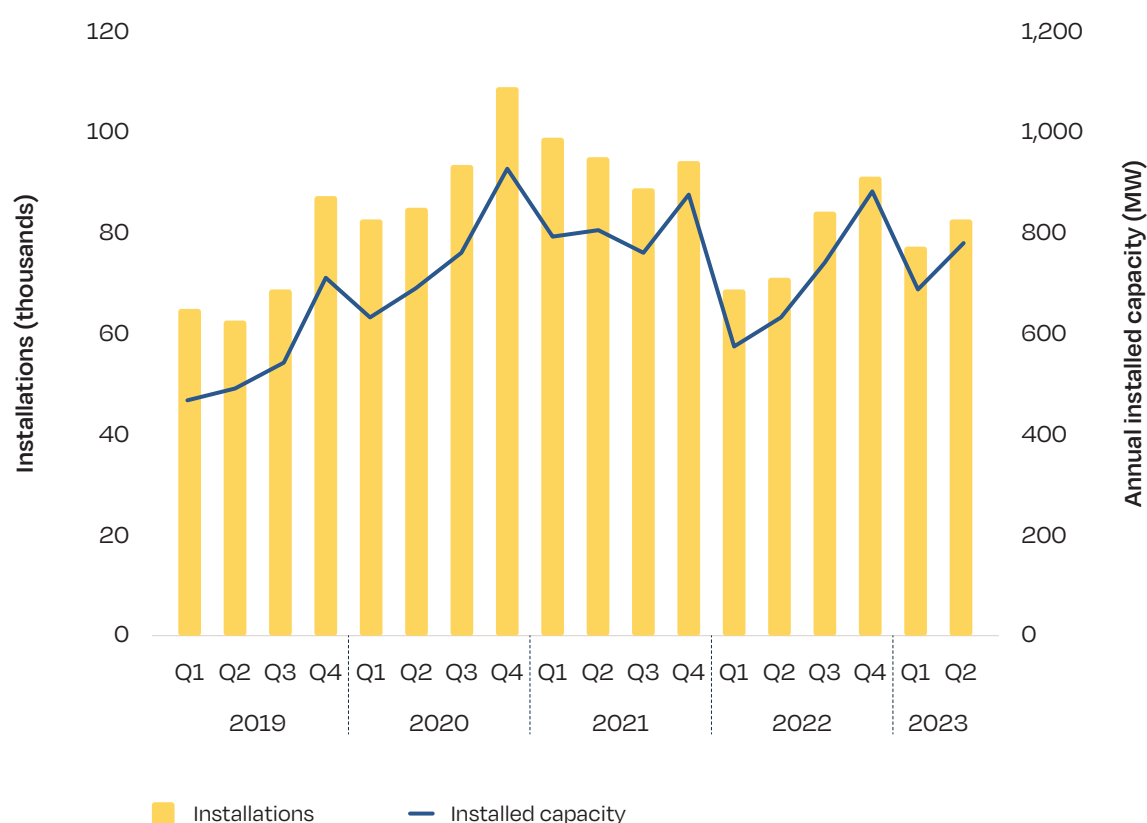
In the residential area, there is a need to unlock investment for lower income earners in public and private housing, including amongst renters. The commercial solar industry is growing significantly off a low base, but there is enormous potential for solar PV on office buildings, apartments and large public buildings.

Concerning large-scale solar PV, there is a need to unlock barriers to deployment, including significant delays in planning approvals processes.

Solar PV will play a critical role in Australia reaching its target of 82% renewables for 2030, with the condition that governments unlock critical barriers to deployment.

Author: Steve Blume, Secretary, Smart Energy Council & Connor Price SEC Policy Analyst.

FIGURE GW 7.1 AUSTRALIA SMALL-SCALE SOLAR PV INSTALLATIONS 2019-2023



SOURCE: Clean Energy Regulator (CER).

© SOLARPOWER EUROPE 2024

8. South Africa

Overview of PV developments

The South African Solar PV landscape has changed dramatically in the last few years since first crossing the 1 GW annual installed capacity threshold in 2020. Regulatory reforms enacted in 2022 have resulted in unprecedented growth in the private utility-scale project development sector, while energy security concerns have driven the uptake of residential solar PV and BESS systems.

The unbundling of South Africa's vertically integrated national utility, ESKOM, into three distinct units, namely Generation, Transmission and Distribution, further indicates that the electricity supply industry is rapidly changing in South Africa. The unbundling will drive private sector participation and competition in generation. Private sector participation in transmission infrastructure development is currently under discussion to unlock solar PV projects in high renewable potential areas suffering from grid connection constraints.

Against this backdrop, the South African solar PV market has grown significantly in 2023 with the total

installed capacity increasing by approximately 3.3 GW. As a result, South Africa now operates a solar fleet larger than 8 GW, across all market segments.

National solar PV targets

South African national energy procurement is guided by the Department of Mineral Resources and Energy (DMRE) through the Integrated Resource Plan (IRP) 2019. A Draft IRP 2023 revision was released for public comment in January 2024. The draft version has 2 planning horizons, from 2025 to 2030, and from 2031 to 2050. The first horizon is focused primarily on energy security, while the second horizon proposes a few energy scenarios for South Africa, ranging from a balance of existing coal, nuclear and renewable generation technologies to renewable dominated plans.

Until the IRP 2023 revision is formally adopted by the DMRE after a public participation programme, the 2019 IRP allocations will remain in force. The current solar PV targets are 8.3 GW or 10.52% of the total installed capacity, for 2030. A total of 4.3 GW has been procured to date across six bidding rounds of the Renewable Energy Independent Power Producer Procurement Program (REIPPPP) of which 2.2 GW is



SCATEC Kenhardt Project, Northern Cape province, 540 MW solar coupled with 225 MW/ 1,140 MWh battery storage.

© SCATEC

4 GW-scale markets / continued

already operational. A seventh REIPPPP bidding round for 1.8 GW of Solar PV capacity was released in December 2023.

Public procurement of Battery Energy Storage Systems (BESS) took off in 2023 with 2 procurement rounds released. The BESS procurement is aimed specifically at easing network congestion in high solar penetration areas and allowing more solar generation to be connected in the future. The first round for 513 MW of BESS at specific substations in the Northern Cape was released in March. Five preferred bidders were announced in November 2023. A second round of procurement for 615 MW/2,460 MWh of BESS capacity was released to the market in December 2023.

Drivers of solar growth

Demand for solar PV in South Africa has market segment specific drivers. Residential demand for solar PV and BESS has been predominantly driven by the electricity generation shortfall, and the ensuing power cuts from loadshedding. While C&I demand has traditionally been driven by OPEX control and sustainability concerns, utility-scale demand has been driven both by the growth of offsite generation for bilateral PPAs, and by the emergence of energy aggregators and traders, a new feature in the South African market.

South Africa experienced 6,907 hours of loadshedding in 2023, the worst year on record and an 82.9% increase compared to 2022. As such, the residential demand for solar PV and BESS was driven up to meet the emergency electricity needs for many South African households. The South African residential market's installed capacity increased by approximately 330 MW in 2023, while approximately 450 MWh of home batteries were added in 2023, bringing the total residential battery fleet to 1.4 GWh.

The uptake of residential solar PV systems has unfortunately been limited to higher income households. As a result, the South African government supported the rollout of financing schemes targeting middle income households. One such scheme is the Energy Bounce Back Scheme (EBBS) supported by the National Treasury. The EBBS provides a 20% first loss guarantee to participating banks to de-risk lending to

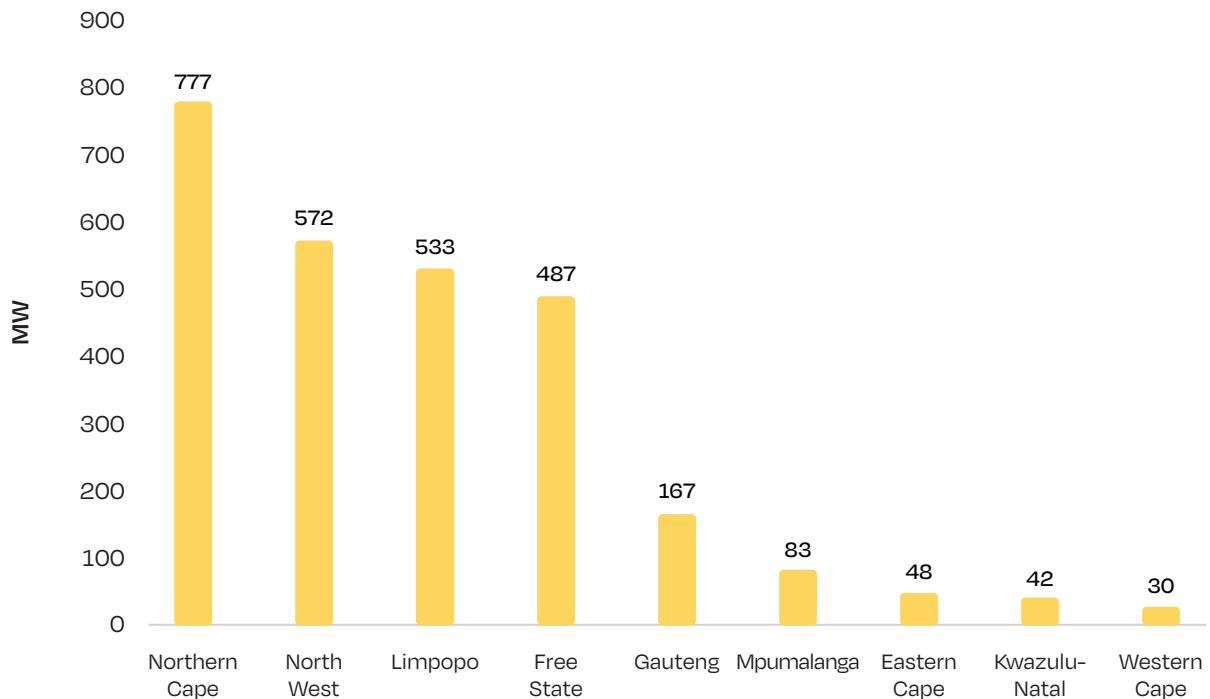
households and businesses investing in solar rooftop solutions. Various commercial banks launched products linked to the EBBS in Q4 2023 after it was launched in August 2023.

Last year, energy resilience became an increasingly important driver of demand in the C&I market segment. Commercial banking members of the South African Photovoltaic Industry Association (SAPVIA) reported that up to 85% of C&I projects funded in 2023 combined rooftop solar PV with BESS, a significant shift from previous years. BESS coupled with C&I solar offer many advantages to customers beyond loadshedding resilience – self-consumption optimisation, peak demand management and peak load shifting to name a few.

The greatest driver for overall solar PV demand in South Africa has been the regulatory reforms enacted in December 2022, removing the licence threshold for private sector projects. Apart from the 10 MW threshold removal, projects are no longer required to apply for ministerial determination to construct and operate a pure generation facility in South Africa. This reform, coupled with the demand from large-scale industrial energy users for more sustainable electricity sources, has resulted in the unprecedented growth of the private PPA market in South Africa, ultimately driving the development of utility-scale solar PV projects. All solar PV projects installed in South Africa larger than 100 kW are required to register with the National Energy Regulator of South Africa (NERSA) prior to construction. In 2023, 384 solar PV projects with a total capacity of more than 2.7 GW were registered with NERSA. The provincial distribution of registered projects is displayed in Figure GW 8.1.

Electricity Wheeling is a uniquely South African term referring to the practice of off-site generation, predominantly in rural high renewable production areas, "wheeling" energy across the existing transmission and distribution networks to industrial off-takers. The practice involves a bilateral PPA or could involve energy aggregators or traders facilitating the transactions. In 2023, 2.1 GW of wheeling-specific solar PV projects were registered with NERSA, representing 79% of all registered projects by capacity and 86% of all projects larger than 10 MW. This shows a clear trend in the market towards larger-scale private sector projects.

FIGURE GW 8.1 SUM OF CAPACITY BY PROVINCE, BY SAPVIA



SOURCE: SAPVIA.

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Unfortunately, wheeling does not guarantee energy security for off-takers. Since off-takers are often embedded with a distribution network, they are exposed to the risk of the distribution network being de-energised due to loadshedding, thus negating the ability to wheel energy from the IPP.

Challenges

Within an integrated network of distribution entities, statutory bodies and regulatory organisations aimed at streamlining installation and equipment standards, technical interconnection requirements and approval processes, the speed of regulatory reform has been far slower than the market uptake of solar PV systems. Undercapacity within these entities to address the required regulatory reforms has been a challenge for the South African solar PV market for much of the last decade. This has resulted in approximately 45% of the residential solar PV systems not being registered with the respective

distribution system operator where the system is located. Registrations and technical compliance levels are significantly higher in the C&I market segment.

The greatest challenge still facing the utility-scale sector is the lack of grid-connection capacity in high renewable potential areas combined with the slow pace of expected transmission infrastructure rollout. In 2023, the high solar potential areas of South Africa, in the Northern, Western and Eastern Cape provinces, respectively, had exhausted all available grid-connection capacity. The next best areas – namely the Free State and the North-West provinces – are expected to exhaust all available capacity within the next 2 to 3 years. The ESKOM transmission entity estimates that approximately 14,000 km of new transmission lines are needed by 2032 to integrate approximately 53 GW of expected renewable projects. This represents 43% of the existing 33,000 km of transmission infrastructure. This infrastructure is expected to be rolled out in a phased approach with 79% of the planned rollout to take place from 2028 to 2032.

4 GW-scale markets / continued

Finally, skill shortages across both semi-skilled and skilled sectors remain a barrier to the solar PV market's growth rate in South Africa. Additional uncertainties regarding the medium-term sustainability of residential solar PV demand are hampering the establishment of medium- to long-term skill development interventions for residential installations. The large rooftop and utility-scale market segments are resorting to direct recruitment of university skills or external expert skills from more established markets to meet the skill requirements of the sector.

Outlook for the years 2024-2028

The energy supply industry in South Africa is undergoing structural changes, initiated in 2017, resulting in opportunities for solar PV deployment while addressing some of the challenges listed above.

The unbundling of the state-owned, vertically-integrated utility ESKOM into three units and the establishment of an independent transmission system operator presents an opportunity for solar PV to play a significant role in the energy generation sector in South Africa. The work of establishing a wholesale electricity market is gaining momentum, with a target date of 1st April 2026 set for the introduction of the market. This will be followed by a five-year transition period to a fully functioning wholesale day-ahead, balancing, and ancillary services market. The expected cost decreases in solar generation and utility-scale BESS

technology will enable the development of bankable projects to participate in the wholesale market, even if the projects are in less desirable solar resource areas due to grid capacity constraints.

Unfortunately, South Africa will run the risk of loadshedding in the short and medium term until the addition of new generation capacity from various sources is systematically matched with appropriate system adequacy. Thus, loadshedding will continue to play a significant role in the solar PV market, as has been the case in driving demand and uptake of residential solar PV and BESS. However, the residential market must adapt and prepare for an increasingly likely future with lower levels of loadshedding by introducing alternative revenue streams to new installations. Residential incentives including feed-in tariffs, additional tax incentives and the integration of PV and batteries with the expected rise of electric vehicles will become the focus for driving residential demand.

Finally, the decarbonisation of supply chains in response to Nationally Determined Contribution (NDC) targets or international trade requirements, such as the recently introduced EU Carbon Border Adjustment Mechanism (CBAM), will drive further uptake of solar PV in the C&I sector, whether through on-site or off-site solar generation.

Authors: *De Wet Taljaard*, Technical Specialist: Solar Energy, South African PV Industry Association (SAPVIA).



6.8 MW, Boardwalk Inkwazi, Richard's Bay, South Africa.

© SolarEFF

9. United Arab Emirates

The United Arab Emirates (UAE) has emerged as a pioneer in developing solar PV energy, driven by its ambitious goals for a sustainable energy future and a commitment to diversifying its energy mix.

Overview of PV development up to 2023

The UAE has firmly established itself as a leader in solar PV development by completing numerous large-scale projects in 2023. Most notably, the world's largest single-site solar power plant, Dubai's 2 GW Al Dhafra solar PV project, was commissioned in November 2023. This achievement is a testament to the country's "relentless efforts to enhance the amount of renewable energy in the entire power mix" and its commitment to a sustainable energy future. The UAE's growth in the solar PV sector is further shown by its reputation as one of the region's fastest-growing utility-scale markets,

with an annual installation rate of around 2 GW of renewable energy capacity. In 2023, 2.6 GW of new capacity was connected to the grid. This brought total installations to 5.3 GW by the end of last year, according to IRENA (see Fig. GW 9.1).

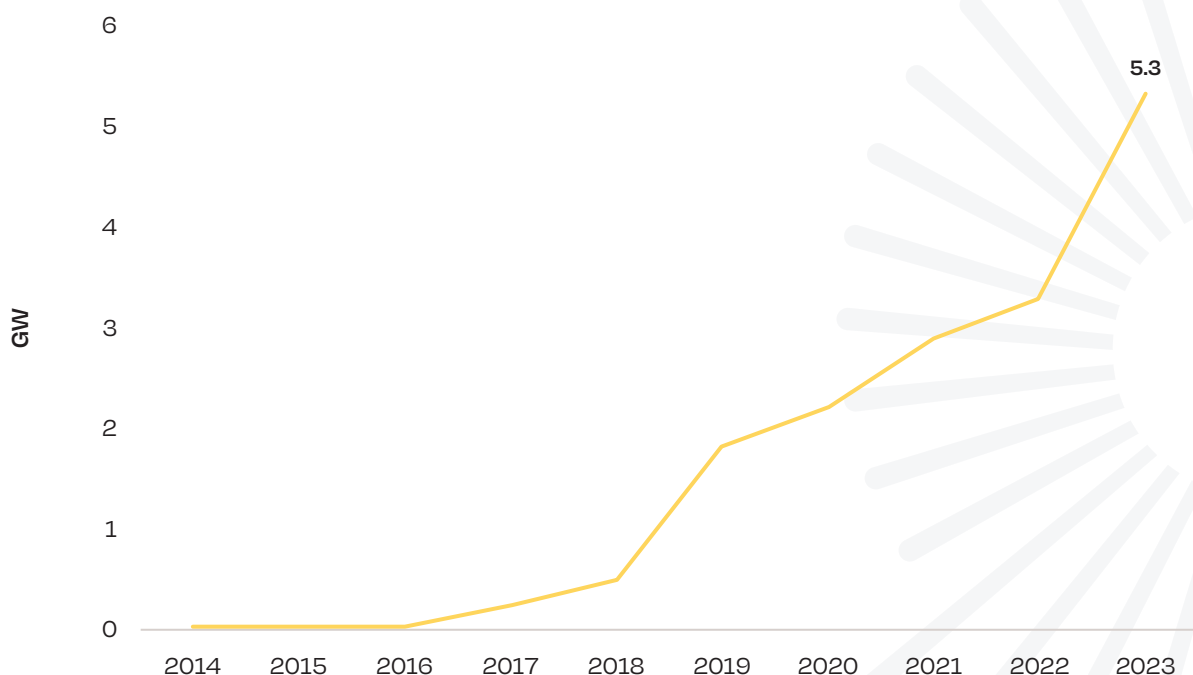
National Solar PV and RE targets

The UAE has set ambitious national targets for solar PV deployment, serving as a driving force behind its solar energy development. The Dubai Clean Energy Strategy 2050 aims to produce 75% of Dubai's energy from renewable sources by 2050, with interim targets of 7% for 2020 and 25% for 2030. Similarly, Abu Dhabi's Plan envisions installing 5.6 GW solar PV capacity by 2026. These targets underscore the UAE's long-term vision and commitment to renewable energy adoption.

Drivers for Solar Growth

The UAE's drive for solar growth in 2023 was fuelled by its ambitious renewable energy targets and a combination of large-scale projects and supportive policies.

FIGURE GW 9.1 UAE CUMULATIVE SOLAR PV CAPACITY, 2014-2023, BY MESIA



SOURCE: IRENA (2024).

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4 GW-scale markets / continued

Monumental endeavours, such as the Al Dhafra Solar PV Independent Power Producer (IPP) project, the world's largest single-site solar power plant with a 2 GW capacity and 4 million bifacial solar panels, and the Mohammed bin Rashid Al Maktoum (MBR) Solar Park in Dubai, with a planned 5 GW capacity, position the UAE as a global leader in solar energy production. These projects are expected to generate enough electricity to power hundreds of thousands of homes while offsetting millions of tonnes of carbon emissions annually. Simultaneously, the UAE has promoted distributed and rooftop installations through initiatives like the Shams Dubai Solar Program, a net-metering scheme facilitating over 5,000 residential and commercial systems with a combined 200 MW capacity by 2023. The UAE has also implemented BIPV and floating solar PV projects to maximise surface utilisation for solar power generation and established regulations to enable broader deployment of distributed systems, including rooftop installations.

Challenges for the market

The UAE's solar PV market faces multifaceted challenges requiring strategic interventions to ensure sustainable growth and development. From a regulatory standpoint, the absence of a comprehensive legal framework governing distributed generation outside the Dubai emirate has created uncertainties, hindering investment and streamlining solar project development. Another significant challenge lies in integrating solar power into the existing grid infrastructure. Careful planning and

coordination are essential to address technical issues such as grid stability, voltage fluctuations, and the management of intermittent energy supply.

Moreover, attracting and fostering foreign investment and international partnerships is vital for achieving the UAE's ambitious target of generating 70% of its electricity from renewables for 2050. Continuous innovation, cost optimisation, and strategic positioning will be crucial for sustained success in this evolving landscape.

Outlook for the years 2024–2028

The UAE's solar PV market outlook for 2024–2028 is highly promising, driven by its strong commitment to renewable energy expansion. Key projects like the Al Dhafra Solar PV Plant and the MBR Solar Park indicate significant growth potential, with around 2 GW of renewable energy installations expected annually. Technological advancements, including bifacial modules and potential transitions to N-type cells, promise efficiency gains and lower LCOE post-2025. The market is projected to grow at a CAGR of 12%, reaching 3.72 billion kWh by 2028, fuelled by favourable policies and a solid commitment to renewable energy. The UAE's focus on human capital development and innovation further strengthens its position as a global hub for sustainable development and economic diversification.

Authors: *Tannishtha Das* and *Hinde Liepmannsohn*, Middle East Solar Industry Association (MESIA).



Al Dhafra PV2 Solar Project, Abu Dhabi, United Arab Emirates.

© Hitachi Energy

10. South Korea

Overview of PV developments

2023 was again a challenging year for the South Korean solar industry. The solar market has been in decline since 2020, when it saw over 5 GW of new installations. Only 2.7 GW of new solar capacity was installed in 2023, and market growth is expected to remain difficult in the coming years.

National Targets for Solar PV

Previous South Korean governments have steadily increased their renewable energy targets. In 2017, renewable energy targets were increased from 7% to 20% for 2030 under the Implementation Plan for Renewable Energy 3020. In 2019, the 3rd Energy Master Plan further introduced a target of 30–35% renewable energy for 2040. In 2020, the 9th Basic Plan for Electricity Supply and Demand (BPESD) further increased the target share of renewable energy to 42% for 2034. South Korea has transformed its energy mix to meet the needs of its people, prioritising a clean and safe environment.

However, the current government's energy policy, led by President Yoon Suk-Yeol since May 2022, shows a

different trend. It aims to create an energy mix which includes an expansion of nuclear energy to at least 30% by 2030, assuming normal operation of both existing nuclear power plants and those under construction. Under the government's plan, renewable energy penetration targets will be set on the basis of actual supply conditions, whereby the optimal shares of different electricity generation sources, such as PV and offshore wind, will be contingent on those factors. The phasing out of coal will be approached by taking into account supply and demand conditions and the state of the grid, and the use of carbon-free generation sources will take technical conditions into account. Finally, the main objectives for the electricity grid will be timely construction, grid stabilisation measures in combination with renewable energy generation, efficiency-oriented redesign and the establishment of an advanced power system.

The current government has lowered the Renewable Portfolio Standard (RPS) targets from 14.5% in 2023 and 17% in 2024, to 13% and 13.5% in 2023 and 2024 respectively (Fig. GW 10.1). The previous 25% RPS target for 2026 has been delayed beyond 2030 as the renewable energy penetration targets in the 10th BPESD have been revised downwards to 21.6% for 2030. This is 8.6% lower than the National Greenhouse Gas Reduction Target confirmed in 2021.

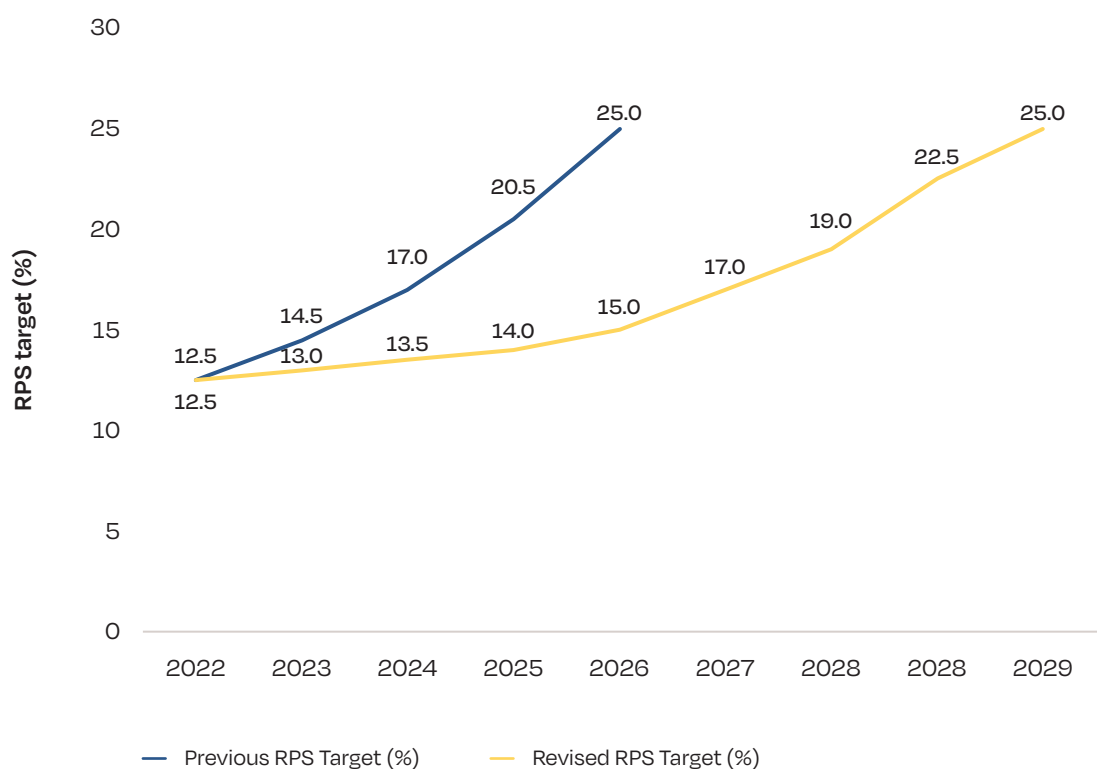


PV on vehicle loading area at Hyundai Motors, Ulsan, South Korea.

© Korea Solar Construction Association

4 GW-scale markets / continued

FIGURE GW 10.1 SOUTH KOREA'S ANNUAL RENEWABLE PORTFOLIO STANDARDS (RPS) TARGETS 2022-2030, BY KSCA



SOURCE: Korea Solar Construction Association.

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While the target for renewable energy generation has been reduced, the target for nuclear power generation has been increased from 25.0% to 32.8% for 2030. This trend is expected to be more concretely reflected in the 11th BPESD, which is expected to be finalised in 2024, with plans to increase the share of nuclear power. In this context, the outlook for renewables is even more negative. Additionally, the current government has drastically cut the 2024 budget for renewables, decreasing funds supporting several sectors, from electricity industry infrastructure to renewable energy deployment and feed-in tariff schemes.

New opportunities in the BIPV market

In South Korea, a mandatory zero-energy building scheme has been in place since 2020, which will be extended to the private sector from 2024. In 2023, public buildings of 500 m² or more and public apartment buildings of 30 units or more will be

required to be certified to ZEB level 5 (between 20% and 40% energy independence). In 2025, these standards will be extended to the private sector. This mandatory ZEB system is expected to be a good growth driver for the BIPV market.

Outlook

South Korea's solar installations have been declining from a peak of 5.5 GW in 2020 to 2.7 GW in 2023, and demand is expected to stabilise in the next years in the range of 2.0 to 2.5 GW per annum. This expected stagnation is attributable to the government's policy actions, such as reducing the renewable energy target to 21.6% for 2030, abolishing the RPS scheme, introducing an auction system, and capping the system marginal price at a fixed rate. In addition, while South Korea does not currently have an auction system in place, the government is considering introducing one for renewable energy, which could

lead to the abolition of the RPS in the future. The main reason for this is the cost of the existing RPS.

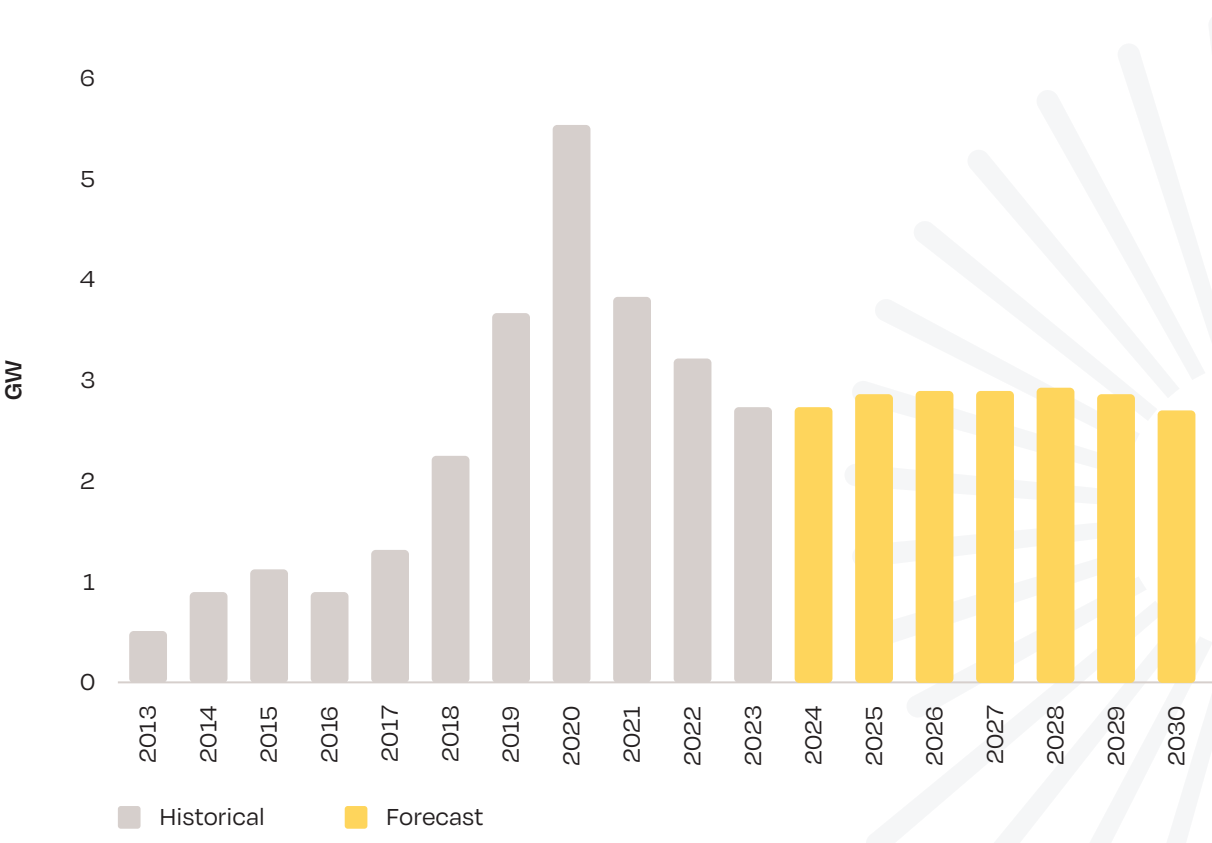
South Korea heavily relies on exports to bolster its economy. Attaining RE100, a global initiative where companies commit to using 100% renewable energy for their operations, holds crucial significance for enhancing Korea's export competitiveness. Thanks to this, a substantial surge in Korean companies' demand for renewable energy to meet this target is anticipated. Nonetheless, the Korean government perceives the current cost of PV generation too high compared to other nations, and only aims to improve it in the future. Consequently, the solar market is currently dwindling, posing a challenge in achieving RE100. This discrepancy underscores a misalignment between market demand and governmental policies.

South Korea's solar power generation is far from grid parity, therefore growth will be limited to subsidy-driven projects. Unless the cost of solar power in South Korea is brought down to global levels, it will be difficult to expand demand, and in the worst case, demand could stagnate below 2.0 GW. Under such conditions, the Export-Import Bank of Korea projects that the market will remain below the 3 GW range in the 2024-2030 period (see Fig. GW 10.2).

South Korea signed the pledge at COP28 and pledged its commitment. As a result, the country agreed to triple its renewable energy generation by 2030. There is also speculation that the renewable energy penetration target may be increased in the 11th BPESD, due to be released this year. Hopefully, South Korea's solar market will grow again.

Author: *Donggun Lim*, Professor, Korea National University of Transportation.

FIGURE GW 10.2 SOUTH KOREA SOLAR PV MARKET SCENARIOS 2024-2030, BY THE EXPORT-IMPORT BANK OF KOREA



SOURCE: Export-Import bank of Korea.

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11. Türkiye

Overview of PV development up to 2023

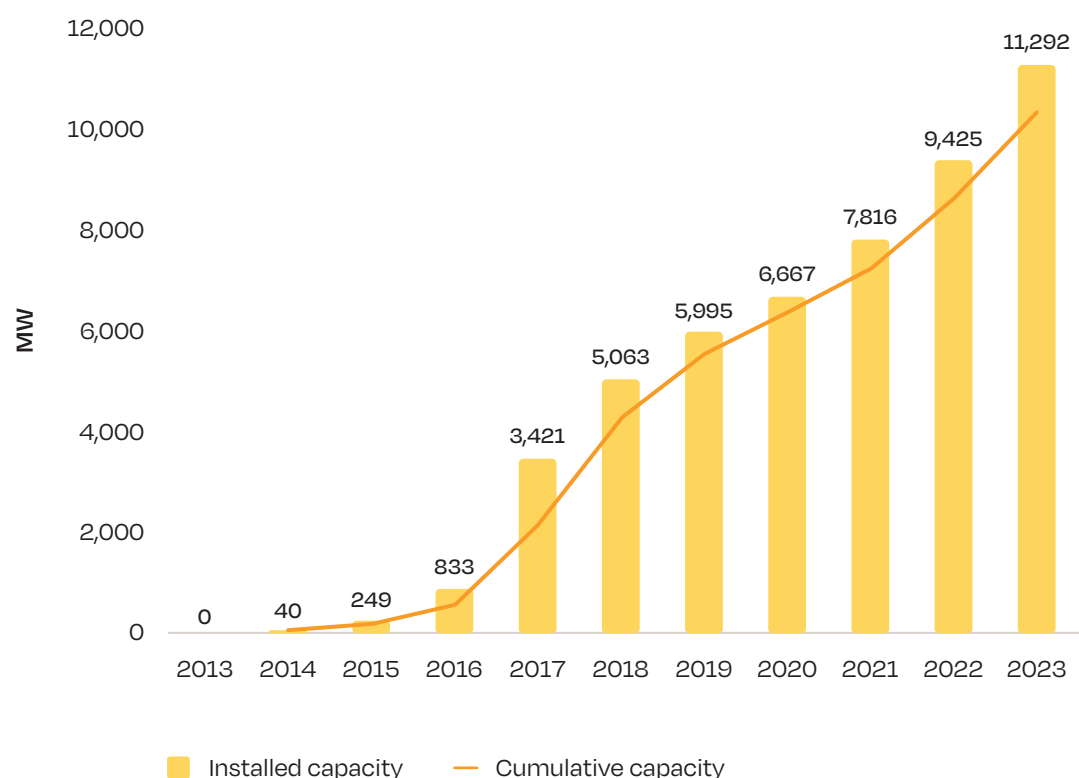
By the end of 2023, total installed power capacity in Türkiye had increased by 2,859 MW, reaching 106,668 MW. Out of it, 1,867 MW of new solar power plants were commissioned, bringing the total solar PV installed capacity to over 11 GW (Licenced + Unlicensed). Current developments regarding PV solar power plants in Türkiye have expanded to different types such as rooftop solar power plants (SPP), land SPP for self-consumption, hybrid SPP and storage SPP. PV module production in Türkiye expanded from 3 factories to 70 within 10 years. Existing solar module manufacturers have a total annual production capacity of 25-30 GW, two companies undertaking cell development activities have reached a total

capacity of 1.8 GW, and there is an increasing number of cell investors. Approximately 15,000 people are employed in solar module factories, 8,000 people in the sub-industry of module factories, nearly 5,000 people in the construction and aluminium sector, and 15,000 people in the installation sector. The Turkish solar PV sector currently employs between 40,000 and 45,000 FTEs. Turkey's solar energy sector provides employment for 150,000 people directly and 200,000 people indirectly.

National Targets for Solar PV

Türkiye's National Energy Plan aims to increase solar energy capacity to 52.9 GW by 2035 and, according to its 12th Development Plan, it will reach 30 GW by the end of 2028. To reach this capacity, 3.4 GW of solar power needs to be added in Türkiye every year through

FIGURE GW 11.1 TÜRKİYE'S SOLAR ENERGY INSTALLED CAPACITY DEVELOPMENT AND ANNUAL CUMULATIVE VALUES, BY GÜNDER



SOURCE: GÜNDER.

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the 2024-2028 period. The 12th Development Plan will provide incentives supporting clean energy production and energy efficiency, green transformation of the industry, sustainable transportation, circular economy and green infrastructure. The 2nd Energy Efficiency Strategy Document and the 2nd National Energy Efficiency Action Plans, covering the years 2024-2030, plan to maximise energy efficiency, increase the share of renewable energy and prioritise solar and wind energy for irrigation. Renewable energy production plants can be established without a zoning plan in the areas of seas, dam lakes, artificial lakes and natural lakes declared as renewable energy resource areas by the Ministry of Energy and Natural Resources. This excludes reservoirs and wetlands providing drinking water, as well as shores and coastlines within the scope of the Coastal Law.

Drivers for solar growth

While electricity production in Türkiye increased by 36% between 2013 and 2023, the share of renewable energy in electricity production increased from 29% to 42%. In response to the current legislation in Türkiye promoting electricity generation, especially from solar energy, momentum has surged for hybrid and storage

facilities, alongside agricultural SPP and floating SPP plans. Following the climate crisis and the increase in energy prices, the sales of solar energy modules installed on the roofs of homes and workplaces broke a record and increased by more than 50% in 2023. Solar power plants established for self-consumption are exempt from obtaining a production licence or establishing a company, provided that they are on residential roofs, and it is now possible to generate extra income by selling excess electricity at the consumer price through monthly net metering. Studies are increasingly conducted to identify the obstacles to local initiatives in Türkiye and to increase the number of renewable energy cooperatives that have a positive impact on local development.

Utility-scale vs. distributed and rooftop developments

Several recent regulatory developments are encouraging PV deployment in different rooftop, distributed and utility-scale applications. The gradual transition to Nearly Zero Energy Buildings (nZEB), with improved energy efficiency and a portion of their energy provided by renewable energy sources, became mandatory on 1 January 2023. In March 2022, the capacity limit for PV electricity production facilities in non-commercial areas such as apartments, sites,



96 MW, Hilvan, Sanliurfa, Türkiye.

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4 GW-scale markets / continued

mosques, farms and villas increased from 10 kW to 25 kW. With the regulatory amendment published in the issue dated 23 July 2022, the Ministry of Environment, Urbanisation, and Climate Change has introduced flexibility regarding solar energy installations on buildings in areas without zoning plans. It has been established that installations intended for self-consumption and agricultural irrigation, covering a surface area of up to 125 m², as well as rooftop SPP installations with a maximum height of 150 cm from ground level, are exempt from requiring a building licence and usage permit. The Public and Municipality Renewable Energy Project (KAYEP) plans to install solar power plants on open car parks and roofs of institutions in order to meet all or part of the electrical energy consumption of central government buildings (subject to self-consumption) under suitable conditions. According to Ember, the country's potential rooftop solar capacity is at least 120 GW, while the annual potential for rooftop solar electricity generation is 148 TWh, corresponding to 45% of Türkiye's total electricity consumption in 2022.¹⁰

Türkiye's floating solar power potential is approximately 80 GW. The first floating solar power plant, the Kuzova Floating Solar Power Plant built on Keban Dam, started to produce 1.8 GWh of electrical energy annually. The aim is to expand floating solar power projects throughout the country.

Challenges for the market

Stringent financial conditions delay the financing of renewable energy projects. There is a need to create a structure that no longer requires appropriate financing packages supported by public banks and state subsidies by pioneering the establishment of financing models such as low-interest loan models

and tax incentives. As the number of licenced solar & storage projects grows and investments continue, the sector's primary goal is to expand capacities to maintain those investment processes. One of the topics that primarily affects the solar sector regards the EU Carbon Border Adjustment Mechanism (CBAM) Regulation adopted on 1 October 2023, in the context of the European Green Deal. The regulation will enter into force on 1 January 2026. In this pilot implementation process, hydrogen, electricity, iron and steel, fertiliser, cement and aluminium were included in the scope of CBAM in the first stage.

Outlook for the years 2024-2028

GÜNDER, Türkiye's Solar Energy Association, continues to provide services to ensure the proper development of the renewable energy sector and contribute to employment, in a world where investment strategies for the energy transition are rapidly being developed. We contribute to projects that promote R&D, combating climate change, qualified workforce, gender equality and energy cooperatives. GÜNDERMYM provides the opportunity to manage employment correctly and benefit from occupational safety at the highest level both in the country and in the European Union within the scope of the Europass Certificate for National Qualifications. The organisation was authorised as the examination center of the solar energy sector on 1 March 2023. METU-GÜNAM continues to lead Türkiye in contributing to new technologies and applications by carrying out R&D and innovation projects through numerous national and Horizon European projects. One of these projects, the SolarHub project, aims to strengthen the solar energy innovation ecosystems of Greece and Türkiye.

Author: Turkish Solar Energy Association-GÜNDER.

¹⁰ Ember Climate (2023): Türkiye can expand solar by 120 GW through rooftops.

12. Taiwan

Solar PV Deployment in Taiwan: A Promising Path Ahead

Taiwan's journey towards renewable energy, particularly solar PV deployment, has seen remarkable progress in recent years. With growing concerns over climate change and energy security, Taiwan has set ambitious targets to increase its share of renewable energy in the power mix.

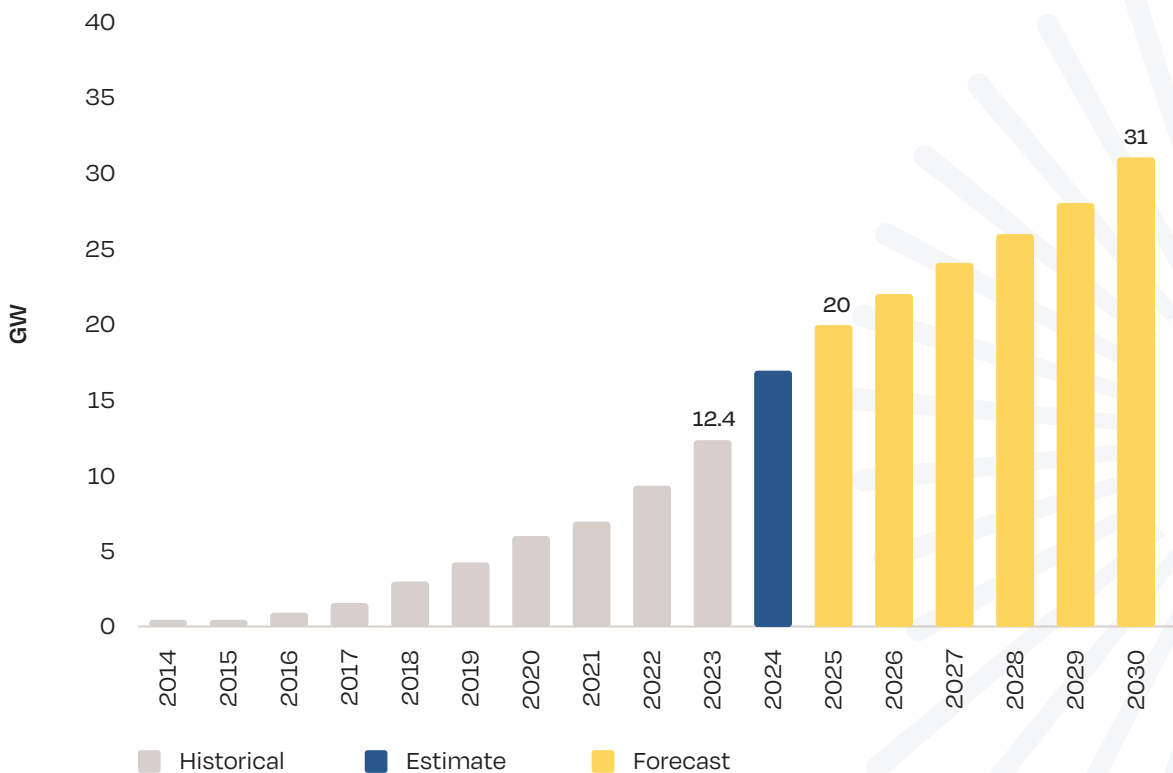
Taiwan's solar PV industry has experienced significant growth over the past decade, driven by favourable government policies, technological advancements, and increasing environmental awareness. In 2023, 2.7 GW of new capacity were added to the country's solar operating fleet, a 33.1% increase compared to the year before. By the end of 2023, Taiwan had installed over 12.4 GW of solar PV capacity, with utility-scale installations

dominating the market. The island's abundant sunshine, particularly in the southern regions, has provided a conducive environment for solar energy development.

Drivers for solar growth

In alignment with its renewable energy goals, Taiwan has set ambitious targets for solar PV deployment. The government aims to reach 20 GW of installed solar capacity by 2025 and 31 GW by 2030, accounting for 4.6% of the country's electricity generation. Considering the 12.4 GW total capacity at the end of 2023, the 2025 solar target would mean installing 7.6 GW of new capacity in the next two years (see Fig. GW 12.1). Taiwan's government had previously stated in June 2022 that the 2025 target had been extended to October 2026. Furthermore, there are long-term aspirations to achieve a carbon-neutral energy sector by 2050, with solar PV playing a crucial role in this transition.

FIGURE GW 12.1 CUMULATIVE INSTALLED CAPACITY OF PV IN TAIWAN 2010-2030, BY BUREAU OF ENERGY, MINISTRY OF ECONOMIC AFFAIRS



SOURCE: Bureau of Energy, Ministry of Economic Affairs

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4 GW-scale markets / continued

Several factors have contributed to the rapid expansion of solar PV in Taiwan. Government incentives, such as feed-in tariffs and tax credits, have stimulated investment in solar projects. Additionally, public support for renewable energy initiatives and the declining costs of PV technology have made solar power increasingly competitive compared to traditional fossil fuels. Moreover, concerns about air pollution and the desire to reduce reliance on imported energy sources have further incentivised the adoption of solar energy.

Utility-scale vs. Distributed and Rooftop Developments

Taiwan distinguishes itself from other countries by predominantly using ground-mounted solar photovoltaic systems. Initially, the country promoted rooftop solar photovoltaic systems through FIT mechanisms. However, ground-mounted systems offer greater generation capacity. Government FIT incentives also have attracted developers to invest in ground-mounted systems, which now account for over 35% of total installations, a significant increase from previous years. Despite this, rooftop solar photovoltaic systems, which have a smaller environmental impact, continue to be strongly promoted by government policies.

Challenges for the Market

Despite the significant progress, the solar PV market in Taiwan faces several challenges. Grid integration issues, such as intermittency and variability, pose operational challenges for utility-scale solar projects. Additionally, land constraints and environmental concerns may limit the expansion of ground-mounted installations, necessitating innovative solutions for space utilisation. Furthermore, uncertainties surrounding policy support and regulatory frameworks could affect investor confidence and project viability in the long term.

Outlook for the Years 2024-2028

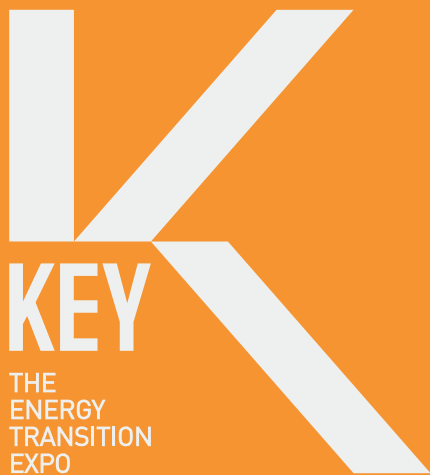
Looking ahead, the future of solar PV in Taiwan appears promising. Continued advancements in technology, including the development of high-efficiency solar panels and energy storage solutions, are expected to drive further cost reductions and enhance the competitiveness of solar energy. Moreover, ongoing efforts to modernise the grid infrastructure and improve grid flexibility will facilitate the integration of large-scale renewable energy projects. With supportive government policies and growing public awareness of the benefits of solar energy, Taiwan is well-positioned to achieve its renewable energy targets and emerge as a leader in the global clean energy transition.

Authors: *ShuYu Yang*, PV Industrial Analyst, ITRI; *Daniel Lee*, Taiwan Photovoltaic Industry Association (TPVIA).



Floating PV on the Kaohsiung Reservoir in Taiwan.

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13. Saudi Arabia

Saudi Arabia has achieved tremendous progress in solar PV development through 2023, motivated by its objective to diversify its energy mix.

Overview of PV development up to 2023

In 2023, Saudi Arabia's newly installed solar capacity reached approximately 1.9 GW. Ongoing projects under construction surpassed 8 GW by the end of 2023, with an additional 13 GW in the development pipeline. Notable initiatives include a 52 MW rooftop PV project portfolio. In terms of cumulative capacity, the solar fleet reached 2.2 GW by the end of 2023, according to IRENA (see Fig. GW 13.1).

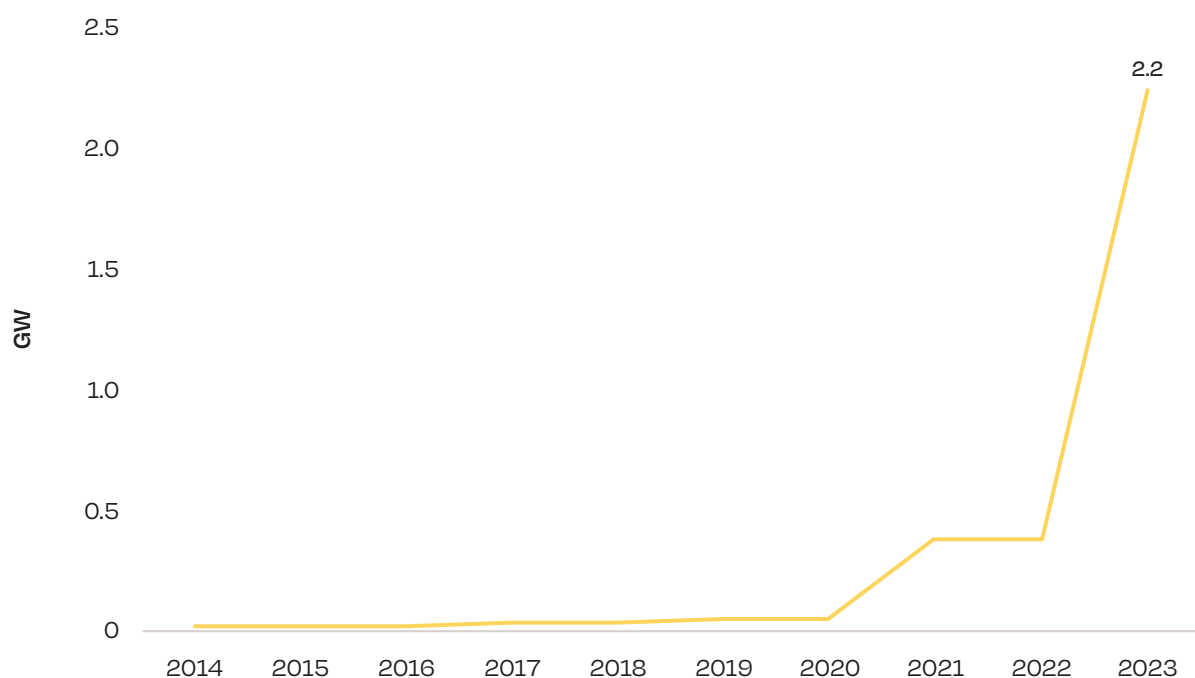
Between 2015 and 2023, Saudi Arabia's renewable power capacity surged from 0.02 GW to 3 GW, with solar PV accounting for 82.6% of the total in 2023. Under the current market projections, it is accelerating the growth of its renewable energy capacity and is

expected to increase at a compound annual growth rate (CAGR) of 40.1% from 2023 to 2030 to reach 31.5 GW, according to GlobalData. Since 2022, the country has added 2.1 GW of renewable power capacity, marking a 300% increase from the cumulative renewable capacity addition of 700 MW during 2012-22. Saudi Arabia's abundant solar radiation and strategic focus on solar and wind power underpin its remarkable progress in the solar PV sector.

National Solar PV and RE Targets

Saudi Arabia has set ambitious national targets for solar PV development from 2024 to 2028 under its National Renewable Energy Program (NREP). The Renewable Energy Project Development Office (REPDO) aims to increase local content requirements (percentage of materials, labour, services, and other inputs sourced from within the country) for PV projects from 17-19% to 33-35% by 2024-25 and 40-45% by 2028 and beyond. Saudi Arabia targets 58.7 GW of renewable energy generation for 2030, with a

FIGURE GW 13.1 SAUDI ARABIA CUMULATIVE SOLAR PV CAPACITY 2014-2023, BY MESIA



SOURCE: IRENA (2024).

© SOLARPPOWER EUROPE 2024

staggering 40 GW coming from solar PV, reflecting its strategy to develop a local renewable manufacturing ecosystem. These ambitious targets, coupled with Saudi Arabia's abundant solar radiation, position solar PV as a primary focus in its drive to diversify energy sources and ensure long-term supply security.

Drivers for Solar Growth

Saudi Arabia's solar growth from 2024 to 2028 is propelled by abundant solar resources, with irradiation levels ranging between 2,000 and 2,500 kWh/m²/year, making solar energy economically attractive. Supportive government policies like the NREP aim to decrease dependency on fossil fuels, with targets set to increase renewable energy's share in the total energy mix. Declining solar module prices and versatile applications offer lucrative investment opportunities, supported by net-metering regulations incentives for private sector investment. Various procurement mechanisms, including competitive auctions and state-owned projects, are accelerating solar expansion.

Utility-scale vs. distributed and rooftop developments

In 2023, Saudi Arabia made significant strides in utility-scale and distributed rooftop solar developments, albeit

with a greater focus on utility-scale projects. On the utility-scale front, the country updated its Renewable Energy and Energy Efficiency Development Plan in Q4 2019, aiming to install 5.6 GW of utility-scale solar PV by 2030 to diversify the electricity generation mix. Regarding distributed and rooftop developments, the Saudi NREP targeted 9.5 GW of installed renewable energy capacity by 2023. This target, however, has been largely underachieved, with residential rooftop developments not being largely economically viable for households in Riyadh based on the current electricity tariffs. It is estimated that the aggregate residential solar power capacity stands at around 400 MW in the capital, Riyadh. While utility-scale developments took precedence, Saudi Arabia made progress on both fronts in 2023, aligning with its strategy to maximise economic benefits from the remaining hydrocarbon reserves while expanding renewable power capacity domestically.

Challenges for the market

The solar PV market in Saudi Arabia faces several critical challenges to its sustainable growth that must be addressed. Firstly, grid integration and stability become increasingly difficult as the share of intermittent solar PV increases in the energy mix. Secondly, large-scale solar projects require significant land resources, which can lead



© Sakaka Solar Energy Co.

4 GW-scale markets / continued

to land use conflicts. Additionally, solar panel production and maintenance can consume substantial amounts of water, a scarce resource in the region. Moreover, despite Saudi Arabia's solid industrial infrastructure, solar PV manufacturing is still in its infancy. As there is no local solar value chain, the country is largely reliant on the import of modules, inverters and BOS components from abroad. Finally, the market is experiencing a transformative impact from ongoing technological advancements and the continual reduction in the cost of solar technology, which needs to be effectively managed. Addressing these challenges through innovative solutions and strategic planning is crucial for the sustainable growth of the solar PV market in Saudi Arabia.

Outlook for the years 2024-2028

The outlook for the PV market in Saudi Arabia from 2024 to 2028 appears highly promising. The Saudi

Arabia renewable energy market size reached approximately 1.5 GW in 2023. Looking forward, IMARC Group expects the market to reach 29.1 GW by 2032, exhibiting a CAGR of 38.97% during 2024-2032.

This growth will be fuelled by several factors, including increasing government initiatives to support solar energy adoption through monetary incentives, subsidies, and promoting solar rooftops. Moreover, Saudi Arabia's latest solar tender saw significant increases in bid prices due to rising PV equipment costs, with the lowest bid at 0.05563 SAR/MWh (0.01366 EUR/MWh), a 43% increase from the previous record. Despite the recent challenges, Saudi Arabia's NREP and its 40 GW solar PV capacity target for 2030 will continue to create significant opportunities for the market.

Authors: *Tannishtha Das* and *Hinde Liepmannsohn*, Middle East Solar Industry Association (MESIA).

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14. United Kingdom

Overview of PV developments

2023 was an extraordinary year for solar energy in the UK. According to official figures, 15.7 GW of cumulative solar capacity was in place at the end of the year. However, the industry considers this to be a significant underestimate, with Solar Media Ltd suggesting that the figure was 17.6 GW, up 1.7 GW on 2022. The difference is largely due to underestimating the burgeoning commercial-scale rooftop market, which is hardly accounted for in official statistics.

The combination of high gas prices, the need for energy security and the response to climate change saw 190,000 smaller-scale installations connect to

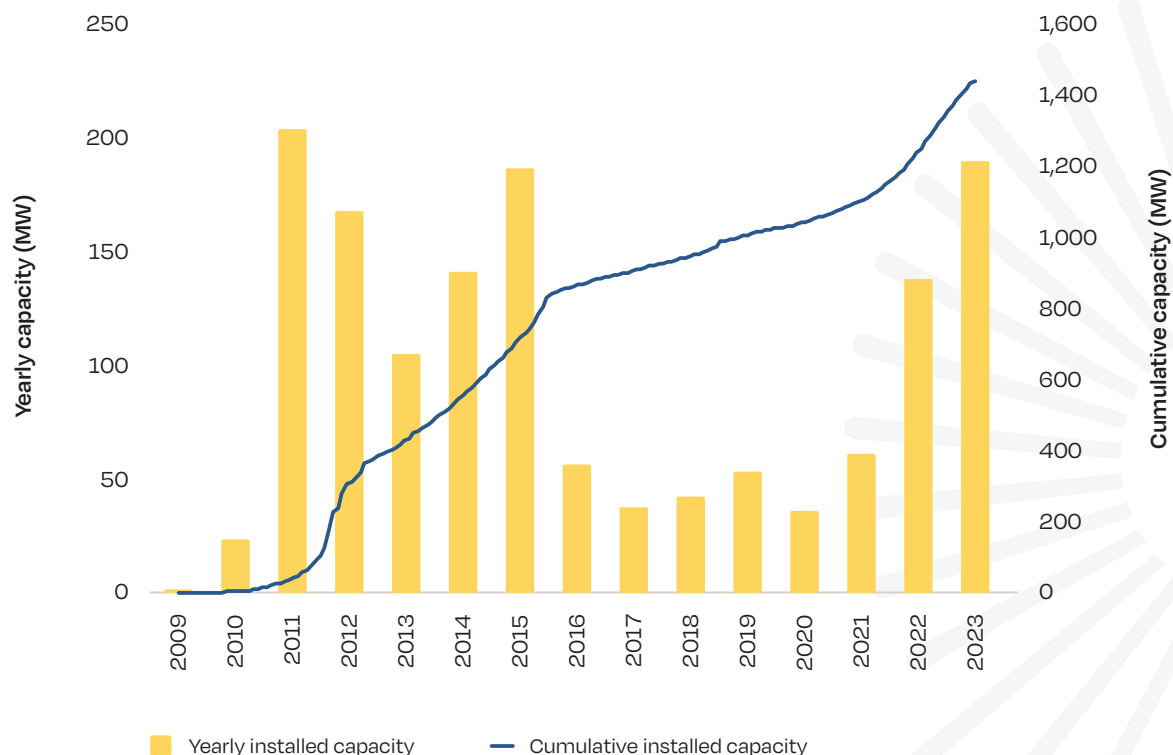
the grid in 2023, the greatest since 2011 (see Fig. GW 14.1). Around 13,000 sub-50 kW installations are being made each month now, according to data from the Microgeneration Certification Scheme (MCS). 2023 set a post-subsidy record for smaller-scale PV installations below 50 kW.

At utility scale, EDF's 500 MW Longfield project, which once built will be the largest solar farm in the country, secured consent from the government in June 2023.

National Targets for solar PV

The government targets 70 GW of solar capacity for 2035. How this will be achieved is the subject of the joint Government and industry Solar Taskforce, which was established in May 2023 was co-chaired by Solar Energy UK chief executive Chris Hewett and two ministers, Graham Stuart MP and latterly Andrew Bowie MP.

FIGURE GW 14.1 YEARLY INSTALLATIONS BELOW 50 kW 2009-2023, BY MCS



SOURCE: MCS.

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4 GW-scale markets / continued

Its five key themes will be grid access, communications, rooftop, skills development and the supply chain. Both Ministers have given the solar and energy storage sector much support – a marked change in attitude from the brief premiership of Liz Truss. At the time of writing, however, the Solar Roadmap remains to be published, as the general election has led to its postponement. It will presumably be revised under the new government.

In contrast with the current government target, the opposition Labour Party has targeted 50 GW of solar capacity for 2030, which is slightly more ambitious in terms of growth than the 2035 goal and is significantly above the current market trajectory. Solar Energy UK's manifesto backs the same 50 GW goal, plus 30 GW of energy storage.

Drivers for solar growth

2023 saw significant policy announcements, including the designation of solar farms of 50 MW or more as 'critical national priority' infrastructure. Following SEUK lobbying, the government also rectified an anomaly concerning residential battery energy storage systems: tax-free when installed alongside a new PV system, VAT still applied when retrofitted to an existing one.

In November, 'full expensing' was made permanent, allowing companies to deduct the cost of PV and energy storage from their profits before paying corporation tax.

Perhaps most critically, the government has acknowledged the paramount importance of resolving long delays in connecting renewable energy projects to the grid. In November, Chancellor of the Exchequer Jeremy Hunt agreed to implement almost every measure outlined in the independent Winner Review of the UK's power networks, intended to secure billions of pounds of investment and deliver clean, green and cheap energy. The chancellor also announced an exemption from the Electricity Generator Levy, introduced in the wake of the energy price crisis, for new renewable energy projects.

Challenges for the market

But there are headwinds, with some Conservative MPs opposed to the use of agricultural land and some parts of the media taking an increasingly strident tone against solar farms. Approvals for solar farms over 50 MW are also being delayed.



9 MW Emberton solar farm, Buckinghamshire, United Kingdom.

© Hollie Blaydes

To address local community concerns about large-scale solar's biodiversity impacts, in the spring of 2023, Solar Energy UK released its first biodiversity monitoring report, followed by a second in March 2024. Solar Habitat 2024 demonstrates that solar farms can be havens for wildlife, with brown hares and skylarks – considered vulnerable species in the UK – among the most common species.

In the area of supply chain sustainability and transparency, Solar Energy UK is proud to collaborate with SolarPower Europe to implement the Solar Stewardship Initiative, intended to ensure high standards of supply chain transparency across Europe. In parallel, the promotion of high installation standards and consumer protection has been enhanced through a partnership with MCS, while joint

activities with the insurance sector improved the RC62 document, so that the commercial PV rooftop technology and practice changes are properly reflected in the requirements placed on commercial property with solar.

Outlook for 2024

Looking forward, the pipeline of new solar farms and battery storage projects has grown exponentially. By the end of the year, 13 GW of solar farm capacity had received planning permission, with 1.7 GW under construction. Over 23 GW of battery energy storage systems were either in construction or had been approved. We expect that 2024 should be a record year for both sectors.

Author: *Gareth Simkins*, Senior Communications Adviser, Solar Energy UK.



Residential solar, South Croydo, London, UK.

© Gareth Simkins

15. Chile

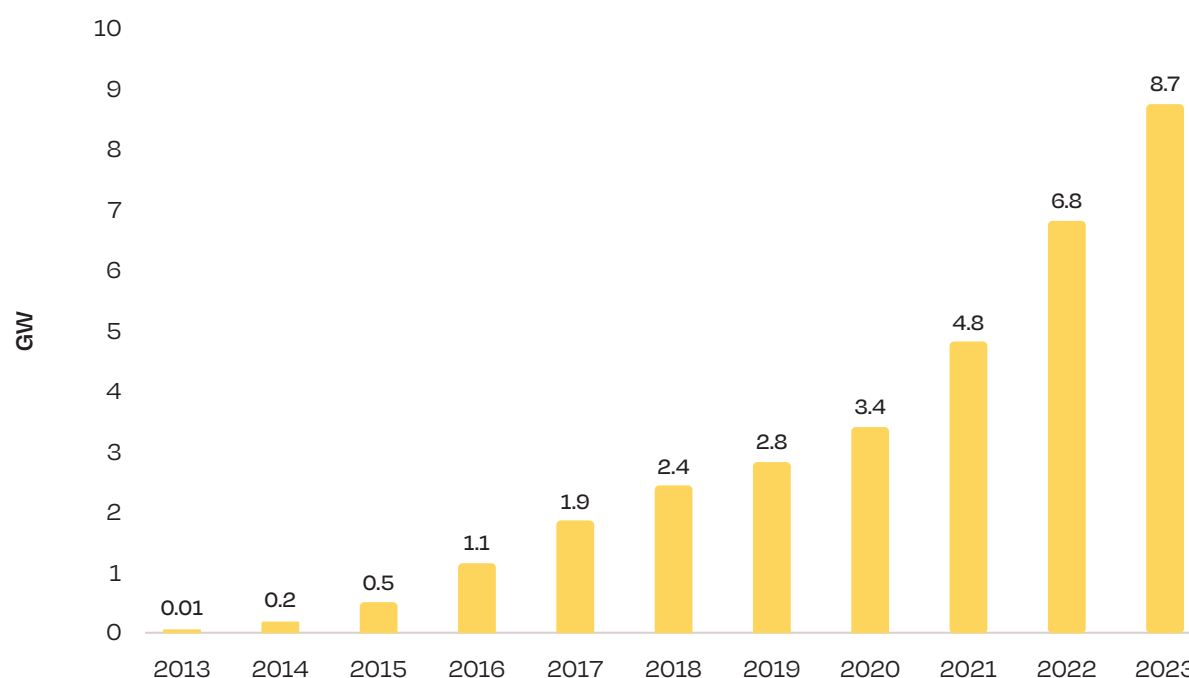
Chile commissioned its first 1 MW on-grid PV power plant in 2013. By the end of 2023, the installed capacity of PV technology has risen dramatically to reach 9 GW, out of the country's 33 GW total electrical capacity. Solar energy now accounts for 27% of the total installed capacity, up from 17% in 2022, and stands as the first source of electricity generation in Chile, producing 17.7 TWh. In 2023, almost 2 GW of solar power capacity was installed, with an additional 4.8 GW currently under construction. Figure GW 15.1 shows the evolution of the total installed PV capacity over the past decade.

In terms of distributed generation, Chilean regulation distinguishes between two categories:

- Medium Size Distributed Generation (MSDG): facilities between 300 kW and 9 MW connected to medium voltage distribution feeders
- Rooftop: facilities below 300 kW of installed capacity

In 2023, medium-scale distributed generation (MSDG) constituted 2.9 GW of the total operating capacity, whereas rooftop installations accounted for 220 MW.

FIGURE GW 15.1 CUMULATIVE SOLAR CAPACITY 2013-2023, BY ACESOL



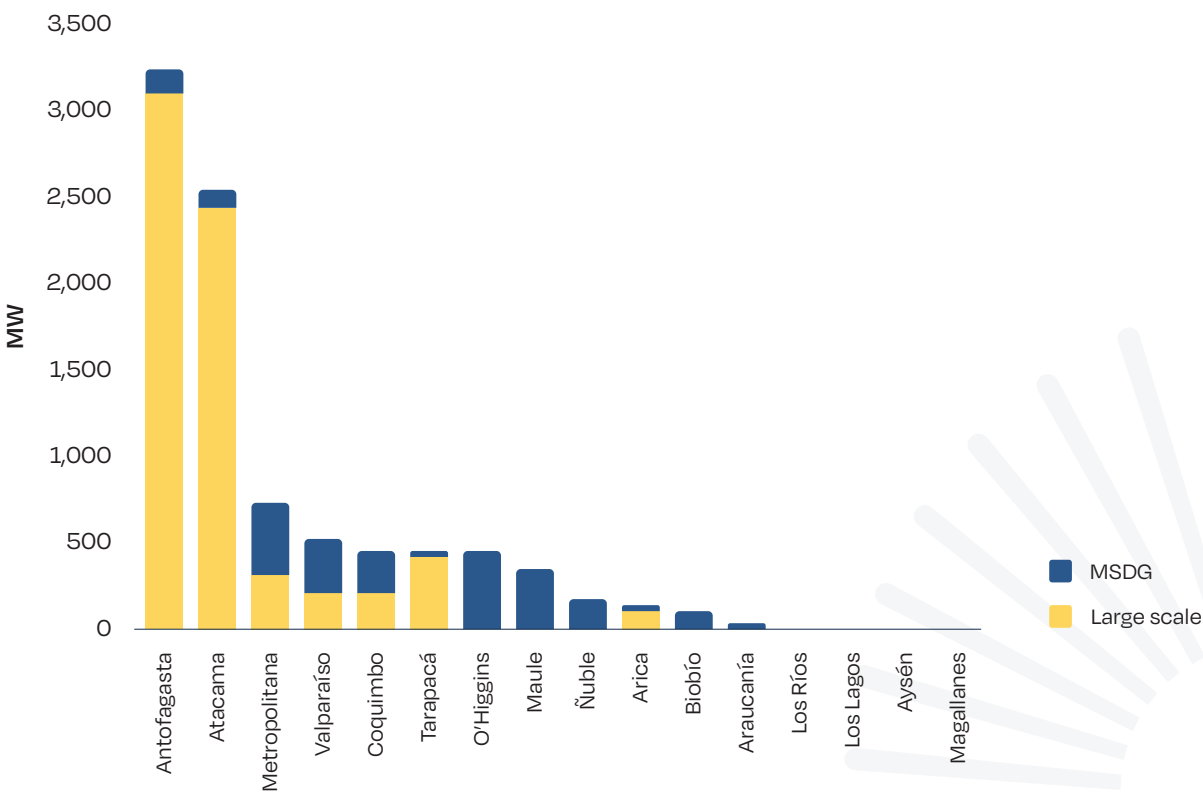
NOTE: Totals include large-scale, net billing and MSDG capacity.
SOURCE: ACESOL.

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The installed solar capacity by region, arranged from north to south (left to right), is displayed in Figure GW 15.2. Most utility-scale projects are located in the sunnier northern part of the country. In contrast, medium-scale distributed generation (MSDG) facilities are predominantly located near the major cities in the central region. This distribution has led to worsening congestion on transmission lines, a problem that has seen a significant increase in 2023.

To solve the congestion problem, at least partially, the Ministry of Energy presented the Energy Transition Bill to Congress in 2023. This bill aims to promote installing battery storage systems and introduce significant modifications to the transmission expansion process. This bill is expected to be approved during the first term of 2024.

FIGURE GW 15.2 GEOGRAPHIC DISTRIBUTION OF SOLAR INSTALLED CAPACITY IN CHILE, BY ACESOL



SOURCE: ACESOL.

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4 GW-scale markets / continued

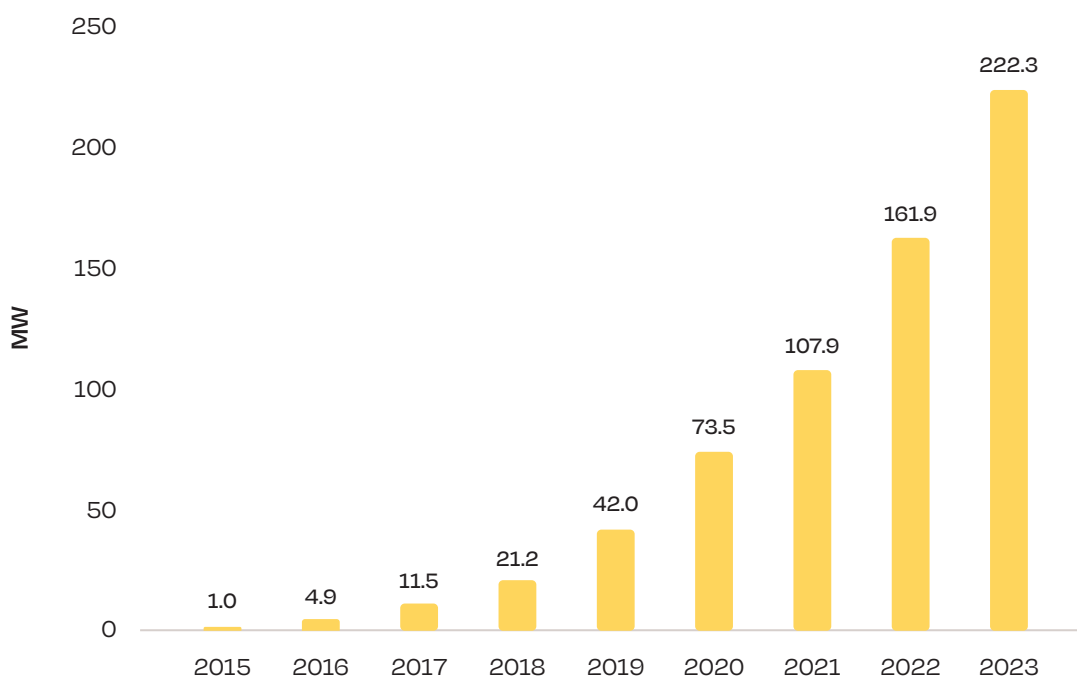
In 2013, Chile implemented a net-billing scheme to promote rooftop installations. Figure GW 15.3 shows the evolution of rooftop installations over the last decade. Although the number of installations has increased, the 220 MW of cumulative installed capacity remains a limited volume compared to Chile's potential. This is primarily due to ongoing regulatory barriers within the electric distribution segment.

Chile is dedicated to diminishing its reliance on fossil fuels and swiftly transitioning to a fully renewable energy portfolio, aiming for 100% renewable sources as fast as possible. Solar energy is anticipated to

maintain its critical role throughout this transition. However, achieving this objective entails addressing three prominent challenges. Firstly, there's a need for a significant revision of the wholesale energy market to encourage the efficient adoption of renewable energy technologies and energy storage solutions. Secondly, a large expansion of the transmission infrastructure is necessary to accommodate new solar capacity. Lastly, a substantial overhaul of the distribution sector is imperative to unleash the potential for decentralised power generation and rooftop installations across the country.

Author: David Rau, Vice President of the Asociación Chilena de Energía Solar (ACESOL)

FIGURE GW 15.3 CUMULATIVE ROOFTOP CAPACITY 2015-2023, BY ACESOL



SOURCE: ACESOL.

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16. Mexico

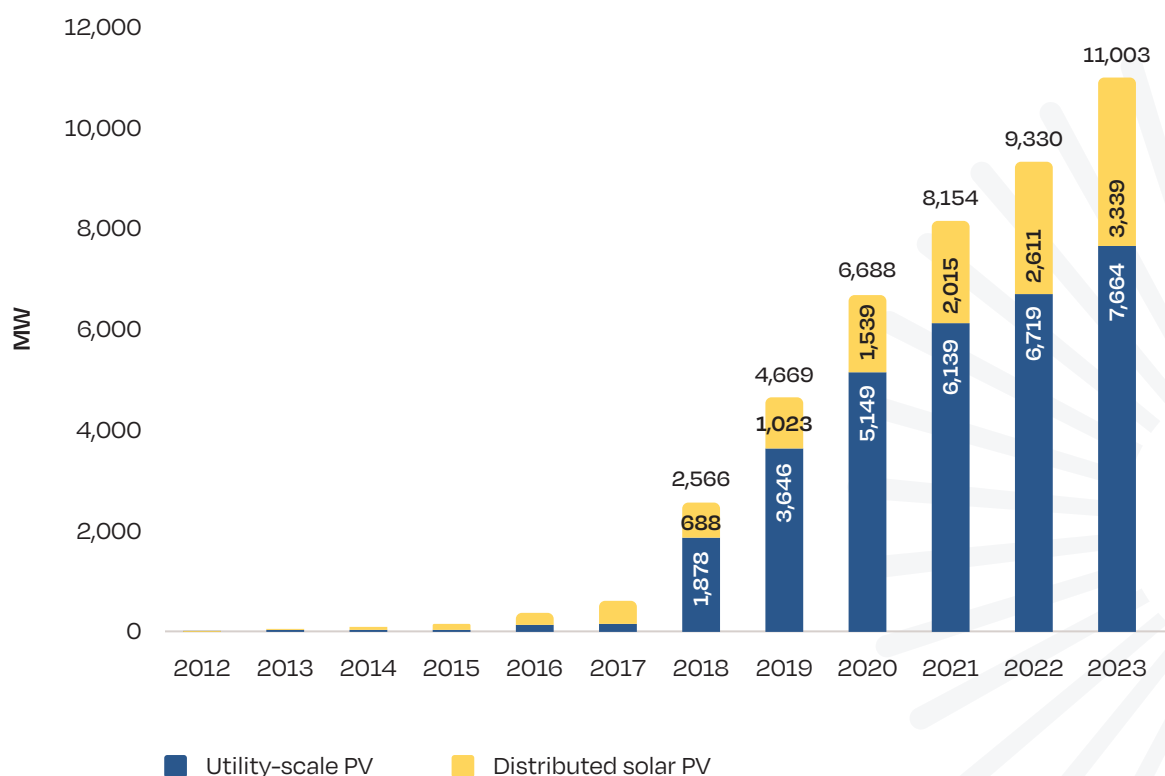
Overview of PV developments

In 2023, the Mexican solar PV market installed 1.67 GW, a 42% increase compared to the 1.18 GW added in 2022, 14% more than the 1.47 GW added in 2021, and 17% less than the 2.02 GW installed in 2020. Total capacity operating in the country reached 11 GW, which constitutes 18% growth from 2022. This is about 18 times the installed capacity in 2017 (0.62 GW). Out of the total installed capacity, utility-scale PV represents 70% (7.66 GW) and distributed solar PV 30% (3.34 GW).

Utility-scale PV contributed 945 MW of added capacity in 2023, a significant increase from 2022 (580 MW), although less than in the previous three years (0.99 GW in 2021, 1.5 GW in 2020, 1.77 GW in 2019). In contrast, distributed solar PV reached a new record of 728 MW, 22% more than in 2022 (596 MW), 53% more than 2021 (476 MW), and higher than 2020 and 2019 (516 MW and 335 MW respectively).

Looking at the whole electricity system, utility-scale PV capacity represents 8.5% of total installed capacity (90.1 GW), while 63% of installations belong to fossil-fuel based technologies. In terms of power generation, fossil-fuel based technologies provided 77%, an increase compared to 2022 (74%). Meanwhile, solar PV showed a slight increase in 2023, contributing to 5.3% of power generation, compared to 4.9% in 2022 and 5.3% in 2021.

FIGURE GW 16.1 MEXICO TOTAL SOLAR PV CAPACITY 2012-2023, BY ASOLMEX



SOURCE: ASOLMEX.

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4 GW-scale markets / continued

Renewable energy and solar PV targets

Before 2013, electricity supply in Mexico was carried out under a vertically integrated monopoly scheme, operated and owned by the State through the Federal Electricity Commission (CFE). After the energy reform of 2013, which included the unbundling and restructuring of CFE, a competitive electricity market was introduced for generation and supply as well as open access to transmission and distribution grids.

In 2015, the Energy Transition Act (LTE) established clean energy goals in electricity generation: 25% for 2018, 30% for 2021, and 35% for 2024. To promote investment in renewable energy, two main mechanisms were implemented: Clean Energy Certificates (CEC) and long-term energy auctions held annually by the Independent System Operator (ISO), which secured nearly 4.67 GW of utility-scale PV capacity between 2015 and 2017, with almost 90% of it starting operation before the end of 2023.

Since December 2018, energy policy changes aimed at favouring CFE have resulted in regulatory paralysis

and hindered new private investment in renewable energy. These changes included cancelling long-term energy auctions and stalling the progress of the CEC mechanism. Consequently, Mexico failed to achieve its clean energy target for 2021 and is not expected to reach the 2024 target either. By 2024, the share of clean energy could drop below 23%, far from the 35% established in the LTE.

At COP27 in November 2022, Mexico announced an updated nationally determined contribution (NDC) target, aiming for a 35% reduction in greenhouse gas emissions by 2030, up from the previous goal of 22%. Studies indicate that Mexico would need to add at least 30 GW of new renewable energy capacity to achieve this updated NDC, comprising 20 GW of utility-scale PV and 10 GW of wind.

One year later in November 2023, at COP28, Mexico joined the Global Renewables and Energy Efficiency Pledge, which aims to triple the world's installed renewable energy generation capacity to at least 11,000 GW by 2030.



60.8 MW, Aguascalientes, Mexico.

© Canadian Solar

Challenges

Despite multiple announcements by Mexico regarding its commitment to the global renewable agenda, the path Mexico will take towards 2030 remains uncertain. While it is necessary to significantly increase investments in new renewable generation capacity, the urgency for the expansion and modernisation of transmission networks is even greater. In 2017, network failures or lack of transmission infrastructure explained 47% of all Energy Emergency Alerts. By 2022, these same conditions accounted for 97% of those alerts.

Decentralised energy solutions, such as distributed solar PV, self-supply projects, microgrids, and battery energy storage, are expected to become the primary alternative to increase the supply of clean energy since expanding transmission capacity will require several years.

A new regulation regarding energy storage is currently being discussed. This proposal primarily focuses on empowering the ISO to mandate the integration of

energy storage systems for new projects to address the variability of renewable energy. However, the concept of variability is not clearly defined, and thus the requirements will depend on the ISO's discretion. Considering the limited operational flexibility of Mexico's electrical system, it appears that the regulation should encompass all the services that energy storage could provide and establish remuneration mechanisms to incentivise investments in these technologies.

Outlook

Mexico not only needs more clean energy to meet the expected growth in demand but also to seize the historic opportunity presented by the process of supply chain relocation, known as nearshoring, in the coming years. In order to foster new investment in the sector, long-term stability of regulations, transparency in permitting process management and technical operation, as well as continuous dialogue between authorities and sector participants, are essential.

Author: Nelson R. Delgado Contreras, Director General, Mexican Association of Solar Energy (Asolmex).



820 MW Villanueva solar park, Viesca, Coahuila, Mexico.

© Kino Energía

17. Switzerland

PV development up to 2023

In 2023, solar PV experienced its fourth consecutive year of market growth exceeding 40%. According to Swissolar's estimates, new installations in 2023 exceeded 1.5 GW and solar power production was around 5 TWh or 9% of total electricity consumption. The main trigger for this boom was the Energy Act, which came into force in 2018 and created a reliable framework for the expansion of renewable energies after years of uncertainty.

National targets

Switzerland does not yet have any binding targets for the expansion of solar PV. The current Energy Act sets a target of 11.4 TWh of annual electricity production from renewable energies (excluding hydropower) for the year 2035, which corresponds to around 20% of the current consumption. To date, hydropower has played a decisive role in electricity supply, annually producing around 40 TWh. This contribution is currently increasing significantly due to the rapid melting of glaciers.

In September 2023, the parliament passed a revision of the Energy Act, setting a much more ambitious target. By 2035, the goal is to generate 35 TWh of electricity from renewable sources excluding hydropower, almost tripling the existing target. Of this, approximately 30 TWh is expected to be generated from solar power. Additionally, a target of 45 TWh of non-hydro RES has been set for 2050. This increase in renewables aims to compensate for the loss of nuclear power, estimated at around 23 TWh annually from the four reactors still in operation, and to meet the rising demand for electricity due to decarbonisation efforts. There will be a public vote on this law via referendum on June 9, 2024.

Drivers for solar growth

Since 2018, a nationwide investment subsidy has been in place, covering approximately 20% of the costs for PV installations, contingent on their size. The subsidy is nearly double for systems without self-consumption. The reason for this is that since a substantial share of self-consumption was the only way to economically operate a PV installation, roofs on buildings with low or no power consumption were not used to deploy PV. With the new subsidy scheme, these buildings are now being used. For systems of



Façade PV in the Swiss alps, Switzerland.

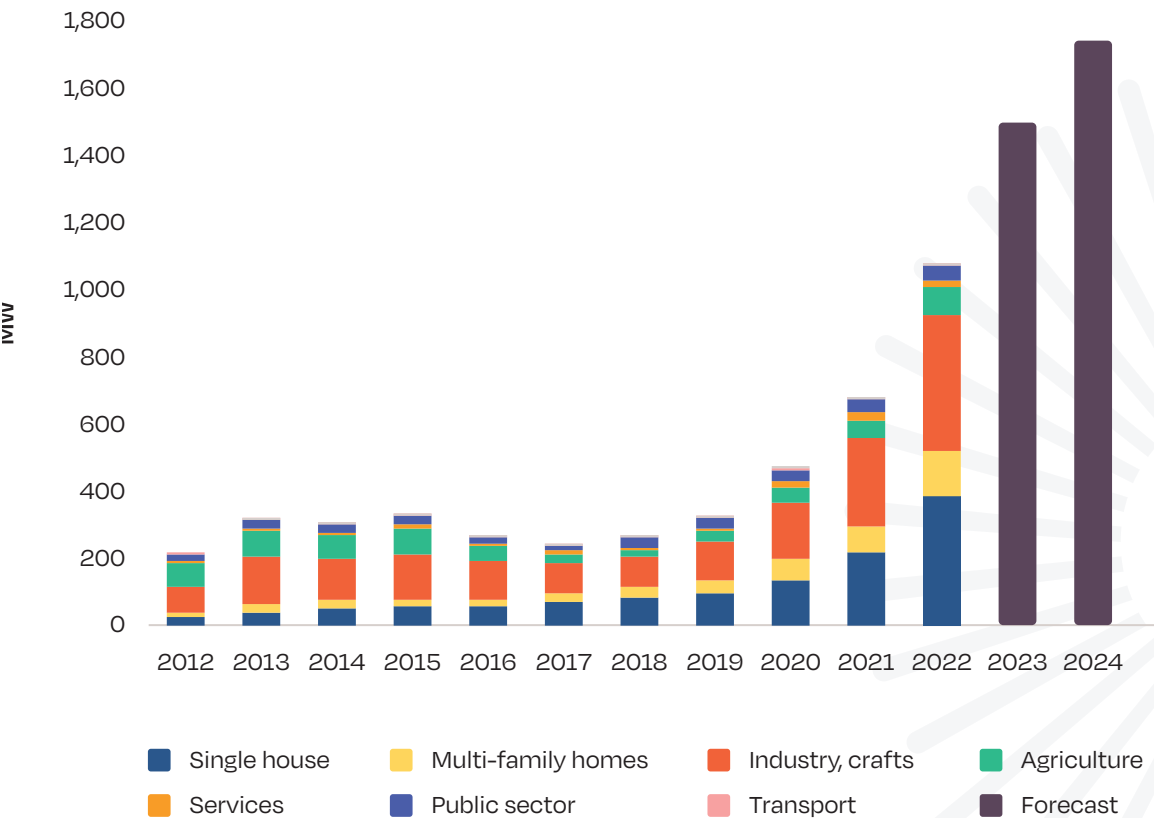
© 3S Swiss Solar Systems

150 kW or larger, an auction system is in place. Additionally, in 2018, self-consumption communities were introduced as a mechanism to facilitate the sale of solar power to neighbours through private lines. This initiative fosters demand in emerging residential areas. Like in many other European countries, the conflict in Ukraine also resulted in a significant hike in electricity prices, sparking a surge in solar installations, especially on industrial and commercial rooftops. Often, these systems are operated through on-site PPAs, ensuring a substantial portion of self-consumption. By contrast, off-site PPAs have seen limited adoption in Switzerland thus far. Lastly, an important catalyst to solar growth is the requirement for new buildings in most Swiss cantons to self-generate the electricity they consume.

Utility-scale vs. distributed and rooftop developments

In Switzerland, solar systems have predominantly been installed on buildings, with approximately 14% of the capacity integrated into the buildings (BIPV). Residential and commercial sectors each contribute roughly half of the installed capacity. An urgent parliamentary resolution adopted in 2022 opened the door to constructing large-scale solar plants in the Alps, potentially increasing winter production. However, significant challenges exist in implementing these projects, and it remains uncertain how many plants will be constructed by 2025 – which is the legal deadline for those projects.

FIGURE GW 17.1 ANNUAL PV MARKET SEGMENTATION IN SWITZERLAND, 2013-2022, BY SWISSOLAR



SOURCE: Swissolar.

© SOLARPOWER EUROPE 2024

4 GW-scale markets / continued

Challenges for the market

The shortage of skilled labour in the Swiss construction industry presents a significant challenge for the solar sector. Despite these obstacles, the industry has managed to navigate through substantial market growth. However, meeting the demands of this growth requires a considerable increase in specialised workforce. To address this, Swissolar will introduce 2-year and 3-year apprenticeship programs for solar installers starting from August 2024. Additionally, the association offers various further training courses for career changers.

The authorisation process for solar installations remains restrictive in many cases, particularly in protected town centres. Agri-PV projects are only permitted if they enhance agricultural crops, and planned large-scale Alpine plants have faced opposition from environmental organisations. Those large-scale plants usually also require power lines to

be reinforced, which is also associated with lengthy authorisation procedures.

There is a growing need for clarification with building insurers. Due to recent extreme hail events heightened requirements for module hail resistance are being discussed. The rise in facade systems, which can significantly contribute to winter power supply, has brought fire protection concerns to the forefront. A joint document by Swissolar and insurance companies offers initial clarification, but standardised fire tests will be necessary moving forward.

Currently, there is no standardised regulation for grid-fed electricity prices. Purchase prices vary widely among the approximately 630 distribution grid operators. With the new Electricity Act, these prices will be tied to quarterly average electricity market prices, posing uncertainties for PV system operators due to anticipated low summer electricity prices.



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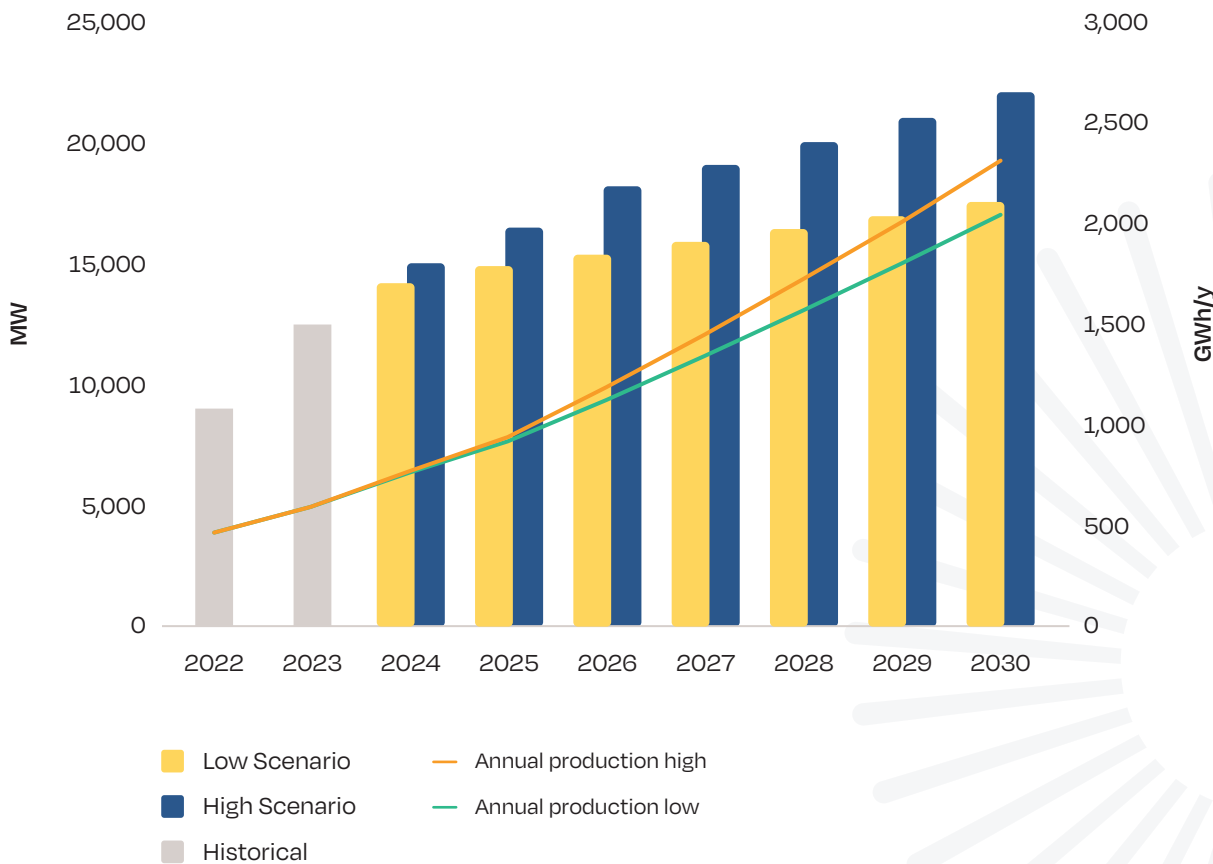
Outlook for the years 2024-2028

In 2024, solar power is projected to meet over 10% of Switzerland's electricity needs. The future trajectory hinges significantly on the outcome of the referendum on the new renewables target on 9th June 2024. A favourable vote could pave the way for an annual expansion of over 2 GW, facilitating the attainment of the targets outlined in the new Energy Act. A key catalyst is the proposed regulation to incentivise self-consumption, particularly through 'local electricity communities' modelled after Austria. These communities can leverage the public electricity

grid at a local level at a reduced tariff, promoting the alignment of production and solar power generation and minimising unnecessary grid expansions. Electromobility plays a pivotal role in this context, serving as an intermediate storage solution via bidirectional charging. Additionally, heat pumps, currently the predominant heating system in Switzerland for new constructions and heating replacements, further bolster the utilisation of photovoltaics.

Author: David Stickelberger, Deputy Managing Director, Swissolar.

FIGURE GW 17.2 ANNUAL SOLAR PV MARKET IN SWITZERLAND, BY SWISSOLAR



SOURCE: Swissolar.

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18. Pakistan

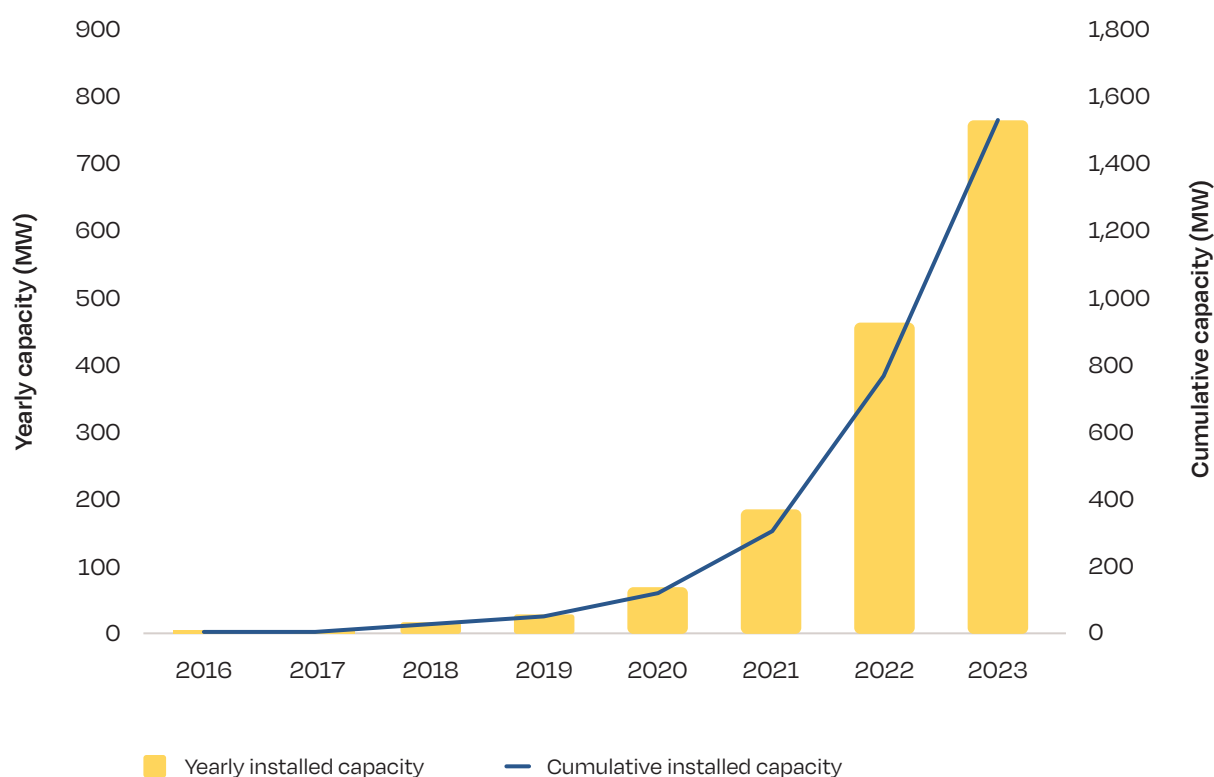
Overview of PV Developments in Pakistan

Pakistan's journey towards solar energy development has been marked by remarkable progress, positioning the country as a significant player in the global renewable energy sector, demonstrating substantial growth since 2021 and highlighting the nation's commitment to clean energy sources. According to industry experts, in 2022, 2.8 GW of PV modules were imported into Pakistan. In 2023, about 5 GW of modules were imported, in spite of import controls. For 2024, the prediction is for up to 12 GW of PV modules import.

Pakistan has witnessed a remarkable expansion in distributed grid-connected solar capacity. By the end of 2023, an additional 0.76 GW of distributed on-grid solar capacity was added under the national net-metering scheme, almost doubling the cumulative distributed installed capacity, and reinforcing the country's path towards large-scale solar adoption (see Fig. GW 18.1).

Looking ahead, Pakistan aims to integrate another 14 GW of solar and wind capacity into the grid over the next decade, according to the Alternative Energy Development Board (AEDB). These ambitious targets underscore Pakistan's determination to leverage its abundant solar potential to meet growing energy demands sustainably.

FIGURE GW 18.1 YEARLY AND CUMULATIVE INSTALLED CAPACITY UNDER NET-METERING SCHEME, BY PSA



SOURCE: PPIB.

© SOLARPOWER EUROPE 2024

While utility-scale installations continue to dominate Pakistan's solar landscape, there has been a notable shift towards decentralised rooftop solar systems. The growing adoption of rooftop solar, particularly in urban areas, reflects consumers' desire for energy independence and resilience, as well as efforts to mitigate rising electricity prices.

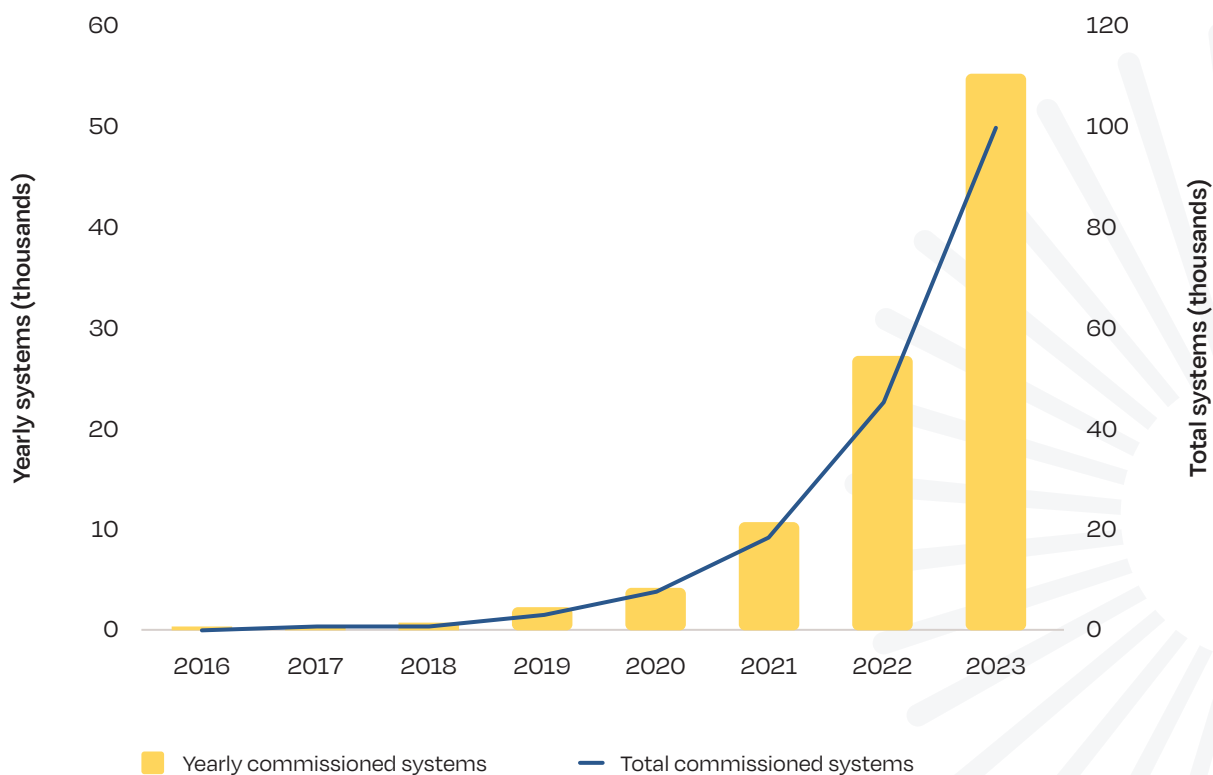
Today, renewable energy sources currently contribute a mere 5% of Pakistan's electricity generation mix, with off-grid renewables playing a negligible role in overall energy consumption. Despite this, local discussions often minimise Pakistan's obligation to shift towards renewables, citing the country's relatively low greenhouse gas emissions. However, such viewpoints overlook the critical imperative for Pakistan to embrace renewable energy sources.

Drivers to solar growth

With approximately 25% of the population still lacking access to the grid, Pakistan possesses vast untapped solar potential. The country, with a young population exceeding 230 million, recognises solar energy as a crucial solution to address its energy needs. The active involvement of the vibrant private sector, particularly in rural areas where grid access is limited, is evident in deploying solar solutions. Initiatives targeting solarisation of tube wells and rural electrification projects are gaining traction, unlocking opportunities for decentralised energy generation and distribution.

To catalyse solar growth, the Pakistani government has implemented various policy measures and initiatives. The establishment of regulatory bodies like the AEDB and the National Energy Efficiency and Conservation Authority

FIGURE GW 18.2 YEARLY AND TOTAL COMMISSIONED SYSTEMS UNDER NET-METERING SCHEME, BY PSA



SOURCE: Pakistan Solar Association (PSA).

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4 GW-scale markets / continued

(NEECA) underscores the government's commitment to promoting renewable energy. The net-metering scheme introduced in 2015 and other incentive programmes, such as subsidised financing through a Long-Term Financing Facility by the State Bank of Pakistan, and Government Building Solarisation schemes launched across various government departments, have encouraged solar adoption among households and businesses. Additionally, the development of large-scale solar projects, exemplified by the 100 MW Quaid-e-Azam Solar Park in Bahawalpur, highlights the government's efforts to facilitate utility-scale solar deployment.

Solar energy plays a pivotal role in addressing Pakistan's energy challenges, including frequent blackouts and heavy reliance on fossil fuels. By diversifying the energy mix, solar contributes to enhanced energy security and

resilience. Moreover, solar initiatives support ongoing electrification efforts, extending access to reliable electricity to underserved communities. Concerns about climate change further underscore the urgency of transitioning to renewable energy, with solar serving as a key tool in mitigating greenhouse gas emissions.

Amid increasing demand due to rising power bills, the prices of PV modules in Pakistan have sharply declined in recent times, increasing solar cost-competitiveness and affordability. According to distributors, the already record-low prices have further decreased by 30% since Q4 2023, with modules of different brands available at under 13 EUR cent per Watt, a significant decline compared to above 26 EUR cent per Watt in 2022. This trend is expected to continue, facilitating wider adoption of solar energy in Pakistan.



500 kW, Chand Bagh School, Punjab, Pakistan.

© Hadron Solar

Challenges to Solar and Renewable growth

Pakistan faces several obstacles hindering widespread adoption of renewables:

- Political and economic instability dissuades foreign investment, exacerbating financial constraints and making renewable energy projects scarce and costly.
- Lack of regulatory clarity regarding the integration of renewables into the evolving energy market plans by the National Electric Power Regulatory Authority (NEPRA).
- Existing long-term energy procurement contracts with conventional non-RE generation undermine incentives for transitioning to renewables due to the structure of the contracts with capacity charges upon non-usage.
- Fiscal constraints due to subsidies on non-renewable sources limit the government's capacity to invest in renewables.

To overcome these challenges, fiscal policies and incentives play a pivotal role. Broader stakeholder engagement is needed to identify specific actions that can facilitate the adoption of renewables. Measures such as tax incentives, subsidies for renewable energy projects, and regulatory reforms can incentivise

investment in clean energy infrastructure and drive sustainable energy transitions. Pakistan must prioritise renewable energy development as a cornerstone of its sustainable development agenda. By addressing barriers to adoption and leveraging fiscal policies effectively, Pakistan can accelerate its transition towards a greener, more resilient energy future.

The government's response to the growing demand for solar energy has been mixed. While there are plans to decrease the rate of solar power exported to the national grid, reports of imposing a fixed tax on solar electricity have been dismissed as baseless. However, the prevailing system of solar net metering is being scrutinised, with concerns about its impact on the sector's investment dynamics. Overall, a comprehensive review process involving input from market experts and stakeholders is essential to ensure sustainable growth in the solar energy sector.

In conclusion, Pakistan's solar energy landscape presents immense opportunities for growth and development. With the right policies, investments, and regulatory frameworks in place, Pakistan can accelerate its transition towards a sustainable energy future while reaping the economic and environmental benefits of solar energy adoption.

Author: *Waqas Moosa*, Pakistan Solar Association (PSA).



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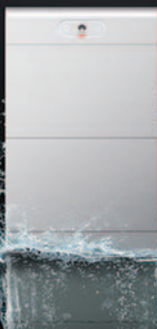


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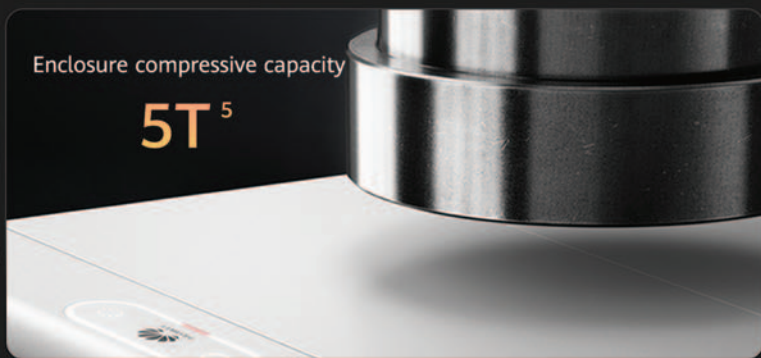
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⁵ Severe impact affects the stability of the ESS. Avoid impact whenever possible.

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